

PREFACE

This book contains the Daily Practice Problems (DPPs) designed for the aspirants JEE(Main+Advanced). It is a collection of problems (Physics, Chemistry & Mathematics in separate booklets) from multiple topics to understand the application of concepts learned in theory. Each DPP is kind of a timed test with marking scheme and prescribed time to be spent on each problem. It is according to the latest pattern of JEE(Advanced) and serves as a great tool for the students to simulate examination conditions at home. It enables a student to practice time management while solving a problem which helps him/her to better prepare for the target exam.

Every effort has been taken to keep this book error free, however any suggestions to improve are welcome at smdd@resonance.ac.in.



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DPP

DAILY PRACTICE PROBLEMS

PHYSICS

TARGET: JEE (Main + Advanced) 2022

COURSE : VIKAAAS (01JA)

DPPs - B1 to B63

INDEX

S. No.	Particulars	Page No.
1.	DPP No. B1	04 – 07
2.	DPP No. B2	07 – 09
3.	DPP No. B3	09 – 11
4.	DPP No. B4	12 – 14
5.	DPP No. B5	15 – 17
6.	DPP No. B6	17 – 19
7.	DPP No. B7	19 – 21
8.	DPP No. B8	21 – 24
9.	DPP No. B9	24 – 26
10.	DPP No. B10	27 – 28
11.	DPP No. B11	29 – 31
12.	DPP No. B12	31 – 34
13.	DPP No. B13	34 – 36
14.	DPP No. B14	36 – 39
15.	DPP No. B15	39 – 41
16.	DPP No. B16	41 – 44
17.	DPP No. B17	44 – 46
18.	DPP No. B18	46 – 48
19.	DPP No. B19	49 – 51
20.	DPP No. B20	51 – 55
21.	DPP No. B21	55 – 57
22.	DPP No. B22	52 – 54
23.	DPP No. B23	57 – 60
24.	DPP No. B24	60 – 62
25.	DPP No. B25	62 – 65
26.	DPP No. B26	65 – 66
27.	DPP No. B27	67 – 69
28.	DPP No. B28	69 – 71
29.	DPP No. B29	71 – 74
30.	DPP No. B30	74 – 76
31.	DPP No. B31	76 – 78



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PAGE NO.-2

32.	DPP No. B32	79 – 81
33.	DPP No. B33	81 – 83
34.	DPP No. B34	84 – 85
35.	DPP No. B35	86 – 88
36.	DPP No. B36	88 – 90
37.	DPP No. B37	90 – 92
38.	DPP No. B38	92 – 94
39.	DPP No. B39	94 – 97
40.	DPP No. B40	97 – 99
41.	DPP No. B41	99 – 101
42.	DPP No. B42	102 – 104
43.	DPP No. B43	104 – 106
44.	DPP No. B44	106 – 108
45.	DPP No. B45	109 – 111
46.	DPP No. B46	112 – 114
47.	DPP No. B47	114 – 116
48.	DPP No. B48	116 – 118
49.	DPP No. B49	118 – 120
50.	DPP No. B50	120 – 123
51.	DPP No. B51	123 – 125
52.	DPP No. B52	125 – 128
53.	DPP No. B53	128 – 130
54.	DPP No. B54	130 – 132
55.	DPP No. B55	132 – 134
56.	DPP No. B56	135 – 137
57.	DPP No. B57	137 – 140
58.	DPP No. B58	140 – 142
59.	DPP No. B59	143 – 145
60.	DPP No. B60	146 – 148
61.	DPP No. B61	148 – 151
62.	DPP No. B62	151 – 152
63.	DPP No. B63	153 – 155
	Answers	156

NOTE : ✎ Marked questions are recommended for Revision.

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DPP No. : B1 (JEE-MAIN)

Total Marks : 60

Max. Time : 40 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.20

(3 marks 2 min.)

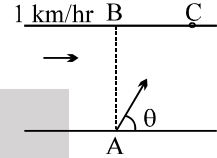
[60, 40]

1. A swimmer crosses the river along the line making an angle of 45° with the direction of flow. Velocity of the river water is 5 m/s. Swimmer takes 12 seconds to cross the river of width 60 m. The velocity of the swimmer with respect to water will be:

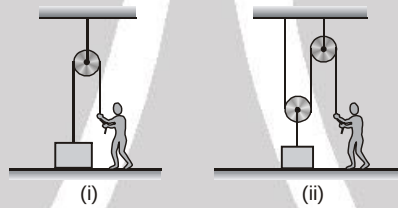
(A) 10 m/s (B) 5 m/s (C) $5\sqrt{5}$ m/s (D) $5\sqrt{2}$ m/s

2. A river is flowing with a speed of 1 km/hr. A swimmer wants to go to point 'C' starting from 'A'. He swims with a speed of 5 km/hr, at an angle θ w.r.t. the river. If $AB = BC = 400$ m. Then the value of θ is:

(A) 37° (B) 53°
(C) 60° (D) 45°



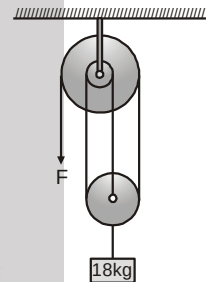
3. In the figure shown, a person wants to raise a block lying on the ground to a height h . In both the cases if time required is same then in which case he has to exert more force. Assume pulleys and strings light.



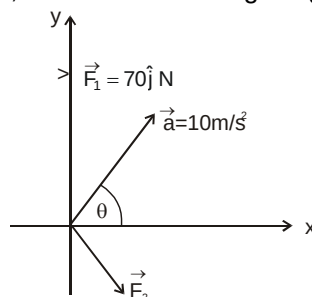
(A) (i) (B) (ii) (C) same in both (D) Cannot be determined

4. In the figure, at the free end of the light string, a force F is applied to keep the suspended mass of 18 kg at rest. Then the force exerted by the ceiling on the system (assume that the string segments are vertical and the pulleys are light and smooth) is: ($g = 10 \text{ m/s}^2$)

(A) 60 N
(B) 120 N
(C) 180 N
(D) 240 N

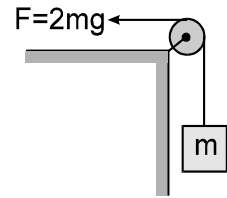


5. A particle of mass $m = 5 \text{ kg}$, is momentarily at rest at $x = 0$ at $t = 0$. It is acted upon by two forces $\vec{F}_1$ and $\vec{F}_2$. It is given that $\vec{F}_1 = 70 \hat{j}$ N and $\vec{F}_2$ is unknown. The particle experiences a constant acceleration $\vec{a}$, in the direction as shown. What third force, $\vec{F}_3$, is required to make the acceleration of the particle zero? (Note: $\sin\theta = 4/5$, $\cos\theta = 3/5$, and $\tan\theta = 4/3$. Neglect gravity.)

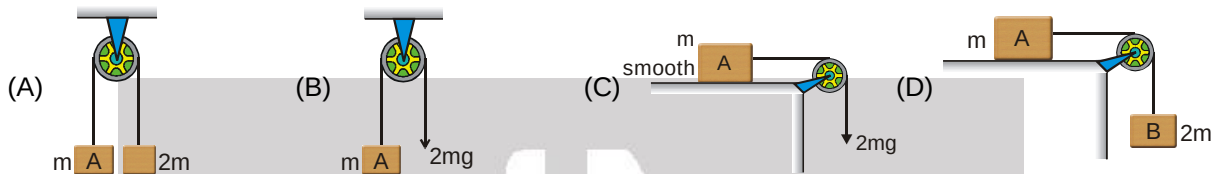


(A) $30\hat{i} + 40\hat{j}$ (B) $-(30\hat{i} + 40\hat{j})$ (C) $40\hat{i} + 30\hat{j}$ (D) $-(40\hat{i} + 30\hat{j})$

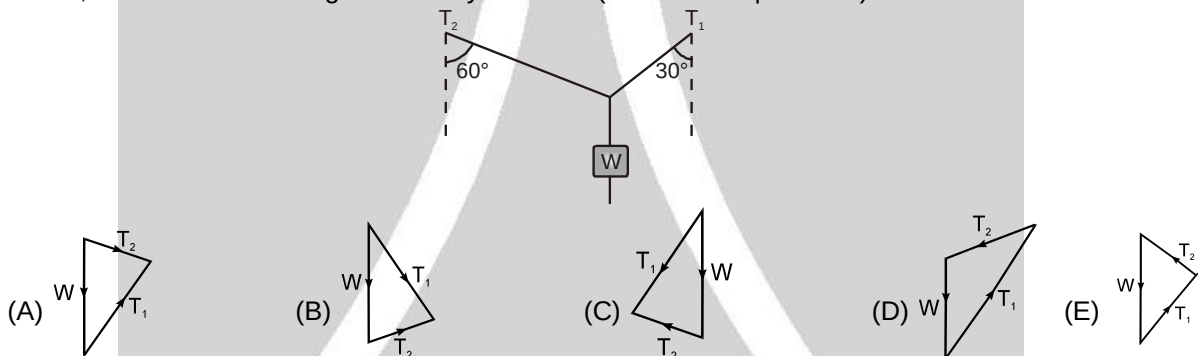
6. The block of mass 'm' equal to 100 kg is being pulled by a horizontal force $F = 2mg$ applied to a string as shown in figure (Take $g = 10 \text{ m/s}^2$). The pulley is massless and is fixed at the edge of an immovable table. What is the value of force exerted by the supporting cable (rod) on the pulley (in Newton)
- (A) $3mg$ (B) $4mg$
(C) $2\sqrt{2}mg$ (D) $2mg$



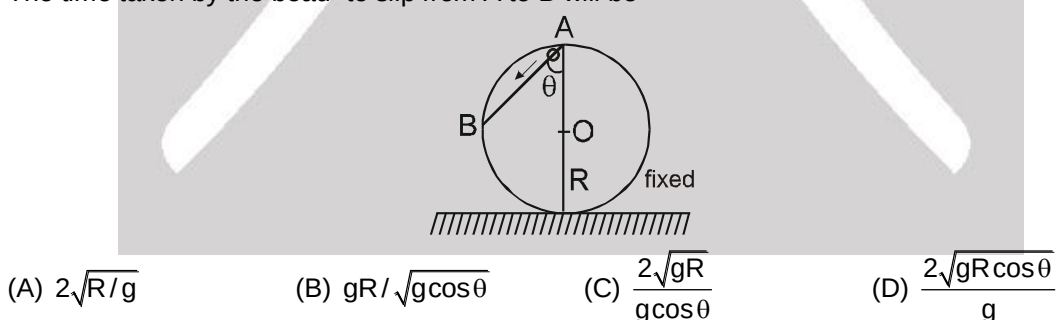
7. In which of the following cases the magnitude of acceleration of the block A will be maximum (Neglect friction, mass of pulley and string)



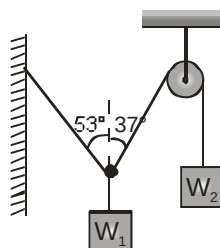
8. A weight W is supported by two strings inclined at 60° and 30° to the vertical. The tensions in the strings are T_1 & T_2 as shown. If these tensions are to be determined in terms of W using a triangle of forces, which of these triangles should you draw? (block is in equilibrium)



9. A frictionless wire is fixed between A and B inside a sphere of radius R . A bead slips along the wire. The time taken by the bead to slip from A to B will be



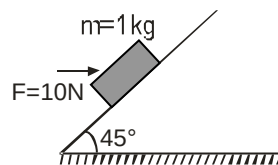
10. Two weights W_1 & W_2 in equilibrium and at rest, are suspended as shown in figure. Then the ratio $\frac{W_1}{W_2}$ is :



- (A) $5/4$ (B) $4/5$ (C) $8/5$ (D) none of these

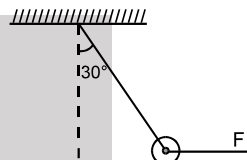
11. A uniform thick string of length 5 m is resting on a horizontal frictionless surface. It is pulled by a horizontal force of 5 N from one end. The tension in the string at 1m from the force applied is:
 (A) zero (B) 5 N (C) 4 N (D) 1 N

12. A body of mass 1 kg lies on smooth inclined plane. The block of mass m is given force $F = 10$ N horizontally as shown. The magnitude of net normal reaction on the block is: ($g = 10\text{m/s}^2$)



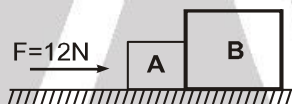
- (A) $10\sqrt{2}$ N (B) $\frac{10}{\sqrt{2}}$ N
 (C) 10 N (D) none of these

13. A heavy particle of mass 1kg is suspended from a massless string attached to a roof. A horizontal force F is applied to the particle such that in the equilibrium position the string makes an angle 30° with the vertical. The magnitude of the force F equals



- (A) 10 N (B) $10\sqrt{3}$ N (C) 5 N (D) $\frac{10}{\sqrt{3}}$ N

14. Two blocks A and B of masses 4 kg and 8 kg respectively are placed on a smooth plane surface. A force F of 12 N is applied on A as shown. Find the force of contact (in N) between A and B?

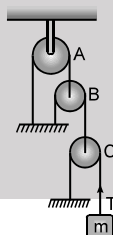


- (A) 8 N (B) 6 N (C) 10 N (D) None of these

15. For motion of an object along the x-axis, the velocity v depends on the displacement x as $v = 3x^2 - 2x$, then what is the acceleration at $x = 2\text{m}$.

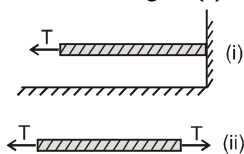
- (A) 48 ms^{-2} (B) 80 ms^{-2} (C) 18 ms^{-2} (D) 10 ms^{-2}

16. Ideal Pulleys A, B, C are connected to the mass as shown in figure. Tension in the rope connecting A to the ceiling is :



- (A) mg (B) $4mg$ (C) $8mg$ (D) none of these

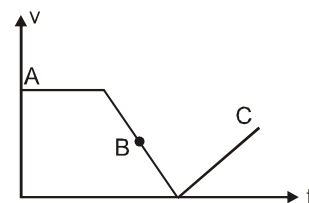
17. In the figure (i) an extensible string is fixed at one end and the other end is pulled by a tension T . In figure (ii) another identical string is pulled by tension ' T ' at both the ends. The ratio of elongation in equilibrium of string in (i) to the elongation of string in (ii) is



- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 0

18. The velocity time graph of a body is shown in Fig. It indicates that:

(A) At B force is zero
 (B) at B there is a force but towards motion
 (C) At B there is a force which opposes motion
 (D) None of the above is true



19. A traveller while in a uniformly moving horizontal train throws a ball vertically up (with respect to himself) in the air. The ball will return-
- (A) In his hand
 (B) Ahead in the direction of motion of the train
 (C) Trail behind
 (D) Deflected sideways

20. As shown in figure, mass $M = 10$ gms is placed on an inclined plane. In order to keep it at rest the value of m will be-

(A) 5 gm
 (B) $10\sqrt{3}$ gm
 (C) 0.10 gm
 (D) $1.0\sqrt{3}$ gm

DPP No. : B2 (JEE-MAIN)

Total Marks : 54

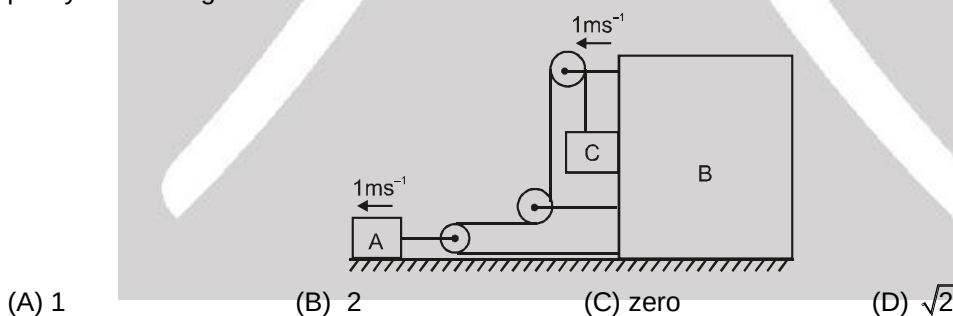
Single choice Objective ('-1' negative marking) Q.1 to Q.18

Max. Time : 36 min.

(3 marks 2 min.)

[54, 36]

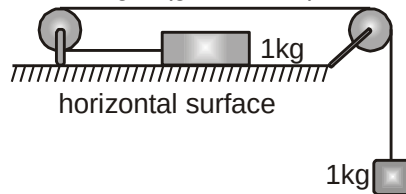
1. The velocities of A and B are shown in the figure. Find the speed (in ms^{-1}) of block C. (Assume that the pulleys and string are ideal.)



2. Drops of water fall at regular intervals from roof of a building of height $H = 16\text{m}$, the first drop striking the ground at the same moment as the fifth drop detaches itself from the roof. The distances between separate drops in air as the first drop reaches the ground are :
- (A) 1m, 5m, 7m, 3m
 (B) 1m, 3m, 5m, 7m
 (C) 1m, 3m, 7m, 5m
 (D) None of the above
3. Two men of masses m and $m/2$ starts climbing up on two massless strings fixed at the ceiling with acceleration g and $g/2$ respectively. The ratio of tensions in the two strings will be
- (A) 2 : 1
 (B) 4 : 1
 (C) 4 : 3
 (D) 8 : 3



4. Consider the system as shown in the figure. The pulley and the string are light and all the surfaces are frictionless. The tension in the string is ($g = 10 \text{ m/s}^2$).



- (A) 0 N (B) 1 N (C) 2 N (D) 5 N

5. The engine of a car produces acceleration 4 m/s^2 in the car. If this car pulls another car of same mass. What will be the acceleration produced-

- (A) 1 m/s^2 (B) 1.5 m/s^2 (C) 2 m/s^2 (D) 4 m/s^2

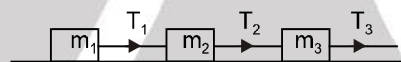
6. A uniform thick rope of length 5m is kept on frictionless surface and a force of 5N is applied to one of its end. Find tension in the rope at 1m from this end-

- (A) 1N (B) 3N (C) 4N (D) 5N

7. A force 10 N acts on a body of mass 20 kg for 10 sec. Change in its momentum is-

- (A) 5 kg m/s (B) 100 kg m/s
(C) 200 kg m/s (D) 1000 kg m/s

8. Three blocks of masses m_1 , m_2 and m_3 are connected by massless strings as shown on a frictionless table. They are pulled with a force $T_3 = 40 \text{ N}$. If $m_1 = 10 \text{ kg}$, $m_2 = 6 \text{ kg}$ and $m_3 = 4 \text{ kg}$, the tension T_2 will be-



- (A) 20 N (B) 40 N (C) 10 N (D) 32 N

9. A light string passes over a frictionless pulley. To one of its ends a mass of 6 kg is attached and to its other end a mass of 10 kg is attached. The tension in the string will be

- (A) 50 N (B) 75 N
(C) 100 N (D) 150 N

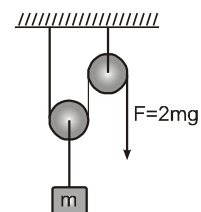


10. A force of 5 Newton acts horizontally on a body of weight 9.8 Newton. What is the acceleration produced in m s^{-2} ? ($g = 9.8 \text{ m/s}^2$)

- (A) 0.51 (B) 1.46 (C) 49.00 (D) 5.00

11. In the shown mass pulley system, pulleys and string are massless. The one end of the string is pulled by the force $F = 2mg$. The acceleration of the block will be

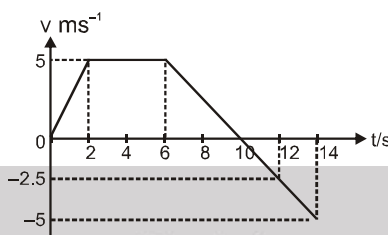
- (A) $g/2$ (B) 0
(C) g (D) $3g$



12. An athlete runs with a speed 12 ms^{-1} . Determine the longest jump he can undertake. ($g = 10 \text{ m/s}^2$)

- (A) 14.4 (B) 12 m (C) 12.1 m (D) 16.2 m

13. A person standing on the floor of an elevator drops a coin. The coin reaches the floor of the elevator in a time t_1 if the elevator is stationary and in time t_2 if it is moving with constant velocity. Then-
- (A) $t_1 = t_2$ (B) $t_1 < t_2$
 (C) $t_1 > t_2$ (D) $t_1 < t_2$ or $t_1 > t_2$
14. The variation of velocity of a particle moving along a straight line is shown in Fig. The distance travelled by the particle in 12 s is :



- (A) 37.5 (B) 32.5 (C) 35.0 (D) None of these
15. If the relation between the range R and time of flight T of a projectile is given as $R = 5T^2$, the value of angle of projection is-
- (A) 45° (B) 15° (C) 60° (D) 90°
16. The adjoining curve represents the velocity time graph of a particle, its acceleration values along OA, AB and BC in m./sec.^2 are respectively-
-
- (A) 1, 0, -0.5 (B) 1, 0, 0.5 (C) 1, 1, 0.5 (D) 1, 0.5, 0
17. A boat P is moving at 40 km/hr and another boat Q is moving at 20 km/hr. Which one of the following is not a possible value for their relative velocity-
- (A) 10 km/hr (B) 20 km/hr (C) 30 km/hr (D) 40 km/hr
18. A balloon of gross weight W newton descends with an acceleration $f \text{ m/s}^2$. The weight that must be thrown out in order to give balloon an equal upward acceleration will be:
- (A) Wf/g (B) $2Wf/g$ (C) $2Wf/(g+f)$ (D) $W(g+f)/f$

DPP No. : B3 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 3 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.9

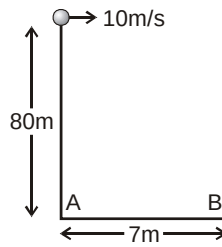
(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

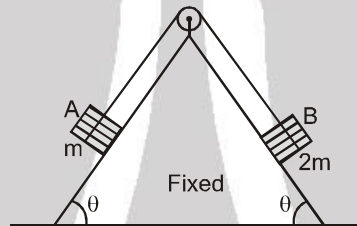
1. A particle moves in a straight line such that it moves in the same direction but its speed decreases uniformly with time to zero in time interval $T = 10\text{s}$. It travels a total distance ' S ' = 12.5 m then the velocity (in m/s) at any intermediate time $t = 6\text{s}$ is
- (A) 1 (B) 2 (C) 3 (D) 4

2. A ball is projected horizontally from top of a 80 m deep well with velocity 10 m/s. Then particle will fall on the bottom at a distance of (all the collisions with the wall are elastic) :

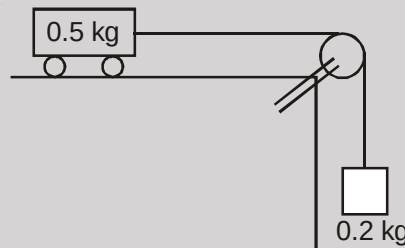


- (A) 5 m from A (B) 5 m from B (C) 3 m from A (D) 2 m from B

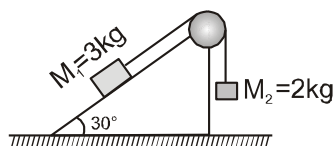
3. For the system shown in figure, the tension in the string is T (inclined surfaces are frictionless). $\frac{m}{2}$ mass is shifted from A to B or B to A as per given in options. Then :



- (A) Tension in the string will increase when mass is shifted from A to B.
 (B) Tension in the string will increase when mass is shifted from B to A.
 (C) acceleration of blocks will increase when mass is shifted from A to B.
 (D) acceleration of blocks will increase when mass is shifted from B to A.
4. A cart of mass 0.5 kg is placed on a smooth surface and is connected by a string to a block of mass 0.2 kg. At the initial moment the cart moves to the left along a horizontal plane at a speed of 7 m/s. (Use $g = 9.8 \text{ m/s}^2$)



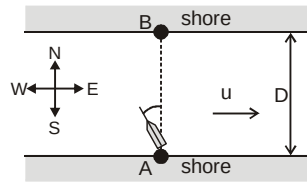
- (A) The acceleration of the cart is $\frac{2g}{7}$ towards right.
 (B) The cart comes to momentary rest after 2.5 s.
 (C) The distance travelled by the cart in the first 5s is 17.5 m.
 (D) The velocity of the cart after 5s will be same as initial velocity.
5. A block of mass $M_1 = 3 \text{ kg}$ on a smooth fixed inclined plane of angle 30° is connected by a cord over a small frictionless pulley to a second block of mass 2 kg hanging vertically. The tension T in the cord and the acceleration a of the blocks are :



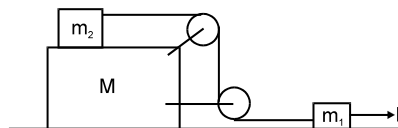
- (A) $a = 1 \text{ m/s}^2$ (B) $a = 2 \text{ m/s}^2$, (C) $T = 9 \text{ N}$ (D) $T = 18 \text{ N}$

COMPREHENSION

Two ports, A and B, on a North-South line are separated by a river of width D . The river flows east with speed u . A boat crosses the river starting from port A. The speed of the boat relative to the river is v . Assume $v = 2u$.



6. Suppose the boat wants to cross the river from A to the other side in the shortest possible time, then :
 (A) direction of the velocity of boat relative to river should be along north
 (B) direction of the velocity of boat relative to river should be 30° west of north
 (C) after crossing the river the boat is $D/4$ distance far from the port B
 (D) after crossing the river the boat is $D/2$ distance far from the port B
7. Suppose the boat wants to cross the river directly on a line from A to B, then :
 (A) the direction of the velocity of boat relative to river should be 30° west of north.
 (B) the direction of the velocity of boat relative to river should be 60° west of north.
 (C) the time taken by boat to cross the river is $\frac{D}{\sqrt{3}u}$
 (D) the time taken by boat to cross the river is $\frac{D}{2u}$
8. The river is flowing with a speed of 10 m/s. and V is the velocity of boat with respect to river. Then choose the correct options
 (A) If $V = 20$ m/s. then minimum possible drift of the boat, while crossing the river is zero
 (B) If $V = 20$ m/s. then minimum possible drift of the boat, while crossing the river is non zero.
 (C) If $V = 5$ m/s. then minimum possible drift of the boat, while crossing the river is zero.
 (D) If $V = 5$ m/s. then minimum possible drift of the boat, while crossing the river is non zero.
9. A particle is moving along a straight line. Its velocity varies as $v = 6 - 2t$ where v is in m/s and t in seconds. If the difference between distance covered and magnitude of displacement in first 4 seconds is $2n$, then find n .
10. **Match the following :**
 Three blocks of masses m_1 , m_2 and M are arranged as shown in figure. All the surfaces are frictionless and string is inextensible. Pulleys are light. A constant force F is applied on block of mass m_1 . Pulleys and string are light. Part of the string connecting both pulleys is vertical and part of the strings connecting pulleys with masses m_1 and m_2 are horizontal.



- (P) Acceleration of mass m_1 (1) $\frac{F}{m_1}$
 (Q) Acceleration of mass m_2 (2) $\frac{F}{m_1 + m_2}$
 (R) Acceleration of mass M (3) zero
 (S) Tension in the string (4) $\frac{m_2 F}{m_1 + m_2}$
- (A) P-1, Q-1, R-1, S-3 (B) P-2, Q-2, R-3, S-4
 (C) P-2, Q-4, R-3, S-1 (D) P-2, Q-2, R-3, S-3

DPP No. : B4 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

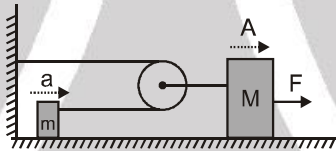
1. For a particle moving along x-axis, the acceleration a of the particle in terms of its x-coordinate x is given by $a = -9x$, where x is in meters and a is in m/s^2 . Take acceleration, velocity and displacement in positive x-direction as positive. The initial velocity of particle at $x = 0$ is $u = +6 \text{ m/s}$. The velocity of particle at $x = 2 \text{ m}$ will be :

(A) $+6\sqrt{2} \text{ m/s}$ (B) $-6\sqrt{2} \text{ m/s}$ (C) 72 m/s (D) 0

2. A stone is projected from level ground at $t = 0 \text{ sec}$ such that its horizontal and vertical components of initial velocity are 10 m/s and 20 m/s respectively. Then the instant of time at which tangential and normal components of acceleration of stone are same is : (neglect air resistance) $g = 10 \text{ m/s}^2$.

(A) $\frac{1}{2} \text{ sec}$ (B) 2 sec (C) 3 sec (D) 4 sec .

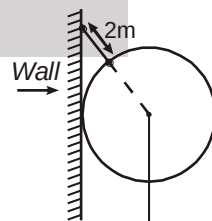
3. Find the acceleration of blocks in Fig. the pulley and the strings are mass less :



- (A) The acceleration of block of mass M is $A = \frac{F}{M + 4m}$
 (B) The acceleration of block of mass m is $a = \frac{2F}{(M + 4m)}$
 (C) The acceleration of block of mass M is $A = \frac{2F}{M + 4m}$
 (D) The acceleration of block of mass m is $a = \frac{4F}{(M + 4m)}$

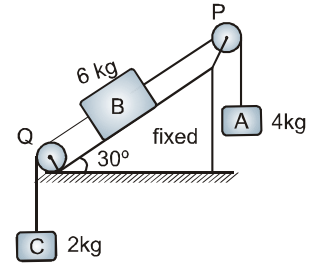
4. A uniform sphere of weight ' w ' and radius $3m$ is being held by a string of length $2m$. attached to a frictionless wall as shown in the figure.

- (A) Tension in the string will be $5w/4$.
 (B) Tension in the string will be $3w/4$.
 (C) Normal reaction by the wall on the sphere will be $3w/4$.
 (D) Normal reaction by the wall on the sphere will be $5w/4$.

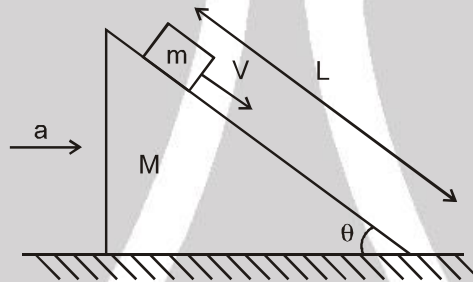


5. In the system shown in figure wedge is fixed. All the contact surfaces are frictionless. All the pulleys are light and strings are light and inextensible. Then : [Take $g = 10 \text{ m/s}^2$]

- (A) Magnitude of acceleration of the each block is $\frac{5}{6} \text{ m/s}^2$.
 (B) Tension in the string connecting block A and block B is $\left(\frac{130}{3}\right) \text{ N}$.
 (C) Tension in the string connecting block B and block C is $\frac{55}{3} \text{ N}$.
 (D) Force exerted by string on pulley Q is $\frac{55}{3} \text{ N}$.



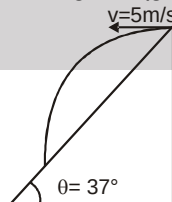
6. A wedge of mass M is pushed with a constant acceleration of $a = g \tan \theta$ along a smooth horizontal surface and a block of mass m is projected down the smooth incline of the wedge with a velocity V relative to the wedge.



- (A) The time taken by the block to cover distance L on the incline plane is $\frac{L}{V}$
 (B) The time taken by the block to cover distance L on the incline plane is $\sqrt{\frac{2L}{g \sin \theta}}$
 (C) The normal reaction between the block and wedge is $mg \sec \theta$
 (D) The horizontal force applied on the wedge to produce acceleration a is $(M + m) g \tan \theta$.

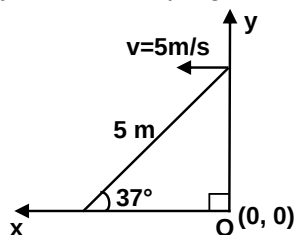
COMPREHENSION :

A particle is projected horizontally with a speed $v = 5 \text{ ms}^{-1}$ from the top of a plane inclined at an angle $\theta = 37^\circ$ to the horizontal as shown in figure. ($g = 10 \text{ m/s}^2$)



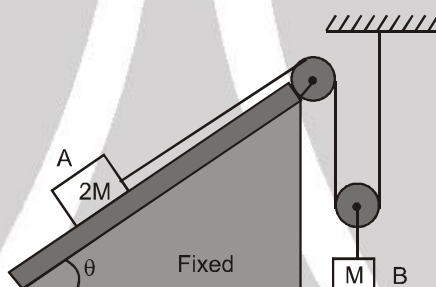
7. Choose the correct option(s)
- (A) the particle strikes the plane at $\frac{75}{16} \text{ m}$ from the point of projection
 (B) the particle strikes the plane at $\frac{65}{16} \text{ m}$ from the point of projection
 (C) the time taken by particle to hit the plane is 3 seconds
 (D) the time taken by particle to hit the plane is $\frac{3}{4}$ seconds

8. With respect to the coordinate system shown (length of the incline plane = 5m).

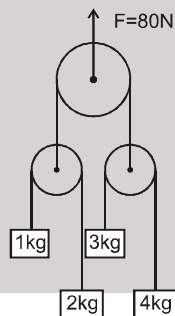


- (A) the coordinate of the point at which the particle strikes the inclined plane is $\left(\frac{15}{4}, \frac{3}{16}\right)$
 (B) the coordinate of the point at which the particle strikes the inclined plane is $\left(\frac{15}{4}, \frac{3}{8}\right)$
 (C) the speed of projection (projected horizontally) so that the particle hits the coordinate (4, 0) is $4\sqrt{5/3}$ m/s
 (D) Time taken by the particle to hit the coordinate (4,0) is 1 second.

9. If the acceleration of the block A is n . Assume all pulleys are massless and frictionless. All surfaces are smooth. Then n is : ($\sin\theta = 1/4$)



10. In the diagram shown in figure, all pulleys are smooth and massless and strings are light. Match the blocks in column-I with their motion in column-II



Column I

- (P) 1 kg block
 (Q) 2 kg block
 (R) 3 kg block
 (S) 4 kg block
 (A) P-1, Q-1, R-1, S-3
 (C) P-2, Q-4, R-3, S-1

Column II

- (1) will remain stationary
 (2) will move down
 (3) will move up
 (4) has acceleration 4 m/s^2
 (B) P-3, Q-1, R-2, S-2
 (D) P-1, Q-4, R-2, S-2

DPP No. : B5 (JEE-MAIN)

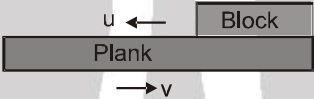
Total Marks : 57

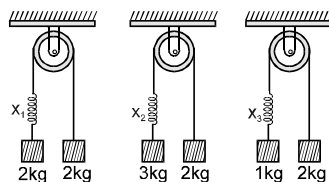
Single choice Objective ('-1' negative marking) Q.1 to Q.19

Max. Time : 38 min.

(3 marks 2 min.)

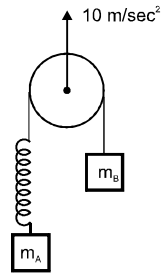
[57, 38]

1. The velocity acquired by a body moving with uniform acceleration is 30 ms^{-1} in 2s and 60 ms^{-1} in 4s. The initial velocity is :
 (A) zero (B) 2 ms^{-1} (C) 3 ms^{-1} (D) 10 ms^{-1}
2. A man can swim in still water with a speed of 2 ms^{-1} . If he wants to cross a river of water current speed $\sqrt{3} \text{ ms}^{-1}$ along shortest possible path, then in which direction should he swim ?
 (A) At an angle 120° to the water current (B) At an angle 150° to the water current
 (C) At an angle 90° to the water current (D) None of these
3. A plank is moving on ground with a velocity v , and a block is moving on the plank with respect to it with a velocity u w.r.t. plank as shown in Fig. What is the velocity of block with respect to ground ?

 (A) $v - u$ towards right (B) $v - u$ towards left
 (C) u towards right (D) None of these
4. A body covered a distance of L m along a curved path of a quarter circle. The ratio of distance to displacement is
 (A) $\frac{\pi}{2\sqrt{2}}$ (B) $\frac{2\sqrt{2}}{\pi}$ (C) $\frac{\pi}{\sqrt{2}}$ (D) $\frac{\sqrt{2}}{\pi}$
5. A particle is moving with velocity 5 m/s towards east and its velocity changes to 5 m/s north in 10 sec. Find the average acceleration in 10 seconds.
 (A) $\sqrt{2} \text{ N} - \text{W}$ (B) $\frac{1}{\sqrt{2}} \text{ N} - \text{W}$ (C) $\frac{1}{\sqrt{2}} \text{ N} - \text{E}$ (D) $\sqrt{2} \text{ N} - \text{E}$
6. An object is released from some height. Exactly after one second, another object is released from the same height. The distance between the two objects exactly after 2 seconds of the release of second object will be :
 (A) 4.9 m (B) 9.8 m (C) 19.6 m (D) 24.5 m
7. At what other angle of elevation, the range of a shell fired from a gun will be the same as that for an angle of elevation $6\pi/36$?
 (A) $30\pi/36$ (B) $\pi/12$ (C) $\pi/3$ (D) no other angle
8. Same spring is attached with 2 kg, 3 kg and 1 kg blocks in three different cases as shown in figure. If x_1 , x_2 and x_3 be the extensions in the spring in these cases then (Assume all the blocks to move with uniform acceleration)



- (A) $x_1 = 0, x_3 > x_2$ (B) $x_2 > x_1 > x_3$ (C) $x_3 > x_1 > x_2$ (D) $x_1 > x_2 > x_3$

9. In the system shown in figure at some instant acceleration of pulley was 10 m/sec^2 in upward direction. At that instant spring was in its natural length. If acceleration of block A and B are a_A and a_B at that instant then :
(Given $m_A = 10 \text{ kg}$, $m_B = 20 \text{ kg}$, $g = 10 \text{ m/sec}^2$, pulley is light and smooth)



- (A) $a_A = \frac{10}{3} \text{ m/sec}^2$ upward $a_B = \frac{10}{3} \text{ m/sec}^2$ downward
(B) $a_A = a_B = 5 \text{ m/sec}^2$ upward
(C) $a_A = a_B = 10 \text{ m/sec}^2$ downward
(D) None of these

10. Initially the spring is undeformed. Now the force 'F' is applied to 'B' as shown. When the displacement of 'B' w.r.t. 'A' is 'x' towards right in some time then the relative acceleration of 'B' w.r.t. 'A' at that moment is:

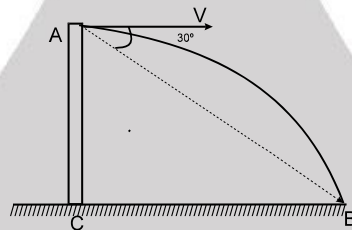


- (A) $\frac{F}{2m}$ (B) $\frac{F - kx}{m}$ (C) $\frac{F - 2kx}{m}$ (D) none of these

11. A body is projected with a velocity u . If the maximum height attained by it is half of its horizontal range, then its range must be :

- (A) $4u/5g^2$ (B) $4g/5u^2$ (C) $4u^2/5g$ (D) $4u^2/5g^2$

12. An object is thrown horizontally from a point 'A' from a tower and hits the ground $3s$ later at B. The line from 'A' to 'B' makes an angle of 30° with the horizontal. The initial velocity of the object is :
(take $g = 10 \text{ m/s}^2$)

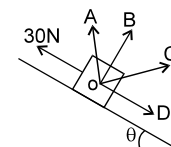


- (A) $15\sqrt{3} \text{ m/s}$ (B) 15 m/s (C) $10\sqrt{3} \text{ m/s}$ (D) $25/\sqrt{3} \text{ m/s}$

13. A police party is reaching a dacoit in a jeep which is moving at a constant speed v . The dacoit is on a motorcycle. When he is at a distance x from the jeep, he accelerates from rest at a constant rate α . Which of the following relations is true if the police is able to catch the dacoit ?

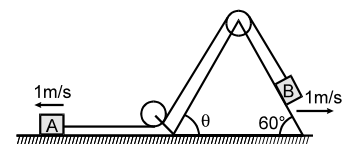
- (A) $v^2 \leq \alpha x$ (B) $v^2 \leq 2\alpha x$ (C) $v^2 \geq 2\alpha x$ (D) $v^2 \geq \alpha x$

14. A body of mass 10 kg lies on a rough inclined plane of inclination $\theta = \sin^{-1} \frac{3}{5}$ with the horizontal. When a force of 30 N is applied on the block parallel to & upward the plane, the total reaction by the plane on the block is nearly along :



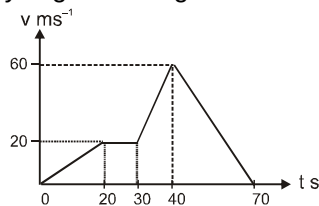
- (A) OA (B) OB (C) OC (D) OD

15. A system is shown in the figure. Block A is moving with 1 m/s towards left. Wedge is moving with 1 m/s towards right. Then speed of the block B will be :

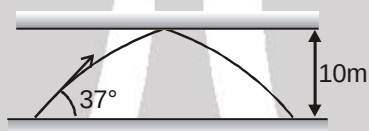


- (A) 1 m/s (B) 2 m/s
(C) $\sqrt{3} \text{ m/s}$ (D) none of these

16. A particle moves along a circle with a uniform speed v . After it has made an angle of 60° , the change in its speed and velocity will be
 (A) $v\sqrt{2}$, v (B) $\frac{v}{\sqrt{2}}$, 0 (C) $v\sqrt{3}$, v (D) 0, v
17. The velocity-time graph of a body is given in Fig. The maximum acceleration in ms^{-2} is :



- (A) 4 (B) 3 (C) 2 (D) 1
18. A ball is shot in a long hall having a roof at a height of 10 m with 25 m/s at an angle of 37° with the floor. The ball lands on the floor at a distance of from the point of projection. (Assume elastic collisions if any)



- (A) 40 m (B) 60 m (C) $45/4$ m (D) 20 m
19. An aeroplane is flying in a horizontal direction at 600 km/hr at a height of 6 km. and is advancing towards a point which is exactly over a target. At that instant the pilot releases a ball which on descending the earth strikes the target, the falling ball appears-
 (A) To the pilot in the aeroplane, as falling along a parabolic path.
 (B) To a person standing near the target, as falling exactly vertical.
 (C) To a person standing near the target as describing a parabolic path.
 (D) To the pilot sitting in the aeroplane as falling in a zigzag path.

DPP No. : B6 (JEE-ADVANCED)

Total Marks : 38

Max. Time : 25 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.7

(4 marks 2 min.) [20, 10]

Subjective Questions ('-1' negative marking) Q.8

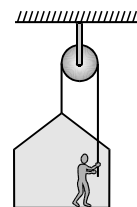
(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.9

(8 marks 6 min.) [08, 06]

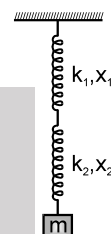
1. A particle is projected up the inclined such that its component of velocity along the incline is 10 m/s. Time of flight is 2 sec and maximum height above the incline is 5 m. Then velocity of projection will be:
 (A) 10 m/s (B) $10\sqrt{2}$ m/s (C) $5\sqrt{5}$ m/s (D) none of these
2. On a particular day, wind offers a horizontal acceleration 7.5 m/s^2 . A particle is projected at a speed of 10 m/s at $\theta = 37^\circ$ with the vertical against the wind. The distance (in meters) from the point of projection where the particle will strike the horizontal ground is. $\tan 37^\circ = \frac{3}{4}$ and $g = 10.0 \text{ m/s}^2$
 (A) 0 (B) 1 (C) 2.5 (D) 3.5

3. A painter is applying force himself to raise him and the box with an acceleration of 5 m/s^2 by a massless rope and pulley arrangement as shown in figure. Mass of painter is 100 kg and that of box is 50 kg . If $g = 10 \text{ m/s}^2$, then:
- (A) tension in the rope is 1125 N
 (B) tension in the rope is 2250 N
 (C) force of contact between the painter and the floor is 375 N
 (D) none of these

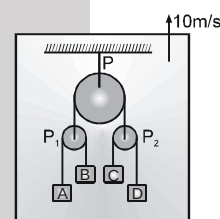


4. Two springs are in a series combination and are attached to a block of mass ' m ' which is in equilibrium. The spring constants and the extensions in the springs are as shown in the figure. Then the force exerted by the spring on the block is :

- (A) $\frac{k_1 k_2}{k_1 + k_2} (x_1 + x_2)$ (B) $k_1 x_1 + k_2 x_2$
 (C) $k_1 x_1$ (D) None of these



5. A lift goes up with velocity 10 m/s . A pulley P is fixed to the ceiling of the lift. To this pulley other two pulleys P_1 & P_2 are attached. P_1 moves up with velocity 30 m/s . A moves up with velocity 10 m/s . D is moving downwards with velocity 10 m/s at same instant of time. (Assume that all velocities are relative to the ground) Then choose the correct option/s :
- (A) The velocity of B is 50 m/s . in upward direction, at that instant.
 (B) The velocity of B is 10 m/s . in downward direction, at that instant.
 (C) The velocity of C is 10 m/s . in downward direction, at that instant.
 (D) The velocity of C is 50 m/s . in upward direction, at that instant.



Comprehension :

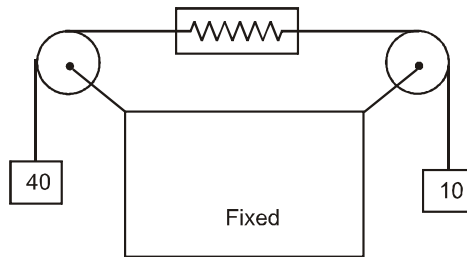
A man can swim at a speed of 3 Km h^{-1} in still water. he wants to cross a 500 m wide river flowing at 2 km h^{-1} . He keeps himself always at an angle of 120° with the river flow while swimming .

6. Choose the correct options

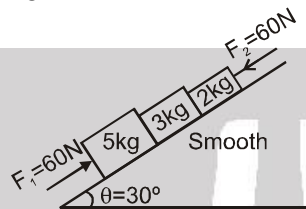
- (A) The time taken to cross the river is $\frac{1}{3\sqrt{3}} \text{ h}$
 (B) The time taken to cross the river is $\frac{1}{2\sqrt{3}} \text{ h}$
 (C) The drift of the man along the direction of flow, when he arrives at the opposite bank is $\frac{1}{6\sqrt{3}} \text{ km}$
 (D) The drift of the man along the direction of flow, when he arrives at the opposite bank is $\frac{1}{3\sqrt{3}} \text{ km}$

7. A boy standing on the ground and a girl on the boat (drafting with the river velocity) observes the man. Then choose the correct option(s)
- (A) The girl observes that the man is moving with a speed which is less than 3 km/hr .
 (B) The girl observes that the man is moving with a speed which is greater than 3 km/hr .
 (C) The boy observes that the man is moving with a speed which is less than 3 km/hr .
 (D) The girl observes that the man is moving with a speed which is equal to 3 km/hr .

8. If the reading of the spring balance (massless) is $10x$ (in Newton) then find x . ($g = 10 \text{ m/s}^2$)



9. In the diagram shown in figure, match the following : ($g = 10 \text{ m/s}^2$)



Column-I

- (P) Acceleration of 2 kg block (in m/s^2)
 (Q) Acceleration of 3 kg block (in m/s^2)
 (R) Normal reaction between 2 kg and 3 kg (in N)
 (S) Normal reaction between 3 kg and 5 kg (in N)
 (A) P-2, Q-1, R-4, S-2
 (C) P-1, Q-2, R-2, S-3

Column-II

- (1) 5
 (2) 50
 (3) 45
 (4) 60
 (B) P-1, Q-1, R-4, S-4
 (D) P-3, Q-3, R-2, S-2

DPP No. : B7 (JEE-ADVANCED)

Total Marks : 50

Max. Time : 31 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.10

(4 marks 2 min.) [32, 16]

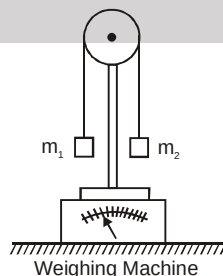
Subjective Questions ('-1' negative marking) Q.11

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.12

(8 marks 6 min.) [08, 06]

1. Two masses m_1 and m_2 which are connected with a light string, are placed over a frictionless pulley. This set up is placed over a weighing machine, as shown. Three combination of masses m_1 and m_2 are used, in first case $m_1 = 6 \text{ kg}$ and $m_2 = 2 \text{ kg}$, in second case $m_1 = 5 \text{ kg}$ and $m_2 = 3 \text{ kg}$ and in third case $m_1 = 4 \text{ kg}$ and $m_2 = 4 \text{ kg}$. Masses are held stationary initially and then released. If the readings of the weighing machine after the release in three cases are W_1 , W_2 and W_3 respectively then :

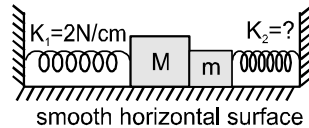


- (A) $W_1 > W_2 > W_3$ (B) $W_1 < W_2 < W_3$ (C) $W_1 = W_2 = W_3$ (D) $W_1 = W_2 < W_3$

2. A bullet with muzzle velocity $10\sqrt{10} \text{ m/s}$ is to be shot at a target 50 m away in the same horizontal line. The height above the target must the rifle be aimed so that the bullet will hit the target is.

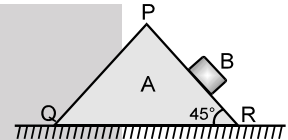
- (A) $50 \tan 15^\circ$ (B) $40 \tan 15^\circ$ (C) $\frac{50}{\sqrt{3}}$ (D) $50\sqrt{3}$

3. Two blocks of mass M and m , are used to compress two different massless springs as shown. The left spring is compressed by 3 cm, while the right spring is compressed by an unknown amount. The system is at rest, and all surfaces fixed and smooth. Which of the following statements are true?



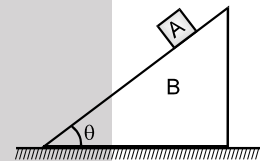
- (A) The force exerted on block of mass m by the right spring is 6 N to the left.
 (B) The force exerted on block of mass m by the right spring is impossible to determine.
 (C) The net force on block of mass m is zero.
 (D) The normal force exerted by block of mass m on block of mass M is 6 N.

4. A block B of mass 0.6 kg slides down the smooth face PR of a wedge A of mass 1.7 kg which can move freely on a smooth horizontal surface. The inclination of the face PR to the horizontal is 45° . Then:



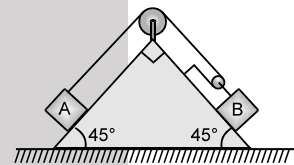
- (A) the acceleration of A is $3g/20$
 (B) the vertical component of the acceleration of B is $23g/40$
 (C) the horizontal component of the acceleration of B is $17g/40$
 (D) none of these

5. In the figure shown, A & B are free to move. All the surfaces are smooth. ($0 < \theta < 90^\circ$)



- (A) the acceleration of A will be more than $g \sin \theta$
 (B) the acceleration of A will be less than $g \sin \theta$
 (C) normal force on A due to B will be more than $mg \cos \theta$
 (D) normal force on A due to B will be less than $mg \cos \theta$

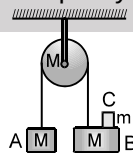
6. Two blocks A and B of mass 10 kg and 40 kg are connected by an ideal string as shown in the figure. Neglect the masses of the pulleys and effect of friction in the pulley and between the blocks and the inclines. (wedge is fixed)



- (A) The acceleration of block A is $5/\sqrt{2} \text{ m/s}^2$
 (B) The acceleration of block B is $5/(2\sqrt{2}) \text{ m/s}^2$
 (C) The tension in the string is $125/\sqrt{2} \text{ N}$
 (D) The tension in the string connecting A is $150/\sqrt{2} \text{ N}$.

COMPREHENSION

For the following system shown assume that pulley is frictionless, string is massless (m remains on M)



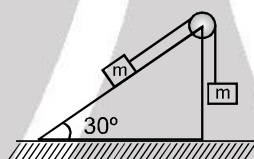
7. Choose the correct option(s)

- (A) Acceleration of block A is $\frac{mg}{2M+m}$
 (B) Acceleration of block A is $\frac{2mg}{2M+m}$
 (C) Normal reaction on C is $\frac{2Mmg}{2M+m}$
 (D) the force on the ceiling is $\frac{(6M+5m)Mg}{2M+m}$

8. If we cut the string after some time then (after cutting the string)

- (A) acceleration of C will be equal to g
 (B) acceleration of B will be greater than g
 (C) normal reaction on C becomes zero
 (D) the separation between B and C widens as they fall.

9. The block C is removed from B after some time :
 (A) A and B stop as soon as C is removed
 (B) A and B will continue to move with a constant and equal speed.
 (C) the tension in the string becomes zero.
 (D) the separation (vertical) between A and B increases at a constant rate.
10. For the given system, choose the **INCORRECT** option(s).
 (A) The tension in the string connecting the blocks is $\left(\frac{6M+5m}{2}\right)\left(\frac{Mg}{2M+m}\right)$
 (B) The tension in the string connecting the blocks is $\left[\left(\frac{6M+5m}{2}\right)\left(\frac{Mg}{2M+m}\right) - Mg\right]$
 (C) The acceleration of block C is $\frac{mg}{2M+m}$
 (D) The acceleration of block B is $\frac{mg}{2M+m}$
11. Two blocks of equal mass 'm' are connected by massless string and pulley as shown in figure. Now the blocks are left to move. Then the force exerted by pulley on the clamp is $\frac{3\sqrt{3}mg}{k}$. Find the value k ?



12. Match the following

Column I

- (a) Instantaneous speed
 (b) Instantaneous velocity
 (c) Average velocity
 (d) Average speed

Column II

- (P) is a vector quantity
 (Q) Its magnitude can decrease with time
 (R) Will remain constant for a particle moving uniformly in a circle
 (S) Does not depend on the initial and final position only but depends on the motion in between

DPP No. : B8 (JEE-MAIN)

Total Marks : 57

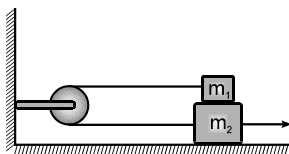
Single choice Objective ('-1' negative marking) Q.1 to Q.19

Max. Time : 38 min.

(3 marks 2 min.)

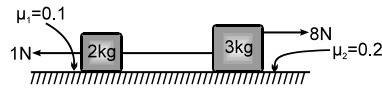
[57, 38]

1. A force F accelerates a block of mass m_1 . The coefficient of friction between the contact surfaces is μ . The acceleration of m_1 will be:

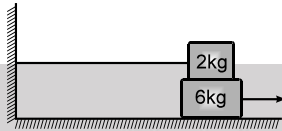


- (A) $\mu g - \frac{F - 2m_1g}{m_1 + m_2}$ (B) $\frac{m_1 + m_2}{F - 2\mu m_1g}$ (C) $\frac{F - 2\mu m_1g}{m_1 + m_2} - \mu g$ (D) $F - (m_1 + m_2)\mu g$

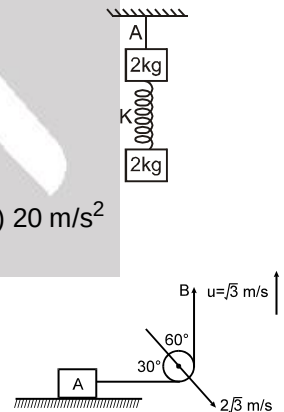
2. In the shown arrangement if f_1 , f_2 and T be the frictional forces on 2 kg block, 3 kg block & tension in the string respectively, then their values are:



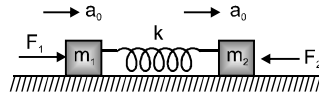
- (A) 2 N, 6 N, 3.2 N
(B) 2 N, 6 N, 0 N
(C) 1 N, 6 N, 2 N
(D) data insufficient to calculate the required values.
3. With reference to the figure shown, if the coefficient of friction at all the surfaces is 0.42, then the force required to pull out the 6.0 kg block with an acceleration of 1.50 m/s^2 will be:



- (A) 36 N
(B) 24 N
(C) 84 N
(D) 51 N
4. A block is placed on an inclined plane and has to be pushed down. The angle of inclination of the plane is:
(A) equal to angle of repose
(B) more than angle of repose
(C) less than the angle of repose
(D) equal to angle of friction
5. A perfectly straight portion of a uniform rope has mass M and length L . At end A of the segment, the tension in the rope is T_A and at end B it is T_B ($T_B > T_A$). Neglect effect of gravity and no contact force acts on the rope in between points A and B. The tension in the rope at a distance $L/5$ from end A is
(A) $T_B - T_A$
(B) $(T_A + T_B)/5$
(C) $(4T_A + T_B)/5$
(D) $(T_A - T_B)/5$
6. A spring balance hanging in steady state inside a moving train shows a reading of double the actual mass of an object. Then the acceleration of the train is :
(A) $\sqrt{5} g$
(B) g
(C) $2g$
(D) $\sqrt{3} g$
7. Two blocks of mass 2 kg are connected by a massless ideal spring of spring constant $K = 10 \text{ N/m}$. The upper block is suspended from roof by a light string A. The system shown is in equilibrium. The string A is now cut, the acceleration of upper block just after the string A is cut will be ($g = 10 \text{ m/s}^2$)
(A) 0 m/s^2
(B) 10 m/s^2
(C) 15 m/s^2
(D) 20 m/s^2
8. A system is shown in the figure. End B of string is moving upwards with $\sqrt{3} \text{ m/s}$. Pulley is moving with speed $2\sqrt{3} \text{ m/s}$ in direction shown in the figure. Then velocity of the block A will be:
(A) $\sqrt{3} \text{ m/s}$
(B) 3 m/s
(C) $3 + 2\sqrt{3} \text{ m/s}$
(D) $3 + \sqrt{3} \text{ m/s}$
9. A train is moving in the north at a speed 10 m/sec . Its length is 150 m . A parrot is flying parallel to the train in the south with a speed of 5 m/s . The time taken by the parrot to cross the train will be-
(A) 12 sec.
(B) 8 sec.
(C) 15 sec.
(D) 10 sec.



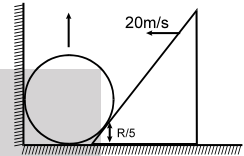
10. Two blocks of masses m_1 and m_2 connected with a light spring of spring constant k are acted by forces F_1 and F_2 on a frictionless horizontal surface. Find the spring force at this instant provided that at this instant the acceleration of both the blocks are same :



- (A) $\frac{F_1 m_2 - F_2 m_1}{(m_1 + m_2)}$ (B) $\frac{F_1 - F_2}{2}$ (C) $\frac{m_1 F_1 + m_2 F_2}{(m_1 + m_2)}$ (D) $\frac{F_1 m_2 + F_2 m_1}{m_1 + m_2}$

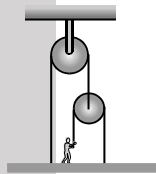
11. A sphere of radius R is in contact with a wedge. The point of contact is $R/5$ from the ground as shown in the figure. Wedge is moving with velocity 20 m/s , then the velocity of the sphere at this instant will be:

- (A) 20 m/s (B) 15 m/s
(C) 5 m/s (D) 10 m/s



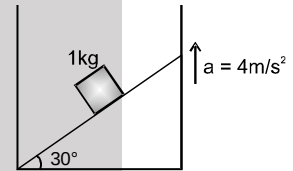
12. In the figure shown with what force must the man pull the rope to hold the plank in position? Weight of the man is 60 kg f . Neglect the weights of plank rope and pulley.

- (A) 15 kg f (B) 30 kg f
(C) 60 kg f (D) none of these



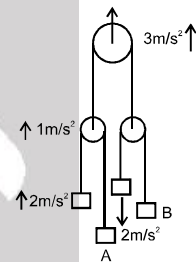
13. A wedge is moving with an acceleration $a = 4 \text{ m/s}^2$ vertically up as shown in figure. What is acceleration of block of mass 1 kg w.r.t. wedge & normal reaction by wedge on block respectively. All surfaces are smooth. Choose correct pair : ($g = 10 \text{ m/s}^2$)

- (A) $7 \text{ m/s}^2, 7\sqrt{3} \text{ N}$ (B) $5 \text{ m/s}^2, 5\sqrt{3} \text{ N}$
(C) $4.9 \text{ m/s}^2, 4.9\sqrt{3} \text{ N}$ (D) $10 \text{ m/s}^2, 10\sqrt{3} \text{ N}$



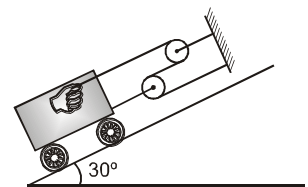
14. Acceleration of pulleys and blocks are as shown in the figure. All pulleys & strings are massless & frictionless magnitude of a_A and a_B are :

- (A) $a_A = 0, a_B = 7$
(B) $a_A = 0, a_B = 5$
(C) $a_A = 0, a_B = 12$
(D) $a_A = 5, a_B = 7$



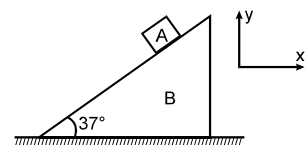
15. A trolley is being pulled up an incline plane by a man inside the trolley. He applied a force of 500 N . If the mass of the man is 50 kg and trolley is 150 kg , then the acceleration of the trolley will be : ($g = 10 \text{ m/s}^2$)

- (A) 5 m/s^2 (B) 1 m/s^2
(C) 1.5 m/s^2 (D) 2.5 m/s^2



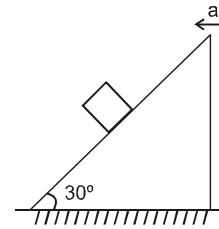
16. In the figure shown the acceleration of A is, $\vec{a}_A = 15\hat{i} + 15\hat{j}$ then the acceleration of B is: (A remains in contact with B)

- (A) $6\hat{i}$ (B) $-15\hat{i}$
(C) $-10\hat{i}$ (D) $-5\hat{i}$



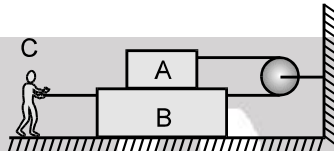
17. For no friction to act on a block kept on an accelerating wedge as shown in figure, acceleration a of the wedge should be -

- (A) $\frac{g}{\sqrt{3}}$ (B) $g\sqrt{3}$
(C) $\frac{g}{2}$ (D) $g\frac{\sqrt{3}}{2}$



18. In the figure $m_A = m_B = m_C = 60$ kg. The co-efficient of friction between C and ground is 0.5, B and ground is 0.3, A & B is 0.4. C is pulling the string with the maximum possible force without moving. Then tension in the string connected to A will be:

- (A) 120 N (B) 60 N (C) 100 N (D) zero



19. Which of the following statement is correct?
(A) Two forces on the body which happen to be equal and opposite, constitute an action reaction pair.
(B) The mutual actions of two bodies upon each other are always equal and directed to contrary parts.
(C) Newton's second law is valid in all frames.
(D) Pseudo forces are applicable in inertial frames.

DPP No. : B9 (JEE-ADVANCED)

Total Marks : 46

Single choice Objective ('-1' negative marking) Q.1 to Q.2

One or more than one options correct type ('-1' negative marking) Q.3 to Q.9

Subjective Questions ('-1' negative marking) Q.10

Match the Following (no negative marking) Q.11

Max. Time : 29 min.

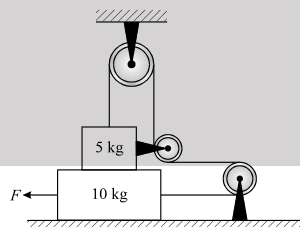
(3 marks 2 min.) [06, 04]

(4 marks 2 min.) [28, 14]

(4 marks 5 min.) [04, 05]

(8 marks 6 min.) [08, 06]

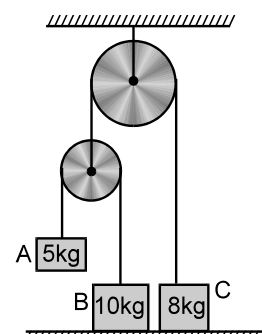
1. Figure shows a system of two blocks of 10 kg and 5 kg mass, connected by ideal strings and pulleys. Here ground is smooth and friction coefficient between the two blocks is $\mu = 0.5$. A horizontal force F is applied on lower block as shown. The minimum value of F required to start sliding between the blocks is : (Take $g = 10 \text{ m/s}^2$)



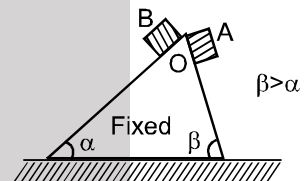
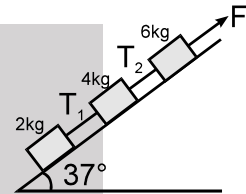
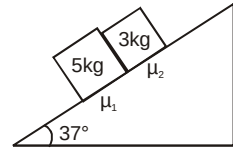
- (A) 12.5 N (B) 25 N (C) 50 N (D) 100 N

2. In the following arrangement the system is initially at rest. The 5 kg block is now released. Assuming the pulleys and string to be massless and smooth. The acceleration of block 'C' is (Take $g = 9.8 \text{ m/s}^2$)

- (A) 0.5 m/s^2
(B) 0.6 m/s^2
(C) 0.7 m/s^2
(D) 0.8 m/s^2

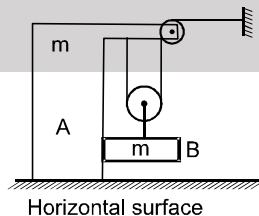


3. Two blocks of masses 5 kg and 3 kg are placed in contact over an inclined surface of angle 37° , as shown. μ_1 is friction coefficient between 5 kg block and the surface of the incline and similarly, μ_2 is friction coefficient between the 3 kg block and the surface of the incline. After the release of the blocks from the inclined surface,
- (A) if $\mu_1 = 0.5$ and $\mu_2 = 0.3$ then 5 kg block exerts 3 N force on the 3 kg block
 (B) if $\mu_1 = 0.5$ and $\mu_2 = 0.3$ then 5 kg block exerts 8 N force on the 3 kg block
 (C) if $\mu_1 = 0.3$ and $\mu_2 = 0.5$ then 5 kg block exerts 1 N force on the 3 kg block.
 (D) if $\mu_1 = 0.3$ and $\mu_2 = 0.5$ then 5 kg block exerts no force on the 3 kg block.
4. Three blocks of masses 2 kg, 4 kg and 6 kg arranged as shown in figure connected by string on a frictionless incline of 37° . A force of 120 N is applied upward along the incline to the uppermost block. The cords are light. The tension T_1 and T_2 in the strings are [$g = 10 \text{ m/s}^2$]
- (A) $T_1 = 8 \text{ N}$ (B) $T_2 = 36 \text{ N}$
 (C) $T_1 = 20 \text{ N}$ (D) $T_2 = 60 \text{ N}$
5. Two inclined frictionless tracks of different inclinations meet at O from where two blocks A and B of different masses are released as shown in figure. Then :
- (A) Both blocks will reach the bottom at the same time
 (B) Block A will reach the bottom earlier than block B
 (C) Both blocks will reach the bottom with same speed
 (D) Block B will reach the bottom with a higher speed than block A
6. A 20 kg block is initially at rest. A minimum of 80 N horizontal force is required to set the block in motion. After the motion, a horizontal force of 60 N is applied to keep the block moving with constant speed. The coefficient of static friction and kinetic friction is-
- (A) $\mu_k = 0.3$ (B) $\mu_s = 0.3$ (C) $\mu_k = 0.4$ (D) $\mu_s = 0.4$



COMPREHENSION :

Figure shows an arrangement of pulleys and two blocks. All surfaces are frictionless. All pulleys and strings are massless. All strings are smooth and massless.



7. Choose the correct option(s)
- (A) Acceleration of block A is $\frac{2g}{9}$
 (B) Acceleration of block A is $\frac{g}{9}$
 (C) Normal reaction between A and ground is $\frac{17mg}{9}$
 (D) Normal reaction between A and ground is $2mg$

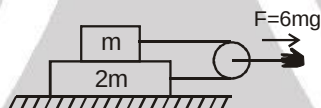
8. Choose the correct option(s)

- (A) Normal reaction between A and B is $\frac{2mg}{9}$
- (B) The tension in the string connected to the fixed wall is $\frac{4mg}{9}$
- (C) The acceleration of block B is in vertically downward direction
- (D) The acceleration of block B is making an angle $\tan^{-1}(2)$ with the vertical

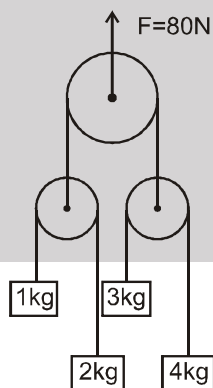
9. Now suppose the string connected to the fixed wall is cut, then

- (A) acceleration of A changes while that of B remains unchanged
- (B) normal reaction between A and B becomes zero
- (C) normal reaction between A and ground decreases by a factor of $\frac{17}{9}$
- (D) acceleration of block B decreases by a factor of $\frac{\sqrt{5}}{9}$

10. A block of mass m is placed on top of a block of mass $2m$ which in turn is placed on fixed horizontal surface. The coefficient of friction between all surfaces is $\mu = 1$. A massless string is connected to each mass and wraps halfway around a massless and frictionless pulley, as shown. The pulley is pulled by horizontal force of magnitude $F = 6mg$ towards right as shown. If the magnitude of acceleration of pulley is $\frac{X}{2} \text{ m/s}^2$, fill the value of X . (Take $g = 10 \text{ m/s}^2$)



11. In the diagram shown in figure, all pulleys are smooth and massless and strings are light. Match the blocks in column-I with their motion in column-II.



Column I

- (A) 1 kg block
- (B) 2 kg block
- (C) 3 kg block
- (D) 4 kg block

Column II

- (p) will remain stationary
- (q) will move down
- (r) will move up
- (s) has acceleration 5 m/s^2

DPP No. : B10 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

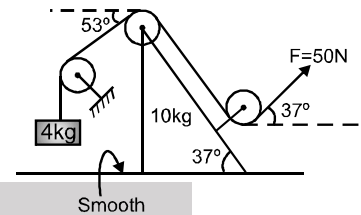
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

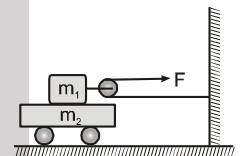
Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

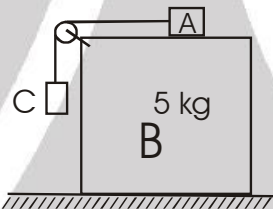
1. Wedge of 10 kg is free to move on horizontal surface. At the given instant, acceleration of wedge is (string and pulleys are ideal)
- (A) 2 m/s^2 towards right
(B) 2 m/s^2 towards left
(C) 1 m/s^2 toward left
(D) 1 m/s^2 toward right



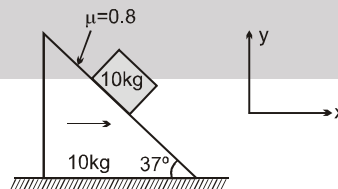
2. In figure the block M_1 has a mass 2kg, the trolley M_2 is 10 kg. Friction coefficient between M_1 and M_2 is 0.5 while the trolley moves frictionlessly on the floor. The the maximum F for which there is no slippage between M_1 and M_2 is (Assume ideal pulley and string)
- (A) 4N (B) 6N (C) 8N (D) 10N



3. All the blocks shown in the figure are at rest. The pulley is smooth and the strings are light. Coefficient of friction at all the contacts is 0.2. A frictional force of 10 N acts between A and B. The block A is about to slide on block B. The normal reaction and frictional force exerted by the ground on the block B is.

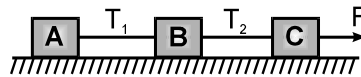


- (A) The normal reaction exerted by the ground on the block B is 110N
(B) The normal reaction exerted by the ground on the block B is 50 N
(C) the frictional force exerted by the ground on the block B is 20N
(D) the frictional force exerted by the ground on the block B is zero
4. A wedge is moving rightwards on which a block of mass 10kg is placed on it. Friction coefficient between the wedge and the block is 0.8. [take $g = 10 \text{ m/s}^2$]. Select correct alternative(s) among the following options.



- (A) If wedge is moving with constant velocity then friction acting on block is 64N.
(B) If wedge is moving with constant velocity then acceleration of block is zero.
(C) If wedge is moving with $= 2 (\hat{i}) \text{ m/s}^2$ then friction acting on block is 44N.
(D) If wedge is moving with $= 10 (\hat{i}) \text{ m/s}^2$ then friction is 20N, downward on the block along the inclined.
5. Suppose a body that is acted on by only two forces, is accelerated. For this situation, mark the incorrect statement(s) :
- (A) The body cannot move with constant speed
(B) The velocity can never be zero
(C) The sum of two forces cannot be zero
(D) The two forces must act in the same line

6. Three blocks are connected by light strings as shown in figure and pulled by a force $F = 60 \text{ N}$. If $m_A = 10 \text{ kg}$, $m_B = 20 \text{ kg}$ and $m_C = 30 \text{ kg}$, then :

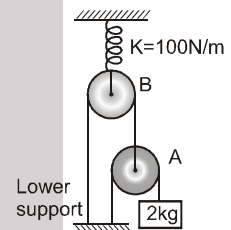


- (A) acceleration of the system is 2 m/s^2 (B) $T_1 = 10 \text{ N}$
 (C) $T_2 = 30 \text{ N}$ (D) $T_1 = 20 \text{ N}$ & $T_2 = 40 \text{ N}$

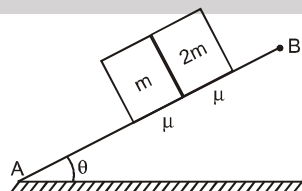
COMPREHENSION :

A projectile is thrown with velocity v making an angle θ with the horizontal. It just crosses the tops of two poles, each of height h , after 1s and 3s , respectively.

7. choose the correct option(s)
 (A) the time of flight of projectile is 4 sec .
 (B) the time of flight of projectile is 5 sec .
 (C) the maximum height attained by projectile is $2g$
 (D) the maximum height attained by projectile is $4g$
8. Which of the following is/are correct :
 (A) the height h of the poles is $\frac{5g}{2}$ (B) the height h of the poles is $\frac{3g}{2}$
 (C) the distance between the poles is $2v\cos\theta$ (D) the range of the projectile is $4v\cos\theta$
9. A mass 2kg is suspended as shown in figure. The system is in equilibrium in vertical plane. Assume pulleys to be massless. The force constant of the spring is 100 N/m . If the extension in the spring is given by $\frac{x}{10} \text{ m}$. Find x ?
 ($g = 10 \text{ m/s}^2$)



10. Two blocks of mass m and $2m$ are slowly just placed in contact with each other on a rough fixed inclined plane as shown. Initially both the blocks are at rest on inclined plane. The coefficient of friction between either block and inclined surface is μ . There is no friction between both the blocks. Neglect the tendency of rotation of blocks on the inclined surface. Column I gives four situations. Column II gives condition under which statements in column I are true. Match the statements in column I with corresponding conditions in column II.



Column I

- (A) The magnitude of acceleration of both blocks are same if
 (B) The normal reaction between both the blocks is zero if
 (C) The net reaction exerted by inclined surface on each block make same angle with inclined surface (AB) if
 (D) The net reaction exerted by inclined surface on block of mass $2m$ is double that of net reaction exerted by inclined surface on block of mass m if

Column II

- (p) $\mu = 0$
 (q) $\mu > 0$
 (r) $\mu > \tan\theta$
 (s) $\mu < \tan\theta$

DPP No. : B11 (JEE-MAIN)

Total Marks : 60

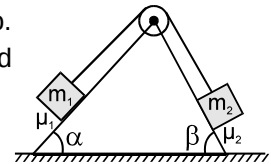
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

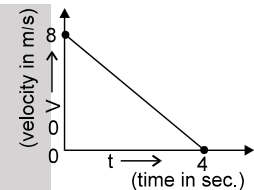
(3 marks 2 min.)

[60, 40]

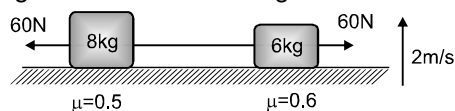
1. Two blocks of masses m_1 and m_2 connected by a string are placed gently over a fixed inclined plane, such that the tension in the connecting string is initially zero. The coefficient of friction between m_1 and inclined plane is μ_1 ; between m_2 and the inclined plane is μ_2 . The tension in the string shall continue to remain zero if
- (A) $\mu_1 > \tan \alpha$ and $\mu_2 < \tan \beta$ (B) $\mu_1 < \tan \alpha$ and $\mu_2 > \tan \beta$
 (C) $\mu_1 > \tan \alpha$ and $\mu_2 > \tan \beta$ (D) $\mu_1 < \tan \alpha$ and $\mu_2 < \tan \beta$



2. A block of mass 2 kg is given a push for a moment horizontally and then the block starts sliding over a horizontal plane. The graph shows the velocity-time graph of the motion. The co-efficient of sliding friction between the plane and the block is:
- (A) 0.02 (B) 0.20
 (C) 0.04 (D) 0.40

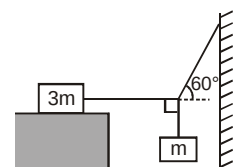


3. A body of mass m is kept on a rough fixed inclined plane of angle of inclination $\theta = 30^\circ$. It remains stationary. Then magnitude of force acting on the body by the inclined plane is equal to:
- (A) mg (B) $mg \sin \theta$ (C) $mg \cos \theta$ (D) none of these
4. A particle of mass 5 kg is moving on rough fixed inclined plane with constant velocity of 5 m/s as shown in the figure. Find the friction force acting on a body by plane.
- (A) 25 N (B) 20 N
 (C) 30 N (D) none of these
5. A block of mass 4 kg is kept on ground. The co-efficient of friction between the block and the ground is 0.80. An external force of magnitude 30 N is applied parallel to the ground. The resultant force exerted by the ground on the block is:
- (A) 40 N (B) 30 N (C) 0 N (D) 50 N
6. Two blocks of masses 8kg and 6kg are connected with a string & placed on a rough horizontal surface. Surface itself is accelerating up with 2 m/s^2 . Two forces 60 N each are acting on the two blocks as shown. Friction coefficient for 8kg is 0.5 & that for 6 kg is 0.6. Tension in the string is : ($g = 10 \text{ m/s}^2$)



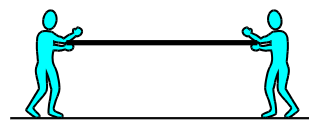
- (A) 60 N (B) 12 N (C) 43.2 N (D) 16.8 N

7. A mass m is supported as shown in the figure by ideal strings connected to a rigid wall and to a mass $3m$ at rest on a fixed horizontal surface. The string connected to larger mass is horizontal, that connected to smaller mass is vertical and the one connected to wall makes an angle 60° with horizontal. Then the minimum coefficient of static friction between the larger mass and the horizontal surface that permits the system to remain in equilibrium in the situation shown is:

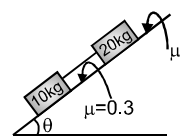


- (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{3\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2}$

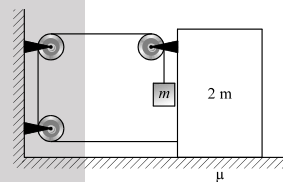
8. Two persons, pull each other through a massless rope in 'tug of war' game. Who will win?
 (A) one whose weight is more
 (B) one who pulls the rope with a greater force
 (C) one who applies more friction force (shear force) on ground
 (D) one who applies more normal force (compressive force) on ground



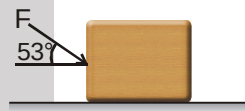
9. Value of θ is increased gradually from $\theta = 0$. At $\theta = \tan^{-1} \left(\frac{1}{2} \right)$ both the blocks just start slipping. Then value of μ_2 is: ($g = 10 \text{ m/s}^2$)
 (A) 0.5 (B) 0.4 (C) 0.6 (D) 0.3



10. In the system shown in figure the friction coefficient between ground and bigger block is μ . There is no friction between both the blocks. The string connecting both the block is light; all three pulleys are light and frictionless. Then the minimum limiting value of μ so that the system remains in equilibrium is



- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{2}$
11. A block of mass 20 kg is acted upon by a force $F = 30 \text{ N}$ at an angle 53° with the horizontal in downward direction as shown. The coefficient of friction between the block and the horizontal surface is 0.2. The friction force acting on the block by the ground is ($g = 10 \text{ m/s}^2$)
 (A) 40.0 N (B) 30.0 N (C) 18.0 N (D) 44.8 N



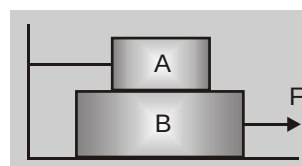
12. A uniform rope of length ℓ lies on a table. If the coefficient of friction is μ then the maximum length ℓ_1 of the part of this rope which can overhang from the edge of the table without sliding down is
 (A) $\frac{\ell}{\mu}$ (B) $\frac{\ell}{\mu + 1}$ (C) $\frac{\mu \ell}{1 + \mu}$ (D) $\frac{\mu \ell}{1 - \mu}$

13. A lift is moving downwards with an acceleration equal to the acceleration due to gravity. A body of mass M kept on the floor of the lift is pulled horizontally, if the coefficient of friction is μ , then the frictional resistance offered by the body is
 (A) Mg (B) μMg (C) $2\mu Mg$ (D) zero

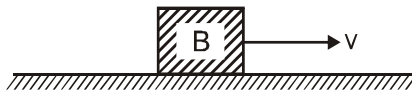
14. A block of metal is lying on the floor of a bus. The maximum acceleration which can be given to the bus so that the block may remain at rest, will be-
 (A) μg (B) $2 \mu g$ (C) $\mu^2 g$ (D) μg^2

15. A 60 kg body is pushed horizontally with just enough force to start it moving across a floor and the same force continues to act afterwards. The coefficient of static friction and sliding friction are 0.5 and 0.4 respectively. The acceleration of the body is :
 (A) 6 m/s^2 (B) 4.9 m/s^2 (C) 3.92 m/s^2 (D) 1 m/s^2

16. Two block A and B placed on a plane surface as shown in the figure. The mass of block A is 100 kg and that of block B is 200 kg. Block A is tied to a stand and block B is pulled by a force F . If the coefficient of friction between the surfaces of A and B is 0.2 and the coefficient of friction between B and the plane is 0.3 then for the motion of B the minimum value of F will be-
 (A) 700 N (B) 1050 N (C) 900 N (D) 1100 N

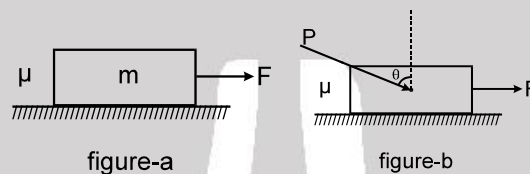


17. A block B is pushed momentarily along a horizontal surface with an initial velocity v . If μ is the coefficient of sliding friction between B and the surface, block B will come to rest after a time :

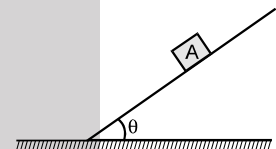
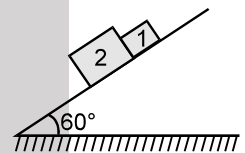


- (A) $\frac{v}{g\mu}$ (B) $\frac{g\mu}{v}$ (C) $\frac{g}{v}$ (D) $\frac{v}{g}$

18. The coefficient of friction between the block and the horizontal surface is μ . The block moves towards right under action of horizontal force F (figure - a). Sometime later another force P is applied to the block making an angle θ (such that $\tan\theta = \mu$) with vertical as shown in (figure - b). After application of force P , the acceleration of block shall



- (A) increase
(B) decrease
(C) remains same
(D) information insufficient for drawing inference.
19. In the figure shown if friction co-efficient of block 1 and 2 with inclined plane is $\mu_1 = 0.5$ and $\mu_2 = 0.4$ respectively, then find out the correct statement.
(A) both block will move together
(B) both block will move separately
(C) there is a non-zero contact force between two blocks
(D) none of these
20. A block A is kept on a rough inclined plane. Initially $\theta = 0$. The co-efficient of kinetic friction between A and the inclined plane is $\sqrt{3}$. The angle θ is increased slowly. When θ becomes 60° , then:
(A) A slides down with constant velocity
(B) A does not slide
(C) A slides down with constant non-zero acceleration
(D) A slides down with constant acceleration.



DPP No. : B12 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

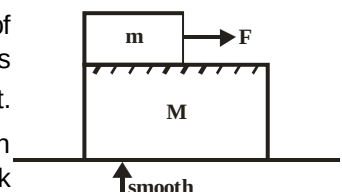
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

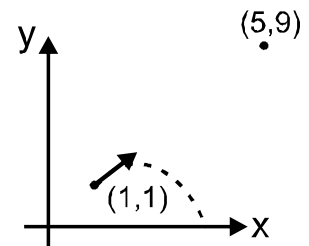
(8 marks 6 min.) [08, 06]

1. A plank having mass M is placed on smooth horizontal surface. Block of mass m is placed on it coefficient of friction between block and plank is $\mu_0 + kx$, where k is constant and x is relative displacement of block w.r.t. plank. A force F is applied on block where $F = at$, where $a = 10$; t is in second. Find t_0 when relative motion will occur between block and plank (use $g = 10 \text{ m/s}^2$).

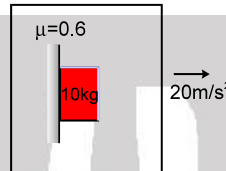


- (A) $\mu_0 M + \frac{\mu_0 M^2}{m}$ (B) $\mu_0 m + \frac{\mu_0 M^2}{m}$ (C) $\mu_0 m + \frac{\mu_0 m^2}{M}$ (D) $\mu_0 M + \frac{\mu_0 m^2}{M}$

2. A particle is projected from a point (1,1) aiming towards a point (5,9). It fell on ground along x axis after 1 sec. If it falls on x-axis then the x-co-ordinate of that point is ($g = 10 \text{ m/s}^2$)
- (A) 1 (B) 2
(C) 3 (D) 4



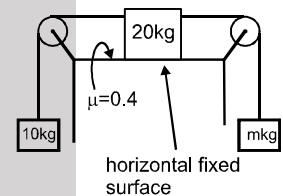
3. A car is accelerating on a horizontal road with acceleration $= 20 \text{ m/s}^2$. A box that is placed inside the car, of mass $m = 10 \text{ kg}$ is put in contact with the vertical wall as shown. The friction coefficient between the box and the wall is $\mu = 0.6$.



- (A) The acceleration (with respect to ground) of the box will be 20 m/sec^2
 (B) The friction force acting on the box will be 100 N
 (C) The contact force between the vertical wall and the box will be $100 \sqrt{5} \text{ N}$
 (D) The net contact force between the vertical wall and the box is only of electromagnetic in nature.

4. Value(s) of m for which system remains at rest (pulleys and strings are ideal) [$g = 10 \text{ m/s}^2$]

- (A) 1 kg
 (B) 2 kg
 (C) 18 kg
 (D) 20 kg

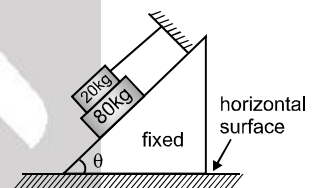


5. The system shown is in limiting equilibrium. The coefficient of friction for all contact surfaces is $\frac{1}{4}$.

(A) $\tan \theta = \frac{3}{8}$

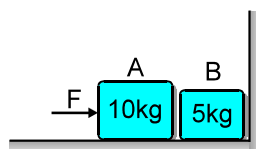
(B) Tension in the string $= \left(\frac{100}{3} g \sin \theta \right) \text{ N}$

- (C) Net frictional force on 80 kg block is $(80 g \sin \theta) \text{ N}$
 (D) Force exerted by 20 kg block on 80 kg block is $(20 g \cos \theta)$

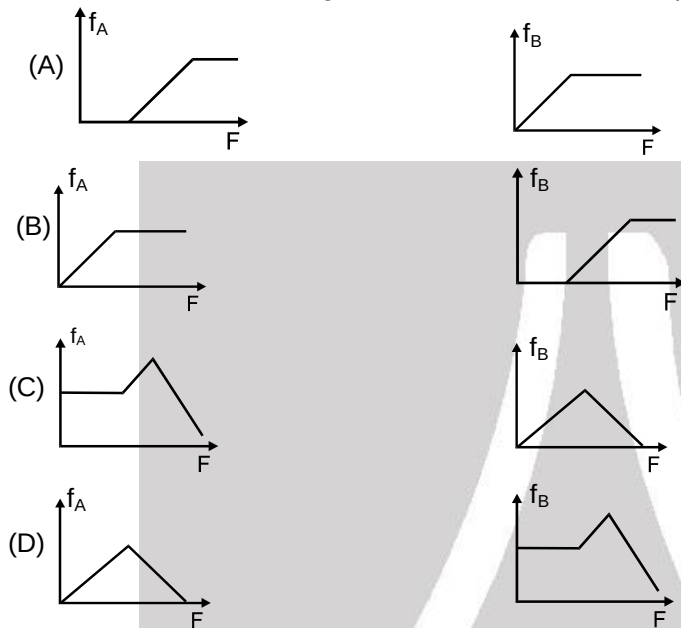


COMPREHENSION

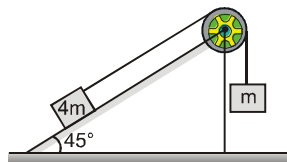
Two bodies A and B of masses 10 kg and 5 kg are placed very slightly separated as shown in figure. The coefficient of friction between the floor and the blocks is $\mu = 0.4$. Block A is pushed by an external force F . The value of F can be changed. When the welding between block A and ground breaks, block A will start pressing block B and when welding of B also breaks, block B will start pressing the vertical wall –



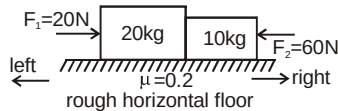
6. Choose the correct option(s)
 (A) If $F = 20$ N, the normal reaction between A and B is zero.
 (B) If $F = 45$ N, the normal reaction between A and B is 5 N
 (C) If $F = 65$ N, the block B presses the wall
 (D) If $F = 70$ N, the normal reaction by wall on the block B will be 10 N.
7. The force of friction acting on A and B varies with the applied force F according to curve :



8. If the vertical wall is removed and the force F is applied then choose the correct option(s)
 (A) If $F = 90$ N, the normal reaction between A and B will be 50 N
 (B) The minimum value of F , so that A and B just start moving together is 60 N
 (C) If $F = 120$ N the accelerations of the blocks will be 4 m/s^2
 (D) If $F = 120$ N the accelerations of the blocks will be $\frac{80}{15} \text{ m/s}^2$
9. The masses $4m$ and m are connected by a light string passing over a frictionless pulley fixed at inclined plane of inclination 45° as shown in figure. The coefficient of friction between $4m$ and inclined plane is $\frac{1}{\sqrt{2}}$. When two blocks are released the frictional force on block of mass $4m$ has magnitude $(x\sqrt{2} - 1)mg$ then the value of x is



10. Two blocks of masses 20 kg and 10 kg are kept on a rough horizontal floor. The coefficient of friction between both blocks and floor is $\mu = 0.2$. The surface of contact of both blocks are smooth. Horizontal forces of magnitude 20 N and 60 N are applied on both the blocks as shown in figure. Match the statement in column-I with the statements in column-II.



Column-I

- (P) Frictional force acting on block of mass 10 kg
 (Q) Frictional force acting on block of mass 20 kg
 (R) Normal reaction exerted by 20 kg block on 10 kg block
 (S) Net force on system consisting of 10 kg block and 20 kg block
 (A) P-1, Q-1, R-2, S-3
 (C) P-2, Q-2, R-1, S-1

Column-II

- (1) has magnitude 20 N
 (2) has magnitude 40 N
 (3) is zero
 (4) has magnitude 60 N
 (B) P-1, Q-1, R-3, S-3
 (D) P-2, Q-2, R-3, S-4

DPP No. : B13 (JEE-ADVANCED)

Total Marks : 43

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.8

(4 marks 2 min.) [28, 14]

Subjective Questions ('-1' negative marking) Q.9

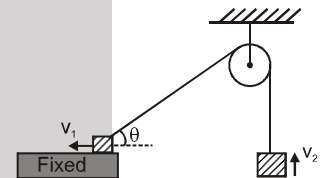
(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. The relation between v_1 and v_2 is :

- (A) $v_1 \sin \theta = v_2$
 (B) $v_1 \cos \theta = v_2$
 (C) $v_2 \cos \theta = v_1$
 (D) $v_2 \sin \theta = v_1$

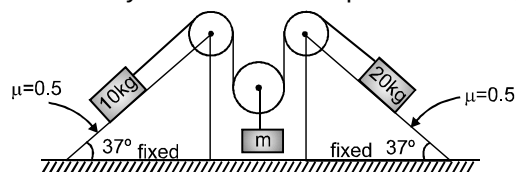


2. A variable force $F = 10t$ is applied to block B placed on a smooth surface. The coefficient of friction between A & B is 0.5. (t is time in seconds. Initial velocities are zero)

- (A) block A starts sliding on B at $t = 5$ seconds
 (B) the heat produced due to friction in first 5 seconds is 312.5J
 (C) the heat produced due to friction in 5 seconds is $(625/8)$ J
 (D) acceleration of A at 10 seconds is 5 m/s^2 .



3. In given arrangement, 10 kg and 20 kg blocks are at rest on two fixed inclined. All strings and pulleys are ideal. value(s) of m for which system remain in equilibrium are: ($g = 10 \text{ m/s}^2$)

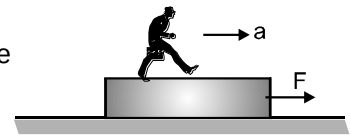


- (A) $m = 6 \text{ kg}$ (B) $m = 3 \text{ kg}$ (C) $m = 8 \text{ kg}$ (D) $m = 20 \text{ kg}$

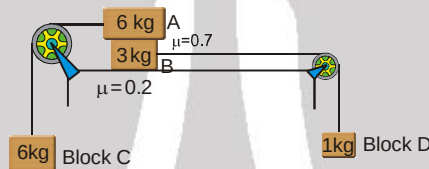
4. A plank of mass M is placed on a rough horizontal surface. A man of mass m walks on the plank with an acceleration ' a ' while the plank is also acted upon by a horizontal force F whose magnitude and direction can be adjusted to keep the plank at rest :

The coefficient of friction between plank and surface is μ . Choose the correct options.

- (A) The maximum value of F to keep plank at rest is $ma + \mu(M + m)g$
 (B) The minimum value of F to keep the plank at rest is $ma - \mu(M + m)g$
 (C) The direction of friction on ground due to plank will always be forward
 (D) If $F = ma$ then friction on man due to plank is also F and friction between plank and ground will be zero



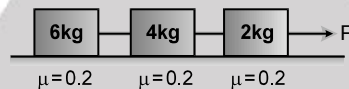
5. An arrangement of the masses and pulleys is shown in the figure. Strings connecting masses A and B with pulleys are horizontal and all pulleys and strings are light. Friction coefficient between the surface and the block B is 0.2 and between blocks A and B is 0.7. The system is released from rest. (use $g = 10 \text{ m/s}^2$)



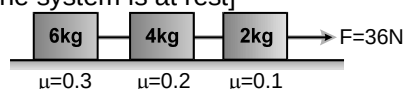
- (A) The magnitude of acceleration of the system is 2 m/s^2 and there is no slipping between block A and block B.
 (B) The magnitude of friction force between block A and block B is 42 N.
 (C) Acceleration of block C is 1 m/s^2 downwards.
 (D) Tension in the string connecting block B and block D is 12 N.

COMPREHENSION

Three blocks of masses 6 kg, 4 kg & 2 kg are pulled on a rough surface by applying a constant force F . The values of coefficient of friction between blocks & surface are shown in the figure.



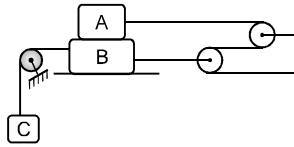
6. If $F = 20 \text{ N}$, then
 (A) the tension in the string connecting 4 kg & 2 kg is 16 N
 (B) the tension in the string connecting 6 kg & 4 kg is 8 N
 (C) the frictional force on 6 kg block is 8 N
 (D) If force is increased slightly above 20N, the 2kg block starts sliding
7. If $F = 36 \text{ N}$, then
 (A) the acceleration of 4 kg block will be 3 m/s^2
 (B) the acceleration of 4 kg block will be 1 m/s^2
 (C) the frictional force on 6 kg block will be 12 N
 (D) the tension in the string connecting 6kg and 4kg blocks will be 18 N
8. Now consider the following situation with blocks having different coefficients of friction as shown in figure. Blocks are pulled by applying $F = 36 \text{ N}$ force for $t = 0$ to $t = 15$ seconds and then the force is removed at $t = 15 \text{ sec}$. [Initially the system is at rest]



Choose the correct option(s) [for time $t = 15 \text{ sec}$ to $t = t_0$ (when all blocks stop)]

- (A) Blocks move with different accelerations and at least one of the strings get slacked before all blocks stop.
 (B) magnitude of acceleration of 4kg block is $7/3 \text{ m/s}^2$
 (C) The tension in the string connecting 6kg and 4kg blocks is 32 N
 (D) all the blocks will stop in $\frac{30}{7}$ seconds after removing the force

9. Block A of weight 500 N and block B of weight 700 N are connected by rope pulley system as shown. The largest weight C that can be suspended without moving block A and B is W. The coefficient of friction for all plane surfaces of contact is 0.3. The pulleys are ideal. Find $\frac{W}{90}$.



10. A particle is performing rectilinear motion on x axis, such that its x- coordinate varies with time as $x = \frac{t^3}{3} - 2t^2 + 3t + 5$ (m) where t is in second. In Column-I time instant is given and Column-II describes the motion of particle at particular instant. Match the proper entry from Column-II to Column-I.

Column-I	Column-II
(A) $t = \frac{1}{2}$ sec	(p) Particle is moving in positive x-direction and speeding up.
(B) $t = \frac{3}{2}$ sec	(q) Particle is moving in negative x-direction and slowing down.
(C) $t = \frac{5}{2}$ sec	(r) Particle is moving in negative x-direction and speeding up.
(D) $t = 4$ sec	(s) Particle is moving in positive x-direction and slowing down.
	(t) Acceleration and velocity of particle have same direction

DPP No. : B14 (JEE-MAIN)

Total Marks : 60

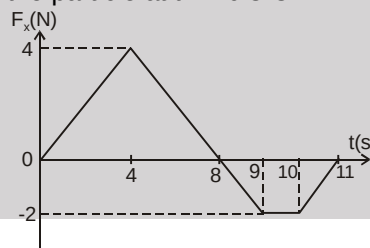
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

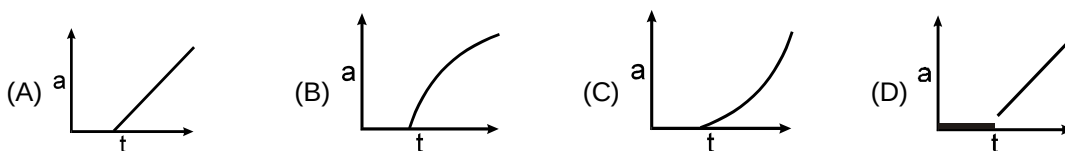
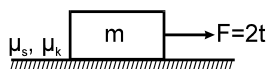
(3 marks 2 min.)

[60, 40]

1. A 2 kg toy car can move along x axis. Graph shows resultant force F_x , acting on the car which begins at rest at time $t = 0$. The velocity of the particle at $t = 10$ s is :

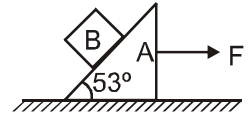


- (A) $-i$ m/s (B) $-1.5 i$ m/s (C) $6.5 i$ m/s (D) $13 i$ m/s
2. A force $F = 2t$ (where t is time in seconds) is applied at $t = 0$ sec to the block of mass m placed on a rough horizontal surface. The coefficient of static and kinetic friction between the block and surface are μ_s and μ_k respectively. Which of the following graphs best represents the acceleration vs time of the block. ($\mu_s > \mu_k$)



3. For the arrangement shown in figure a force F is applied on block A in horizontal direction as shown in figure so that its acceleration is 8 m/sec^2 . What is the acceleration of block B with respect to block A.

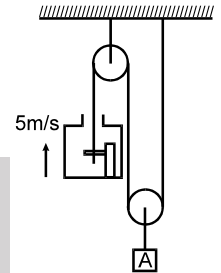
(Friction coefficient is 0.5 at all surfaces and take $g = 10 \text{ m/sec}^2$, $\tan 37^\circ = \frac{3}{4}$)



- (A) 10 m/sec^2 (B) $8\sqrt{3} \text{ m/sec}^2$ (C) $2\sqrt{41} \text{ m/sec}^2$ (D) $\frac{22}{5} \text{ m/sec}^2$

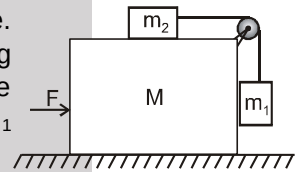
4. A motor is fixed inside a box which is moving upwards with velocity 5 m/s . String is winding at the rate 3 m/s . Then the velocity of block A will be:

- (A) 2.5 m/s downwards
(B) 5 m/s downwards
(C) 1 m/s downwards
(D) 2 m/s downwards



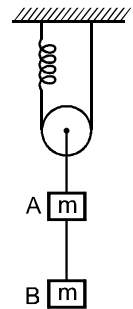
5. A large cubical shaped block of mass M rests on a fixed horizontal surface. Two blocks of mass m_1 and m_2 are connected by a light inextensible string passing over a light pulley as shown. Neglect friction everywhere. Then the constant horizontal force of magnitude F that should be applied to M so that m_1 and m_2 do not move relative to M is :

- (A) $F = \frac{m_2}{m_1} (m_1 + m_2 + M)g$ (B) $F = \frac{m_1}{m_2} (m_1 + m_2 + M)g$
(C) $F = \frac{m_1}{m_2} (m_1 + M)g$ (D) $F = \frac{m_2}{m_1} (m_1 + M)g$



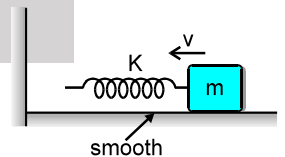
6. Two mass A and B, each of mass m , are initially in equilibrium as shown in figure. The acceleration of block A just after string between A and B is cut will be : (g = acceleration due to gravity).

- (A) $2g$ downward
(B) $2g$ upward
(C) g downward
(D) g upward

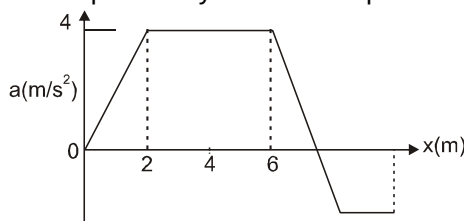


7. A block is attached with a spring and is moving towards a fixed wall with speed v as shown in figure. As the spring reaches the wall, it starts compressing. The work done by the spring on the wall during the process of compression is :

- (A) $\frac{1}{2} mv^2$ (B) mv^2
(C) Kmv (D) zero



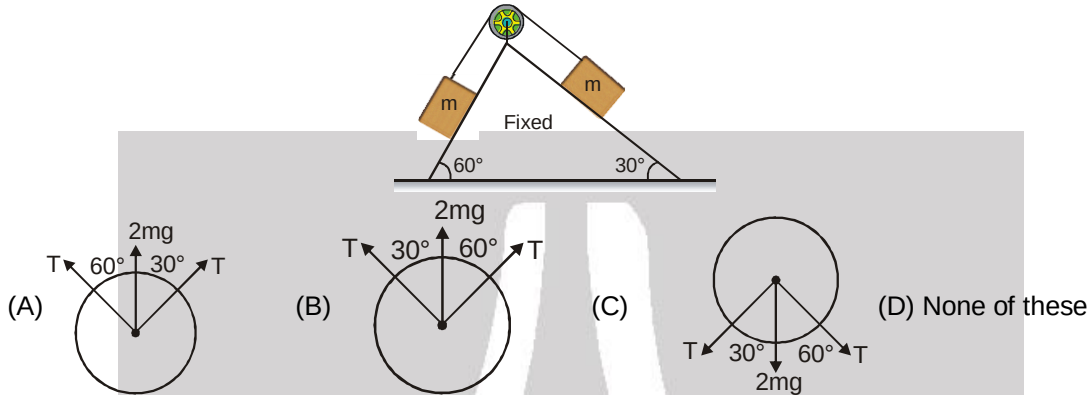
8. Graph shows the acceleration of a 3 kg particle as an applied force moves it from rest along x axis. The total work done by the force on the particle by the time the particle reaches $x = 6 \text{ m}$, is equal to



- (A) 20 J (B) 30 J (C) 40 J (D) 60 J

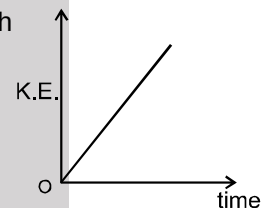
9. Inside a horizontally moving box, an experimenter (who is stationary relative to box) finds that when an object is placed on a smooth horizontal table and is released, it moves with an acceleration of 10 m/s^2 . In this box if 1 kg body is suspended with a light string, the tension in the string in equilibrium position. (w.r.t. experimenter) will be. (Take $g = 10 \text{ m/s}^2$)
 (A) 10 N (B) $10\sqrt{2} \text{ N}$ (C) 20 N (D) zero

10. Two blocks of same mass are attached from a string which goes over a smooth pulley mass of pulley is $2m$ then which diagram will represents force exerted by the axle of pulley on the pulley.



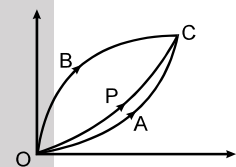
11. A body moves along a straight line and the variation of its kinetic energy with time is linear as depicted in the figure below :

- Then force acting on the body is :
 (A) directly proportional to velocity
 (B) inversely proportional to velocity
 (C) zero
 (D) constant



12. A particle is constrained to move from initial point 'O' to final point 'C' along three different smooth horizontal tracks namely OBC, OPC & OAC. If the particle moves under the influence of an external force F such that the initial and the final speeds are same, then:

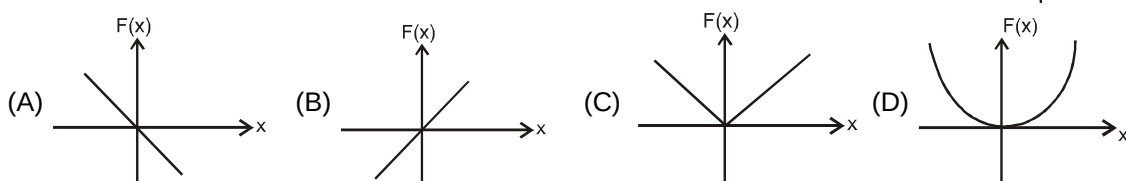
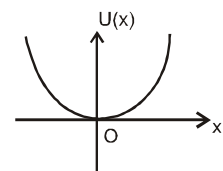
- (A) force F is conservative
 (B) force F cannot be conservative
 (C) there exists no closed path along which line integral of force F is zero
 (D) there necessarily exists a path along which line integral of force F is zero.

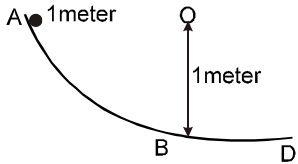
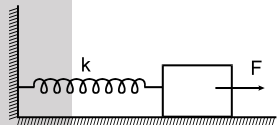
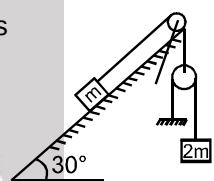


13. The work done by kinetic friction on a body :

- (A) is always negative (B) is always zero
 (C) may be +ve, -ve or zero (D) is always positive

14. Figure shows a plot of potential energy function $U(x) = kx^2$ where x = displacement and k = constant. Identify the correct conservative force function $F(x)$



15. In the track shown in figure section AB is a quadrant of a circle of 1 metre radius. A block is released at A and slides without friction until it reaches B. After B it moves on a rough horizontal floor and comes to rest at distance 3 metres from B. What is the coefficient of friction between floor and body ?
 (A) $1/3$ (B) $2/3$
 (C) $1/4$ (D) $3/8$
- 
16. The total work done on a particle is equal to the change in its kinetic energy.
 (A) always
 (B) only if the forces acting on the body are conservative
 (C) only if the forces acting on the body are gravitational
 (D) only if the forces acting on the body are elastic.
17. A ship of mass 3×10^7 kg, initially at rest is pulled by a force of 5×10^4 N through a distance of 3 m. Assuming that the resistance due to water is negligible, the speed of the ship is:
 (A) 1.5 m/s (B) 60 m/s (C) 0.1 m/s (D) 5 m/s
18. A block attached to a spring, pulled by a constant horizontal force, is kept on a smooth surface as shown in the figure. Initially, the spring is in the natural state. Then the maximum positive work that the applied force F can do is : [Given that spring does not break]
 (A) $\frac{F^2}{K}$ (B) $\frac{2F^2}{K}$ (C) ∞ (D) $\frac{F^2}{2K}$
- 
19. A body has kinetic energy E when projected at an angle of projection for maximum range. Its K.E. at the highest point of its path will be :
 (A) E (B) $\frac{E}{\sqrt{2}}$ (C) $\frac{E}{2}$ (D) zero
20. In the given figure inclined surface and pulleys are smooth. Strings and pulleys are massless. Acceleration of mass m is :
 (A) $\frac{7g}{9}$ (B) $\frac{7g}{11}$ (C) $\frac{7g}{18}$ (D) $\frac{7g}{20}$
- 

DPP No. : B15 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

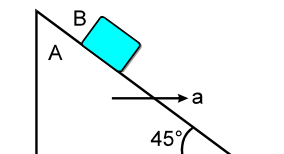
1. If the coefficient of friction between A and B is μ , the maximum horizontal acceleration of the wedge A for which B will remain at rest w.r.t the wedge is :

(A) μg

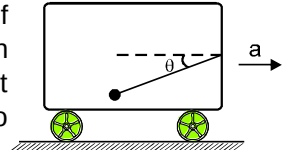
(B) $g \left(\frac{1+\mu}{1-\mu} \right)$

(C) $\frac{g}{\mu}$

(D) $g \left(\frac{1-\mu}{1+\mu} \right)$



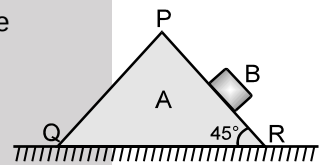
2. In the figure shown a particle of mass m is tied to a string and the other end of string is tied to the side wall of a carriage, which is accelerating horizontally with constant acceleration a . Assuming the string remains taut and the particle is not in contact with the base of carriage. Then : (Particle is at rest with respect to carriage)



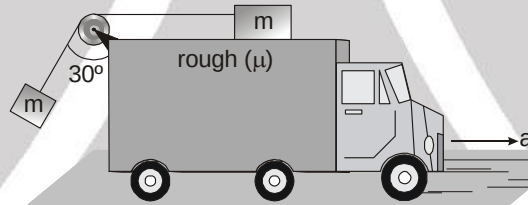
- (A) The angle θ that string makes with the horizontal is $\tan^{-1} \left(\frac{g}{a} \right)$
- (B) The angle θ that string makes with the horizontal is $\tan^{-1} \left(\frac{a}{g} \right)$
- (C) Tension in the string is $m\sqrt{g^2 + a^2}$
- (D) Tension in the string is $m\sqrt{g^2 - a^2}$

3. A block B of mass 0.6 kg slides down the smooth face PR of a wedge A of mass 1.7 kg which can move freely on a smooth horizontal surface. The inclination of the face PR to the horizontal is 45° . Then:

- (A) the acceleration of A is $3g/20$
- (B) the vertical component of the acceleration of B is $23g/40$
- (C) the horizontal component of the acceleration of B is $17g/40$
- (D) none of these



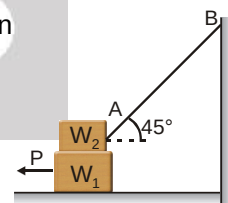
4. In the figure a truck is moving on a horizontal surface with acceleration a . Two blocks of equal masses m are supported on the truck as shown in figure. Given that when the block at the top surface is just about to slide, other block remains hanging at 30° from the vertical. In this system.



- (A) $a = \frac{g}{\sqrt{3}}$
- (B) $T = \frac{2}{\sqrt{3}} mg$
- (C) $\mu = \frac{5 - \sqrt{3}}{3\sqrt{3}}$
- (D) $T = \frac{\sqrt{3}}{2} mg$

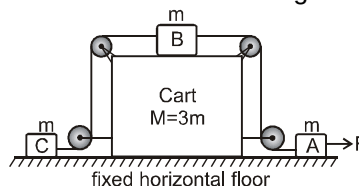
5. In the arrangement shown, $W_1 = 200 \text{ N}$, $W_2 = 100 \text{ N}$, $\mu = 0.25$ for all surfaces in contact. The block W_1 just slides under the block W_2 .

- (A) A pull of 50 N is to be applied on W_1
- (B) A pull of 90 N is to be applied on W_1
- (C) Tension in the string AB is 10 N
- (D) Tension in the string AB is $20\sqrt{2} \text{ N}$

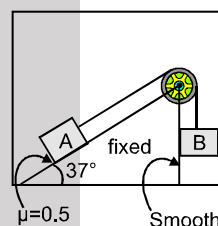


COMPREHENSION

A block B is placed over a cart which in turn lies over a smooth horizontal floor. Block A and block C are connected to block B with light inextensible strings passing over light frictionless pulleys fixed to the cart as shown. Initially the blocks and the cart are at rest. All the three blocks have mass m and the cart has mass M ($M = 3m$). Now a constant horizontal force of magnitude F is applied to block A towards right.



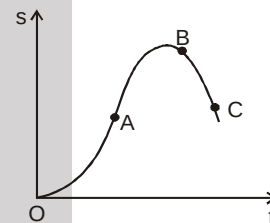
6. Assuming friction to be absent everywhere, choose the correct statement(s)
 (A) The acceleration of cart at the shown instant is zero
 (B) The acceleration of cart at the shown instant is $F/4m$
 (C) The acceleration of block B is $F/3m$
 (D) The acceleration of block C is $F/4m$.
7. Assuming friction to be absent everywhere, choose the correct statement(s)
 (A) Tension in the string connecting block B and C is $F/3$.
 (B) Tension in the string connecting block B and C is $2F/3$.
 (C) Tension in the string connecting block A and B is $2F/3$.
 (D) Tension in the string connecting block A and B is $F/3$.
8. Let the coefficient of friction between block B and cart is μ ($\mu > 0$) and friction is absent everywhere else. Then the values of force F applied to block A such that there is no relative acceleration between block B and cart is
 (A) μmg (B) $2 \mu mg$ (C) $3 \mu mg$ (D) $4 \mu mg$
9. Two blocks A and B are arranged in a lift as shown in figure. Lift is moving upward with a constant velocity 2m/sec . The pulley is ideal and string is ideal. Find the acceleration of block B with respect to ground.
 (Take $m_A = 10\text{ kg}$, $m_B = 5\text{ kg}$ and $g = 10\text{m/s}^2$)



10. The displacement-time graph of a body acted upon by some forces is shown in the figure. For this situation match the entries of column I with the entries of column II.

- Column I**
- (A) For OA, the total work done by all forces together
 (B) For OA, the work done by few of the acting forces
 (C) For AB, the work done by few of the acting forces
 (D) For BC, the work done by few of the acting forces.

- Column II**
- (p) always positive
 (q) always negative
 (r) can be positive
 (s) can be zero
 (t) can be negative



DPP No. : B16 (JEE-MAIN)

Total Marks : 60

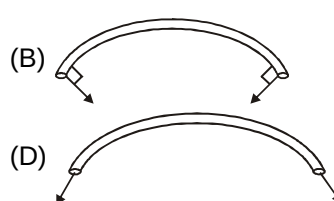
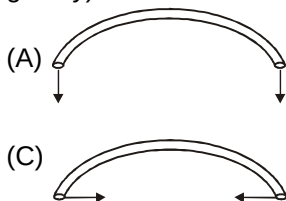
Max. Time : 40 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.20

(3 marks 2 min.)

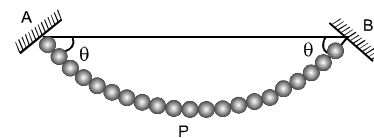
[60, 40]

1. A man holds a thin stick at its two ends and bends it in arc, like a bow without a string. Which of the following figures correctly show that directions of the forces exerted by him on the stick? (Neglect gravity)



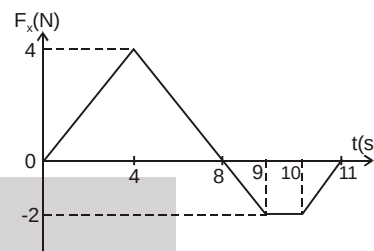
2. A flexible chain of mass M hangs between two fixed points A and B at the same level. The inclination of the chain with the horizontal at the two points of support is θ . The tension at the mid-point of the chain is :

(A) $\frac{Mg \cot \theta}{2}$ (B) $Mg \cot \theta$ (C) zero (D) $\frac{Mg}{2} (\sin \theta + \cos \theta)$



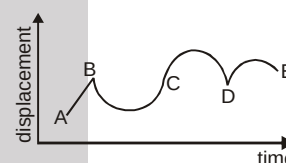
3. A 2 kg toy car can move along x axis. Graph shows resultant force F_x , acting on the car which begins at rest at time $t = 0$. The velocity of the car at $t = 10$ s is :

(A) -1 m/s
(B) -1.5 m/s
(C) 6.5 m/s
(D) 13 m/s



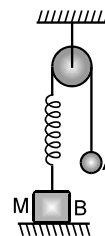
4. Figure shows the displacement of a particle moving along the x-axis as a function of time, the force acting on a particle is zero from

(A) AB (B) BC
(C) CD (D) DE



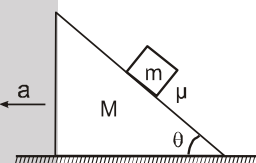
5. In the Figure, the ball A is released from rest when the spring is at its natural length. For the block B, of mass M to leave contact with the ground at some stage, the minimum mass of A must be:

(A) $2M$
(B) M
(C) $M/2$
(D) A function of M and the force constant of the spring.

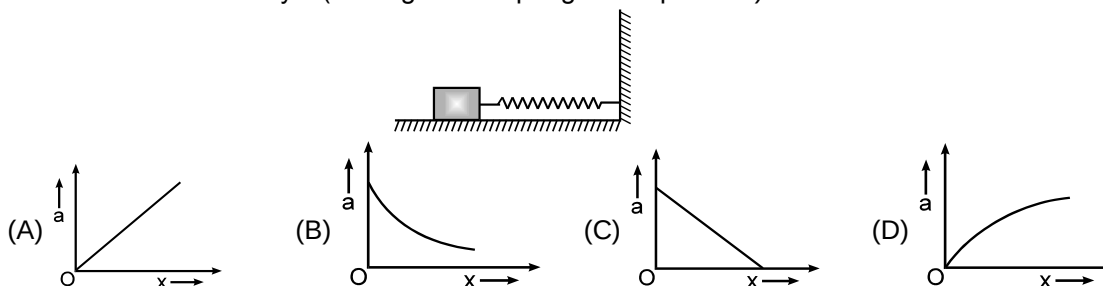


6. A block of mass m is at rest relative to the stationary wedge of mass M . The coefficient of friction between block and wedge is μ . The wedge is now pulled horizontally with acceleration 'a' as shown in figure. Then the minimum magnitude of 'a' for the friction between block and wedge to be zero is :

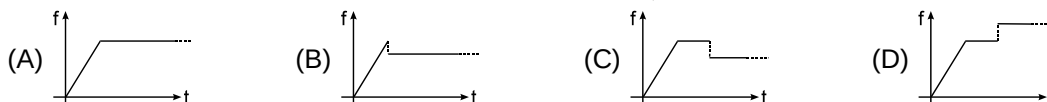
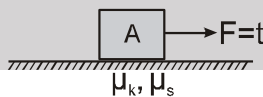
(A) $g \tan \theta$ (B) $\mu g \tan \theta$
(C) $g \cot \theta$ (D) $\mu g \cot \theta$



7. A light spring is compressed and placed horizontally between a vertical fixed wall and a block free to slide over a smooth horizontal table top as shown in the figure. The system is released from rest. The graph which represents the relation between the magnitude of acceleration 'a' of the block and the distance 'x' travelled by it (as long as the spring is compressed) is



8. A 20 kg body is pushed horizontally with just enough force to start it moving across a floor and the same force continues to act afterwards. The coefficient of static friction and sliding friction are 0.7 and 0.6 respectively. The acceleration of the body is :
 (A) 6 m/s^2 (B) 4.9 m/s^2 (C) 3.92 m/s^2 (D) 1 m/s^2
9. A particle moves in east with velocity of 15 m/s for 2 sec. then moves northward with 5 m/s for 8 sec. then average velocity of the particle is-
 (A) 1 m/s (B) 5 m/s (C) 7 m/s (D) 10 m/s
10. An aeroplane is flying at height 'h' with horizontal velocity u, the velocity of a dropped packet on reaching ground will be-
 (A) $\sqrt{u^2 + 2gh}$ (B) $\sqrt{2gh}$ (C) $2gh$ (D) $\sqrt{u^2 - 2gh}$
11. The angle between the vector $-\hat{i} + \hat{j}$ and $2\hat{i} + 3\hat{j}$ is :
 (A) $\cos \theta = \frac{1}{5}$ (B) $\sin \theta = \sqrt{\frac{25}{26}}$ (C) $\tan \theta = \sqrt{5}$ (D) none of these
12. The component of vector $2\hat{i} + 3\hat{j}$ along vector $-\hat{j} + 5\hat{i}$ is :
 (A) $\frac{7}{\sqrt{13}}$ (B) $\frac{7}{\sqrt{26}}$ (C) $\frac{13}{\sqrt{13}}$ (D) none of these
13. The resultant of two vectors of magnitudes 3 units and 4 units is 1 unit . What is the value of their dot product ?
 (A) -12 units (B) -7 units (C) -1 unit (D) zero
14. Which of the following is correct about kinetic friction ?
 (A) always acts in opposite direction of the applied force
 (B) total work done by this force on the system may be zero
 (C) direction of friction is always opposite to direction of velocity
 (D) it can increase velocity.
15. A force $F = t$ is applied to a block A as shown in figure, where t is time in seconds. The force is applied at $t = 0$ seconds when the system was at rest. Which of the following graph correctly gives the frictional force between A and horizontal surface as a function of time t. [Assume that $\mu_s > \mu_k$].



16. The minimum work done required to accelerate a truck on a horizontal road from rest to speed v
 (A) is less than that required to accelerate it from v to $2v$.
 (B) is equal than that required to accelerate it from v to $2v$.
 (C) is more than that required to accelerate it from v to $2v$.
 (D) may be any one of the above since it depends on the force acting on the truck and the distance over which it acts.

17. A 20 kg body is pushed with just enough force to start it moving across a floor and the same force continues to act afterwards. The co-efficient of static and kinetic friction are 0.6 & 0.2 respectively. The acceleration of the body is:
 (A) 6 m/s^2 (B) 1 m/s^2 (C) 2 m/s^2 (D) 4 m/s^2
18. An object is moving along a straight line path from P to Q under the action of a force $\vec{F} = (4\hat{i} - 3\hat{j} + 2\hat{k}) \text{ N}$. If the co-ordinate of P & Q in metres are (3, 2, -1) & (2, -1, 4) respectively. Then the work done by the force is:
 (A) -15 J (B) +15 J (C) 1015 J (D) $(4\hat{i} - 3\hat{j} + 2\hat{k})$
19. The work done by all the forces (external and internal) on a system equals the change in -
 (A) total energy (B) kinetic energy (C) potential energy (D) none of these
20. A body moves a distance of 10 m long a straight line under the action of force of 5N. If the work done is 25 joules, the angle which the force makes with the direction of motion of the body is -
 (A) 0° (B) 30° (C) 60° (D) 0°

DPP No. : B17 (JEE-ADVANCED)

Total Marks : 41

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.3

(3 marks 2 min.) [09, 06]

One or more than one options correct type ('-1' negative marking) Q.4 to Q.8

(4 marks 2 min.) [20, 10]

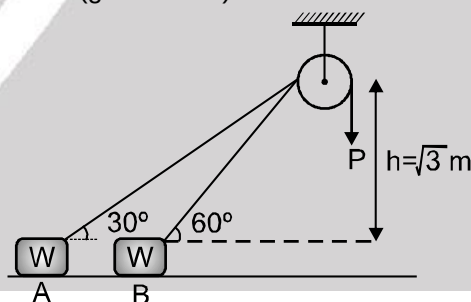
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

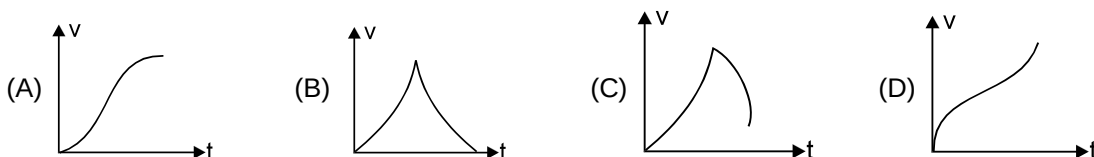
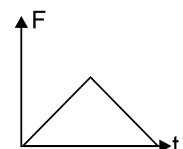
Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

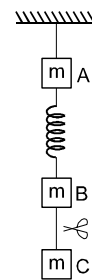
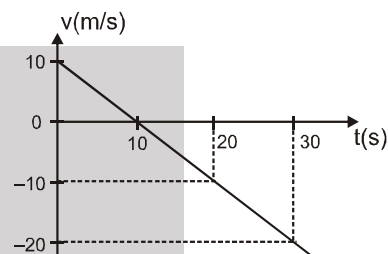
1. A block of weight W is dragged across the horizontal floor from A to B by the constant vertical force P acting at the end of the rope. The work done on the block by the force $P = (\sqrt{3} + 1) \text{ N}$ is, assuming that block does not lift off the floor. ($g = 10 \text{ m/s}^2$)



- (A) 4 (B) 8 (C) 12 (D) None of these
2. The force acting on a particle moving along a straight line varies with time as shown in the diagram. Which of the following graphs is best representative of its speed and time graphs. Initial velocity of the particle is zero.

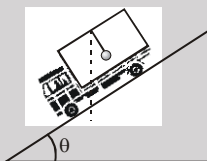


3. Which of the following quantities is/are not a frame dependent
 (A) Kinetic energy (B) Work done by a force on a particle
 (C) Potential energy (D) None of these
4. The potential energy of a particle is given by formula $U = 100 - 5x + 100x^2$, U and x are in SI units. If mass of the particle is 0.1 kg then magnitude of its acceleration
 (A) At 0.05 m from the origin is 50m/s^2
 (B) At 0.05 m from the mean position is 100m/s^2
 (C) At 0.05 m from the origin is 150m/s^2
 (D) At 0.05 m from the mean position is 200m/s^2
5. The velocity-time plot for a particle moving on a straight line is shown in Fig. in the time interval 0 to 30 seconds.
 (A) The particle has a constant acceleration
 (B) The particle has never turned around
 (C) The particle has zero displacement
 (D) The average speed in the interval 0 to 10 s is the same as average speed in the interval 10 to 20 seconds
6. System shown in figure is in equilibrium and at rest. The spring and strings are massless. Now the string between B and C is cut. Choose the correct option(s) just after cut. ($g = 10\text{ ms}^{-2}$).
 (A) acceleration of A is zero
 (B) acceleration of B is g in upward direction
 (C) acceleration of C is g in downward direction
 (D) none of these



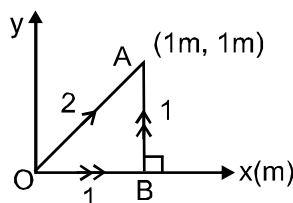
COMPREHENSION

A van accelerates uniformly down an inclined hill going from rest to 30 m/s in 6 s . During the acceleration, a toy of mass $m = 0.1\text{ kg}$ hangs by a light string from the van's ceiling. The acceleration is such that string remains perpendicular to the ceiling. (Take $g = 10\text{ m/s}^2$)

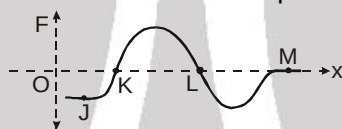


7. choose the correct option(s) :
 (A) The angle θ of the incline is 30° (B) The angle θ of the incline is 60°
 (C) The tension in string is $\frac{\sqrt{3}}{2}\text{ N}$ (D) The tension in string is 1 N
8. Choose the correct statement(s)
 (A) The friction force on van is 0
 (B) The friction force on van is $mg\cos\theta$
 (C) The direction of pseudo force on bob as seen by the driver is along up the inclined plane.
 (D) The direction of pseudo force on bob as seen by the driver is vertically upward.

9. A particle is moved under a force $\vec{F} = x \hat{j}$ N (x in meter) from O to A by two paths : path 1 is OBA and path 2 is straight line OA. If w_1 and w_2 are work done by the force $\vec{F}$ in two paths respectively, then find $\frac{w_1}{w_2}$.



10. A particle moving along x-axis is being acted upon by one dimensional conservative force F. In the F–x curve shown, four points J, K, L, M are marked on the curve. Column II gives different type of equilibrium for the particle at different positions. Column I gives certain positions on the force position graphs. Match the positions in Column-I with the corresponding nature of equilibrium at these positions.

**Column I**

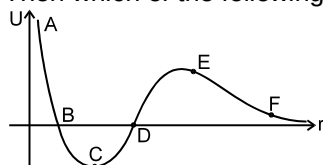
- (A) Point J is position of
(B) Point K is position of
(C) Point L is position of
(D) Point M is position of

Column II

- (p) Neutral equilibrium
(q) Unstable equilibrium
(r) Stable equilibrium
(s) No equilibrium

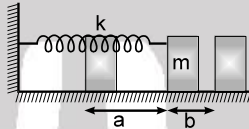
DPP No. : B18 (JEE-ADVANCED)**Total Marks : 43****Max. Time : 27 min.****Single choice Objective ('-1' negative marking) Q.1****(3 marks 2 min.) [03, 02]****One or more than one options correct type ('-1' negative marking) Q.2 to Q.8****(4 marks 2 min.) [28, 14]****Subjective Questions ('-1' negative marking) Q.9****(4 marks 5 min.) [04, 05]****Match the Following (no negative marking) Q.10****(8 marks 6 min.) [08, 06]**

1. A moving particle is acted upon by several forces $F_1, F_2, F_3, \dots$ etc. One of the force is chosen, say F_2 , then which of the following statement about F_2 will be true.
(A) Work done by F_2 will be negative if speed of the particle decreases.
(B) Work done by F_2 will be positive if speed of the particle increases
(C) Work done by F_2 will be equal to the work done by other forces if speed of the particle does not change
(D) If F_2 is a conservative force, then work done by all other forces will be equal to change in potential energy due to force F_2 when speed remains constant.
2. The given plot shows the variation of U, the potential energy of interaction between two particles with the distance separating them, r. Then which of the following statements are correct. :



- (A) B and D are equilibrium points
(B) C is a point of stable equilibrium
(C) The force of interaction between the two particles is attractive between points C and D and repulsive between points D and E on the curve.
(D) The force of interaction between the particles is repulsive between points E and F on the curve.

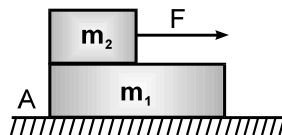
3. A particle is revolving in a circle with increasing its speed uniformly. Which of the followings are variable
 (A) centripetal acceleration (B) Tangential acceleration
 (C) Angular acceleration (D) None of these
4. One of the forces acting on a particle is conservative then which of the following statement(s) are true about this conservative force
 (A) Its work is zero when the particle moves exactly once around any closed path.
 (B) Its work equals the change in the kinetic energy of the particle
 (C) Then that particular force must be constant.
 (D) Its work depends on the end points of the motion, not on the path between.
5. The spring is compressed by a distance a and released. The block again comes to rest when the spring is elongated by a distance b . During this



- (A) work done by spring on the block $= \frac{1}{2} k (a + b)^2$
 (B) work done by spring on the block $= \frac{1}{2} k (a^2 - b^2)$
 (C) co-eff. of friction $= \frac{k(a - b)}{2mg}$
 (D) co-eff. of friction $= \frac{k(a + b)}{2mg}$
6. A particle starts from rest & moves in a circle of radius 3m. Angular acceleration of the particle varies as $\alpha = 4t \text{ rad/sec}^2$. Then:
 (A) particle is moving in a uniform circular motion
 (B) particle will travel a distance of 2 m in 1 sec
 (C) angle between the velocity & the acceleration vector will be 45° after 1 sec
 (D) at 1 sec, speed of the particle will be 6 m/sec.

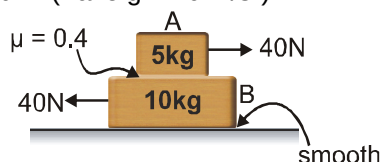
COMPREHENSION

Slab A is resting on a frictionless floor. Its mass is 35 kg. Another block of mass 7 kg is resting on it as shown in the diagram. The co-efficient of static friction between the block and slab is 0.5, while kinetic friction is 0.4. If a force of F N is applied to m_2 , : ($g = 10 \text{ m s}^{-2}$)



7. The values of force to cause m_2 to move with respect to m_1 is:
 (A) 72 N (B) 42 N (C) 35 N (D) none
8. In the above problem, if $m_1 = 10\text{kg}$, $m_2 = 40 \text{ kg}$, and applied force is 40N. Then
 (A) Acceleration of the block with respect to m_1 will be zero
 (B) Acceleration of the block with respect to m_1 will be 0.5m/s^2
 (C) Acceleration of the block with respect to ground will be 0.8m/s^2
 (D) Acceleration of the block with respect to ground will be zero

9. Two block A and B are arranged on a smooth horizontal plane as shown in figure. If two forces of equal magnitude act on the block as shown in figure. If the ratio of acceleration of block A and block B is $n : 1$ (where n is an integer). Then find n . (Take $g = 10 \text{ m/s}^2$)



10. Figure shows four situations in which a small block of mass ' m ' is released from rest (with respect to smooth fixed wedge) as shown in figure. Column-II shows work done by normal reaction with respect to an observer who is stationary with respect to ground till block reaches at the bottom of inclined wedge, match the appropriate column (Assume that there is infinite friction between block and floor of cabin) :

Column-I	Column-II
(A)	(p) Positive
(B)	(q) Negative
(C)	(r) equal to mgh in magnitude
(D)	(s) equal to zero
	(t) equal to $\sqrt{2} \text{ mgh}$ in magnitude

DPP No. : B19 (JEE-MAIN)

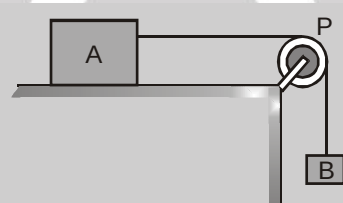
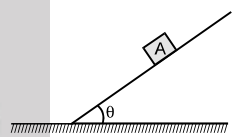
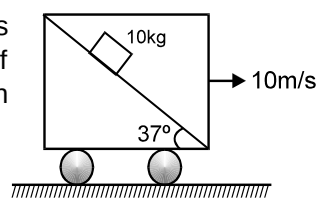
Total Marks : 60

Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

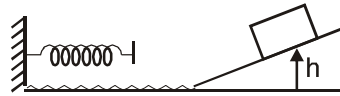
(3 marks 2 min.)

[60, 40]

1. If $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ & $\vec{B} = 3\hat{i} - 2\hat{j} + \hat{k}$, then the area of parallelogram formed with $\vec{A}$ and $\vec{B}$ as the sides of the parallelogram is :
 (A) $\sqrt{3}$ (B) $8\sqrt{3}$ (C) 64 (D) 0
2. Two forces of magnitudes P & $\sqrt{3}P$ act at right angles to each other. Their resultant makes an angle θ with P . Which of the following gives the correct value of θ ?
 (A) 30° (B) 45° (C) 60° (D) none of these
3. If the angle between two vectors is 60° , then $\frac{\vec{A} \cdot \vec{B}}{|\vec{A} \times \vec{B}|}$ is :
 (A) $\frac{1}{\sqrt{3}}$ (B) $\sqrt{3}$ (C) 1
 (D) cannot be determined because values of $|\vec{A}|$ and $|\vec{B}|$ are not given.
4. The blocks A and B are arranged as shown in the figure. The pulley is frictionless. The mass of A is 10 kg. The coefficient of friction between block A and horizontal surface is 0.20. The minimum mass of B to start the motion will be-

 (A) 2 kg (B) 0.2 kg (C) 5 kg (D) 10 kg
5. A block A is kept on a rough inclined plane. Initially $\theta = 0$. The co-efficient of kinetic friction between A and the inclined plane is $\sqrt{3}$. The angle θ is increased slowly. When θ becomes 60° , then:
 (A) A slides down with constant velocity
 (B) A does not slide
 (C) A slides down with constant non-zero acceleration
 (D) A slides down with constant acceleration.

6. The momentum of a body is increased by 50%. The K.E of the body will be increased by -
 (A) 50% (B) 125% (C) 330% (D) 400%
7. A block of mass 10 kg is released on a fixed wedge inside a cart which is moved with constant velocity 10 m/s towards right. Take initial velocity of block with respect to cart zero. Then work done by normal reaction (with respect to ground) on block in two seconds will be: ($g = 10 \text{ m/s}^2$).
 (A) zero (B) 960 J
 (C) 1200 J (D) none of these




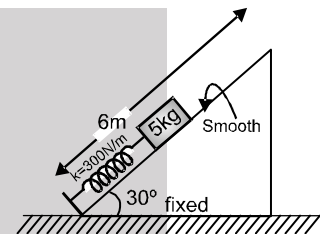
8. A block of mass m starts at rest at height h on a frictionless inclined plane. The block slides down the plane, travels across a rough horizontal surface with coefficient of kinetic friction μ , and compresses a spring with force constant k a distance x before momentarily coming to rest. Then the spring extends and the block travels back across the rough surface, sliding up the plane. The block travels a total distance d on rough horizontal surface. The correct expression for the maximum height h' that the block reaches on its return is:



- (A) $mgh' = mgh - \mu mgd$
 (B) $mgh' = mgh + \mu mgd$
 (C) $mgh' = mgh + \mu mgd + kx^2$
 (D) $mgh' = mgh - \mu mgd - kx^2$

9. A block of mass 5 kg is released from rest when compression in spring is 2m. Block is not attached with the spring and natural length of the spring is 4m. Maximum height of block from ground is :
 ($g = 10 \text{ m/s}^2$)

- (A) 5.5 m
 (B) 4.5 m
 (C) 6 m
 (D) 7.5 m

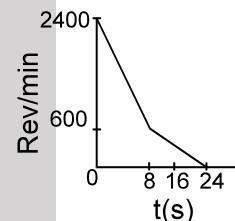


10. A particle moves along a circle of radius R with a constant angular speed ω . Its displacement (only magnitude) in time t will be

- (A) ωt
 (B) $2R \cos \omega t$
 (C) $2R \sin \omega t$
 (D) $2R \sin \frac{\omega t}{2}$

11. A table fan rotating at a speed of 2400 rpm is switched off and the resulting variation of the rpm with time is shown in the figure. The total number of revolutions of the fan before it come to rest is :

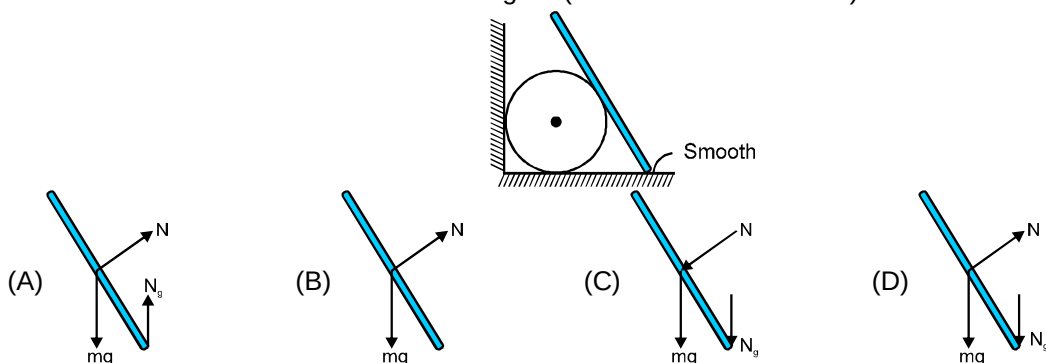
- (A) 420
 (B) 280
 (C) 190
 (D) 16800



12. A body of mass m is moving in a circle of radius r with a constant speed v . The force on the body is mv^2/r and is directed towards the centre. What is the work done by the force in moving the body half the circumference of the circle.

- (A) $\frac{mv^2}{r} \times \pi r$
 (B) $\frac{1}{2} mv^2$
 (C) mv^2
 (D) zero

13. F.B.D. of the rod of mass M as shown in figure (All surfaces are smooth).



14. Which of the following is not the unit of length :

- (A) micron
 (B) light year
 (C) angstrom
 (D) radian

15. A force $\vec{F} = -k(\hat{y}\hat{i} + \hat{x}\hat{j})$, where k is a positive constant, acts on a particle moving in the xy plane. Starting from the origin, the particle is taken along the positive x -axis to the point $(a, 0)$ and then parallel to the y -axis to the point (a, a) . The total work done by the force on the particle is
 (A) $-2ka^2$ (B) $2ka^2$ (C) $-ka^2$ (D) ka^2
16. A stone of mass 0.5 kg tied with a string of length 1 metre is moving in a horizontal circular path with a speed of 4 m/sec . The tension acting on the string in newton is :
 (A) 2 (B) 8 (C) 0.2 (D) 0.8
17. A mass is supported on a frictionless horizontal surface. It is attached to a string and rotates about a fixed centre at an angular velocity ω_0 . If the length of the string and angular velocity are doubled, the tension in the string which was initially T_0 is now -
 (A) T_0 (B) $T_0/2$ (C) $4T_0$ (D) $8T_0$
18. A heavy mass is attached to a thin wire and is whirled in a vertical circle. The wire is most likely to break.
 (A) When the mass is at the height point of the circle
 (B) When the mass is at the lowest point of the circle
 (C) When the wire is horizontal
 (D) At an angle of $\cos^{-1}(1/3)$ from the upward vertical
19. An engine exerts a force $\vec{F} = (20\hat{i} - 3\hat{j} + 5\hat{k})\text{N}$ and moves with velocity $\vec{v} = (6\hat{i} + 20\hat{j} - 3\hat{k})\text{ m/s}$. The power of the engine (in watt) is :
 (A) 45 (B) 75 (C) 20 (D) 10
20. The formula for centripetal acceleration in a circular motion is.
 (A) $\vec{\alpha} \times \vec{r}$ (B) $\vec{\omega} \times \vec{v}$ (C) $\vec{\alpha} \times \vec{v}$ (D) $\vec{\omega} \times \vec{r}$

DPP No. : B20 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

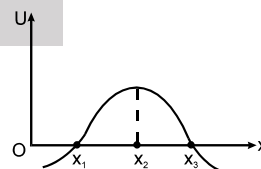
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

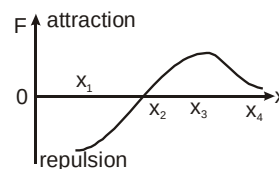
(8 marks 6 min.) [08, 06]

1. In the figure shown the potential energy U of a particle is plotted against its position ' x ' from origin. Then which of the following statement is correct. A particle at:
 (A) x_1 is in stable equilibrium (B) x_2 is in stable equilibrium
 (C) x_3 is in stable equilibrium (D) none of these



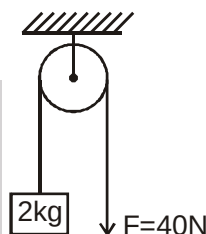
2. A particle is moving in a circle. Choose the correct option.
 (A) The resultant force on the particle must be towards the centre.
 (B) The resultant force may be towards the centre.
 (C) The direction of the angular acceleration and the angular velocity must be the same.
 (D) The cross product of the tangential acceleration and the angular velocity will be zero.

3. The diagram given shows how the net interaction force (conservative) between two particles A and B is related to the distance between them varies from x_1 to x_4 . Then:



- (A) potential energy of the system increases from x_1 to x_2 .
 (B) potential energy of the system increases from x_2 to x_3 .
 (C) potential energy of the system increases from x_3 to x_4 .
 (D) KE. increases from x_1 to x_2 and decreases from x_2 to x_3 .

4. A block of mass 2 kg is hanging over a smooth and light pulley through a light string. The other end of the string is pulled by a constant force $F = 40$ N. At $t = 0$ the system is at rest as shown. Then in the time interval from $t = 0$ to $t = \frac{2}{\sqrt{10}}$ seconds,

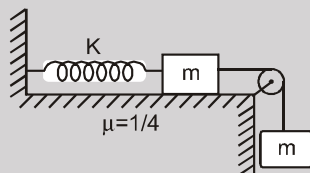


pick up the correct statement (s) : ($g = 10 \text{ m/s}^2$)

- (A) tension in the string is 40 N
 (B) work done by gravity is -20 J
 (C) work done by tension on block is 80 J
 (D) None of these
5. There are two massless springs A and B of spring constant K_A and K_B respectively and $K_A > K_B$. If W_A and W_B be denoted as work done on A and work done on B respectively, then
 (A) If they are compressed to same distance, $W_A > W_B$
 (B) If they are compressed by same force (upto equilibrium state) $W_A < W_B$
 (C) If they are compressed by same distance, $W_A = W_B$
 (D) If they are compressed by same force (upto equilibrium state) $W_A > W_B$

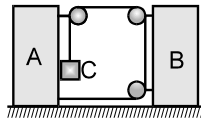
COMPREHENSION

Consider the system shown below, with two equal masses m and a spring with spring constant K . The coefficient of friction between the left mass and horizontal table is μ and the pulley is frictionless. The string connecting both the blocks is massless and inelastic. The system is held with the spring at its unstretched length and then released.



6. The extension in spring when the masses come to momentary rest for the first time may be
 (A) $\frac{3}{2} \frac{mg}{K}$ (B) $\frac{mg}{2K}$ (C) $3 \frac{mg}{K}$ (D) $\frac{2mg}{K}$
7. The values of μ for which the system remains at rest once it has stopped for the first time is
 (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$
8. If the string connecting both the masses is cut just at the instant both masses came to momentary rest for the first time, then maximum compression of spring during resulting motion is x_0 and maximum speed of block is v_0 (Take $\mu = 1/4$)
 (A) $x_0 = \frac{2mg}{3K}$ (B) $x_0 = \frac{mg}{K}$ (C) $v_0 = \left(\sqrt{\frac{3m}{2K}} \right) \cdot g$ (D) $v_0 = \left(\sqrt{\frac{3m}{4K}} \right) \cdot g$

9. In the arrangement shown in Figure mass of blocks A, B and C is 18.5 kg, 8 kg and 1.5 kg respectively. All the surfaces are smooth. System is released from rest at $t = 0$ & pulleys are light & frictionless. If acceleration of block C $\frac{150\sqrt{n}}{41}$. Then find n. (where n is an integer) ($g = 10 \text{ m/s}^2$)



10. Motion of particle is described in column-I. In column-II, some statements about work done by forces on the particle from ground frame is given. Match the particle's motion given in column-I with corresponding possible work done on the particle in some time interval given in column-II.

Column-I

- (A) A particle is moving in horizontal circle
(B) A particle is moving in vertical circle with uniform speed
(C) A particle is moving in air (projectile motion without any air resistance) under gravity
(D) A particle is attached to roof of moving train on inclined surface.

Column-II

- (p) Total work done by all the forces may be positive
(q) Total work done by all the forces may be negative
(r) Total work done by all the forces must be zero
(s) Total work done by gravity may be positive.

DPP No. : B21 (JEE-MAIN)

Total Marks : 60

Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

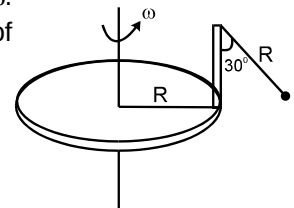
(3 marks 2 min.)

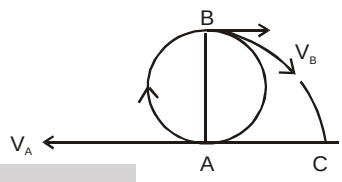
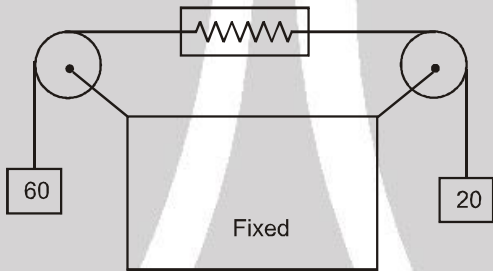
[60, 40]

1. The relation between an angular velocity, the position vector and linear velocity of a particle moving in a circular path is.
(A) $\vec{\omega} \times \vec{r} = \vec{v}$ (B) $\vec{\omega} \cdot \vec{r} = \vec{v}$ (C) $\vec{r} \times \vec{\omega} = \vec{v}$ (D) $\vec{\omega} \cdot \vec{r} = \vec{v}$
2. A particle of mass m under the action of a force F moves in a circular path of radius ' r ' with a constant speed. Its speed will be.
(A) $\sqrt{rF/m}$ (B) $\sqrt{mrF}$ (C) $\sqrt{F/r}$ (D) F/mr
3. In a simple pendulum, the breaking strength of the string is double the weight of the bob. The bob is released from rest when the string is horizontal. The string breaks when it makes an angle θ with the vertical.
(A) $\theta = \cos^{-1} \frac{1}{3}$ (B) $\theta = 60^\circ$ (C) $\theta = \cos^{-1} \frac{2}{3}$ (D) $\theta = 0$
4. A disc of radius R has a light pole fixed perpendicular to the disc at the circumference which in turn has a pendulum of length R attached to its other end as shown in figure. The disc is rotated with a constant angular speed ω . The string is making an angle 30° with the rod. Then the angular speed ω of disc is:

(A) $\left(\frac{\sqrt{3}g}{R}\right)^{1/2}$
(C) $\left(\frac{g}{\sqrt{3}R}\right)^{1/2}$

(B) $\left(\frac{\sqrt{3}g}{2R}\right)^{1/2}$
(D) $\left(\frac{2g}{3\sqrt{3}R}\right)^{1/2}$



5. A hollow sphere has radius 6.4 m. Minimum velocity required by a motor cyclist at bottom to complete the circle will be :
 (A) 17.7 m/s (B) 10.2 m/s (C) 12.4 m/s (D) 16.0 m/s
6. A motorcycle is going on an overbridge of radius R. The driver maintains a constant speed. As the motorcycle is ascending on the overbridge, the normal force on it :
 (A) Increases (B) Decreases (C) Remains the same (D) Fluctuates
7. A body is tied up by a string of length ℓ and rotated in vertical circle at minimum speed. When it reaches at highest point string breaks and body moves on a parabolic path in presence of gravity according to fig. In the plane of point A, value of horizontal range AC will be -
 (A) $x = \ell$ (B) $x = 2\ell$ (C) $x = \sqrt{2\ell}$ (D) $x = 2\sqrt{2\ell}$
- 
8. Find the reading of the spring balance (massless)
- 
- (A) 600N (B) 300N (C) 200N (D) 400N
9. The time in which a force of 2 N produces a change in momentum of $0.4 \text{ kg}\cdot\text{ms}^{-1}$ in the body is :
 (A) 0.2 s (B) 0.02 s (C) 0.5 s (D) 0.05 s
10. A boat moves relative to water with a velocity which is n times the river flow velocity. At what angle to the stream direction must be boat move to minimize drifting?
 (A) $\frac{\pi}{2}$ (B) $\sin^{-1}\left(\frac{1}{n}\right)$ (C) $\frac{\pi}{2} + \sin^{-1}\left(\frac{1}{n}\right)$ (D) $\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{n}\right)$
11. A car with a vertical wind shield moves along in a rain storm at speed of 40 km/hr. The rain drops fall vertically with a terminal speed of 20 m/sec. The angle at which the rain drops strike the wind shield is-
 (A) $\tan^{-1}\left(\frac{5}{9}\right)$ (B) $\tan^{-1}\left(\frac{9}{5}\right)$ (C) $\tan^{-1}\left(\frac{3}{2}\right)$ (D) $\tan^{-1}\left(\frac{2}{3}\right)$
12. A person is standing in an elevator. In which situation he finds his weight less than actual when :
 (A) The elevator moves upward with constant acceleration
 (B) The elevator moves downward with constant acceleration
 (C) The elevator moves upward with uniform velocity
 (D) The elevator moves downward with uniform velocity.
13. A projectile is fired at an angle of 45° with the horizontal. Elevation angle of the projectile at its highest point as seen from the point of projection is :
 (A) 60° (B) $\tan^{-1}\left(\frac{1}{2}\right)$ (C) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (D) 45°
14. A particle has initial velocity $(2\vec{i} + 3\vec{j})$ and acceleration $(0.3\vec{i} + 0.2\vec{j})$. The magnitude of velocity after 10 seconds will be :
 (A) $3\sqrt{2}$ units (B) $5\sqrt{2}$ units (C) 5 units (D) 9 units

15. A car of mass m starts from rest and accelerates so that the instantaneous power delivered to the car has a constant magnitude P_0 . The instantaneous velocity of this car is proportional to :
- (A) $t^2 P_0$ (B) $t^{1/2}$ (C) $t^{-1/2}$ (D) $\frac{t}{\sqrt{m}}$
16. A particle is projected up the inclined such that its component of velocity along the incline is 10 m/s. Time of flight is 2 sec and maximum height above the incline is 5 m. Then velocity of projection will be:
- (A) 10 m/s (B) $10\sqrt{2}$ m/s (C) 5 m/s (D) none of these
17. The constant quantity in uniform circular motion is.
- (A) linear speed (B) centripetal force (C) acceleration (D) momentum
18. A wheel is of diameter 1m. If it makes 30 revolutions/sec., then the linear speed of a point on its circumference will be.
- (A) 30π m/s (B) π m/s (C) 60π m/s (D) $\pi/2$ m/s
19. The angular velocity of the second's needle in watch is.
- (A) $\frac{\pi}{30}$ (B) 2π (C) π (D) $\frac{60}{\pi}$
20. An object of mass 5kg falls from rest through a vertical distance of 20 m and reaches a velocity of 10 m/s. How much work is done by the push of the air on the object? ($g = 10 \text{ m/s}^2$) -
- (A) 350 J (B) 750 J (C) 200 J (D) 300 J

DPP No. : B22 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

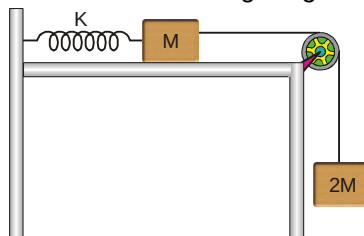
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. A force which varies with time t as $\vec{F} = (3t\hat{i} + 5\hat{j})\text{N}$ acts on a body due to which its position varies with time t as $\vec{s} = (2t^2\hat{i} - 5\hat{j})$ where t is in seconds. Work done by this force in initial 2s is:
- (A) 23 J (B) 32 J (C) zero (D) can't be obtained
2. Two blocks, of masses M and $2M$, are connected to a light spring of spring constant K that has one end fixed, as shown in figure. The horizontal surface and the pulley are frictionless. The blocks are released from rest when the spring is not deformed. The string is light. Choose the incorrect option



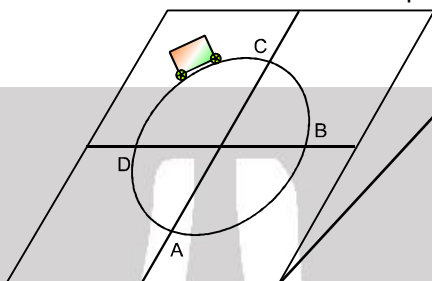
- (A) Maximum extension in the spring is $\frac{4Mg}{K}$.
- (B) Maximum kinetic energy of the system is $\frac{2M^2g^2}{K}$
- (C) Maximum energy stored in the spring is four times that of maximum kinetic energy of the system.
- (D) When kinetic energy of the system is maximum, energy stored in the spring is $\frac{4M^2g^2}{K}$

3. Which of the following relations are always true?

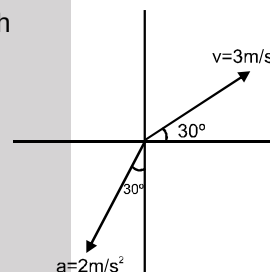
$\vec{v}$ = velocity, $\vec{a}$ = acceleration, $K = \frac{1}{2}mv^2$ = Kinetic energy

(A) $\frac{dK}{dt} = m\vec{v} \cdot \vec{a}$ (B) $\frac{d|\vec{v}|}{dt} = \frac{\vec{a} \cdot \vec{v}}{|\vec{v}|}$ (C) $\frac{d|\vec{v}|}{dt} = \left| \frac{d\vec{a}}{dt} \right|$ (D) $|\Delta\vec{v}| = \left| \int_{t_1}^{t_2} \vec{a} dt \right|$

4. A car is moving along a circle with constant speed on an inclined plane as shown in diagram. Then friction force on car will be in horizontal direction at least at one point :



- (A) in portion 'AB' including point A and B (B) in portion 'BC' including point B and C
(C) in portion 'CD' including point C and D (D) in portion 'DA' including point D and A
5. Initial velocity and acceleration of a particle are as shown in the figure. Acceleration vector of particle remain constant. Then radius of curvature of path of particle :
- (A) is 9m initially
(B) is $\frac{9}{\sqrt{3}}$ m initially
(C) will have minimum value of $\frac{9}{8}$ m
(D) will have minimum value $\frac{3}{8}$ m

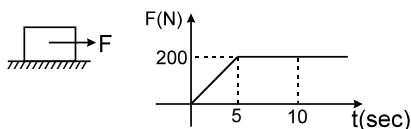


COMPREHENSION

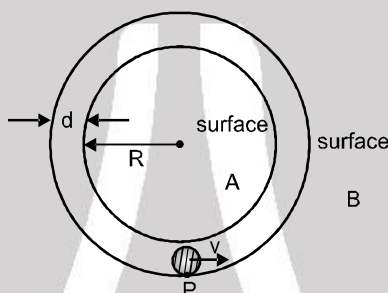
The velocity of a block of mass 2 kg moving along x-axis at any time t is given by $v = 20 - 10t$ (m/s) where t is in seconds and v is in m/s. At time $t = 0$, the block is moving in positive x-direction.

6. The block starts at $t = 0$ and at $t = t_1$ it covers a distance 25m. Then
(A) $t_1 = 3$ s
(B) $t_1 = 2$ s
(C) The work done by net force on the block in time $t = 0$ to $t = t_1$ is -300 J
(D) The work done by net force on the block in time $t = 0$ to $t = t_1$ is -200 J
7. Chose the correct statement(s)
(A) At $t = 3$ sec. net force (magnitude) on the block is 20 N
(B) At $t = 3$ sec. net force (magnitude) on the block is 30 N
(C) The power due to net force on block at $t = 3$ sec. is 200 W
(D) The power due to net force on block at $t = 3$ sec. is 300 W
8. Choose the correct option(s)
(A) The speed of block at $t = 2$ sec. is zero
(B) The speed of block at $t = 3$ sec. is 10 m/sec
(C) The kinetic energy of block at $t = 3$ sec. is 100 J
(D) The kinetic energy of block at $t = 2$ sec. is zero

9. A 20 kg block is originally at rest on a horizontal surface for which the coefficient of friction is 0.6. If a horizontal force F is applied such that it varies with time as shown in figure. Determine speed of block at 10 sec.



10. A small spherical ball of mass m is projected from lowest point (point P) in the space between two fixed, concentric spheres A and B (see figure). The smaller sphere A has a radius R and the space between the two spheres has a width d . The ball has a diameter very slightly less than d . All surfaces are frictionless. Speed of ball at lowest point is v . N_A and N_B represent magnitudes of the normal reaction force on the ball exerted by the spheres A and B respectively. Match the value of v given in column-I with corresponding results in column-II.



Column-I

- (A) $v = \sqrt{gR}$
 (B) $v = \sqrt{2gR}$
 (C) $v = \sqrt{3gR}$
 (D) $v = \sqrt{5gR}$

Column-II

- (p) maximum value of $N_A = 0$
 (q) minimum value of $N_B = 0$
 (r) maximum value of $N_B = 6 mg$
 (s) maximum value of $N_B = 4 mg$
 (t) maximum value of $N_B = 2 mg$

DPP No. : B23 (JEE-MAIN)

Total Marks : 60

Single choice Objective ('-1' negative marking) Q.1 to Q.20

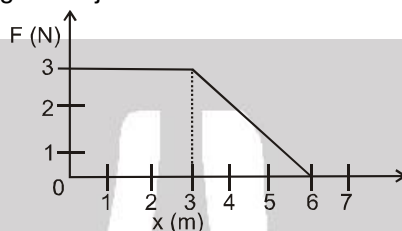
Max. Time : 40 min.

(3 marks 2 min.)

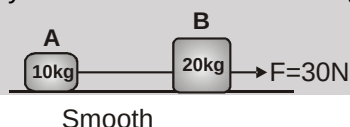
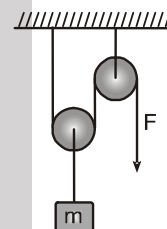
[60, 40]

1. Two particles whose masses are 10 kg and 30 kg and their position vectors are $\hat{i} + \hat{j} + \hat{k}$ and $-\hat{i} - \hat{j} - \hat{k}$ respectively would have the combined centre of mass at -
- (A) $-\frac{(\hat{i} + \hat{j} + \hat{k})}{2}$ (B) $\frac{(\hat{i} + \hat{j} + \hat{k})}{2}$ (C) $-\frac{(\hat{i} + \hat{j} + \hat{k})}{4}$ (D) $\frac{(\hat{i} + \hat{j} + \hat{k})}{4}$
2. Where will be the centre of mass on combining two masses m and M ($M > m$) :
- (A) towards m (B) towards M (C) between m and M (D) anywhere
3. Centre of mass is a point
- (A) Which is geometric centre of a body
 (B) From which distance of particles are same
 (C) Where the whole mass of the body is supposed to concentrated
 (D) Which is the origin of reference frame

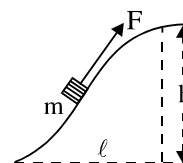
4. The work done by all the conservative forces (external and internal) on a system equals -
 (A) change in total energy (B) change in kinetic energy
 (C) Negative of change in potential energy (D) change in Potential energy
5. A body of mass m kg is lifted by a man to a height of one meter in 30 sec. Another man lifts the same mass to the same height in 60 sec. The work done by them are in the ratio -
 (A) 1 : 2 (B) 1 : 1 (C) 2 : 1 (D) 4 : 1
6. A force F acting on an object varies with distance x as shown here. The force is in N and x is in m. The work done by the force in moving the object from $x = 0$ to $x = 6$ m is

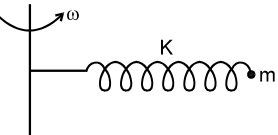
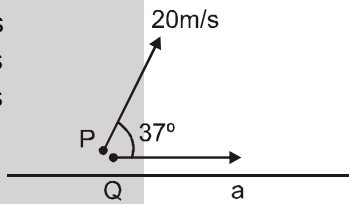
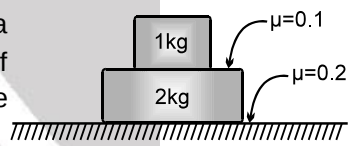
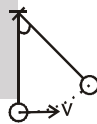
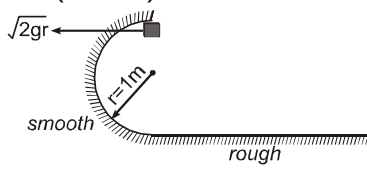


- (A) 18.0 J (B) 13.5 J (C) 4.5 J (D) 9.0 J
7. Two objects moving along the same straight line are leaving point A with an acceleration a , $2a$ and velocity $2u$, u respectively at time $t = 0$. The distance moved by the object with respect to point A when one object overtakes the other is :
 (A) $\frac{6u^2}{a}$ (B) $\frac{6u^2}{a}$ (C) $\frac{4u^2}{a}$ (D) none of these
8. In the shown mass pulley system, pulleys and string are massless. The one end of the string is pulled by the force $F = 2mg$. The acceleration of the block will be
 (A) $g/2$
 (B) 0
 (C) g
 (D) $3g$
9. Two blocks A and B are connected by a massless string (shown in figure) A force of 30 N is applied on block B. The distance travelled by centre of mass in 2s starting from rest is :



- (A) 1m (B) 2m (C) 3m (D) none of these
10. A body of mass m was slowly hauled up the hill as shown in figure by a force F which at each point was directed along a tangent to the trajectory. Find the work performed by this force, if the height of the hill is h , the length of its base ℓ , and the coefficient of friction k .
 (A) $mg h$ (B) $mg (h + k\ell)$
 (C) $mg (h - k\ell)$ (D) None of these
11. Two bodies of masses m and $4m$ are moving with equal linear momentum. The ratio of their kinetic energies is:
 (A) 1 : 4 (B) 4 : 1 (C) 1 : 1 (D) 1 : 2



12. A particle of mass m is fixed to one end of a light spring of force constant k and unstretched length ℓ . The system is rotated about the other end of the spring with an angular speed ω in gravity free space. The increase in length of the spring is (assume no oscillation) :
- (A) $\frac{m\omega^2\ell}{k}$ (B) $\frac{m\omega^2\ell}{k-m\omega^2}$ (C) $\frac{m\omega^2\ell}{k+m\omega^2}$ (D) none of these
- 
13. A ball suspended by a thread swings in a vertical plane so that its acceleration in the extreme position and lowest position are equal in magnitude. Angle θ of thread deflection in the extreme position will be:
- (A) $2 \tan^{-1} \frac{1}{2}$ (B) $\tan^{-1} \frac{1}{2}$ (C) $\tan^{-1} \sqrt{2}$ (D) $\tan^{-1} 2$
14. A particle is moving along a circular path of radius $r = \frac{5}{\pi}$ m with a uniform speed 5 ms^{-1} . What is the magnitude of average acceleration (in m/s^2) during the interval in which particle completes half revolution?
- (A) 10 (B) 5 (C) π (D) None of these
15. A particle P is projected with speed 20 m/s at angle 37° from horizontal as shown in figure. At the same instant Q starts from same point and moves with uniform acceleration towards right. Path of the particle P w.r.t. Q is straight line. Acceleration of Q is :
- (A) 40 m/s^2 (B) 20 m/s^2 (C) $\frac{20}{3} \text{ m/s}^2$ (D) $\frac{40}{3} \text{ m/s}^2$
- 
16. Both the blocks shown in the given arrangement are given together a horizontal velocity towards right. If a_{cm} be the subsequent acceleration of the centre of mass of the system of blocks, then a_{cm} equals (before sliding stops at all surfaces) :
- (A) 0 m/s^2 (B) $5/3 \text{ m/s}^2$ (C) $7/3 \text{ m/s}^2$ (D) 2 m/s^2
- 
17. As shown in the given figure the ball is given sufficient velocity at the lowest point to complete the circle. Length of string is 1 m . Find the tension in the string, when it is at 60° with vertical position. (Mass of ball = 5 kg)
- (A) 160 N (B) 180 N (C) 200 N (D) 225 N
- 
18. A body is given a velocity $\sqrt{2gr}$ at the highest point of a half circular smooth track which is joined by a rough horizontal track whose co-efficient of friction is $\mu = 0.5$. Then the distance travelled by particle before it stops on horizontal track is: ($r = 1 \text{ m}$)
- (A) 1 m (B) 4 m (C) 6 m (D) none of these
- 

19. A particle of mass 100 g is thrown vertically upwards with a speed of 5 m/s. The work done by the force of gravity during the time the particle goes up is :
 (A) -0.5 J (B) -1.25 J (C) 1.25 J (D) 0.5 J
20. A particle of mass m is executing a uniform motion along a circular path of radius r . If the magnitude of its linear momentum is p , the radial force acting on the particle will be.
 (A) pmr (B) rm/p (C) mp^2/r (D) p^2/mr

DPP No. : B24 (JEE-ADVANCED)

Total Marks : 43

Max. Time : 30 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.7

(4 marks 2 min.) [24, 12]

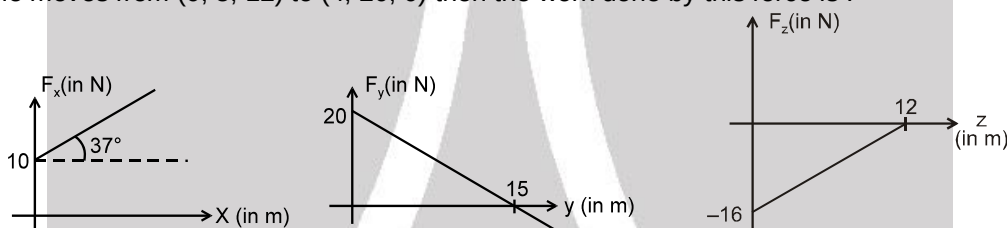
Subjective Questions ('-1' negative marking) Q.8 to Q.9

(4 marks 5 min.) [08, 10]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

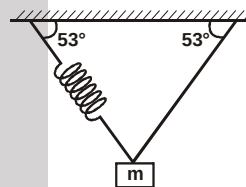
1. The components of a force acting on a particle are varying according to the graphs shown. When the particle moves from $(0, 5, 12)$ to $(4, 20, 0)$ then the work done by this force is :



- (A) 192 J (B) $400/3 \text{ J}$ (C) 0 (D) None of these

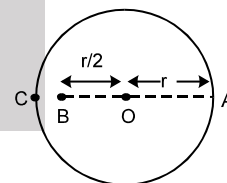
2. A block is attached with an ideal spring and string, and is in equilibrium as shown in Figure. The acceleration of the block just after breaking only the string is a_1 and just after breaking only the spring is a_2 . Then

- (A) $a_1 = \frac{5g}{8}$ (B) $a_2 = \frac{3g}{5}$
 (C) $a_1 = \frac{g}{2}$ (D) $a_2 = g$



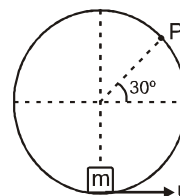
3. A particle is revolving in a circle of radius r and centre at 'O' with uniform angular velocity ω . Choose the correct option(s) :

- (A) the ratio of angular velocity at A about O and about B is 3 : 2.
 (B) the angular velocity at A about all points O, B and C is same.
 (C) the angular velocity at A about C is $\omega/2$.
 (D) the velocity at A and C are same.

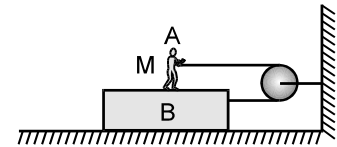


4. A horizontal cylinder is fixed, its inner surface is smooth and its radius is R . A small block is initially at the lowest point. the minimum velocity that should be given to the block at the lowest point, so that it can just cross the point P is u then

- (A) If the block moves anti clockwise then $u = \sqrt{3.5gR}$
 (B) If the block moves anti clockwise then $u = \sqrt{3gR}$
 (C) If the block moves clockwise then $u = \sqrt{3.5gR}$
 (D) If the block moves clockwise then $u = \sqrt{5gR}$

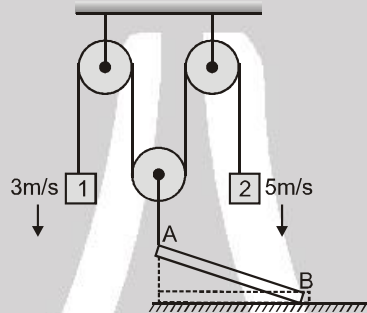


5. As shown in the figure, M is a man of mass 60 kg standing on a block of mass 40 kg kept on ground. The co-efficient of friction between the feet of the man and the block is 0.3 and that between B and the ground is 0.1. If the person pulls the string with 100 N force, then :
- (A) B will slide on ground
 (B) A and B will move together with acceleration 1 m/s^2
 (C) the friction force acting between A & B will be 40 Nt
 (D) the friction force acting between A & B will be 180 Nt

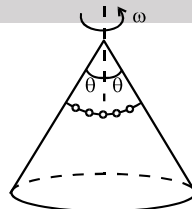


COMPREHENSION

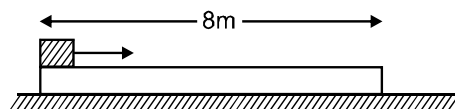
A meter stick AB of length 1 meter rests on a frictionless floor in horizontal position with end A attached to the string as shown. Assume that string connecting meter stick with pulley always remains vertical.



6. If blocks 1 and 2 are given constant speeds as shown then the distance moved by end A and B in the period for which speed of B is less than A.
- (A) $S_B = \left(\frac{\sqrt{2}+1}{\sqrt{2}} \right) \text{ m}$ (B) $S_B = \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) \text{ m}$ (C) $S_A = \frac{1}{\sqrt{2}} \text{ m}$ (D) $S_A = \frac{1}{2} \text{ m}$
7. If t represent the time taken to cover the distance (S_A and S_B) as in above part and s represent the displacement of centre of mass of rod in time t then :
- (A) $s = \sqrt{\frac{2+\sqrt{2}}{4}} \text{ m}$ (B) $t = \left(\frac{\sqrt{2}-1}{4\sqrt{2}} \right) \text{ sec}$ (C) $t = \frac{1}{4\sqrt{2}} \text{ sec}$ (D) $s = \sqrt{\frac{2-\sqrt{2}}{4}} \text{ m}$
8. A chain of mass ' M ' = $2\pi \text{ kg}$ and radius ' r ' = 1 m and $\theta = 45^\circ$, $\omega = 5 \text{ rad/sec}$. is placed onto a cone of semi vertical angle θ . Cone rotated with angular velocity ω . The tension in the chain if it does not slide on the cone is ' n ' (where n is integer) . Find n , (take $g = 10 \text{ m/s}^2$)

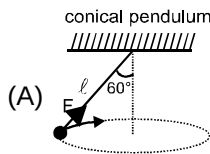


9. A small block is projected with some speed on a large block of same mass & of length 8m as shown in figure. Initially larger block is at rest. Friction is present only between the blocks. When the relative motion ends between the blocks, the small block is at the rightmost edge of the larger block. What is the displacement (in m) of the lower block w.r.t. earth in this time ?



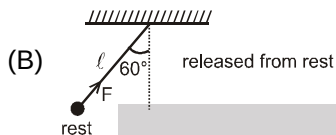
10. A particle of mass m moves as shown in figures of column-I. In cases A, B and C mass is attached at one end of a light string and the other end of string is fixed. 'a' is the initial acceleration of the particle. Match the column.

Column - I

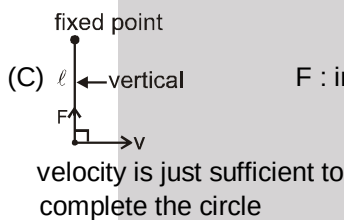


F : initial tension

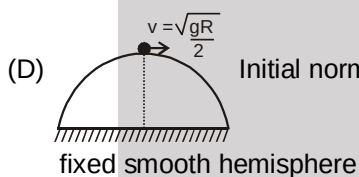
Column - II

(p) $F > mg$ 

F : initial tension

(q) $F < mg$ 

F : initial tension

(r) $a > g$ 

Initial normal reaction is F

(s) $a < g$

DPP No. : B25 (JEE-MAIN)

Total Marks : 60

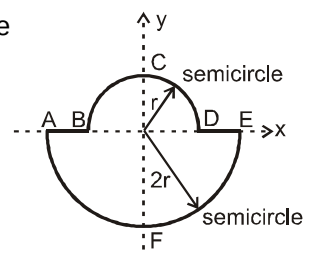
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

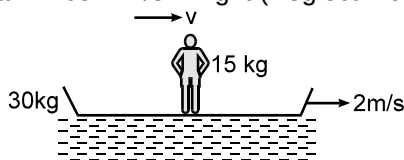
(3 marks 2 min.)

[60, 40]

1. With what angular velocity the earth should spin in order that a body lying at 37° latitude may become weightless.
- (A) $\frac{5}{4} \sqrt{\frac{g}{R}}$ (B) $\frac{25}{16} \sqrt{\frac{g}{R}}$ (C) $\frac{5}{3} \sqrt{\frac{g}{R}}$ (D) $\frac{25}{9} \sqrt{\frac{g}{R}}$
2. A man of mass M stands at one end of a plank of length L which lies at rest on a frictionless surface. The man walks to the other end of the plank. If the mass of plank is $M/3$, the distance that the plank moves relative to the ground is :
- (A) $3L/4$ (B) $L/4$ (C) $4L/5$ (D) $L/3$
3. Two semicircular rings of linear mass densities λ and 2λ and of radius ' R ' each are joined to form a complete ring. The distance of the center of the mass of complete ring from its geometrical centre is :
- (A) $\frac{3R}{8\pi}$ (B) $\frac{2R}{3\pi}$ (C) $\frac{3R}{4\pi}$ (D) none of these
4. A uniform thin rod is bent in the form of closed loop ABCDEFA as shown in the figure. The x-coordinate of the centre of mass of the system is
- (A) $\frac{2r}{\pi}$ (B) $-\frac{6r}{3\pi+2}$ (C) $-\frac{2r}{\pi}$ (D) Zero

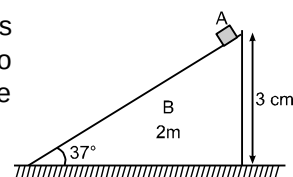
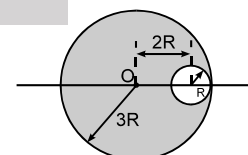
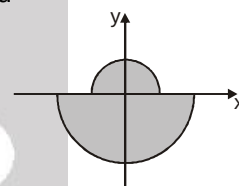
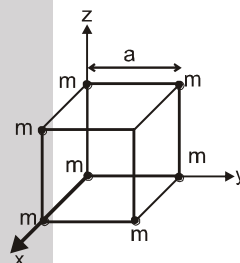


5. In the figure shown the initial velocity of boat (30 kg) + person (15 kg) is 2 m/s. Find velocity of person w.r.t. boat so that velocity of boat will be 1 m/s in right (Neglect friction between boat and water)



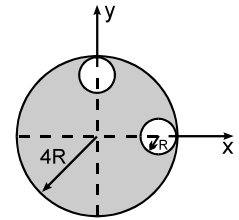
- (A) 3 m/s towards right
(B) 3 m/s towards left
(C) 4 m/s towards right
(D) 4 m/s towards left
6. Three identical metal balls each of radius r are placed touching each other on a horizontal surface such that an equilateral triangle is formed, when centres of three balls are joined. The centre of the mass of system is located is :
- (A) horizontal surface
(B) centre of one of the balls
(C) line joining centres of any two balls
(D) point of intersection of the medians
7. A heavy particle is projected from a point on the horizontal at an angle 45° with the horizontal with a speed of 20 m/s. Then the radius of the curvature of its path at the instant of crossing the same horizontal is _____.
- (A) $10\sqrt{2}$
(B) $40\sqrt{2}$
(C) $20\sqrt{2}$
(D) None of these
8. The motion of the centre of mass of a system of two particles is unaffected by their internal forces :
- (A) irrespective of the actual directions of the internal forces
(B) only if they are along the line joining the particles
(C) only if they are at right angles to the line joining the particles
(D) only if they are obliquely inclined to the line joining the particles.
9. A particle of mass $4m$ which is at rest explodes into four equal fragments. All 4 fragments scattered in the same horizontal plane. Three fragments are found to move with velocity V each as shown in the figure. The total energy released in the process of explosion is :
- (A) $mV^2 (3 - \sqrt{2})$
(B) $2mV^2$
(C) $mV^2 \frac{(3 - \sqrt{2})}{2}$
(D) $mV^2 \frac{(1 + \sqrt{2})}{2}$
-
10. The resultant force on a system of particles is zero.
- (A) The centre of mass of the system must be at rest.
(B) Acceleration of each particle may be in the same direction.
(C) Velocity of each particle may be in the same direction at some instant of time.
(D) If only one particle has initially non-zero velocity then it is possible that all the particles have zero velocity simultaneously, later on.
11. Two elastic bodies P and Q having equal masses are moving along the same line with velocities of 16 m/s and 10 m/s respectively. Their velocities after the elastic collision will be in m/s -
- (A) 0 and 25
(B) 5 and 20
(C) 10 and 16
(D) 20 and 5
12. A body of mass ' m ' moving with a constant velocity V hits another body of the same mass moving with the same velocity V but in the opposite direction and sticks to it. The velocity of the compound body after collision is -
- (A) V
(B) $2V$
(C) $V/2$
(D) 0

13. In an inelastic collision -
 (A) momentum is conserved but kinetic energy is not
 (B) momentum is not conserved but kinetic energy is conserved
 (C) neither momentum nor kinetic energy is conserved
 (D) both the momentum and kinetic energy are conserved
14. A shell is fired from a cannon with a velocity v at an angle θ with the horizontal direction. At the highest point in its path, it explodes into two equal pieces, one retraces its path to the cannon and the speed of the other piece immediately after the explosion is :
 (A) $3v \cos\theta$ (B) $2v \cos\theta$ (C) $\left(\frac{3}{2}\right) v \cos\theta$ (D) $\frac{\sqrt{3}}{2} v \cos\theta$
15. The centre of mass of a non uniform rod of length L whose mass per unit length λ varies as $\lambda = \frac{k \cdot x^2}{L}$ where k is a constant & x is the distance of any point on rod from its one end, is (from the same end)
 (A) $\frac{3}{4} L$ (B) $\frac{1}{4} L$ (C) $\frac{k}{L}$ (D) $\frac{3k}{L}$
16. Seven particles, each of mass m are placed at the seven corners of a cube of side 'a', but one corner is vacant, as shown in figure. The co-ordinates of the centre of mass of the system is :
 (A) $\left(\frac{2a}{7}, \frac{2a}{7}, \frac{2a}{7}\right)$ (B) $\left(\frac{3a}{7}, \frac{3a}{7}, \frac{3a}{7}\right)$
 (C) $\left(\frac{a}{3}, \frac{a}{3}, \frac{a}{3}\right)$ (D) $\left(\frac{5a}{11}, \frac{5a}{11}, \frac{5a}{11}\right)$
17. Two semicircular discs made of same material having radius r and $2r$ are placed as shown in figure. Find out the center of mass of structure
 (A) $\left(0, \frac{-28r}{5\pi}\right)$ (B) $\left(0, \frac{-28r}{15\pi}\right)$
 (C) $\left(0, \frac{-9r}{5\pi}\right)$ (D) $\left(0, \frac{-9r}{15\pi}\right)$
18. In the figure shown find out the distance of centre of mass of a system of a uniform circular plate of radius $3R$ from O in which a hole of radius R is cut whose centre is at $2R$ distance from centre of large circular plate.
 (A) $R/4$ (B) $R/5$
 (C) $R/2$ (D) none of these
19. A particle A of mass m is situated at highest point of wedge B of mass $2m$ is released from rest. Then distanced travelled by wedge B (With respect to ground) when particle A reaches at lowest position. Assume all surfaces are smooth.
 (A) $4/3$ cm (B) $8/3$ cm
 (C) $2/3$ cm (D) none of these



20. From the uniform disc of radius $4R$ two small disc of radius R are cut off. The centre of mass of the new structure will be : (Centre of lower circular cavity lies on x-axis and centre of upper circular cavity lies on y-axis)

- (A) $\hat{i} \frac{R}{5} + \hat{j} \frac{R}{5}$ (B) $-\hat{i} \frac{R}{5} + \hat{j} \frac{R}{5}$
 (C) $-\hat{i} \frac{R}{5} - \hat{j} \frac{R}{5}$ (D) $-\frac{3R}{14}(\hat{i} + \hat{j})$



DPP No. : B26 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.9

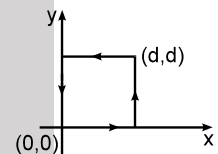
(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. The work done by the force $\vec{F} = A(y^2\hat{i} + 2x^2\hat{j})$, where A is a constant and x & y are in meters around the path shown is:

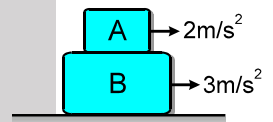
- (A) zero (B) $A d$
 (C) $A d^2$ (D) $A d^3$



2. Choose the correct statement about the centre of mass (CM) of a system of two particles
- (A) The CM lies on the line joining the two particles exactly midway between them
 (B) The CM lies on the line joining them at a point whose distance from each particle is inversely proportional to the mass of that particle
 (C) The CM lies on the line joining them at a point whose distance from each particle is proportional to the square of the mass of that particle
 (D) The CM is on the line joining them at a point whose distance from each particle is proportional to the mass of that particle

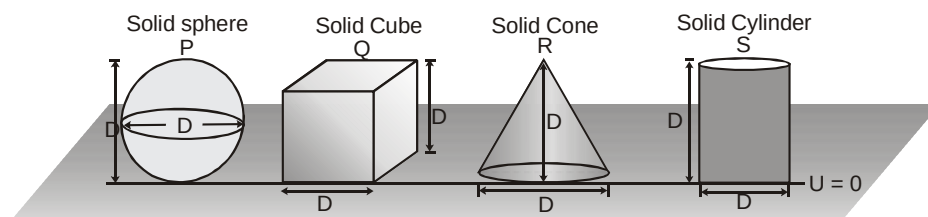
3. Block A is kept on block B as shown in figure. It is known that acceleration of block A is 2 m/s^2 towards right and acceleration of block B is 3 m/s^2 towards right under the effect of unknown forces. Direction of friction force acting on A by B ($\mu_{AB} = 0.3$)

- (A) is necessarily towards right (B) may be towards right
 (C) may be towards left (D) may be zero



4. Two particle P and Q are in motion under earth's gravitational field only. Then:
- (A) their relative acceleration is constant but not zero
 (B) their relative velocity is constant
 (C) their centre of mass has constant velocity
 (D) their centre of mass has constant acceleration.

5. Assuming potential energy 'U' at ground level to be zero.



All objects are made up of same material.

U_P = Potential energy of solid sphere

U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone

U_S = Potential energy of solid cylinder

(A) $U_S > U_P$

(B) $U_Q > U_S$

(C) $U_P > U_Q$

(D) $U_P > U_S$



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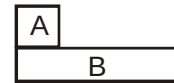
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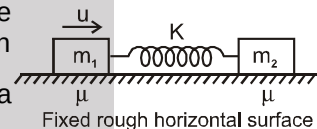
PAGE NO.-65

COMPREHENSION :

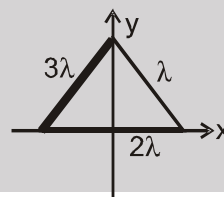
Figure shows a small block A of mass m kept at the left end of a plank B of mass $M = 2m$ and length ℓ . The system can slide on a horizontal road. The system is started towards right with the initial velocity v . The friction coefficients between the road and the plank is $1/2$ and that between the plank and the block is $1/4$.



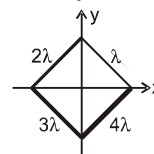
6. Choose the correct option (s)
 (A) Friction on block A acts towards left (B) Friction on block A acts towards right
 (C) acceleration of plank B is $\frac{5g}{8}$ (D) Acceleration of plank B w.r.t. block A is $\frac{3g}{8}$
7. If S_A and S_B represents the displacement of block and plank w.r.t. ground respectively when block separates, then
 (A) $S_A = 4v \sqrt{\frac{\ell}{3g}} - \frac{2}{3}\ell$ (B) $S_A = 4v \sqrt{\frac{\ell}{g}} - \frac{2}{3}\ell$ (C) $S_B = 4v \sqrt{\frac{\ell}{3g}} - \frac{5}{3}\ell$ (D) $S_B = 4v \sqrt{\frac{\ell}{g}} - \frac{1}{3}\ell$
8. If t and w_{fg} represents the time and work done by friction on block A w.r.t. ground when they separate then :
 (A) $t = 4\sqrt{\frac{\ell}{3g}}$ (B) $t = 4\sqrt{\frac{\ell}{g}}$
 (C) $w_{fg} = mgv \sqrt{\frac{\ell}{3g}} - \frac{mg\ell}{6}$ (D) $w_{fg} = mgv \sqrt{\frac{\ell}{3g}} - \frac{5mg\ell}{12}$
9. The blocks of mass $m_1 = 1$ kg and $m_2 = 2$ kg are connected by an ideal spring, rest on a rough horizontal surface. The spring is unstressed. The spring constant of spring is $K = 2$ N/m. The coefficient of friction between blocks and horizontal surface is $\mu = \frac{1}{2}$. Now the left block is imparted a velocity u towards right as shown. The largest value of u (in m/s) such that the block of mass m_2 never moves is (Take $g = 10\text{m/s}^2$)
10. In each situation of column-I a mass distribution is given and information regarding x and y -coordinate of centre of mass is given in column-II. Match the figures in column-I with corresponding information of centre of mass in column-II.

**Column-I****Column-II**

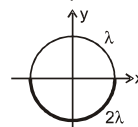
- (A) An equilateral triangular wire frame is made using three thin uniform rods of mass per unit lengths λ , 2λ and 3λ as shown

(p) $x_{cm} \geq 0$

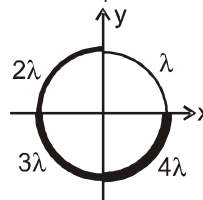
- (B) A square frame is made using four thin uniform rods of mass per unit length lengths λ , 2λ , 3λ and 4λ as shown

(q) $y_{cm} \geq 0$

- (C) A circular wire frame is made of two uniform semicircular wires of same radius and of mass per unit length λ and 2λ as shown

(r) $x_{cm} < 0$

- (D) A circular wire frame is made of four uniform quarter circular wires of same radius and mass per unit length λ , 2λ , 3λ and 4λ as shown

(s) $y_{cm} < 0$

DPP No. : B27 (JEE-MAIN)

Total Marks : 60

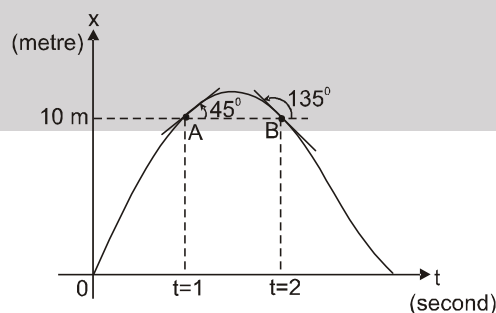
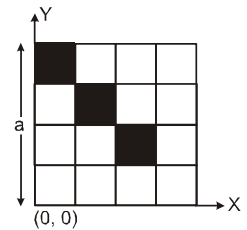
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

(3 marks 2 min.)

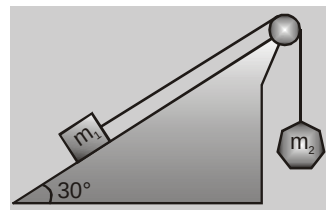
[60, 40]

1. From a uniform square plate the shaded portions are removed as shown in figure. The coordinates of centre of mass of the remaining plate are x, y . Axes and origin are shown in figure.
- (A) $x < \frac{a}{2}, y < \frac{a}{2}$ (B) $x > \frac{a}{2}, y < \frac{a}{2}$
 (C) $x < \frac{a}{2}, y > \frac{a}{2}$ (D) $x > \frac{a}{2}, y > \frac{a}{2}$
2. The breaking tension of a string is 10 N. A particle of mass 0.1 kg tied to it is rotated along a horizontal circle of radius 0.5 metre. The maximum speed with which the particle can be rotated without breaking the string is-
- (A) $\sqrt{5}$ m/sec (B) $\sqrt{50}$ m/sec (C) $\sqrt{500}$ m/sec (D) $\sqrt{1000}$ m/sec
3. Consider a system of two particles having masses m_1 and m_2 . If the particle of mass m_1 is pushed towards the centre of mass particles through a distance d , by what distance would the particle of mass m_2 move so as to keep the mass centre of particles at the original position ?
- (A) $\frac{m_1}{m_1 + m_2} d$ (B) $\frac{m_1}{m_2} d$ (C) d (D) $\frac{m_2}{m_1} d$
4. A particle of mass m_1 is moving with a velocity v_1 and another particle of mass m_2 is moving with a velocity v_2 . Both of them have the same momentum but their different kinetic energies are E_1 and E_2 respectively. If $m_1 > m_2$ then :
- (A) $E_1 < E_2$ (B) $\frac{E_1}{E_2} = \frac{m_1}{m_2}$ (C) $E_1 > E_2$ (D) $E_1 = E_2$
5. A bomb of mass 30 kg at rest explodes into two pieces of masses 18 kg and 12 kg. The velocity of 18 kg mass is 6 ms^{-1} . The kinetic energy of the other mass is :
- (A) 256 J (B) 486 J (C) 524 J (D) 324 J
6. Displacement-time curve of a particle moving along a straight line is shown. Tangents at A and B make angles 45° and 135° with positive x-axis respectively. The average acceleration of the particle during $t = 1$ to $t = 2$ second is :

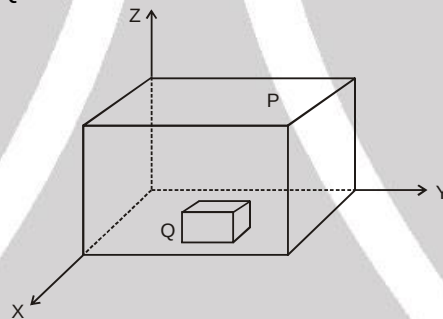
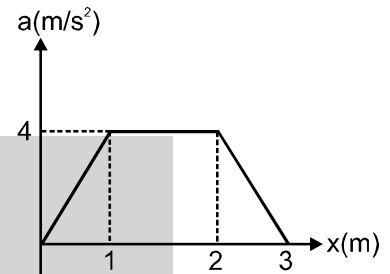


- (A) -2 m/s^2 (B) 1 m/s^2 (C) -1 m/s^2 (D) zero
7. $\int \frac{(x^2 + 1)x}{x^4 - 1} dx =$
- (A) $\ln(\sqrt{x^2 - 1}) + C$ (B) $\frac{1}{2} \ln(\sqrt{x^2 - 1}) + C$
 (C) $\frac{1}{2} \ln(\sqrt{x^2 - 1}) + C$ (D) $\frac{\ln(x^2 + 1)}{2} + C$

8. Two bodies of mass 1 kg and 3 kg have position vector $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$ respectively. The centre of mass of this system has a position vector.
 (A) $-2\hat{i} + 2\hat{k}$ (B) $-2\hat{i} - \hat{j} + \hat{k}$ (C) $2\hat{i} - \hat{j} - 2\hat{k}$ (D) $-\hat{i} + \hat{j} + \hat{k}$
9. A ball moving with velocity 2 m/s collides head on with another stationary ball of double the mass. If the coefficient of restitution is 0.5, then their velocities (in m/s) after collision will be
 (A) 0, 1 (B) 1, 1 (C) 1, 0.5 (D) 0, 2
10. A mass m moving horizontally (along the x-axis) with velocity v collides and sticks to mass of $3m$ moving vertically upward (along the y-axis) with velocity $2v$. The final velocity of the combination is :
 (A) $\frac{1}{4}v\hat{i} + \frac{3}{2}v\hat{j}$ (B) $\frac{1}{3}v\hat{i} + \frac{2}{3}v\hat{j}$ (C) $\frac{2}{3}v\hat{i} + \frac{1}{3}v\hat{j}$ (D) $\frac{3}{2}v\hat{i} + \frac{1}{4}v\hat{j}$
11. Two persons of masses 55 kg and 65 kg respectively, are at the opposite ends of a boat. The length of the boat is 3.0 m and weighs 100 kg. The 55 kg man walks up to the 65 kg man and sits with him. If the boat is in still water the centre of mass of the system shifts by :
 (A) 3.0 m (B) 2.3 m (C) zero (D) 0.75 m
12. Two spheres A and B of masses m_1 and m_2 respectively collide. A is at rest initially and B is moving with velocity v along x-axis. After collision B has a velocity $\frac{v}{2}$ in a direction perpendicular to the original direction. The mass A moves after collision in the direction.
 (A) same as that of B (B) Opposite to that of B
 (C) $\theta = \tan^{-1}(2)$ to the x-axis (D) $\theta = \tan^{-1}(1/2)$ to the x-axis
13. A block of mass 1kg is pushed on a movable wedge of mass 2kg and height $h = 30$ cm with a velocity $u = 6$ m/sec. Before striking the wedge it travels 2 m on a rough horizontal portion. Velocity is just sufficient for the block to reach the top of the wedge. Assuming all surfaces are smooth except the given horizontal part and collision of block and wedge is jerkless, the friction coefficient of the rough horizontal part is :
 (A) 0.125 (B) 0.377 (C) 0.675 (D) 0.45
14. An object is moving in a circle at constant speed v . The magnitude of rate of change of momentum of the object
 (A) is zero (B) is proportional to v
 (C) is proportional to v^2 (D) is proportional to v^3
15. A block of mass $m_1 = 2$ kg on a smooth inclined plane at angle 30° is connected to a second block of mass $m_2 = 3$ kg by a cord passing over a frictionless pulley as shown in fig. The acceleration of each block is- (assume $g = 10$ m/sec²)
 (A) 2 m/sec² (B) 4 m/sec²
 (C) 6 m/sec² (D) 8 m/sec²
16. If kinetic energy is doubled, then its momentum becomes n times, here n is :
 (A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2\sqrt{2}}$



17. A particle of mass m is executing a uniform motion along a circular path of radius r . If the magnitude of its linear momentum is p , the radial force acting on the particle will be.
 (A) pmr (B) rm/p (C) mp^2/r (D) p^2/mr
18. A 1.0kg ball drops vertically into a floor from a height of 25 cm. It rebounds to a height of 4cm. The coefficient of restitution for the collision is -
 (A) 0.16 (B) 0.32 (C) 0.40 (D) 0.56
19. A body initially at rest starts moving in x -direction. Its acceleration a is plotted against x as shown in figure. Find the maximum velocity (in m/s) of the body ?
 (A) 2 (B) 4 (C) 6 (D) 8
20. A hollow cubical box P is moving on a smooth horizontal surface in the x - y plane with constant acceleration of $\vec{a} = 3\hat{i} + 4\hat{j} \text{ m/s}^2$. A block Q of mass 2kg is at rest inside the cubical box as shown in figure. If the coefficient of friction between the surface of the cube P and the block Q is 0.6. Then the force of friction between P and Q is :



- (A) 5 N (B) 8 N (C) 12 N (D) 10 N

DPP No. : B28 (JEE-ADVANCED)

Total Marks : 50

Max. Time : 31 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.10

(4 marks 2 min.) [32, 16]

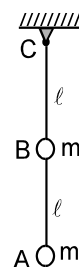
Subjective Questions ('-1' negative marking) Q.11

(4 marks 5 min.) [04, 05]

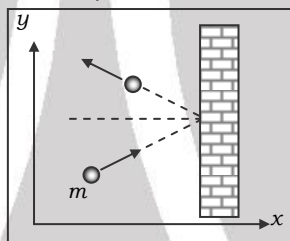
Match the Following (no negative marking) Q.12

(8 marks 6 min.) [08, 06]

1. A chain is held on a frictionless table with $L/4$ hanging over. Knowing total mass of the chain is M and total length is L , the work required to pull hanging part back to the table is:
 (A) $\frac{MgL}{16}$ (B) $\frac{MgL}{8}$ (C) $\frac{MgL}{32}$ (D) $\frac{MgL}{24}$]
2. A weightless rod of length 2ℓ carries two equal masses ' m ', one secured at lower end A and the other at the middle of the rod at B. The rod can rotate in vertical plane about a fixed horizontal axis passing through C. What horizontal velocity must be imparted to the mass at A so that it just completes the vertical circle. :
 (A) $\sqrt{\frac{48g\ell}{5}}$ (B) $\sqrt{\frac{64g\ell}{5}}$
 (C) $\sqrt{\frac{72g\ell}{5}}$ (D) $\sqrt{\frac{120g\ell}{5}}$



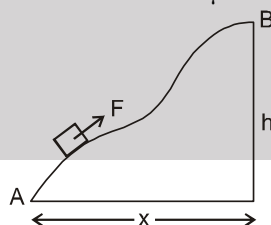
3. A man of mass 40 kg is standing on a block A of mass 140 kg. He pushes another block B of mass 60 kg on the ground, so that they are set in motion. [Assume no friction between the block and the ground. The man does not slide on A]
 (A) Speed of block A is three times that of block B immediately after the interaction
 (B) Speed of block B is three times that of block A immediately after the interaction
 (C) Distance travelled by block B is three times that of block A in the same time
 (D) Distance travelled by block B is nine times that of block A in the same time.
4. If the resultant force on a system of particles is non-zero, then :
 (A) The linear momentum of the system must increase.
 (B) The velocity of the centre of mass of the system must change.
 (C) The distance of the centre of mass may remain constant from a fixed point.
 (D) kinetic energy of all particles must either increase simultaneously or decrease simultaneously.
5. A ball of mass m moving with velocity $\vec{u} = u_x \hat{i} + u_y \hat{j}$ hits a vertical wall of infinite mass as shown in the figure. The ball slips up along the wall for the duration of collision and there is friction between the ball and the wall. Neglect the effect of gravity. Pick up the correct alternative.



- (A) The net impulse of the wall on the ball must not be along the negative x-axis for the duration of collision.
 (B) The collision changes only the x- component of velocity of the ball
 (C) The collision changes only the y- component of velocity of the ball
 (D) The impulse provided by frictional force to the ball for the duration of collision cannot be neglected in comparison to impulse provided by normal reaction.

COMPREHENSION

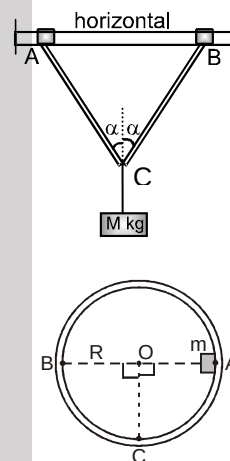
A body of mass m is moved up the plane of varying slope by a tangential force upto height h . The coefficient of friction between the surface and the block is μ .



6. The work done on the block.
 (A) by gravity depends upon the height h (B) by gravity depends upon the force F
 (C) by friction depends upon the speed of block (D) by friction is μmgx , when moved slowly
7. The friction force on the block is
 (A) Kinetic and constant in magnitude (B) Kinetic and variable in magnitude
 (C) Maximum at B when moved slowly (D) Maximum at B
8. If the block is moved up with constant speed,
 (A) the acceleration of the block is zero.
 (B) the acceleration component tangent to the plane is zero.
 (C) the acceleration component tangent to the plane is non zero.
 (D) the acceleration component normal to the surface is non zero.



9. If $\vec{F}$ is such that at any time, $v \approx 0$ (very small) and paths from A to B is non-decreasing in x and y, then work done by force of friction in going from A to B will be :
 (A) Less than work done by $\vec{F}$ (B) Depends on x
 (C) Equal for all such paths (D) Depends on inclination θ from A to B.
10. If whole system is dipped into a liquid of density $d \text{ kg/m}^3$. Then work done by $m\vec{g}$ in going from A to B will :
 (A) decrease in comparison to air
 (B) increase in comparison to air
 (C) depend on 'h'
 (D) be negative of change in gravitational potential energy
11. The system shown is in equilibrium in vertical plane. All the rods and a small supports A & B are light and friction coefficient between supports and horizontal rod is μ . If maximum value of angle α is 37° then find the value of 4μ . Assume light rod (AC and BC) exerts force only along the rod.
12. A fixed hollow cylinder of radius $R = 30\sqrt{3} \text{ m}$ with its axis horizontal is as shown. AB is a horizontal diameter of shown cross-section and C is the lowest point. A small block of mass $m = 1 \text{ kg}$ is released at rest from point A inside the cylinder. There is no friction between block and cylinder. Match the statements in column-I with results in column-II.

**Column-I**

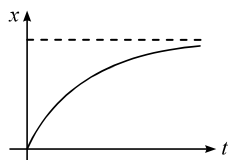
- (A) Magnitude of maximum vertical component of velocity of block in m/s is
 (B) The magnitude of centripetal acceleration (in m/s^2) of block at the instant acceleration of block is vertical, will be
 (C) The magnitude of normal reaction (in N) by cylinder on block at the instant acceleration of block is horizontal, will be
 (D) The magnitude of normal reaction (in N) by cylinder on block at the instant acceleration of block is vertical, will be

Column-II

- (p) 20
 (q) 0
 (r) $10\sqrt{3}$
 (s) 30

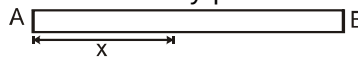
DPP No. : B29 (JEE-MAIN)**Total Marks : 60****Max. Time : 40 min.****Single choice Objective ('-1' negative marking) Q.1 to Q.20****(3 marks 2 min.)****[60, 40]**

1. The displacement x of a particle as a function of time t is shown in following figure. The figure indicates:

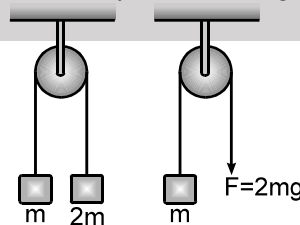
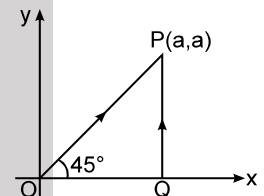
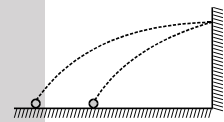


- (A) the particle starts with a certain velocity, but the motion is retarded and finally particle stops
 (B) the velocity of particle is constant throughout
 (C) the acceleration of the particle is constant throughout
 (D) the particle starts with a constant velocity, the motion is accelerated.
2. The principle of conservation of energy implies that -
 (A) the total mechanical energy is conserved (B) the total kinetic energy is conserved
 (C) the total potential energy is conserved (D) sum of all types of energies is conserved

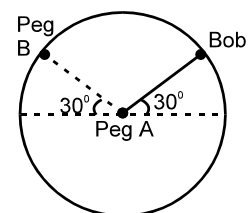
3. The centre of mass of a non uniform rod of length L whose mass per unit length λ varies as $\lambda = kx^2$ where k is a constant & x is the distance of any point on rod from left end A is (from the same end)



- (A) at the centre of the rod (B) is at $x = \frac{3L}{4}$
 (C) is at $x = \frac{4L}{5}$ (D) is at $x = \frac{5L}{6}$
4. A ball kept in a closed box moves in the box making collisions with the walls. The box is kept on a smooth surface. The velocity of centre of mass :
 (A) of the box remains constant (B) of the box plus the ball system remains constant
 (C) of the ball remains constant (D) of the ball relative to the box remains constant
5. A body of mass 1kg strikes elastically with another body at rest and continues to move in the same direction with one fourth the initial velocity. The mass of the other body is -
 (A) 0.6 kg (B) 2.4 kg (C) 3 kg (D) 4 kg
6. A stone is projected from ground and hits a smooth vertical wall after 1 sec. and again falls back on the ground. The time taken by stone to reach the ground after the collision is 3 secs. The maximum height reached by the same stone if the vertical wall were not to be present is. ($g = 10 \text{ m/s}^2$)
 (A) 10 m (B) 12.5 m (C) 15 m (D) 20 m
7. A particle of mass m describes a circle of radius r . The centripetal acceleration of the particle is $4/r^2$. What will be the magnitude of momentum of the particle?
 (A) $2\frac{m}{r}$ (B) $2\frac{m}{\sqrt{r}}$ (C) $4\frac{m}{\sqrt{r}}$ (D) none
8. Which of the following bodies has the largest kinetic energy ?
 (A) Mass 3M and speed V (B) Mass 3M and speed 2V
 (C) Mass 2M and speed 3V (D) Mass M and speed 4V
9. A particle is moved from (0, 0) to (a, a) under a force $F = (3\hat{i} + 4\hat{j})$ from two paths. Path 1 is OP and Path 2 is OQP. Let W_1 and W_2 be the work done by this force in these two paths. Then
 (A) $W_1 = W_2$ (B) $W_1 = 2W_2$
 (C) $W_2 = 2W_1$ (D) $W_2 = 4W_1$
10. Figure shows two pulley arrangements for lifting a mass m . In (a) the mass is lifted by attaching a mass 2 m while in (b) the mass is lifted by pulling the other end with a downward force $F = 2mg$. If f_a & f_b are the accelerations of the two masses then (Assume string is massless and pulley is ideal)

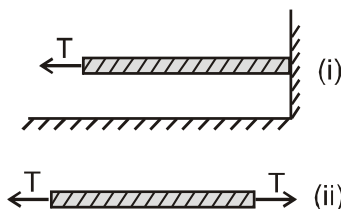


- (A) $f_a = f_b$ (B) $f_a = f_b/2$ (C) $f_a = f_b/3$ (D) $f_a = 2f_b$
11. A bob is attached to one end of a string other end of which is fixed at peg A. The bob is taken to a position where string makes an angle of 30° with the horizontal. On the circular path of the bob in vertical plane there is a peg 'B' at a symmetrical position with respect to the position of release as shown in the figure. If V_c and V_a be the minimum tangential velocity in clockwise and anticlockwise directions respectively, given to the bob in order to hit the peg 'B' then ratio $V_c : V_a$ is equal to :
 (A) 1 : 1 (B) 1 : $\sqrt{2}$ (C) 1 : 2 (D) 1 : 4

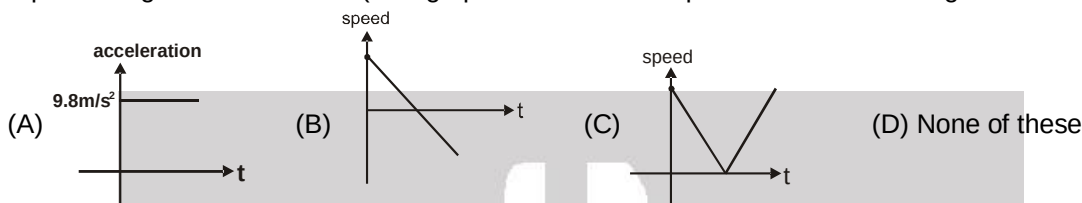


12. In the figure (i) an extensible string is fixed at one end and the other end is pulled by a tension T . In figure (ii) another identical string is pulled by tension ' T ' at both the ends. The ratio of elongation in equilibrium of string in (i) to the elongation of string in (ii) is

(A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) 0



13. A particle is thrown from ground with some initial speed in vertically upward direction, then the graphs representing this motion are : (taking upward direction as positive direction and $g = 9.8 \text{ m/s}^2$)



14. **STATEMENT-1** : For a particle moving along a circular path with increasing speed, the magnitude of acceleration of particle may decrease with time.

STATEMENT-2 : The centripetal acceleration of a particle of mass m moving along a circle of radius R is $\frac{v^2}{R}$ where v is speed of the particle.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True.

15. A stone is released at rest from a height of 45 m above the horizontal level ground. There is horizontal wind blowing due to which stone acquires an additional (in addition to acceleration due to gravity) horizontal acceleration of magnitude 10 m/s^2 . Then the net distance travelled by stone before reaching ground is: (Take $g = 10 \text{ m/s}^2$ and neglect air resistance)

(A) 45 m. (B) $45\sqrt{2}$ m (C) $45\sqrt{3}$ m (D) cannot be determined

16. On a displacement/time graph, two straight lines make angles at 30° & 60° with the time axis. The ratio of the velocities represented by them is:

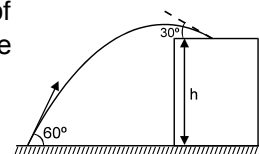
(A) 1 : 2 (B) 1 : 3 (C) 2 : 1 (D) 3 : 1

17. A train is standing on a platform, a man inside a compartment of a train drops a stone. At the same instant train starts to move with constant acceleration. The path of the particle as seen by the person who drops the stone is :

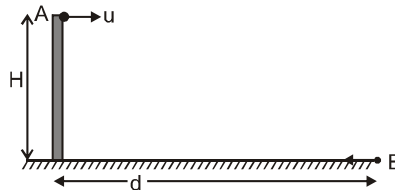
(A) parabola
(B) straight line for sometime & parabola for the remaining time
(C) straight line
(D) variable path that cannot be defined

18. A stone projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of a building of height ' h '. Then the speed of projection of the stone is :

(A) $\sqrt{2gh}$ (B) $\sqrt{6gh}$
(C) $\sqrt{3gh}$ (D) $\sqrt{gh}$

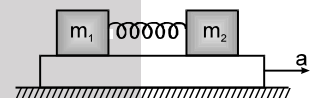


19. Two particles A and B are placed as shown in figure. The particle A on the top of a tower of height H , is projected horizontally with a velocity u and the particle B is projected along the horizontal surface towards the foot of the tower, simultaneously. When particle A reaches at ground, it simultaneously hits particle B. Then the speed of projection of particle B is : (Neglect any type of friction)



- (A) $d\sqrt{\frac{g}{2H}}$ (B) $d\sqrt{\frac{g}{2H}} - u$ (C) $d\sqrt{\frac{g}{2H}} + u$ (D) u

20. Two blocks of masses m_1 and m_2 are connected with a massless unstretched spring and placed over a plank moving with an acceleration 'a' as shown in figure. The coefficient of friction between the blocks and platform is μ .



- (A) spring will be stretched if $a > \mu g$
 (B) spring will be compressed if $a \leq \mu g$
 (C) spring will neither be compressed nor be stretched for $a \leq \mu g$
 (D) spring will be in its natural length under all conditions

DPP No. : B30 (JEE-ADVANCED)

Total Marks : 42

Single choice Objective ('-1' negative marking) Q.1 to Q.2

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

Subjective Questions ('-1' negative marking) Q.9

Match the Following (no negative marking) Q.10

Max. Time : 27 min.

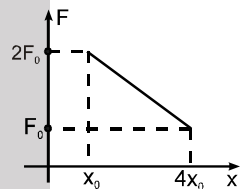
(3 marks 2 min.) [06, 04]

(4 marks 2 min.) [24, 12]

(4 marks 5 min.) [04, 05]

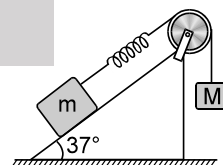
(8 marks 6 min.) [08, 06]

1. A particle of mass m moving along a straight line experiences force F which varies with the distance x travelled as shown in the figure. If the velocity of the particle at x_0 is $\sqrt{\frac{2F_0 x_0}{m}}$, then velocity at $4x_0$ is:



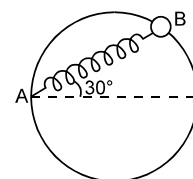
- (A) $2\sqrt{\frac{2F_0 x_0}{m}}$ (B) $2\sqrt{\frac{F_0 x_0}{m}}$ (C) $\sqrt{\frac{F_0 x_0}{m}}$ (D) none of these

2. A block of mass m is attached with a massless spring of force constant k . The block is placed over a fixed rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. The block of mass m is initially at rest. The block of mass M is released from rest with spring in unstretched state. If minimum value of M required to move the block up the plane is $\frac{a}{b}m$. Then find the minimum value of



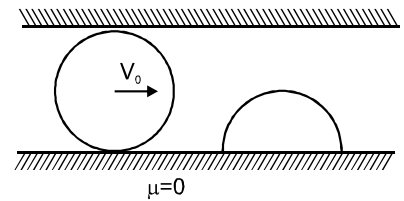
- $a + b$. a, b are integers. (neglect mass of string and pulley and friction in pulley.)
 (A) 8 (B) 9 (C) 11 (D) 5

3. A bead of mass m is attached to one end of a spring of natural length R and spring constant $k = \frac{(\sqrt{3} + 1)mg}{R}$. The other end of the spring is fixed at point A on a smooth vertical ring of radius R as shown in figure. The normal reaction at B just after it is released to move is



- (A) $\frac{mg}{2}$ (B) $\sqrt{3}mg$ (C) $3\sqrt{3}mg$ (D) $\frac{3\sqrt{3}mg}{2}$

4. A smooth hemisphere of mass m and radius R is at rest. A smooth solid sphere of mass $2m$ and radius R moving with velocity V_0 between two horizontal smooth surfaces separated by a distance slightly greater than $2R$ as shown in figure. Solid sphere collides with the hemisphere. If coefficient of restitution is $\frac{1}{2}$, then :

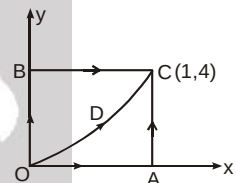


- (A) The speed of hemisphere after collision is V_0
 (B) The speed of solid sphere after collision is $\frac{V_0}{2}$
 (C) The loss in kinetic energy of the system is $\frac{1}{4}mV_0^2$
 (D) The final kinetic energy of hemisphere $\frac{1}{4}$ is th the initial kinetic energy of sphere
5. A smooth sphere A of mass m collides elastically with an identical sphere B at rest. The velocity of A before collision is 8 m/s in a direction making 60° with the line joining the centres at the time of impact. Which of the following is/are possible :
- (A) the sphere A comes to rest after collision
 (B) the sphere B will move with a speed of 8 m/s after collision
 (C) the directions of motion of A and B after collision are at right angles
 (D) the speed of B after collision is 4 m/s.
6. One end of an ideal spring (spring constant k) is attached to a rigid wall and other end to a block kept on a horizontal smooth surface. The spring is always horizontal. The block is released from rest when the spring is elongated by x
- (A) The maximum work done by the spring on the block in the subsequent motion is $\frac{1}{2}kx^2$
 (B) The block is at rest when the spring is in its undeformed state
 (C) The spring always does positive work when the block moves away from the wall.
 (D) The spring does negative work whenever the block slows down.

COMPREHENSION

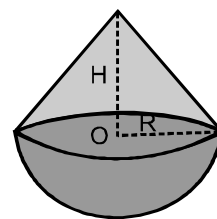
A particle is moved along the different paths OAC, OBC & ODC as shown in the fig. Path ODC is a parabola, $y = 4x^2$. A force $\vec{F} = xy\hat{i} + x^2y\hat{j}$ acts on the particle.

Now answer the following questions :



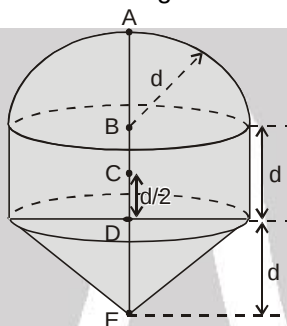
7. The work done by the force $\vec{F}$ on particle :
- (A) on path OAC is 8J
 (B) On path OBC is 2J
 (C) On path ODC is $\frac{19}{3}$ J
 (D) In a loop ODCBO is $\frac{13}{3}$ J
8. The force $\vec{F} = xy\hat{i} + x^2y\hat{j}$
- (A) is conservative
 (B) is non-conservative
 (C) work is non-zero while particle moving on x-axis.
 (D) work is zero while particle moving on y-axis.
9. Two particle of equal mass 1 and 2 are at rest at A and B respectively. they come under the influence of force $\vec{F}$, particle 1 moves on part A to C, whereas, particle 2 moves on part B to C, and they acquire kinetic energies E_1 and E_2 by reaching point C respectively. Then (Ignore work done by all other forces)
- (A) $E_1 < E_2$
 (B) $E_2 < E_1$
 (C) $E_1 = 4E_2$
 (D) $E_2 = 4E_1$

10. A homogenous body consists of right circular conical portion attached to a hemispherical portion of radius $R = \sqrt{3}$ m. The height H of cone if the centre of gravity of the composite body coincides with the centre O of the circular base of the cone is n . then find n (where n is integer)



11. Match the following

Following is a solid object formed by three parts which are a solid hemisphere, solid cylinder and a solid cone. The material of the object is uniform and all the above parts are made up of the same material. The dimensions of the objects are indicated in the figure. The points A, B, C, D, E lie on the common axis of the system as shown in the figure. Point C is the centre of the cylinder.



Column I

- (A) Centre of mass of the whole system lies on segment
 (B) Centre of mass of the system of only hemisphere and cylinder lies on segment
 (C) Centre of mass of the system of only cone and cylinder lies on segment
 (D) Centre of mass of the system of only hemisphere and cone lies on segment

Column II

- (p) AB
 (q) BC
 (r) CD
 (s) DE

DPP No. : B31 (JEE-ADVANCED)

Total Marks : 42

Single choice Objective ('-1' negative marking) Q.1 to Q.2

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

Subjective Questions ('-1' negative marking) Q.9

Match the Following (no negative marking) Q.10

Max. Time : 27 min.

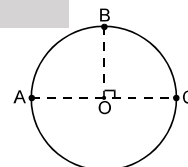
(3 marks 2 min.) [06, 04]

(4 marks 2 min.) [24, 12]

(4 marks 5 min.) [04, 05]

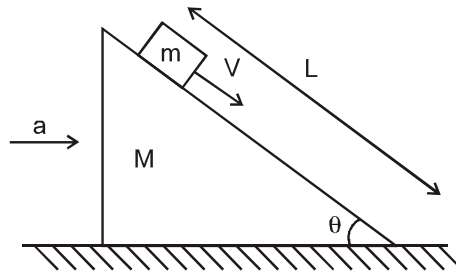
(8 marks 6 min.) [08, 06]

1. Three particle A, B & C move in a circle of radius $r = \frac{1}{\pi}$ m, in anticlockwise direction with speeds 1 m/s, 2.5 m/s and 2 m/s respectively. The initial positions of A, B and C are as shown in figure. The ratio of distance travelled by B and C by the instant A, B and C meet for the first time is
 (A) 3 : 2 (B) 5 : 4 (C) 3 : 5 (D) 3 : 7



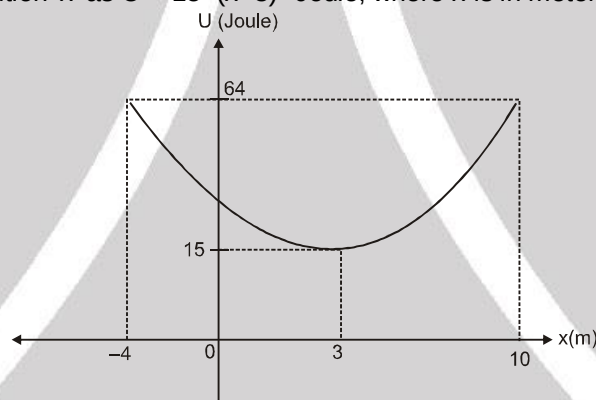
2. A particle is tied to one end of a light inextensible string and is moving in a vertical circle, the other end of string is fixed at the centre. Then for complete motion in circle, which is correct. (air resistance is negligible)
 (A) Acceleration of the particle is directed towards the centre.
 (B) Total mechanical energy of the particle and earth remains constant
 (C) Tension in the string remains constant
 (D) Acceleration of the particle remains constant

3. A wedge of mass M is pushed with a constant acceleration of $a = g \tan \theta$ along a smooth horizontal surface and a block of mass m is projected down the smooth incline of the wedge with a velocity V relative to the wedge.



- (A) The time taken by the block to cover distance L on the incline plane is $\frac{L}{V}$
 (B) The time taken by the block to cover distance L on the incline plane is $\sqrt{\frac{2L}{g \sin \theta}}$
 (C) The normal reaction between the block and wedge is $mg \sec \theta$
 (D) The horizontal force applied on the wedge to produce acceleration a is $(M + m) g \tan \theta$.

4. A single conservative force $F(x)$ acts on a particle that moves along the x -axis. The graph of the potential energy with x is given. At $x = 5\text{m}$, the particle has a kinetic energy of 50J and its potential energy is related to position ' x ' as $U = 15 + (x-3)^2$ Joule, where x is in meter. Then :



- (A) The mechanical energy of system is 69 J .
 (B) The mechanical energy of system is 19 J .
 (C) At $x = 3$, the kinetic energy of particle is minimum
 (D) The maximum value of kinetic energy is 54 J .

5. A particle moves in a circle of radius r with angular velocity $\vec{\omega}$. At some instant its velocity $\vec{v}$ and radius vector with respect to centre of the circle is $\vec{r}$. At this instant centripetal acceleration $\vec{a}_c$ will be :

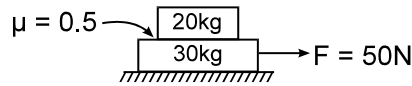
- (A) $\vec{\omega} \times \vec{v}$ (B) $\vec{v} \times \vec{\omega}$ (C) $\vec{\omega} \times (\vec{\omega} \times \vec{r})$ (D) $\vec{v} \times (\vec{r} \times \vec{\omega})$

6. A point moves in the plane so that its tangential acceleration $w_t = a$ & its normal acceleration $w_n = bt^4$, where a & b are positive constants & t is time. At the moment $t = 0$ the point was at rest. Find how the curvature radius R of the point's trajectory & the total acceleration w depend on the distance covered s .

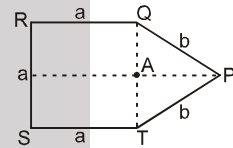
- (A) $R = \frac{a^3}{2bs}$ (B) $R = \frac{a^2 s}{2b}$
 (C) $w = a \sqrt{1 + \left(\frac{4bs^2}{a^3}\right)^2}$ (D) $w = a \sqrt{1 + \left(\frac{4bs}{a^3}\right)^2}$

COMPREHENSION :

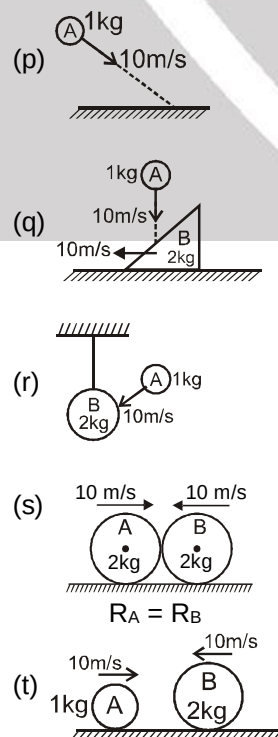
A small block of mass 20 kg rests on a bigger block of mass 30 kg, which lies on a smooth horizontal plane. Initially the whole system is at rest. The coefficient of friction between the blocks is 0.5. A horizontal force $F = 50$ N is applied on the lower block. (Assume 30 kg block is long)



7. For the given system choose the correct options regarding work done by the frictional force in $t = 2$ s
 (A) The work done by the frictional force on upper block is 40J
 (B) The work done by the frictional force on upper block is -20 J
 (C) The work done by the frictional force on lower block is -40 J
 (D) The work done by the frictional force on lower block is 20J
8. For the given system choose the correct options regarding velocity of blocks at $t = 4$ s,
 (A) The velocity of upper block is 4m/s (B) The velocity of lower block is 3m/s
 (C) The velocity of upper block is 6m/s (D) The velocity of lower block is 4m/s
9. A homogeneous plate PQRST is as shown in figure. The centre of mass of plate lies at midpoint A of segment QT. If the ratio of $\left(\frac{b}{a}\right)^2$ is $\frac{m}{n}$ then find minimum value of $m + n$, where a and b are integer (PQ = PT = b ; QR = RS = ST = a)
10. In column-II different situations are shown in which one object collides with the another object. In each case friction is absent and neglect effect of non-impulsive forces. In column-I different direction are given. You have to match the directions for each case in which momentum conservation can be applied on object A or object B or system A & B. (Assume that objects do not bounce off the ground)

**Column-I**

- (A) Along the line of impact
- (B) Perpendicular to line of impact
- (C) In horizontal direction
- (D) In vertical direction

Column-II

DPP No. : B32 (JEE-MAIN)

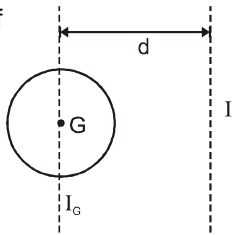
Total Marks : 60

Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

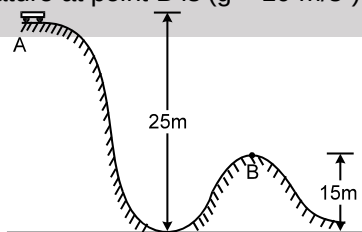
(3 marks 2 min.)

[60, 40]

1. Write the theorem of parallel axes for the given figure, where G is the centre of mass of the body :
- (A) $I = I_G - Md^2$
 (B) $I = I_G + Md^2$
 (C) $I + I_G = Md^2$
 (D) $I_G = I + Md^2$
- 
2. The moment of inertia of a thin scale of length L and mass M about an axis passing through the centre of mass and perpendicular to its length would be
- (A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{4}$ (C) $\frac{ML^2}{3}$ (D) ML^2
3. The moment of inertia of a solid sphere of mass M and radius R about a tangential axis is
- (A) $\frac{5}{3}MR^2$ (B) $\frac{2}{5}MR^2$ (C) $\frac{2}{3}MR^2$ (D) $\frac{7}{5}MR^2$
4. The moment of inertia of solid sphere is 20 kg-m^2 about the diameter. The moment of inertia about any tangent will be :
- (A) 70 kg-m^2 (B) 35 kg-m^2 (C) 50 kg-m^2 (D) 20 kg-m^2
5. A wheel of mass 10 kg has a moment of inertia of 160 kg m^2 about its own axis, the radius of gyration will be :
- (A) 10 m (B) 8 m (C) 6 m (D) 4 m
6. From the theorem of perpendicular axes. If the lamina is in X- Y plane
- (A) $I_x - I_y = I_z$ (B) $I_x + I_z = I_y$ (C) $I_x + I_y = I_z$ (D) $I_y + I_z = I_x$
7. The moment of inertia of a thin ring of mass M and radius R about its diameter is
- (A) $\frac{1}{2}MR^2$ (B) $\frac{3}{2}MR^2$ (C) MR^2 (D) $2MR^2$
8. The rotational kinetic energy of a rigid body of mass m rotating about a fixed axis with angular velocity ω having I as moment of inertia about that axis, is :
- (A) $\frac{1}{2}m\omega^2$ (B) $\frac{1}{2}I\alpha^2$ (C) $\frac{1}{2}I\omega^2$ (D) $\frac{1}{2}mv^2$
9. The moment of inertia of a uniform thin rod of length L and mass M about an axis passing through a point at a distance of L/3 from one of its ends and perpendicular to the rod is
- (A) $\frac{7ML^2}{48}$ (B) $\frac{ML^2}{9}$ (C) $\frac{ML^2}{12}$ (D) $\frac{ML^2}{3}$
10. Two rings have their moments of inertia in the ratio 2 : 1 and their diameters are in the ratio 2 : 1. The ratio of their masses will be
- (A) 2 : 1 (B) 1 : 2 (C) 1 : 4 (D) 1 : 1
11. One circular ring and one circular disc, both are having the same mass and radius. The ratio of their moments of inertia about the axes passing through their centres and perpendicular to their planes. will be
- (A) 1 : 1 (B) 2 : 1 (C) 1 : 2 (D) 4 : 1

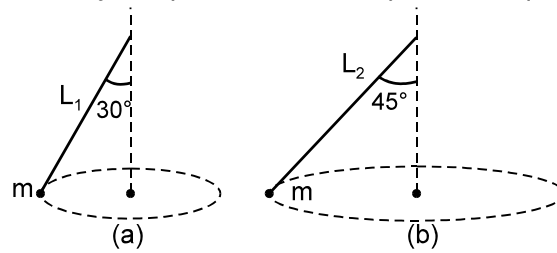


12. A 1.0kg ball drops vertically into a floor from a height of 25 cm. It rebounds to a height of 4cm. The coefficient of restitution for the collision is -
 (A) 0.16 (B) 0.32 (C) 0.40 (D) 0.56
13. If the angle between two vectors is 30° , then $\frac{\vec{A} \cdot \vec{B}}{|\vec{A} \times \vec{B}|}$ is :
 (A) $\frac{1}{\sqrt{3}}$
 (B) $\sqrt{3}$
 (C) 1
 (D) cannot be determined because values of $|\vec{A}|$ and $|\vec{B}|$ are not given .
14. A body of mass 2kg travels according to the law $x(t) = pt + qt^2 + rt^3$ where $p = 3\text{ms}^{-1}$, $q = 4\text{ms}^{-2}$ and $r = 5\text{ms}^{-3}$. The force acting on the body at $t = 2$ seconds is
 (A) 136 N (B) 134 N (C) 158 N (D) 68 N
15. A body with mass 5 kg is acted upon by a force $\vec{F} = (-3\hat{i} + 4\hat{j})\text{N}$. If its initial velocity at $t = 0$ is $\vec{v} = 6\hat{i} - 12\hat{j}\text{ms}^{-1}$, the time at which it will just have a velocity along the y-axis is :
 (A) never (B) 10 s (C) 2 s (D) 15 s
16. A car is moving on a plane inclined at 30° to the horizontal with an acceleration of 9.8 m/s^2 parallel to the plane upward. A bob is suspended by a string from the roof. The angle in degrees which the string makes with the vertical is : (Assume that the bob does not move relative to car) [$g = 9.8\text{ m/s}^2$]
 (A) 20° (B) 30° (C) 45° (D) 60°
17. The density of a non-uniform rod of length 1m is given by $\rho(x) = a(1+bx^2)$ where a and b are constants and $0 \leq x \leq 1$. The centre of mass of the rod will be at
 (A) $\frac{3(2+b)}{4(3+b)}$ (B) $\frac{4(2+b)}{3(3+b)}$ (C) $\frac{3(3+b)}{3(2+b)}$ (D) $\frac{4(3+b)}{3(2+b)}$
18. Figure shows the roller coaster track. Each car will start from rest at point A and will roll with negligible friction. It is important that there should be at least some small positive normal force exerted by the track on the car at all points, otherwise the car would leave the track. With the above fact, the minimum safe value for the radius of curvature at point B is ($g = 10\text{ m/s}^2$) :



- (A) 20 m (B) 10 m (C) 40 m (D) 25 m
19. The angular velocity of a rigid body about any point of that body is same:
 (A) only in magnitude
 (B) only in direction
 (C) both in magnitude and direction necessarily
 (D) both in magnitude and direction about some points, but not about all points.

20. Two particles tied to different strings are whirled in a horizontal circle as shown in figure. The ratio of lengths of the strings so that they complete their circular path with equal time period is:



- (A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{\frac{2}{3}}$ (C) 1 (D) None of these

DPP No. : B33 (JEE-ADVANCED)

Total Marks : 46

Max. Time : 29 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.9

(4 marks 2 min.) [28, 14]

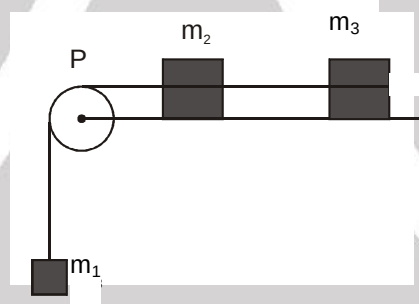
Subjective Questions ('-1' negative marking) Q.10

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.11

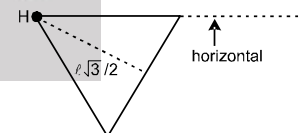
(8 marks 6 min.) [08, 06]

1. A system consists of three masses m_1 , m_2 and m_3 connected by a string passing over a pulley P. The mass m_1 hangs freely and m_2 and m_3 are on a rough horizontal table (the coefficient of friction = μ). The pulley is frictionless and of negligible mass. The downward acceleration of mass m_1 is :
(Assume $m_1 = m_2 = m_3 = m$)



- (A) $\frac{g(1+g\mu)}{9}$ (B) $\frac{2g\mu}{3}$ (C) $\frac{g(1-2\mu)}{3}$ (D) $\frac{g(1-2\mu)}{2}$

2. A rigid equilateral triangular frame made of three identical thin rods (mass = m & length = ℓ) is free to rotate smoothly in vertical plane. Frame is hinged at one of its vertices H. Frame is released from rest from the position shown in figure then select incorrect alternative (s).

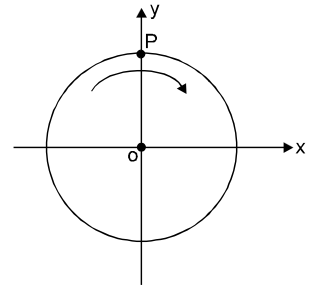


- (A) Net initial torque about point H is $\frac{3}{2} mg\ell$
(B) Initial angular acceleration of the frame is g/ℓ
(C) Initial force of hinge on the frame is $3 mg$
(D) Initial force of hinge on the frame is $\sqrt{3} mg$

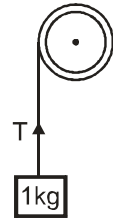
3. Two identical balls each of mass m moving on straight track approaching towards each other with same speed. Final kinetic energy of the two ball system is equal to the total energy loss during collision. E is the total kinetic energy of the balls before collision and E' is after collision and coefficient of restitution is e . Then choose the correct options

- (A) $\frac{E}{E'} = 2$ (B) $e = \frac{1}{2}$ (C) $e = \frac{1}{\sqrt{2}}$ (D) $\frac{E}{E'} = \sqrt{2}$

4. A ring rotates about z axis as shown in figure. The plane of rotation is xy. At a certain instant the acceleration of a particle P (shown in figure) on the ring is $(6\hat{i} - 8\hat{j}) \text{ m/s}^2$. At that instant, choose the correct option(s). (Radius of the ring is 2m)
- (A) angular acceleration of the ring is $-3\text{rad/sec}^2 \hat{k}$
 (B) angular acceleration of the ring is $3\text{rad/sec}^2 \hat{k}$
 (C) the angular velocity of the ring is $-2 \text{ rad/sec} \hat{k}$
 (D) the angular velocity of the ring is $3 \text{ rad/sec} \hat{k}$

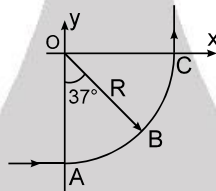


5. A uniform disc of mass 2kg and radius 1m is mounted on an horizontal axle supported on fixed frictionless bearings. A light cord is wrapped around the rim of the disc and mass of 1kg is tied to the free end. If it is released from rest,
- (A) the tension in the cord is 5N
 (B) in first 4 seconds the angular displacement of the disc is 40 rad.
 (C) the work done by the torque on the disc in first 4 sec. is 200J
 (D) the increase in the kinetic energy of the disc in the first 4 seconds is 200J.



COMPREHENSION

A car initially traveling eastwards turns north by traveling in a quarter circular path of radius R metres at uniform speed as shown in figure. The car completes the turn in T second.

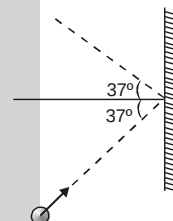


6. Choose the correct option(s) :
- (A) acceleration of car at point B is $\vec{a} = \frac{\pi^2 R}{20T^2} (-3\hat{i} + 4\hat{j}) \text{ m/s}^2$
 (B) acceleration of car at point B $\vec{a} = \frac{\pi^2 R}{20T^2} (-4\hat{i} + 3\hat{j}) \text{ m/s}^2$
 (C) the magnitude of average acceleration of car during T second period is $\frac{\pi R}{\sqrt{2} T^2} \text{ m/s}^2$ towards NW
 (D) the magnitude of average acceleration of car during T second period is $\frac{R}{\sqrt{2} T^2} \text{ m/s}^2$ towards ES
7. Which of the following is correct :
- (A) average velocity of car during T second period is $\frac{\sqrt{2}R}{T} \text{ m/s}$ towards E-N
 (B) average velocity of car during T second period is $\frac{\sqrt{2}R}{T} \text{ m/s}$ towards N-W
 (C) Friction force on car acting during motion is always towards O and has constant magnitude
 (D) Friction force on car acting during motion is always towards O and is constant.

8. Which of the following is/are correct
- (A) The angle between the acceleration at $t = \frac{T}{2}$ second and the velocity vector $\vec{v}_A - \vec{v}_C$ is π (where $\vec{v}_A$ and $\vec{v}_C$ denotes the velocity of car at points A and C respectively)
- (B) The angle between the acceleration and the velocity at $t = \frac{5T}{17}$ second is $\frac{2\pi}{3}$
- (C) The angle between the average velocity over T second duration and the acceleration at $t = \frac{T}{2}$ second is $\frac{\pi}{2}$
- (D) The angle between the displacement during T second period and the velocity at $t = \frac{T}{2}$ is zero

9. Choose the correct option(s) (more than one may correct)
- (A) The coordinate of the car at $t = \frac{2T}{3}$ is $\left(\frac{\sqrt{3}R}{2}, -\frac{R}{2}\right)$
- (B) The ratio of magnitude of average velocity over T second duration and the magnitude of velocity at $t = \frac{13T}{17}$ is equal to $\frac{2\sqrt{2}}{\pi}$
- (C) At $t = \frac{3T}{2}$ second the ratio of speed and magnitude of velocity is unity.
- (D) Magnitude of displacement of car in $t = \frac{T}{3}$ to T second duration is R .

10. A particle moving on a smooth horizontal surface strikes a stationary wall. The angle of strike is equal to the angle of rebound & is equal to 37° and the coefficient of restitution with wall is $e = \frac{1}{5}$. Find the friction coefficient between wall and the particle in the form $\frac{X}{10}$ and fill value of X.:



11. Two identical uniform solid smooth spheres each of mass m each approach each other with constant velocities such that net momentum of system of both spheres is zero. The speed of each sphere before collision is u . Both the spheres then collide. The condition of collision is given for each situation of column-I. In each situation of column-II information regarding speed of sphere(s) is given after the collision is over. Match the condition of collision in column-I with statements in column-II.

Column-I

- (A) Collision is perfectly elastic and head on
- (B) Collision is perfectly elastic and oblique
- (C) Coefficient of restitution is $e = \frac{1}{2}$ and collision is head on
- (D) Coefficient of restitution is $e = \frac{1}{2}$ and collision is oblique

Column-II

- (p) speed of both spheres after collision is u
- (q) velocity of both spheres after collision is different
- (r) speed of both spheres after collision is same but less than u .
- (s) speed of one sphere may be more than u .



DPP No. : B34 (JEE-MAIN)

Total Marks : 60

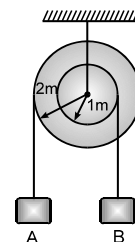
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

(3 marks 2 min.)

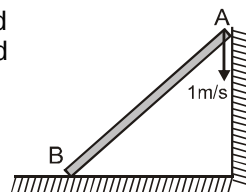
[60, 40]

- A cricket ball of mass 150 g is moving with a velocity of 12 m/sec and is hit by a bat so that the ball is turned back with a velocity of 20 m/sec. The force of blow acts for 0.01 s on the ball. Find the average force exerted by the bat on the ball.
(A) 320 N (B) 480 N (C) 160 N (D) 240 N
- Two points of a rod move with velocities $3v$ & v perpendicular to the rod and in the same direction, separated by a distance ' r '. Then the angular velocity of the rod is:
(A) $\frac{3v}{r}$ (B) $\frac{4v}{r}$ (C) $\frac{5v}{r}$ (D) $\frac{2v}{r}$
- A uniform rod of mass $6M$ and length $6l$ is bent to make an equilateral hexagon. Its M.I. about an axis passing through the centre of mass and perpendicular to the plane of hexagon is:
(A) $5ml^2$ (B) $6ml^2$ (C) $4ml^2$ (D) $ml^2/12$
- The moment of inertia of a door of mass m , length 2ℓ and width ℓ about its longer side is
(A) $\frac{11m\ell^2}{24}$ (B) $\frac{5m\ell^2}{24}$ (C) $\frac{m\ell^2}{3}$ (D) none of these
- A rigid body of mass m rotates with angular velocity ω about an axis at a distance a from the centre of mass G . The radius of gyration about a parallel axis through G is k . The kinetic energy of rotation of the body is
(A) $\frac{1}{2}mk^2\omega^2$ (B) $\frac{1}{2}ma^2\omega^2$ (C) $\frac{1}{2}m(a^2+k^2)\omega^2$ (D) $\frac{1}{2}m(a+k)^2\omega^2$
- A flywheel rotating about a fixed axis has a kinetic energy of 360 joule when its angular speed is 30 rad/sec. The moment of inertia of the wheel about the axis of rotation is
(A) $0.6 \text{ kg} \times \text{m}^2$ (B) $0.15 \text{ kg} \times \text{m}^2$ (C) $0.8 \text{ kg} \times \text{m}^2$ (D) $0.75 \text{ kg} \times \text{m}^2$
- The product of moment of inertia and angular acceleration is
(A) force (B) torque (C) angular momentum (D) rotational kinetic energy
- A particle is at a distance r from the axis of rotation. A given torque τ produces some angular acceleration in it. If the mass of the particle is doubled and its distance from the axis is halved, the value of torque to produce the same angular acceleration is
(A) $\tau/2$ (B) τ (C) 2τ (D) 4τ
- A constant torque of 31.4 N-m is exerted on a pivoted wheel. If angular acceleration of wheel is $4\pi \text{ rad/sec}^2$, then the moment of inertia of the wheel is :
(A) 2.5 kg-m^2 (B) 3.5 kg-m^2 (C) 4.5 kg-m^2 (D) 5.5 kg-m^2
- The rotational kinetic energy of a rigid body of moment of inertia 5 kg-m^2 is 10 joules. The angular momentum about the axis of rotation would be :
(A) 100 joule-sec (B) 50 joule-sec (C) 10 joule-sec (D) 2 joule-sec
- In the pulley system shown, if radii of the bigger and smaller pulley are 2 m and 1m respectively and the acceleration of block A is 5 m/s^2 in the downward direction, then the acceleration of block B will be:
(A) 0 m/s^2
(B) 5 m/s^2
(C) 10 m/s^2
(D) $5/2 \text{ m/s}^2$

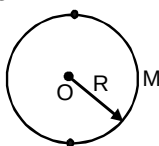


12. A rod of length 1m is sliding in a corner as shown. At an instant when the rod makes an angle of 60° with the horizontal plane, the velocity of point A on the rod is 1m/s. The angular velocity of the rod at this instant is :

(A) 2 rad/s (B) 1.5 rad/s
(C) 0.5 rad/s (D) 0.75 rad/s



13. The moment of inertia of a sphere of density ρ and radius R about an axis passing through its centre is proportional to
(A) R^2 (B) R^3 (C) R^4 (D) R^5
14. A man sitting on a rotating stool with his arms stretched out, suddenly lowers his hands
(A) his angular velocity decreases (B) his moment of inertia decreases
(C) his angular velocity remains constant (D) his angular momentum increases
15. A smooth uniform rod of length L and mass M has identical beads of negligible size, each of mass m , which can slide freely along the rod. Initially the two beads are at the centre of the rod and the system is rotating with angular velocity ω_0 about an axis perpendicular to the rod and passing through the mid point of the rod. There are no external forces. When the beads reach the ends of the rod, the angular velocity of the rod would be
(A) $\frac{M\omega_0}{M+2m}$ (B) $\frac{M\omega_0}{M+4m}$ (C) $\frac{M\omega_0}{M+6m}$ (D) $\frac{M\omega_0}{M+8m}$
16. The angular momentum of particle of mass 0.01 Kg and position vector $\vec{r} = (10\hat{i} + 6\hat{j})$ meter and moving with a velocity $5\hat{i}$ metre/sec about the origin will be
(A) $-0.3\hat{k}$ Joule sec (B) $3\hat{k}$ Joule sec (C) $1/3$ Joule sec (D) $0.03\hat{k}$ Joule sec
17. A uniform heavy disc is rotating at constant angular velocity ω about a vertical axis through its centre and perpendicular to the plane of the disc. Let L be its angular momentum. A lump of plastic is dropped vertically on the disc and sticks to it. Which will be constant
(A) ω (B) ω and L both (C) L only (D) Neither ω nor L
18. The angular momentum of a system of particles is conserved
(A) When no external force acts upon the system
(B) When no external torque acts on the system
(C) When no external impulse acts upon the system
(D) When axis of rotation remains same
19. A disc of mass M and radius R is suspended in a vertical plane by a horizontal axis passing through its centre. After sticking an object of same mass M at its rim the mass is raised to the position of maximum height. Now this unstable system is released. When the object passes the position right below the centre the angular velocity of the system would be



- (A) $\sqrt{\frac{8g}{3R}}$ (B) $\sqrt{\frac{2g}{3R}}$ (C) $\sqrt{\frac{3g}{2R}}$ (D) $\sqrt{\frac{3g}{8R}}$
20. A particle performs uniform circular motion with an angular momentum L . If the frequency of particle's motion is doubled and its kinetic energy is halved, the angular momentum becomes
(A) $2L$ (B) $4L$ (C) $L/2$ (D) $L/4$

DPP No. : B35 (JEE-ADVANCED)

Total Marks : 43

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.8

(4 marks 2 min.) [28, 14]

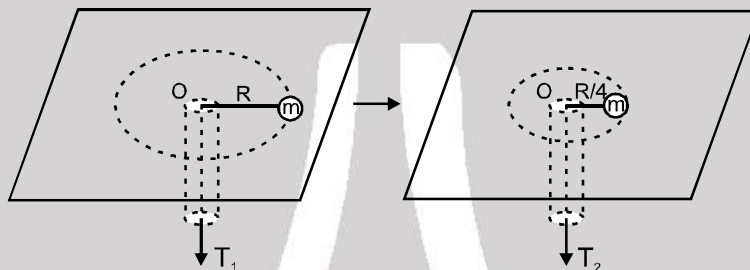
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. A ball of mass m is connected by a light inextensible cord and is rotated in a circle of radius R on a smooth fixed horizontal table. Initially the angular velocity of the ball was ω_0 and pulling force applied was T_1 . Now the pulling force is increased to T_2 , until the radius of rotation of the ball becomes $\frac{R}{4}$. Then

ratio $\frac{T_2}{T_1}$ is :

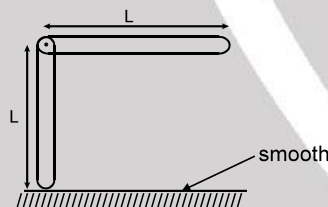
(A) 64

(B) 16

(C) 48

(D) 256

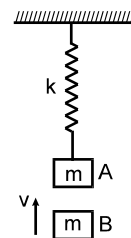
2. Two identical rods are joined at one of their ends by a pin. Joint is smooth and rods are free to rotate about the joint. Rods are released in vertical plane on a smooth surface as shown in the figure. The displacement of the joint and centre of mass of system from its initial position to the final position is (i.e. when the rods lie straight on the ground) :

(A) $S_J = \frac{\sqrt{17}}{4}$ (B) $S_J = \frac{\sqrt{5}L}{2}$ (C) $S_{COM} = \frac{3L}{4}$ (D) $S_{COM} = L$

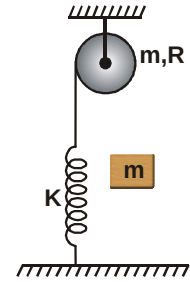
3. Block 'A' is hanging from a vertical spring and is at rest. Block 'B' strikes the block 'A' with velocity ' v ' and sticks to it. And v is such that it just attains its natural length, then choose the correct option(s) :

(A) $v = \sqrt{\frac{6mg^2}{k}}$

(B) $v = \sqrt{\frac{10mg^2}{k}}$

(C) The energy loss during collision is $\frac{3m^2g^2}{2K}$ J(D) The energy loss during collision is $\frac{m^2g^2}{2K}$ J

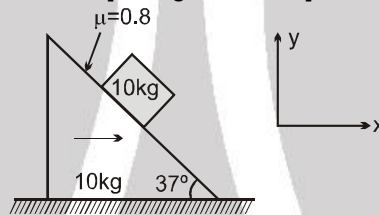
4. A uniform disc of mass m and radius R is free to rotate about its fixed horizontal axis without friction. There is sufficient friction between the inextensible light string and disc to prevent slipping of string over disc. At the shown instant extension in light spring is $\frac{3mg}{K}$, where m is mass of block, g is acceleration due to gravity and K is spring constant.



- (A) Acceleration of block just after it is released is $\frac{4g}{3}$.
- (B) The tension in the string connecting mass m , just after the system is released is mg
- (C) The tension in the string connecting mass m , just after the system is released is $3mg$.
- (D) The tension in the string connecting mass m , just after the system is released is $\frac{7mg}{3}$

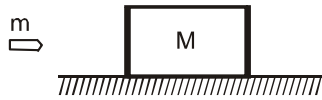
COMPREHENSION

A wedge is moving rightwards on which a block of mass 10kg is placed on it. Friction coefficient between the wedge and the block is 0.8 . [take $g = 10\text{ m/s}^2$].



5. If wedge is moving with constant velocity then
- (A) acceleration of block is zero. (B) acceleration of block is 2 m/s^2 .
- (C) friction force on the block is 60N (D) friction force on the block is 64N
6. If wedge is moving with acceleration of $\vec{a} = 2\hat{i}\text{ m/s}^2$ then
- (A) magnitude of friction force on block is 44N (B) magnitude of friction force on block is 64N
- (C) magnitude of contact force on block is 92N (D) magnitude of contact force on block is 102.2N
7. choose the correct option(s)
- (A) If acceleration of wedge is 7.5 m/s^2 then friction force on block is zero.
- (B) If acceleration of wedge is 10 m/s^2 then friction force on block is acting towards downward direction
- (C) If acceleration of wedge is 3 m/s^2 then friction force on block is acting towards upward direction
- (D) If acceleration of wedge is 3 m/s^2 then friction force on block is acting towards downward direction
8. Consider the length of the incline is 12m . The wedge is moving with an acceleration $a = 2\hat{i}$ and the block is released from the rest then
- (A) The time taken by the block to reach at the bottom of the wedge is 2 sec .
- (B) The normal force applied by the block on the wedge is 92 N .
- (C) The acceleration block w.r.t. wedge is 6 m/s^2 downwards along the incline surface
- (D) The acceleration of block w.r.t. ground is $2\hat{i}\text{ m/s}^2$
9. A block of mass $m = 20\text{ kg}$ is kept at a distance $R = 1\text{m}$ from central axis of rotation of a round turn table (A table whose surface can rotate about central axis). Table starts from rest and rotates with constant angular acceleration, $\alpha = 3\text{ rad/sec}^2$. The friction coefficient between block and table is $\mu = 0.5$. At time $t = \frac{x}{30}$ from starting of motion (i.e. $t = 0$) the block is just about to slip. Find the value of x . ($g = 10\text{ m/s}^2$)

10. A bullet fired with speed u sticks to the block after penetrating some distance 'X' inside the block after that both move together with same velocity, if average force of resistance on bullet by block was P , 'S' is distance moved by bullet during time of collision. 'y' is distance moved by block in duration of collision.



Column-I

- (P) maximum possible heat generated in collision
(Q) Distance 'S' is equal to
(R) Work done by bullet on block
(S) Distance 'y' moved by block in duration of collision

Column-II

- (A) $\left(\frac{M^2 + 2Mm}{2P(M+m)^2} \right) u^2 m$
(B) $\frac{1}{2} \left(\frac{mM}{m+M} \right) u^2$
(C) Py
(D) $\frac{1}{2} \frac{m^2 u^2}{(M+m)^2} \frac{M}{P}$

Codes :

	P	Q	R	S
(A)	3	1	2	4
(B)	2	3	1	4
(C)	4	2	1	3
(D)	2	1	3	4

DPP No. : B36 (JEE-MAIN)

Total Marks : 60

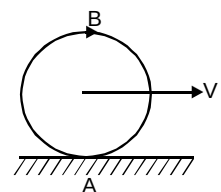
Single choice Objective ('-1' negative marking) Q.1 to Q.20

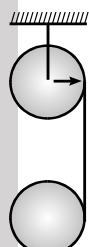
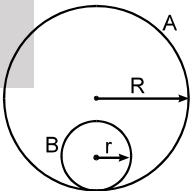
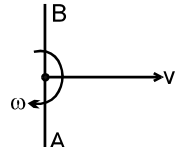
Max. Time : 40 min.

(3 marks 2 min.)

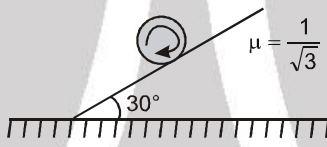
[60, 40]

- A particle moves in a circular path so that its distance travel varies with time t as $s = 3t^2 + 6t$. Then its acceleration at $t = 1$ sec. is (radius of path is 12 m) -
(A) $6\sqrt{5}$ m/s² (B) 6 m/s² (C) 12 m/s² (D) $12\sqrt{3}$ m/s²
- A particle of mass m is rotating in a plane in circular path of radius r . Its angular momentum is L . The centripetal force acting on the particle is
(A) L^2/mr (B) L^2m/r^2 (C) L^2/m^2r^2 (D) L^2/mr^3
- The position of a particle is given by $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k})$ and momentum $\vec{P} = (3\hat{i} + 4\hat{j} - 2\hat{k})$. The angular momentum is perpendicular to
(A) X-axis (B) Y-axis
(C) Z-axis (D) Line at equal angles to all the three axes
- A small ball of radius r rolls down without sliding in a big hemispherical bowl. of radius R . What would be the ratio of the translational and rotational kinetic energies at the bottom of the bowl
(A) 2 : 1 (B) 3 : 2 (C) 4 : 3 (D) 5 : 2
- A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass K . If radius of the ball be R , then the fraction of total energy associated with its rotational energy will be
(A) $\frac{K^2}{R^2}$ (B) $\frac{K^2}{K^2 + R^2}$ (C) $\frac{R^2}{K^2 + R^2}$ (D) $\frac{K^2 + R^2}{R^2}$
- A wheel is rolling uniformly along a level road (see fig.) The speed of transitional motion of the wheel axis is V . What are the speeds of the points A and B on the wheel rim relative to the road at the instant shown in the fig.
(A) $V_A = V$, $V_B = 0$ (B) $V_A = 0$, $V_B = V$
(C) $V_A = 0$, $V_B = 0$ (D) $V_A = 0$, $V_B = 2V$



7. Moment of inertia of a uniform quarter disc of radius R and mass M about an axis through its centre of mass and perpendicular to its plane is :
- (A) $\frac{MR^2}{2} - M \left(\frac{4R}{3\pi} \right)^2$ (B) $\frac{MR^2}{2} - M \left(\sqrt{2} \frac{4R}{3\pi} \right)^2$
 (C) $\frac{MR^2}{2} + M \left(\frac{4R}{3\pi} \right)^2$ (D) $\frac{MR^2}{2} + M \left(\sqrt{2} \frac{4R}{3\pi} \right)^2$
8. A bicyclist comes to a skidding stop in 10 m. During this process, the force on the bicycle due to the road is 200N and is directly opposite to the motion. The work done by the cycle on the road is
 (A) + 2000J (B) - 200J (C) zero (D) - 20,000J
9. A Merry-go-round, made of a ring-like platform of radius R and mass M , is revolving with angular speed ω . A person of mass M is standing on it. At one instant, the person jumps off the round, radially away from the centre of the round (as seen from the round). The speed of the round afterwards is
 (A) 2ω (B) ω (C) $\frac{\omega}{2}$ (D) 0
10. A solid sphere and a solid cylinder having the same mass and radius, roll down the same incline. The ratio of their acceleration will be
 (A) 15 : 14 (B) 14 : 15 (C) 5 : 3 (D) 3 : 5
11. If the radius of the earth is suddenly contracts to half of its present value, then the duration of day will be of
 (A) 6 hours (B) 12 hours (C) 18 hours (D) 24 hours
12. Two identical discs of mass m and radius r are arranged as shown in the figure. If α is the angular acceleration of the lower disc and a_{cm} is acceleration of centre of mass of the lower disc, then relation between a_{cm} , α & r is :
 (A) $a_{cm} = \frac{\alpha}{r}$ (B) $a_{cm} = 2 \alpha r$
 (C) $a_{cm} = \alpha r$ (D) none of these
- 
13. A light string is wrapped around a solid uniform solid cylinder of mass ' m ' and radius ' R '. The string is pulled vertically upward to prevent the centre of mass from falling as the cylinder unwinds the string. Then length of the string unwound when the cylinder has reached a speed ω will be :
 (A) $\frac{R^2 \omega^2}{4g}$ (B) $\frac{R^2 \omega^2}{g}$ (C) $\frac{R^2 \omega^2}{8g}$ (D) none of these
14. A ball of radius r rolls without sliding inside a fixed circular track A. The ratio of the angular velocity of the ball about its centre to the angular velocity of centre of ball about the centre of the track A is :
 (A) 1 (B) $\frac{r}{R-r}$
 (C) $\frac{R-r}{r}$ (D) none of these
- 
15. A solid iron sphere A rolls down an inclined plane, while another hollow sphere B with the same mass and same radius also rolls down the inclined plane. If V_A and V_B are their velocities at the bottom of the inclined plane, then
 (A) $V_A > V_B$ (B) $V_A = V_B$ (C) $V_A < V_B$ (D) $V_A > = < V_B$
16. A uniform rod of length l rotating with an angular velocity ω , while its centre moves with linear velocity $v = \frac{\omega l}{6}$. If the end A of the rod is suddenly fixed, the angular velocity of the rod will be :
 (A) $\frac{3}{4} \omega$ (B) $\frac{\omega}{3}$ (C) $\frac{\omega}{2}$ (D) $\frac{2}{3} \omega$
- 



17. A solid sphere of mass m is situated on a horizontal surface and a tangential force acts at the top of the sphere. If the sphere rolls without slipping then the acceleration of the centre of the sphere would be
 (A) $\frac{5}{3} \frac{F}{m}$ (B) $\frac{3F}{5m}$ (C) $\frac{10F}{7m}$ (D) $\frac{7F}{10m}$
18. Magnitude of angular momentum of a wheel changes from $2L$ to $3L$ in 5 sec. by a constant torque acting opposite to initial direction of rotation. What is the magnitude of the torque?
 (A) $\frac{L}{5}$ (B) L (C) $\frac{2L}{5}$ (D) $\frac{3L}{5}$
19. A ring of mass m and radius R rolls on a horizontal rough surface without slipping due to an applied force ' F '. The friction force acting on ring is : –
 (A) $\frac{F}{3}$ (B) $\frac{2F}{3}$ (C) $\frac{F}{4}$ (D) Zero
20. A disc is rotated about its axis with a certain angular velocity and lowered gently on an inclined plane as shown in Figure, then

 (A) It will rotate at the position where it was placed and then will move downwards
 (B) It will go downwards just after it is lowered
 (C) It will go downwards first and then climb up
 (D) It will climb upwards and then move downwards

DPP No. : B37 (JEE-ADVANCED)

Total Marks : 39

Max. Time : 25 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.7

(4 marks 2 min.) [24, 12]

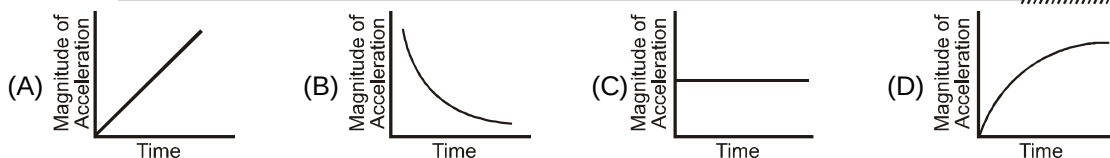
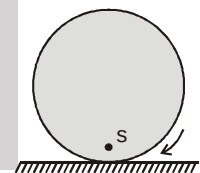
Subjective Questions ('-1' negative marking) Q.8

(4 marks 5 min.) [04, 05]

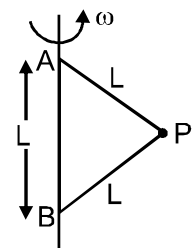
Match the Following (no negative marking) Q.9

(8 marks 6 min.) [08, 06]

1. As shown in figure, S is a point on a uniform disc rolling with uniform angular velocity on a fixed rough horizontal surface. The only forces acting on the disc are its weight and contact forces exerted by horizontal surface. Which graph best represents the magnitude of the acceleration of point S as a function of time



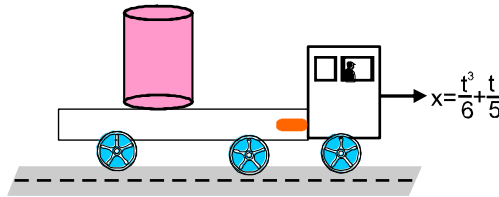
2. A particle P of mass m attached to a vertical axis by two light strings AP and BP of length L each. The separation $AB = L$, P rotates around the axis in a horizontal plane with a constant angular velocity ω . The tension in the strings AP and BP are T_1 and T_2 respectively, then: (take acceleration due to gravity g)



- (A) $T_1 = T_2$
 (B) $T_1 + T_2 = m\omega^2 L$
 (C) $T_1 - T_2 = 2mg$

(D) BP will remain tight only if $\omega \geq \sqrt{\frac{g}{2L}}$

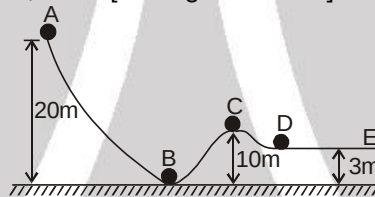
3. A solid cylinder of radius 0.5 m and height 4m is placed on a truck as shown in figure. The coefficient of friction between truck and cylinder is $\mu = 0.3$. The truck is moving such that its displacement (x) varies with time(t) as $x = \frac{t^3}{6} + \frac{t}{5}$ where x is in meters and t in seconds. Then choose the incorrect option/s :
($g = 10 \text{ m/sec.}^2$)



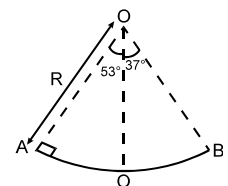
- (A) The cylinder will remain at rest relative to the truck and after sometime, it will slide
(B) The cylinder will remain rest relative to the truck and after sometime, it will topple
(C) The cylinder start sliding and after sometime, it will topple
(D) The cylinder will neither slide nor topple relative to the truck

COMPREHENSION :

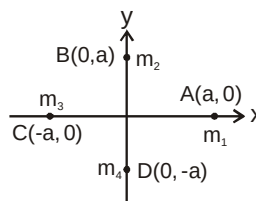
A smooth sphere (mass 10 kg, negligible radius) moves on a smooth curved surface from the point A with a speed of 10 m/s as shown in figure . The sphere reaches the point D passing through point B . If the ground is taken as reference, Then [Take $g = 10 \text{ m/s}^2$]



4. Choose the correct option(s)
(A) The total mechanical energy of the sphere at the point A is zero
(B) The total mechanical energy of the sphere at the point A is 2500 J
(C) The kinetic energy at point B is 2500 J.
(D) The potential energy at point B is 0 J
5. Choose the correct option(s) :
(A) The kinetic energy at C is 1500 J
(B) The kinetic energy at D is 2200 J
(C) The kinetic energy at C is zero
(D) The sphere will go beyond D.
6. Mark out the correct options.
(A) total mechanical energy at the point C is 2500 J.
(B) total mechanical energy at the point B is 2200 J.
(C) Potential energy at the point E is 300 J.
(D) Ratio of velocity at the point B and C is $\sqrt{\frac{5}{3}}$
7. Mark out the correct options.
(A) Ratio of K.E. at point B and A is 5 : 1
(B) Ratio of K.E. at point C to P.E. at point D is 5 : 1
(C) Ratio of K.E. at point D and C is 15 : 22
(D) Potential energy at C is two times of K.E. at A.
8. A section of fixed smooth circular track of radius R in vertical plane is shown in the figure. A block is released from position A and leaves the track at B. The radius of curvature of its trajectory when it just leaves the track at B is $\frac{R}{n}$, then n will be :



9. Four particles of mass m_1, m_2, m_3 and m_4 are situated at points $A(a, 0)$, $B(0, a)$, $C(-a, 0)$ and $D(0, -a)$ respectively as shown. For each situation of column-I, the x and y coordinates of centre of mass of system of given four particles, that is x_{cm} and y_{cm} are given in column-II. Match each situation in column-I with corresponding coordinates of centre of mass as given in column-II.

**Column-I**

- (A) $m_1 \geq m_3, m_4 \geq m_2$
 (B) $m_1 \geq m_3, m_4 < m_2$
 (C) $m_1 < m_3, m_4 \geq m_2$
 (D) $m_1 < m_3, m_4 < m_2$

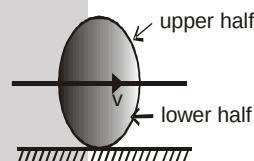
Column-II

- (p) $x_{cm} \geq 0$
 (q) $y_{cm} \leq 0$
 (r) $x_{cm} < 0$
 (s) $y_{cm} > 0$

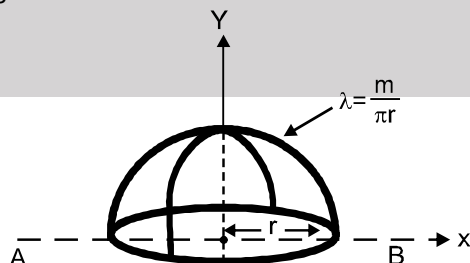
DPP No. : B38 (JEE-ADVANCED)**Total Marks : 46****Single choice Objective ('-1' negative marking) Q.1 to Q.2****One or more than one options correct type ('-1' negative marking) Q.3 to Q.9****Subjective Questions ('-1' negative marking) Q.10****Match the Following (no negative marking) Q.11****Max. Time : 29 min.****(3 marks 2 min.) [06, 04]****(4 marks 2 min.) [28, 14]****(4 marks 5 min.) [04, 05]****(8 marks 6 min.) [08, 06]**

1. Consider a uniform disc of mass 'm' performing pure rolling with velocity 'v' on a fixed rough surface

- (A) Kinetic energy of upper half will be $\frac{3}{8} mv^2$
 (B) Kinetic energy of upper half will be less than $\frac{3}{8} mv^2$
 (C) Kinetic energy of upper half will be more than $\frac{3}{8} mv^2$
 (D) Kinetic energy of upper half will be more than $\frac{3}{4} mv^2$

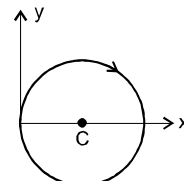


2. Calculate moment of inertia of the system about axis AB shown in figure. Each segment of the system is made of a uniform wire of mass per unit length $\lambda = \frac{m}{\pi r}$, where r is radius of each segment. The axis of full ring is along Y-axis. The axis of the semicircular rings are along x and z-axis. The centres of all rings coincide at the origin.



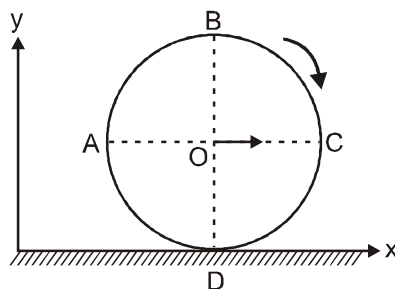
- (A) $\frac{mr^2}{2}$ (B) $\frac{5}{2} mr^2$ (C) $\frac{3}{2} mr^2$ (D) $4 mr^2$

3. A particle revolves in a circle of radius $R = 2m$, in xy plane, in clockwise direction as shown (The centre C of circle lies on x -axis. The y -axis is tangent to the circle). Its centripetal acceleration is of constant magnitude 18 m/s^2 . At time $t = 0$, the x and y coordinates of the particle are $x = R$ and $y = R$. Then at later time 't' the x and y coordinates of particle are :



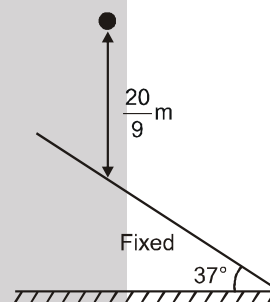
- (A) $x = 2 \cos 3t$ (B) $y = 2 \cos 3t$ (C) $x = 2 (1 + \sin 3t)$ (D) $y = 2 (1 + \sin 3t)$

4. A rigid uniform circular disc rolls without slipping on a horizontal surface with uniform speed along the positive x-direction as shown.



AC and BD are the two diameters and at the instant shown in diagram AC is horizontal and BD is vertical then choose the correct options at the instant shown. (O : centre of the disc)

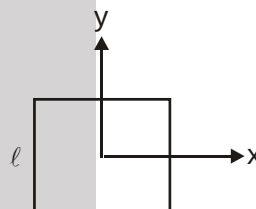
- (A) Sector BOC has greater kinetic energy than sector COD with respect to ground
 (B) Sector ABC has greater kinetic energy than sector ADC with respect to ground
 (C) Sector BOC has same kinetic energy as sector AOB with respect to ground
 (D) Sector BOC has same kinetic energy as sector AOD with respect to the centre of mass frame of disc
5. A ball is dropped on a large smooth inclined plane of angle of inclination 37° , from a height of $\frac{20}{9}$ m above the point of impact. The coefficient of restitution of the impact is $e = \frac{9}{16}$ then [Take $g = 10 \text{ m/s}^2$]
- (A) The velocity of the ball just after the first impact is 5 m/s .
 (B) The maximum distance between the Incline and the ball between the first and second impact is $\frac{9}{16}$ m.
 (C) The time interval between the first and second impact between the ball and incline plane is $\frac{3}{4}$ s.
 (D) The time interval between the first and second impact between the ball and incline plane is 2s.



COMPREHENSION

6. A particle is projected at an angle of 30° w.r.t. horizontal with speed 20 m/s : (use $g = 10 \text{ m/s}^2$) Choose the correct option(s)
- (A) The position vector of the particle after 1s (take point of projection as origin) is $10\sqrt{3} \hat{i} + 5 \hat{j}$.
 (B) The position vector of the particle after 1s (take point of projection as origin) is $5 \hat{i} + \sqrt{3} \hat{j}$.
 (C) The angle between velocity vector and position vector at $t = 1 \text{ s}$. is $\cos^{-1} \left(2\sqrt{\frac{3}{13}} \right)$
 (D) The radius of curvature at the highest point of trajectory is 30 m
7. Let t_1 and t_2 be the time instant at which particle is at half of its maximum height. then :
- (A) average velocity during the time interval $(t_2 - t_1)$ is 10 ms^{-1}
 (B) Average velocity during the time interval $(t_2 - t_1)$ is $10\sqrt{3} \text{ ms}^{-1}$
 (C) Average velocity during the time interval $(t_2 - t_1)$ is in horizontal direction
 (D) Average velocity during the time interval $(t_2 - t_1)$ is vertical direction
8. Let t_0 to be the time after which velocity of the particle will be at 60° to its initial direction of motion, then
- (A) $t_0 = \sqrt{3} \text{ sec}$. (B) $t_0 = 2 \text{ sec}$.
 (C) $v(t = t_0) = 10\sqrt{3} \text{ ms}^{-1}$ (D) $v(t = t_0) = 20 \text{ ms}^{-1}$

9. Let ω be the magnitude of rate of change of angle between velocity vector and horizontal axis, then :
 (A) ω is maximum at the highest point of trajectory
 (B) ω is minimum at the highest point of trajectory
 (C) $\omega(t = 0) = \frac{\sqrt{3}}{4}$ rad/sec.
 (D) $\omega(t = 0) = \frac{\sqrt{3}}{2}$ rad/sec.
10. If a ball is dropped from rest, it bounces from the floor repeatedly. The coefficient of restitution is 0.5 and the speed just before the first bounce is 5ms^{-1} . The total time taken by the ball to come to rest finally is t sec. Then value of $2t$ will be :
11. Four identical rods, each of mass m and length ℓ are joined to form a rigid square frame. The frame lies in the X-Y plane, with its centre at the origin and the sides parallel to the x and y axis. its moment of inertia about :-

**Column I**

- (P) An axis parallel to z-axis and passing through a corner
 (Q) One side
 (R) The x-axis
 (S) The z-axis

Codes :

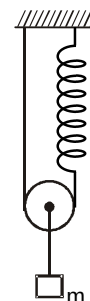
	P	Q	R	S
(A)	3	1	2	4
(B)	2	3	1	4
(C)	4	2	1	3
(D)	4	1	2	3

Column II

- (1) $\frac{5}{3} m\ell^2$
 (2) $\frac{2}{3} m\ell^2$
 (3) $\frac{4}{3} m\ell^2$
 (4) $\frac{10}{3} m\ell^2$

DPP No. : B39 (JEE-MAIN)**Total Marks : 60****Single choice Objective ('-1' negative marking) Q.1 to Q.20****Max. Time : 40 min.****(3 marks 2 min.)****[60, 40]**

1. The system is released from rest with spring initially in its natural length. If mass of the block $m = 10$ kg, and spring constant $k = 100$ N/m, then maximum extension in spring is :
 (A) 1m
 (B) $\frac{1}{2}\text{m}$
 (C) 2m
 (D) 2.5m

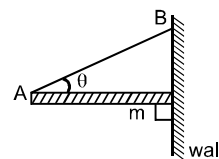


2. A cubical block of mass M and edge a slides down a rough inclined plane of inclination θ with a uniform velocity. The torque of the normal force on the block about its centre has a magnitude -
 (A) zero
 (B) Mga
 (C) $Mga \sin \theta$
 (D) $\frac{1}{2} Mga \sin \theta$

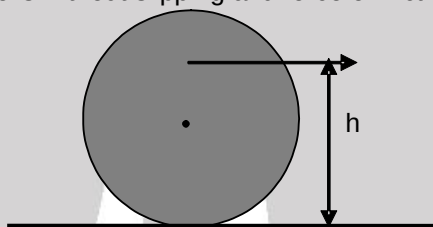


3. Moment of inertia of an ellipse shaped wire of mass 'm', semimajor axis 'a' about an axis passing through the centre and perpendicular to the plane of wire can be:
 (A) ma^2 (B) $1.2 ma^2$ (C) $1.4 ma^2$ (D) $0.6 ma^2$

4. A rod of mass m is supported by string AB and friction due to wall. Then friction force on rod due to wall is : (g = acceleration due to gravity).
 (A) mg upward (B) mg downward
 (C) $\frac{mg}{2}$ upward (D) Data insufficient



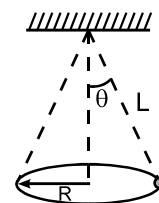
5. A uniform disc (mass m, radius R) is pulled on rough horizontal surface as shown in the figure. What should be height h so that disc rolls without slipping and force of friction is zero.



- (A) $\frac{3R}{4}$ (B) $\frac{3R}{2}$ (C) $\frac{4R}{3}$ (D) $\frac{5R}{3}$
6. A solid sphere of mass m is situated on a horizontal surface and a tangential force acts at the top of the sphere. If the sphere rolls without slipping then the acceleration of the centre of the sphere would be
 (A) $\frac{5}{3} \frac{F}{m}$ (B) $\frac{3F}{5m}$ (C) $\frac{10F}{7m}$ (D) $\frac{10F}{7m}$
7. A disc rolls on a table. The ratio of its K.E. of rotation to the total K.E. is
 (A) $\frac{2}{5}$ (B) $\frac{1}{3}$ (C) $\frac{5}{6}$ (D) $\frac{2}{3}$
8. A thin uniform disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to it with angular velocity ω . Another disc of same radius but of mass $M/4$ is placed gently on the first disc coaxially. The angular velocity of the system will now finally change to:
 (A) $\frac{2\omega}{\sqrt{5}}$ (B) $\frac{2\omega}{5}$ (C) $\frac{3\omega}{5}$ (D) $\frac{4\omega}{5}$
9. A sphere is released on a smooth inclined plane from the top. When it moves down its angular momentum is:
 (A) conserved about every point
 (B) conserved about the point of contact only
 (C) conserved about the centre of the sphere only
 (D) conserved about any point on a line parallel to the inclined plane and passing through the centre of the ball.
10. A sphere rolls without sliding on a rough inclined plane (only mg and contact forces are acting on the body). The angular momentum of the body:
 (A) about centre is conserved
 (B) is conserved about the point of contact
 (C) is conserved about a point whose distance from the inclined plane is greater than the radius of the sphere
 (D) is not conserved about any point.

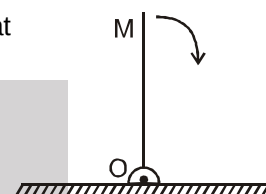
11. A conical pendulum of length L makes an angle θ with the vertical as shown in the figure. The time period will be

(A) $2\pi\sqrt{\frac{L\cos\theta}{g}}$ (B) $2\pi\sqrt{\frac{L}{g\cos\theta}}$
 (C) $2\pi\sqrt{\frac{L\tan\theta}{g}}$ (D) $2\pi\sqrt{\frac{L}{g\tan\theta}}$



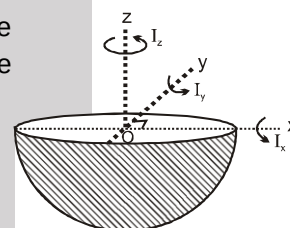
12. A thin uniform rod of mass M and length L is hinged vertically above frictionless pivot point O , as shown in figure and then allowed to fall to the ground. With what speed does the free end of the rod strike the ground?

(A) $\sqrt{\frac{1}{3}gL}$ (B) $\sqrt{gL}$
 (C) $\sqrt{3gL}$ (D) $\sqrt{2gL}$



13. There is a uniform solid hemisphere. On its upper plane x and y axis are drawn which are mutually perpendicular as shown. Z -axis is perpendicular to the upper plane and passing through the centre O . If moment of inertia of the hemisphere about x , y and z -axis are I_x , I_y and I_z respectively then :

(A) $I_z = I_x + I_y$ (B) $I_z = I_x - I_y$
 (C) $I_z = \frac{I_x + I_y}{2}$ (D) $I_z = \frac{I_x - I_y}{2}$



14. A force of $5N$ acts on a 15 kg particle initially at rest. What will be instantaneous power due to the force at the end of 6^{th} second.

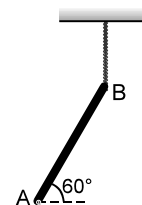
(A) 10 watt (B) 5 watt (C) 20 watt (D) 25 watt

15. A particle of mass m moves in circular path of radius r with centripetal acceleration at any time t is given as $a_c = kt$, then power of resultant force is:

(A) $\frac{mkr}{2}$ (B) $\frac{m\sqrt{kr}}{2}$ (C) $\frac{m(kr)^{3/2}}{2}$ (D) $\frac{kr}{2}$

16. A uniform rod of mass m and length ℓ is attached to smooth hinge at end A and to a string at end B as shown in figure. It is at rest. The angular acceleration of the rod just after the string is cut is:

(A) $\frac{3g}{2\ell}$ (B) $\frac{3g}{4\ell}$ (C) $\frac{5g}{4\ell}$ (D) $\frac{5g}{2\ell}$

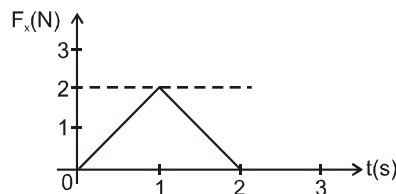


17. A block of mass $m = 3\text{ kg}$ resting on a horizontal frictionless floor is horizontally struck by a $9N$ force that acts for 0.02 sec . After 3 sec it receives a second blow of force $9N$ but in opposite direction which acts for 0.01 sec . The speed of the body after 30 sec . is

(A) 0 (B) 3 cm/sec (C) 90 cm/sec (D) 30 cm/sec

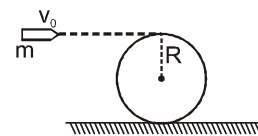
18. The given figure shows a plot of the time dependent force F_x acting on a particle in motion along the x-axis. What is the total impulse delivered by this force to the particle from time $t = 0$ to $t = 2$ second?

(A) 0
(B) 1 kg-m/s
(C) 2 kg-m/s
(D) 3 kg-m/s



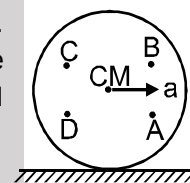
19. A bullet of mass m moving with velocity v_0 just grazes (bullet does not slip on the disc during the contact with the disc) the top of a uniform circular disc of mass M and radius R resting on a rough horizontal surface as shown in figure. Assuming that the disc rolls without slipping, find the velocity of the bullet just after it grazes the disc :

(A) $\frac{4mu}{8m+3M}$ (B) $\frac{8mu}{8m+3M}$ (C) $\frac{8mu}{4m+3M}$ (D) $\frac{4mu}{4m+3M}$



20. A uniform circular disc is in pure rolling on horizontal surface as shown in the figure. At the instant its acceleration of centre of mass is towards right and velocity of the centre of the mass towards right. Among the four points A, B, C, D which could possible have zero acceleration :

(A) A (B) B (C) C (D) D



DPP No. : B40 (JEE-ADVANCED)

Total Marks : 39

Max. Time : 25 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.7

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.8

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.9

(8 marks 6 min.) [08, 06]

1. Two blocks each of mass m are placed on a rough horizontal surface and connected by a massless inelastic string as shown. The coefficient of friction between each block and horizontal surface is μ . The string connecting both the blocks initially has zero tension. The minimum force to be applied on block A to just move the two block system horizontally (without the string getting slack) is :

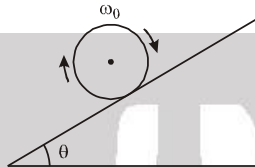
(A) $2\mu mg$ (B) $\frac{2\mu mg}{\sqrt{\mu^2 + 1}}$ (C) $\frac{2\mu mg}{\sqrt{4\mu^2 + 1}}$ (D) $\frac{\mu mg}{\sqrt{\mu^2 + 1}}$



2. In a free space a rifle (fixed) of mass M shoots a bullet of mass m at a stationary block of mass M distance D away from it. When the bullet has moved through a distance d towards the block the centre of mass of the bullet-block system is at a distance of :

(A) $\frac{(D-d)m}{M+m}$ from the block
(B) $\frac{md+MD}{M+m}$ from the rifle
(C) $\frac{2dm+DM}{M+m}$ from the rifle
(D) $(D-d)\frac{M}{M+m}$ from the bullet

3. A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground. Then pickup the correct alternative or alternatives, for motion of the particle before hitting the ground
- (A) Angular momentum of the particle about O is increasing.
 (B) Torque of the gravitational force on the particle about O is decreasing.
 (C) The moment of inertia of the particle about O is decreasing.
 (D) The angular velocity of the particle about O is increasing.
4. When a cylinder is spun and placed somewhere on surface of a rough inclined plane as shown then after release.

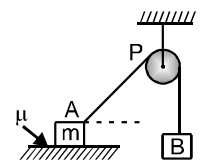


- (A) The cylinder may continue to accelerate down and never be in pure rolling.
 (B) The cylinder may accelerate up for some time then retard and come down in pure rolling.
 (C) The cylinder may accelerate down continuously and pure rolling will start after some time.
 (D) Friction will act on cylinder in upward direction initially and later it will be downward direction.

COMPREHENSION

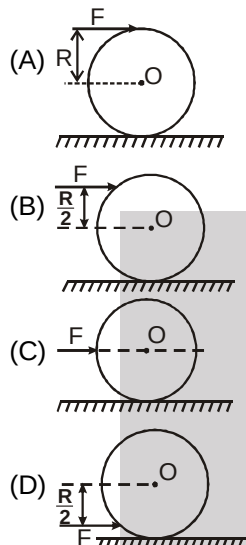
One end of massless inextensible string of length ℓ is fixed and other end is tied to a small ball of mass m . The ball is performing a circular motion in vertical plane. At the lowest position, speed of ball is $\sqrt{20g\ell}$. Neglect any other forces on the ball except tension and gravitational force. Acceleration due to gravity is g .

5. Choose the correct option(s)
- (A) the nature of motion of ball is circular with variable speed
 (B) the nature of motion of ball is circular with constant speed
 (C) At the highest position of ball, tangential acceleration of ball is zero
 (D) At the highest position of ball, tangential acceleration of ball is g
6. Choose the correct option(s)
- (A) During circular motion, minimum value of tension in the string is 10 mg
 (B) During circular motion, minimum value of tension in the string is 15 mg
 (C) During circular motion, maximum value of tension in the string is 15 mg
 (D) During circular motion, maximum value of tension in the string is 21 mg
7. Let a be the magnitude of maximum possible rate of change of speed, and v be the speed when rate of change of speed is maximum. Then :
- (A) $a = 10\text{ ms}^{-2}$ (B) $a = 5\text{ ms}^{-2}$ (C) $v = \sqrt{9g\ell}$ (D) $v = \sqrt{18g\ell}$
8. In figure shown minimum mass of block B (at a particular angle between horizontal and string AP) to just slide the block A on rough horizontal surface is $\frac{m}{2}$ as shown in figure. If μ is the coefficient of friction between block A and ground then $\frac{1}{\mu^2}$ will be :



9. A uniform solid cylinder of mass m and radius R is placed on a rough horizontal surface where friction is sufficient to provide pure rolling. A horizontal force of magnitude F is applied on cylinder at different positions with respect to its centre O in each of four situations of column-I, due to which magnitude of acceleration of centre of mass of cylinder is a . Match the appropriate results in column-II for conditions of column-I.

Column-I



Column-II

(p) Friction force on cylinder will not be zero

(q) $a = \frac{F}{m}$

(r) $a \neq \frac{F}{m}$

(s) the direction of friction force acting on cylinder is towards left

DPP No. : B41 (JEE-ADVANCED)

Total Marks : 42

Single choice Objective ('-1' negative marking) Q.1 to Q.2

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

Subjective Questions ('-1' negative marking) Q.9

Match the Following (no negative marking) Q.10

Max. Time : 27 min.

(3 marks 2 min.) [06, 04]

(4 marks 2 min.) [24, 12]

(4 marks 5 min.) [04, 05]

(8 marks 6 min.) [08, 06]

- A curved road with radius of curvature 200 m is banked with a θ angle of banking equal to $\tan^{-1}(0.2)$. Now if the traffic is at double the speed for which the road is designed, the minimum value of the frictional coefficient needed is ($g = 10 \text{ m/s}^2$)
(A) 0.52 (B) 0.35 (C) 0.94 (D) 0.80
- The potential energy (in SI units) of a particle of mass 2 kg in a conservative field is $U = 6x - 8y$. If the initial velocity of the particle is $\vec{u} = -1.5\hat{i} + 2\hat{j}$ then the total distance travelled by the particle in first two seconds is
(A) 10 m (B) 12 m (C) 15 m (D) 18 m

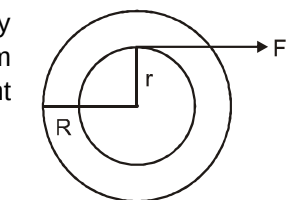
3. A heavy homogeneous cylinder has a mass m and radius R . It is accelerated by a horizontal force F , which is applied through a rope wound around a light drum of radius r attached to the cylinder. The coefficient of static friction is sufficient for the cylinder to roll without slipping :
Choose the correct options :

(A) acceleration of centre of cylinder is $a = \frac{2F}{3mR} (R + r)$

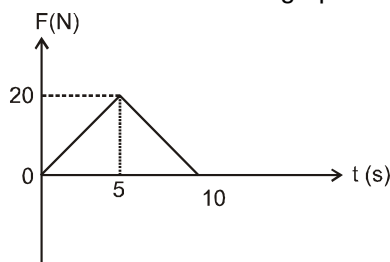
(B) friction assumed to be in the opposite direction of F is $f = \frac{2}{3} \left(\frac{1}{2} - \frac{r}{R} \right) F$

(C) If $\left(\frac{r}{R} \right) < \frac{1}{2}$ a will be greater than $\frac{F}{m}$

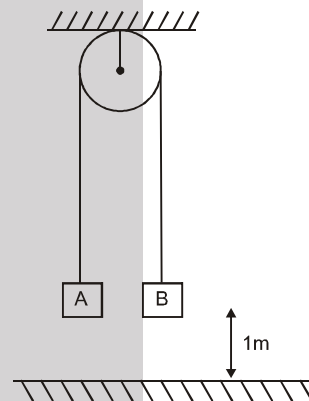
(D) Friction assumed to be in the direction of F is $f = \frac{2}{3} \left(\frac{1}{2} - \frac{r}{R} \right) F$



4. A particles of mass 25 kg, moving at 6 m/s, is acted upon by a force in the opposite direction to the velocity. The variation of force with time is shown in the graph. Then :

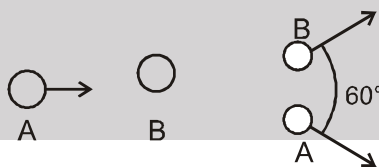


- (A) Its speed will be 2 m/s when the force stops acting.
- (B) its magnitude of average acceleration for the whole time interval in which force acts is $\frac{2}{5} \text{ m/s}^2$
- (C) its magnitude of average acceleration for the whole time interval in which force acts is $\frac{4}{5} \text{ m/s}^2$
- (D) Its direction of motion will be reversed atleast once during the time interval.
5. Two masses A and B of 3kg and 2kg are connected by a long inextensible string which passes over a massless and frictionless pulley. Initially the height of both the masses from the ground is same and equal to 1 metre. When the masses are released, mass A hits the ground and gets stuck to the ground. Consider the length of the string as large, so that the pulley does not obstruct the motion of masses A and B. [$g = 10 \text{ m/s}^2$]
- (A) The impulse A exerts on ground is 6N-S
- (B) The impulse A exerts on ground is 2N-S
- (C) B reaches a maximum height of 2.2 m from the ground.
- (D) B reaches maximum height from the ground 1.2sec after being released.



COMPREHENSION

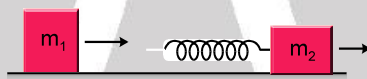
A smooth ball 'A' moving with velocity 'V' collides with another smooth identical ball at rest. After collision both the balls move with same speed with angle between their velocities 60° . No external force acts on the system of balls.



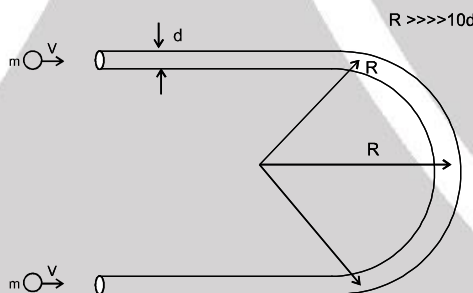
6. Choose the correct option(s)

- (A) The speed of each ball after the collision is $\frac{V}{\sqrt{3}}$
- (B) The speed of each ball after the collision is $\frac{2V}{\sqrt{3}}$
- (C) The magnitude of change in momentum of ball B is $\left(\frac{mV}{\sqrt{3}}\right)$
- (D) The magnitude of change in momentum of ball B is $\left(\frac{2mV}{\sqrt{3}}\right)$

7. Which of the following is/are correct for the given system of balls.
- (A) $e = \frac{1}{2}$
- (B) Loss of energy in collision is $\frac{1}{3}mv^2$
- (C) $e = \frac{1}{3}$
- (D) Loss of energy in collision is $\frac{1}{6}mv^2$
8. Which of the following is/are correct for the given system of balls.
- (A) Ratio of kinetic energy of the system of balls before collision and after collision is 3 : 2.
- (B) Ratio of the kinetic energy of the system of balls before collision and after collision is 3 : 1
- (C) Ratio of kinetic energy of the system of balls after collision to that of energy dissipated is 2:1
- (D) Ratio of the kinetic energy of the system of balls after collision to that of energy dissipated is 1 : 3
9. A block of mass $m_1 = 2$ kg slides on a frictionless table with speed of 10 m/s. In front of it, another block of mass $m_2 = 5$ kg is moving with speed 3 m/s in the same direction. A massless spring of spring constant $k = 1120$ N/m is attached on the backside of m_2 as shown. If the maximum compression of the spring is n cm. (where n is an integer). Find n (Take $g = 10$ m/s²)



10. A U shaped tube of mass $2m$ is placed on a horizontal surface. Two identical spheres each of diameter d (just less than the inner diameter of tube) and mass m enter into the tube with velocity v as shown in the figure taking all collisions to be elastic and all surfaces smooth.



Column-I

Column-II

- | | |
|--|---------------------------|
| (P) The speed of the tube. when spheres are just about to collide inside the tube. | (1) v |
| (Q) The speed of spheres when spheres are just about to collide. | (2) 0 |
| (R) The speed of the spheres when they come out of the tube. | (3) $\frac{\sqrt{3}}{2}v$ |
| (S) The speed of the tube when spheres come out | (4) $v/2$ |

- | | | | | |
|-----|---|---|---|---|
| | P | Q | R | S |
| (A) | 3 | 1 | 2 | 4 |
| (B) | 2 | 3 | 1 | 4 |
| (C) | 4 | 2 | 3 | 1 |
| (D) | 4 | 3 | 2 | 1 |

DPP No. : B42 (JEE-MAIN)

Total Marks : 60

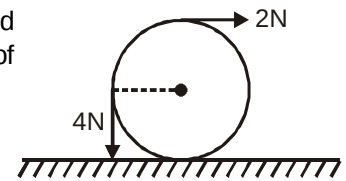
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

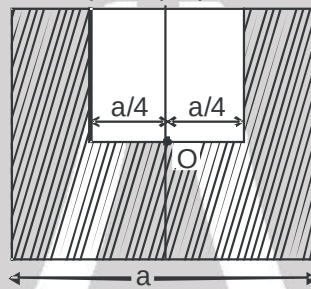
(3 marks 2 min.)

[60, 40]

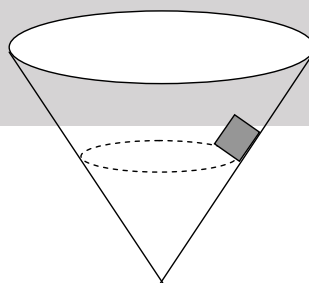
1. A uniform solid disc of mass 1 kg and radius 1m is kept on a rough horizontal surface. Two forces of magnitude 2N and 4N have been applied on the disc as shown in the figure. Linear acceleration of the centre of mass of the disc is if there is no slipping.
- (A) 4 m/s^2
(B) 2 m/s^2
(C) 1 m/s^2
(D) zero



2. A square plate of edge $a/2$ is cut out from a uniform square plate of edge 'a' as shown in figure. The mass of the remaining portion is M. The moment of inertia of the shaded portion about an axis passing through 'O' (centre of the square of side a) and perpendicular to plane of the plate is :



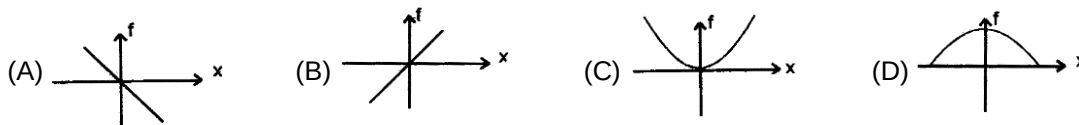
- (A) $\frac{9}{64} Ma^2$ (B) $\frac{3}{16} Ma^2$ (C) $\frac{5}{12} Ma^2$ (D) $\frac{Ma^2}{6}$
3. A ball is thrown vertically up in air where the resistive force can be considered to be constant. If time of ascent is t_1 and time of descent is t_2 . The ratio of velocity of projection to the velocity just before it hits the ground is :
- (A) $\frac{t_2}{t_1}$ (B) $\left(\frac{t_2}{t_1}\right)^2$ (C) $\frac{2t_2}{t_1}$ (D) $\frac{t_2}{2t_1}$
4. Consider a block moving in a horizontal circular path on the inner surface of a cone. The normal force on the block is :



- (A) mg
(B) always greater than mg
(C) may be greater or less than mg
(D) always less than mg
5. A simple pendulum suspended from the ceiling of a stationary trolley has a length ℓ . Its period of oscillation is $2\pi\sqrt{\ell/g}$. What will be its period of oscillation if the trolley moves forward with an acceleration f ?

- (A) $2\pi\sqrt{\frac{\ell}{f-g}}$ (B) $2\pi\sqrt{\frac{\ell}{f+g}}$ (C) $2\pi\sqrt{\frac{\ell}{(f^2+g^2)^{1/2}}}$ (D) $2\pi\sqrt{\frac{\ell}{f^2-g^2}}$

6. The variation of the acceleration (f) of the particle executing S.H.M. with its displacement (x) is represented by the curve



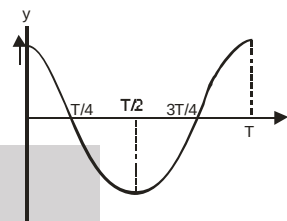
7. The graph in the figure shows how the displacement of a particle describing S.H.M. varies with time. Which one of the following statements is not true ?

(A) the force is zero at time $\frac{3T}{4}$

(B) the velocity is maximum at time $T/2$

(C) the acceleration is maximum at time T

(D) the P.E. = total energy at time $T/2$



8. For a simple harmonic vibrator of frequency n , the frequency of kinetic energy changing completely to potential energy is

(A) $n/2$

(B) n

(C) $2n$

(D) $4n$

9. A particle is executing SHM with an amplitude 4cm. The displacement at which its energy is half kinetic and half potential is

(A) 1 cm

(B) $(2)^{1/2}$ cm

(C) 2 cm

(D) $2(2)^{1/2}$ cm

10. For a particle executing SHM, which of the following statements does not hold good ?

(A) the total energy of the particle always remains the same

(B) the restoring force is always directed towards a fixed point

(C) the restoring force is maximum at the extreme positions

(D) the velocity of the particle is minimum at the centre of motion of the particle

11. The equation of S.H.M. of a particle is $\frac{d^2y}{dt^2} + ky = 0$, where k is a positive constant. The time period of motion is given by

(A) $\frac{2\pi}{\sqrt{k}}$

(B) $\frac{2\pi}{k}$

(C) $2\pi k$

(D) $2\pi\sqrt{k}$

12. The total energy of the body executing S.H.M. is E . Then the kinetic energy when the displacement is half of the amplitude, is

(A) $\frac{E}{2}$

(B) $\frac{E}{4}$

(C) $\frac{3E}{4}$

(D) $\frac{\sqrt{3}}{4}E$

13. A linear harmonic oscillation of force constant 2×10^6 N/m and amplitude 0.01 m has a total mechanical energy of 160 joules. Its -

(A) Maximum potential energy is 100 J

(B) Maximum K.E. is 100 J

(C) Maximum P.E. is 40 J

(D) Minimum P.E. is zero

14. A particle executing S.H.M of amplitude 4 cm and $T = 4$ sec. The time taken by it to move from positive extreme position to half the amplitude is -

(A) 1 sec

(B) $1/3$ sec

(C) $2/3$ sec

(D) $\sqrt{3/2}$ sec

15. The potential energy of a particle executing S.H.M. is 2.5 J, when its displacement is half of amplitude. The total energy of the particle be -

(A) 18 J

(B) 10 J

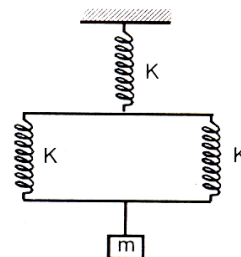
(C) 12 J

(D) 2.5 J



16. One body of mass m is suspended from three springs as shown in figure each spring has spring constant k . If mass m is displaced slightly then time period of oscillation is

- (A) $2\pi\sqrt{\frac{m}{3k}}$ (B) $2\pi\sqrt{\frac{3m}{2k}}$
 (C) $2\pi\sqrt{\frac{2m}{3k}}$ (D) $2\pi\sqrt{\frac{3k}{m}}$



17. One mass m is suspended from a spring. Time period of oscillation is T . Now if spring is divided into n pieces & these are joined in parallel order then time period of oscillation if same mass is suspended.

- (A) $2\pi\sqrt{\frac{n^2m}{k}}$ (B) $2\pi\sqrt{\frac{n^2k}{m}}$ (C) $2\pi\sqrt{\frac{m}{n^2k}}$ (D) $2\pi\sqrt{\frac{k}{n^2m}}$

18. A spring has a certain mass suspended from it and its period for vertical oscillations is T_1 . The spring is now cut into two equal halves and the same mass is suspended from one of the half. The period of vertical oscillation is now T_2 . The ratio of T_2/T_1 is

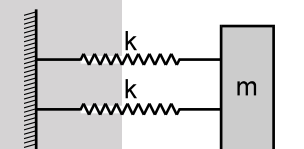
- (A) $1/2$ (B) $1/\sqrt{2}$ (C) $\sqrt{2}$ (D) 2

19. Two objects A and B of equal mass are suspended from two springs of spring constants k_A and k_B if the objects oscillate vertically in such a manner that their maximum kinetic energies are equal, then the ratio of their amplitudes is

- (A) $\frac{k_B}{k_A}$ (B) $\sqrt{\frac{k_B}{k_A}}$ (C) $\frac{k_A}{k_B}$ (D) $\sqrt{\frac{k_A}{k_B}}$

20. A block of mass ' m ' is attached with two springs of spring constant ' k ' is performing SHM on a smooth horizontal surface. One of the spring is cut when the block is at extreme position. Find the ratio of the amplitude of new SHM & old SHM.

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) $\sqrt{2}$



DPP No. : B43 (JEE-ADVANCED)

Total Marks : 38

Single choice Objective ('-1' negative marking) Q.1 to Q.2

One or more than one options correct type ('-1' negative marking) Q.3 to Q.7

Subjective Questions ('-1' negative marking) Q.8

Match the Following (no negative marking) Q.9

Max. Time : 25 min.

(3 marks 2 min.) [06, 04]

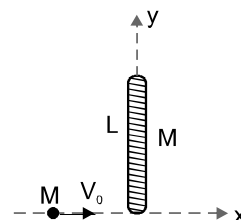
(4 marks 2 min.) [20, 10]

(4 marks 5 min.) [04, 05]

(8 marks 6 min.) [08, 06]

1. A uniform rod of length L and mass M rests on horizontal frictionless table. A particle of equal mass M is moving along the ' x ' axis at a speed V_0 . At time $t = 0$ the particle strikes one end of the rod and sticks to it. Consider (particle + rod) as combined system. The position $\vec{R}_{cm}$ of centre of mass of system as a function of time is :

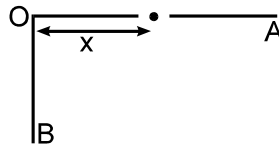
- (A) $\frac{V_0}{2}t\hat{i} + \frac{L}{4}\hat{j}$ (B) $\frac{V_0}{2}t\hat{i} + \frac{L}{2}\hat{j}$ (C) $V_0t\hat{i} + \frac{L}{2}\hat{j}$ (D) $V_0t\hat{i} + \frac{L}{4}\hat{j}$



2. A ball is dropped onto a horizontal surface from a height of 36 cm. After bouncing several times it comes to rest covering a total distance of 100 cm measured in a vertical direction. The percentage loss in kinetic energy after its first impact is

- (A) 36 (B) 64 (C) 53 (D) 96

3. A light rigid rod AOB bent at right angles can freely rotate about OB which is vertical. A bug P is at A and the rod starts rotating with initial angular velocity ω_0 . The bug crawls along the rod towards 'O'. Then during the motion:



- (A) K.E. of the bug decreases
 (B) angular velocity decreases
 (C) angular momentum remains constant.
 (D) both the angular velocity and K.E. of the bug increases.
4. A particle performing S.H.M. undergoes displacement of $\frac{A}{2}$ (where A = amplitude of S.H.M.) in one second. At $t = 0$ the particle was located at either extreme position or mean position. The time period of S.H.M. can be : (consider all possible cases)
 (A) 12s (B) 2.4 (C) 6s (D) 1.2s
5. The acceleration of a particle is, $a = -100x + 50$. It is released from $x = 2$. Here 'a' & 'x' are in S.I. units.
 (A) the particle will perform SHM of amplitude 2 m
 (B) the particle will perform SHM of amplitude 1.5 m
 (C) the particle will perform SHM of time period 0.63 seconds
 (D) the particle will have a maximum velocity 15 m/s

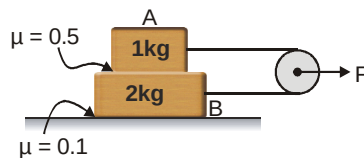
COMPREHENSION

A thin uniform rod having mass m and length 4ℓ is free to rotate about horizontal axis passing through a point distant ℓ from one of its end, as shown in figure. It is released, from the horizontal position as shown:



6. Choose the correct option(s)
- (A) The normal reaction due to hinge at the instant of release is $\frac{4mg}{7}$.
 (B) The normal reaction due to hinge at the instant of release $\frac{\sqrt{2}mg}{7}$.
 (C) The angular velocity of rod when it becomes vertical is $\sqrt{\frac{12g}{7\ell}}$
 (D) The angular velocity of rod when it becomes vertical is $\sqrt{\frac{6g}{7\ell}}$
7. Mark out the correct options :
- (A) Angular acceleration about hinge at this instant is $\frac{3g}{7\ell}$
 (B) Acceleration of center of mass at this instant is $\frac{3}{5}g$
 (C) Net torque about hinge at this instant will be $mg\ell$
 (D) Net torque about hinge at this instant will be $\frac{7}{5}mg\ell$

8. The maximum horizontal force F newton which can be applied such that sliding does not occur between A and B. Then value of F is



9. Consider a system of particles (it may be rigid or non rigid). In the column-I some condition on force and torque is given. Column-II contains the effects on the system. (Letters have usual meaning)

Column-I

Column-II

- | | |
|------------------------------------|--|
| (A) $\vec{F}_{\text{res}} = 0$ | (p) $\vec{P}_{\text{system}}$ will be constant |
| (B) $\vec{\tau}_{\text{res}} = 0$ | (q) $\vec{L}_{\text{system}}$ will be constant |
| (C) External force is absent | (r) total work done by all forces will be zero |
| (D) No nonconservative force acts. | (s) total mechanical energy will be constant. |

DPP No. : B44 (JEE-MAIN)

Total Marks : 60

Max. Time : 40 min.

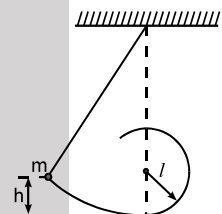
Single choice Objective ('-1' negative marking) Q.1 to Q.20

(3 marks 2 min.)

[60, 40]

1. A peg is placed directly below the pendulum's point of support at a distance $\ell = 0.5\text{m}$ from the lowest point of the pendulum swing as shown in the fig. The minimum height h from which the pendulum can be released such that the string remains taut for at least one full revolution around the peg is

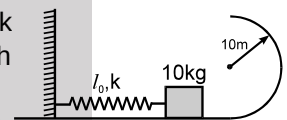
- (A) 0.75 m (B) 0.5 m
(C) 1.25 m (D) 0.2 m



2. In the figure shown find out minimum compression of spring so that 10 kg block completes its circular motion on a track of 10 m radius. All surfaces are smooth and block is not attached to spring.

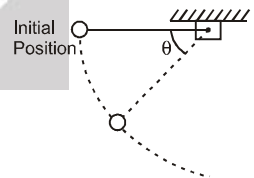
(where $k = 100 \text{ N/m}$)

- (A) 10 m (B) $\sqrt{50} \text{ m}$ (C) $\sqrt{10} \text{ m}$ (D) none of these



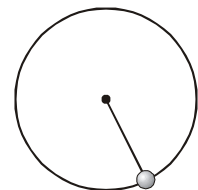
3. The given figure shows a small mass connected to a string, which is attached to a vertical post. If the mass is released from rest when the string is horizontal as shown, the magnitude of the total acceleration of the mass as a function of the angle θ is

- (A) $2g \sin\theta$ (B) $2g \cos\theta$ (C) $g\sqrt{3\cos^2\theta + 1}$ (D) $g\sqrt{3\sin^2\theta + 1}$



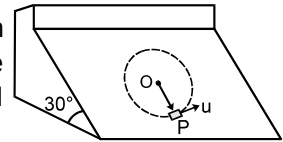
4. A particle is whirled in a vertical circle of radius 1.0 m using a string with one end fixed. If the ratio of maximum and minimum tension in the string is $\frac{5}{3}$, the minimum velocity of the particle is :

- (A) $\sqrt{10} \text{ m/s}$ (B) $\sqrt{50} \text{ m/s}$
(C) 10 m/s (D) $10\sqrt{5} \text{ m/s}$



5. ✎

A particle is attached with a string of length ℓ which is fixed at point O on an inclined plane what minimum velocity should be given to the particle along the incline so that it may complete a circle on inclined plane (plane is smooth and initially particle was resting on the inclined plane.)



- (A) $\sqrt{5g\ell}$ (B) $\sqrt{\frac{5g\ell}{2}}$ (C) $\sqrt{\frac{5\sqrt{3}g\ell}{2}}$ (D) $\sqrt{4g\ell}$

6.

Small blocks A and B connected with a string are rotated with angular velocity ω about point O as shown in the figure. Breaking strength of both the strings is 75N. The maximum angular velocity by which system can be rotated on a horizontal smooth plane.



- (A) 4 rad/sec (B) 5 rad/sec (C) 6 rad/sec (D) $\frac{4}{3}$ rad/sec

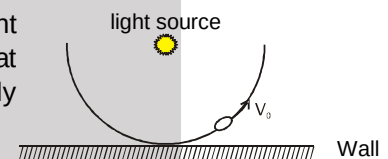
7.

A particle begins to move with a tangential acceleration of constant magnitude 0.6 m/s^2 in a circular path. If it slips when its total acceleration becomes 1 m/s^2 , then the angle through which it would have turned before it starts to slip is :

- (A) $1/3$ rad (B) $2/3$ rad (C) $4/3$ rad (D) 2 rad

8. ✎

An insect moves along a semi-circular track of radius R, with a constant speed V_0 . A point source of light is fixed at the centre of the circle. What is the velocity of shadow of the insect on the wall at time t. (initially insect was at bottom most point of the track)

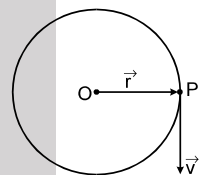


- (A) $V_0 \sin\left(\frac{V_0 t}{R}\right)$ (B) $V_0 \operatorname{cosec}^2\left(\frac{V_0 t}{R}\right)$ (C) $V_0 \sec^2\left(\frac{R}{V_0 t}\right)$ (D) $V_0 \sec^2\left(\frac{V_0 t}{R}\right)$

9. ✎

A particle moves in circle with velocity $\vec{v}$. Let the position vector and angular velocity of particle with respect to centre be $\vec{r}$ and $\vec{\omega}$ respectively

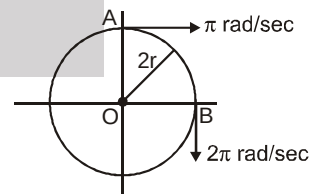
- (A) $\frac{\vec{r} \times \vec{v}}{|\vec{r} \times \vec{v}|} = \frac{\vec{\omega}}{|\vec{\omega}|}$ (B) $\frac{\vec{\omega} \times \vec{v}}{|\vec{\omega} \times \vec{v}|} = \frac{\vec{r}}{|\vec{r}|}$
(C) $\frac{\vec{r} \times \vec{\omega}}{|\vec{r} \times \vec{\omega}|} = \frac{\vec{v}}{|\vec{v}|}$ (D) $\frac{\vec{v} \times \vec{r}}{|\vec{v} \times \vec{r}|} = \frac{\vec{\omega}}{|\vec{\omega}|}$



10.

Two particles A and B are moving in a circle with angular velocities $\omega_A = \pi \text{ rad/sec}$, $\omega_B = 2\pi \text{ rad/sec}$. Initial positions of both the particles are shown in figure. Find the time after which A and B will collide

- (A) $\frac{1}{2}$ sec (B) $\frac{3}{2}$ sec
(C) 1 sec (D) 2 sec



11.

A simple pendulum has some time period T. What will be the % change in its time-period if its amplitude is decreased by 8% ?

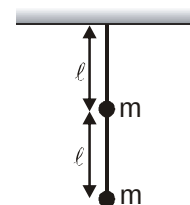
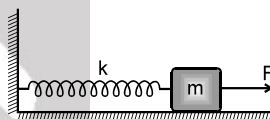
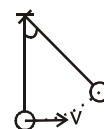
- (A) 8 % (B) 3 %
(C) 1.5% (D) it will remain unchanged

12.

The potential energy of a simple harmonic oscillator when the particle is half way of its end point is (where E is the total energy) -

- (A) $\frac{1}{8}E$ (B) $\frac{1}{4}E$ (C) $\frac{1}{2}E$ (D) $\frac{2}{3}E$

13. A particle executing S.H.M. having amplitude 0.01m & frequency 60Hz. Maximum acceleration of particle is
 (A) $14.4 \pi^2 \text{ m/s}^2$ (B) $144 \pi^2 \text{ m/s}^2$ (C) $1.44 \pi^2 \text{ m/s}^2$ (D) $.144 \pi^2 \text{ m/s}^2$
14. A particle is executing S.H.M. with amplitude 'a' and has maximum velocity 'v'. Its speed at displacement a/2 will be
 (A) $0.866 v$ (B) $v/2$ (C) v (D) $v/4$
15. A thin annular metal disc of inner and outer radii R_1 and R_2 respectively, is freely suspended from a point on its outer circumference. The length of the corresponding equivalent simple pendulum is
 (A) $\frac{R_1^2 + R_2^2}{2R_2}$ (B) $\frac{R_1^2 + 3R_2^2}{2R_2}$ (C) $\frac{3R_1^2 + R_2^2}{R_2}$ (D) $\frac{R_1^2 + 3R_2^2}{R_2}$
16. As shown in the given figure the ball is given sufficient velocity at the lowest point to complete the circle. Length of string is 1m. Find the tension in the string, when it is at 60° with vertical position. (Mass of ball = 5kg)
 (A) 160 N (B) 180 N (C) 200 N (D) 225 N
17. With what angular velocity the earth should spin in order that a body lying at 37° latitude may become weightless.
 (A) $\frac{5}{4} \sqrt{\frac{g}{R}}$ (B) $\frac{25}{16} \sqrt{\frac{g}{R}}$ (C) $\frac{5}{3} \sqrt{\frac{g}{R}}$ (D) $\frac{25}{9} \sqrt{\frac{g}{R}}$
18. A light rod of length 2m is suspended from the ceiling horizontally by means of two vertical wires of equal length tied to its end. One of the wires is made of steel and is of cross-section 0.1 cm^2 . The other wire is of brass of cross-section 0.2 cm^2 . A weight is suspended from a certain point of the rod such that equal stresses are produced in both the wires. The rod remains horizontal in this case also. Find out the position of the load from the steel wire.
 (A) $4/3 \text{ m}$ (B) $2/3 \text{ m}$ (C) 1 m (D) $3/2 \text{ m}$
19. A constant force is applied on a block as shown in the figure initially spring is relaxed and block is at rest, maximum speed of the block in consequent motion is V. If spring is replaced by another spring of spring constant 4k then maximum speed a block will be :
 (A) $V/4$ (B) $2V$ (C) $V/2$ (D) V
20. Two particle of mass m each are fixed to a massless rod of length 2ℓ . The rod is smoothly hinged at one end to a ceiling. It performs oscillation of small angle in vertical plane. The length of the equivalent simple pendulum is
 (A) $\frac{3\ell}{2}$ (B) $\frac{10\ell}{3}$
 (C) $\frac{5\ell}{3}$ (D) None of these



DPP No. : B45 (JEE-ADVANCED)

Total Marks : 46

Max. Time : 29 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.9

(4 marks 2 min.) [28, 14]

Subjective Questions ('-1' negative marking) Q.10

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.11

(8 marks 6 min.) [08, 06]

1. A system is shown in the figure. The time period for small oscillations of the two blocks will be.



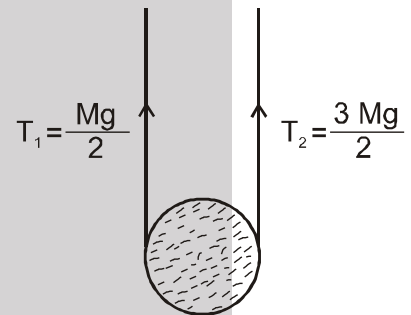
- (A) $2\pi\sqrt{\frac{3m}{k}}$ (B) $2\pi\sqrt{\frac{3m}{2k}}$ (C) $2\pi\sqrt{\frac{3m}{4k}}$ (D) $2\pi\sqrt{\frac{3m}{8k}}$

2. A uniform disc of mass M and radius R is lifted using a string as shown in the figure. Then choose incorrect option(s),

- (A) its linear acceleration is g upward
(B) its linear acceleration is g downward

- (C) its angular acceleration is $\frac{2g}{R}$.

- (D) its rate of change of angular momentum is MgR.



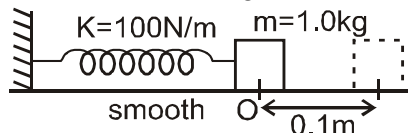
3. A block A of mass 2 kg rests on a horizontal surface. Another block B of mass 1 kg moving at a speed of 1 m/s when at a distance of 16 cm from A, collides elastically with A. The coefficient of friction between the horizontal surface and each of the blocks is 0.2. Then, ($g = 10 \text{ m/s}^2$),

- (A) after collision block B rebounds.
(B) after collision block B comes to rest.
(C) final separation between the blocks is 3 cm.
(D) final separation between the blocks is 5 cm.

4. Which of the following expression represent simple harmonic motion ?

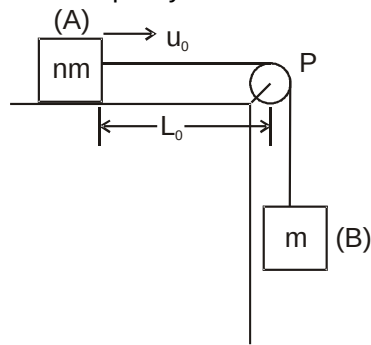
- (A) $x = A \sin(\omega t + \delta)$ (B) $x = B \cos(\omega t + \phi)$
(C) $x = A \tan(\omega t + \phi)$ (D) $x = A \sin \omega t \cos \omega t$

5. A block of mass $m = 1 \text{ kg}$ is placed on a smooth surface and is connected with a spring of spring constant $k = 100 \text{ N/m}$ and another end of spring is connected to a fixed wall as shown. The block is pulled by a distance $A = 0.10 \text{ m}$ from its natural length and released at $t = 0$.



- (A) The maximum speed is after $t = \frac{\pi}{20} \text{ s}$. (B) Time taken to cover first 0.10 m, $t = \frac{\pi}{20} \text{ s}$.
(C) Time taken to cover first 0.05 m, $t = \frac{\pi}{40} \text{ s}$. (D) Time taken to cover first 0.05 m, $t = \frac{\pi}{30} \text{ s}$.

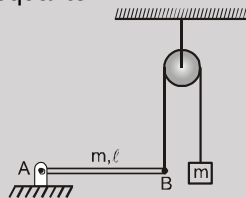
6. In the following figure all surfaces are assumed to be frictionless and pulley is assumed to be ideal. The block 'A' is projected towards the pulley 'P' with an initial velocity u_0 then select correct option :



- (A) the string would become tight at $t = \frac{2u_0}{g}$
- (B) the distance travelled by 'A' before the string is taut is $\frac{u_0^2}{g}$
- (C) the distance travelled by 'B' before string is taut is $\frac{2u_0^2}{g}$
- (D) the common speed of the blocks just after the string is taut is $\left(\frac{n+2}{n+1}\right)u_0$

COMPREHENSION

Uniform rod AB is hinged at the end A in a horizontal position as shown in the figure (the hinge is frictionless, that is, it does not exert any friction force on the rod). The other end of the rod is connected to a block through a massless string as shown. The pulley is smooth and massless. Masses of the block and the rod are same and are equal to 'm'.



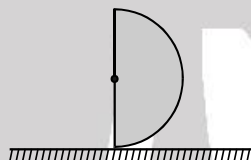
7. Find the correct option(s):
- (A) Then just after release of block from this position, the tension in the thread is $\frac{5mg}{8}$
- (B) Then just after release of block from this position, the angular acceleration of the rod is $\frac{3g}{8l}$
- (C) Then just after release of block from this position, the angular acceleration of the rod is $\frac{5g}{8}$
- (D) Then just after release of block from this position, the magnitude of reaction exerted by hinge on the rod is $\frac{9mg}{16}$.
8. Which of the following is/are correct just after release of the block from initial position.
- (A) force exerted by the pulley on the wall is $2mg$.
- (B) Torque on the rod about point B is $\frac{mg\ell}{8}$
- (C) Force exerted by the pulley on the wall is $\frac{5mg}{4}$
- (D) torque on the rod due to hinge force about point B is $\frac{mg\ell}{4}$

9. If the string attached to block of mass 'm' is cut, then which of these option(s) is/are true for the acceleration of block and rod.

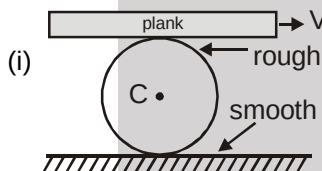
(a_b = acceleration of block, a_r = acceleration of rod)

- (A) $a_b \geq g$, $a_r \geq \frac{2g}{5}$ (B) $a_b > \frac{g}{2}$, $a_r > \frac{5g}{2}$
 (C) $a_b < \frac{3g}{2}$, $a_r < g$ (D) $a_b > \frac{3g}{2}$, $a_r > g$

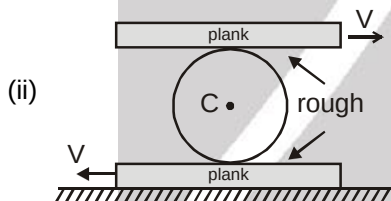
10. A semicircular disc of radius 'r' is released from rest from the position shown. If no slipping occurs between the disc and the horizontal surface, In the expression for the angular velocity ω reached by the disc when its kinetic energy is maximum is $\omega = x \sqrt{\frac{g}{(y\pi - z)r}}$ rad/sec. Then x + y + z will be:



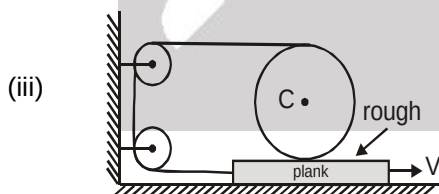
11. There are four arrangements of a cylinder and a plank as shown in the figures. Some surfaces are smooth and some are rough as indicated. There is no slipping at each rough surface. The plank and/or centre of cylinder are given a horizontal constant velocity as shown in each of the situations. Using this information fill in the blanks.



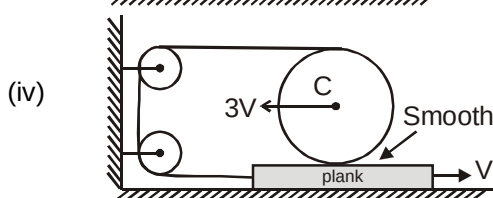
The speed of center of mass of the cylinder is _____.



The angular velocity of the cylinder is _____.



The speed of center of mass of the cylinder is _____.



The angular velocity of the cylinder is _____.

- (a) V (b) V/R (c) 2V/R (d) 4V/R
 (e) cannot be determined from the given information (f) Zero.
 (A) (i) d (ii) b (iii) f (iv) c (B) (i) e (ii) b (iii) f (iv) c
 (C) (i) e (ii) d (iii) f (iv) c (D) (i) e (ii) b (iii) f (iv) a

DPP No. : B46 (JEE-ADVANCED)

Total Marks : 43

Max. Time : 30 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.7

(4 marks 2 min.) [24, 12]

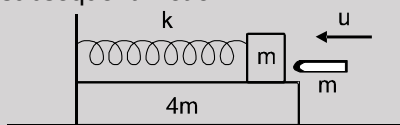
Subjective Questions ('-1' negative marking) Q.8 to Q.9

(4 marks 5 min.) [08, 10]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

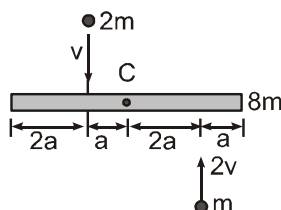
1. Three objects, all of mass M are released simultaneously from the top of an inclined plane of inclination 60° and height H . The objects are described as follows -
 (i) a cube of side R .
 (ii) a solid cylinder of radius R .
 (iii) a hollow cylinder of radius R .
 Assume the cylinder roll down the plane without slipping and cube slides down the plane (coefficient of friction 0.4). Which object(s) reach(s) the bottom of the plane first ?
 (A) I (B) II (C) III (D) I & II
2. A particle of mass m is projected with a velocity V making an angle of 45° with the horizontal. The magnitude of the angular momentum of the projectile about the point of projection when the particle is at its maximum height h is:
 (A) zero (B) $\frac{mV^3}{4\sqrt{2}g}$ (C) $\frac{mV^3}{\sqrt{2}g}$ (D) $m\sqrt{2gh^3}$
3. Two blocks of masses m_1 and m_2 are connected by a massless spring and placed on smooth surface. The spring initially stretched and released. Then :
 (A) the momentum of each particle remains constant separately
 (B) the momenta of both bodies are equal
 (C) the magnitudes of momenta of both bodies are equal
 (D) the mechanical energy of system remains constant
4. A plank of mass $4m$ is placed on a smooth horizontal surface and a spring of force constant k is attached to the plank, whose other end is fixed on a block of mass m placed over the plank. All surfaces are smooth. A bullet of mass m moving with horizontal velocity u strikes the block and gets embedded in it. Choose the correct option for subsequent motion :



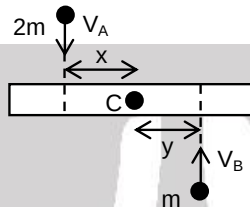
- (A) velocity of plank when spring gets compressed to maximum is $\frac{u}{6}$
 (B) the maximum compression in the spring is $\sqrt{\frac{m}{3k}} u$
 (C) the bullet will have velocity relative to plank at the instant of maximum compression
 (D) the force exerted by spring on plank will reach a maximum of $\sqrt{\frac{mu^2}{6k}}$

COMPREHENSION

A uniform bar of length $6a$ & mass $8m$ lies on a smooth horizontal table. Two point masses m and $2m$ moving in the same horizontal plane with speeds $2v$ and v respectively strike the bar as shown & stick to the bar after collision

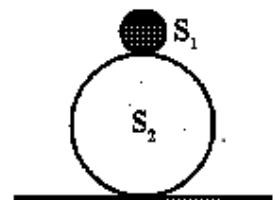


5. Choose the correct option(s)
 (A) Velocity of the centre of mass of the system is zero.
 (B) Angular velocity of the rod about centre of mass of the system is $\frac{v}{5a}$
 (C) Total kinetic energy of the system, just after the collision is $\frac{3}{5}mv^2$
 (D) Total kinetic energy of the system, just after the collision is $\frac{3}{15}mv^2$
6. The velocity of 2 point masses $2m$ and m moving in same horizontal plane are V_A and V_B respectively, strikes the bar as shown

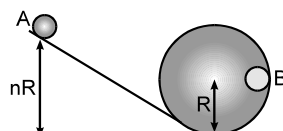


If $V_A = V$, $V_B = 0$. It is given that particles do not stick to rod and comes to rest after collision then choose the correct option(s)

- (A) If $x = a$, angular velocity of rod is $\frac{V}{12a}$ after collision
 (B) If $x = 2a$, angular velocity of rod is $\frac{V}{6a}$ after collision
 (C) If $x = a$, angular velocity of rod is $\frac{V}{6a}$ after collision
 (D) If $x = 2a$ angular velocity of rod is $\frac{V}{12a}$ after collision
7. In above question if $V_A = 2V$ and $V_B = V$ and given that particles sticks to rod after collision. then choose the correct option(s)
 (A) If $x = a$, $y = 2a$; kinetic energy of the system just after the collision is $\frac{3mv^2}{5}$.
 (B) If $x = 2a$, $y = a$; kinetic energy of the system just after the collision is $\frac{27}{22}mv^2$
 (C) If $x = a$, $y = 2a$; then velocity of CM of the system is zero.
 (D) If $x = 2a$, $y = a$; then velocity of cm of the system is zero.
8. S_2 is a fixed rough sphere and S_1 is a uniform solid sphere. S_1 is given a negligible velocity, so that it starts moving on the sphere S_2 . It rolls without slipping. The angle formed with the vertical, by the line joining the centres of S_1 and S_2 when S_1 leaves S_2 is $\alpha = \cos^{-1}\left(\frac{x}{y}\right)$. Then minimum value of $y - x$ will be



9. A metal ball of mass m is put at the point A of a loop track and the vertical distance of A from the lower most point of track is n times the radius R of the circular part. The linear velocity of ball when it rolls to the point B to a height R in the circular track is $\sqrt{\frac{xy(n-1)R}{y}}$. Then $x + y$ will be



10. In Column-I, information about the force(s) acting on a body is mentioned, while in list-II information about the motion of a body is given, Match the entries in list-II with the entries in column-I.

List-I

- (P) A single force through centre of mass
 (Q) Equal and opposite forces separated by non-zero distance
 (R) Equal and opposite forces acting at the same point
 (S) A single force not through centre of mass

List-II

- (1) Rotational motion
 (2) Translational motion
 (3) No motion
 (4) Combined translation & rotation

Codes :

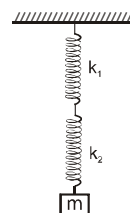
	P	Q	R	S
(A)	1	2	4	3
(B)	2	1	4	3
(C)	2	1	3	4
(D)	4	3	2	1

DPP No. : B47 (JEE-MAIN)**Total Marks : 60****Max. Time : 40 min.****Single choice Objective ('-1' negative marking) Q.1 to Q.20****(3 marks 2 min.)****[60, 40]**

1. Equation of two S.H.M. $x_1 = 5 \sin(2\pi t + \pi/4)$, $x_2 = 5\sqrt{2}(\sin 2\pi t + \cos 2\pi t)$ ratio of amplitude & phase difference will be
 (A) 2 : 1, 0 (B) 1 : 2, 0 (C) 1 : 2, $\pi/2$ (D) 2 : 1, $\pi/2$
2. A particle is executing S.H.M. If at 5cm distance from the mean position the acceleration is 20cm/sec^2 then value of angular frequency will be
 (A) 2 rad/sec (B) 4 rad/sec (C) 10 rad/sec (D) 15 rad/sec
3. The equation of motion of a particle of mass 1 gm is $\frac{d^2x}{dt^2} + \pi^2x = 0$ where x is displacement (in m) from mean position. The frequency of oscillation is (in Hz):
 (A) $\frac{1}{2}$ (B) 2 (C) $5\sqrt{10}$ (D) $\frac{1}{5\sqrt{10}}$
4. A particle of mass 'm' executes SHM according to the equation $\frac{d^2x}{dt^2} + kx = 0$. Its time period will be :
 (A) $\frac{2\pi}{\sqrt{k}}$ (B) $2\pi\sqrt{\frac{m}{k}}$ (C) $2\pi k$ (D) $2\pi\sqrt{k}$
5. A small mass executes linear SHM about O with amplitude a and period T. Its displacement from O at time T/8 after passing through O is:
 (A) $\frac{a}{8}$ (B) $\frac{a}{2\sqrt{2}}$ (C) $\frac{a}{2}$ (D) $\frac{a}{\sqrt{2}}$
6. A particle performs SHM with a time period T and amplitude a. The magnitude of average velocity of the particle over the time interval during which it travels a distance of $\frac{a}{2}$ starting from the extreme position is
 (A) $\frac{a}{T}$ (B) $\frac{2a}{T}$ (C) $\frac{3a}{T}$ (D) $\frac{a}{2T}$

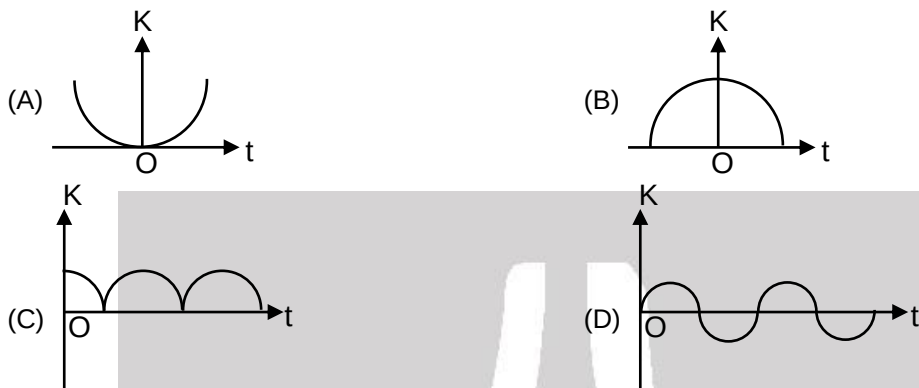


7. For a particle performing S.H.M., if the amplitude of displacement is 'a' and the amplitude of velocity is 'v' the amplitude of acceleration is
 (A) va (B) $\frac{v^2}{a}$ (C) $\frac{v^2}{2a}$ (D) $\frac{v}{a}$
8. In SHM, the phase difference between the displacement and acceleration is:
 (A) 0 (B) $\pi/2$ (C) π (D) 2π
9. The acceleration of a particle moving along x-axis is $a = -100x + 50$. It is released from $x = 2$. Here 'a' and 'x' are in S.I units. The motion of particle will be :
 (A) periodic, oscillatory but not SHM. (B) periodic but not oscillatory.
 (C) oscillatory but not periodic. (D) simple harmonic.
10. The potential energy of a particle, executing simple harmonic motion, at a distance x from the equilibrium position is proportional to
 (A) $\sqrt{x}$ (B) x (C) x^2 (D) x^3
11. The time period for the oscillating system (see fig.) is
 (A) $T = 2\pi\sqrt{\frac{m}{k_1k_2}}$ (B) $T = 2\pi\sqrt{\frac{m}{k_2 + k_1}}$
 (C) $T = 2\pi\sqrt{mk_1k_2}$ (D) none of these
12. A ball is suspended by an inextensible thread of length ℓ from a point on a vertical wall. The ball is displaced away from the wall such that the thread makes a very small angle with it and then left free. Assuming the collision of the ball with the wall to be perfectly elastic, the time period of the resulting oscillations is
 (A) $T = 2\pi\sqrt{\frac{\ell}{g}}$ (B) $T = \pi\sqrt{\frac{\ell}{g}}$ (C) $T = 2\pi\sqrt{\frac{\ell}{2g}}$ (D) $T = 0$
13. A simple pendulum of length 1m is attached to the ceiling of an elevator which is accelerating upward at the rate of 1m/s^2 . Its frequency is approximately
 (A) 2 Hz (B) 1.5 Hz (C) 5 Hz (D) 0.5 Hz
14. The velocity of the particle is given by the expression $v = 2t + 3t^2$ (where v is in ms^{-1} and t is in seconds). find out the distance travelled by the particle from $t = 2$ to $t = 3$ seconds.
 (A) 9 (B) 24 (C) 27 (D) 36
15. A cubical block of mass M and edge a slides down a rough inclined plane of inclination θ with a uniform velocity. The torque of the normal force on the block about its centre has a magnitude -
 (A) zero (B) Mga (C) Mga $\sin \theta$ (D) $\frac{1}{2}$ Mga $\sin \theta$
16. If the period of oscillation of mass M suspended from a spring is one second, then the period of 4M mass suspended from the same spring will be -
 (A) $1/2$ sec (B) $1/4$ sec (C) 2 sec (D) 4 sec
17. The maximum velocity of a harmonic oscillator is α and its maximum acceleration is β . Its time period will be.
 (A) $\frac{2\pi\alpha}{\beta}$ (B) $\frac{2\pi\beta}{\alpha}$ (C) $2\pi\alpha\beta$ (D) $\frac{\pi\beta}{\alpha}$

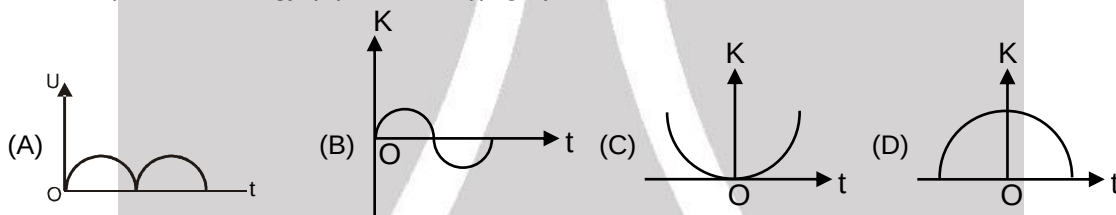


18. Amplitude of harmonic oscillator is a . Position of the particle when velocity of particle is half of maximum velocity will be
 (A) $\frac{\sqrt{3}}{2}a$ (B) $\frac{3}{\sqrt{2}}a$ (C) $a/2$ (D) a

19. In S.H.M., the graph between kinetic energy K and time ' t ' is.



20. In SHM, potential energy (U) V/s. time (t), graph is



DPP No. : B48 (JEE-ADVANCED)

Total Marks : 39

Max. Time : 42 min.

Single choice Objective ('-1' negative marking) Q.1

(3 marks 2 min.) [03, 02]

One or more than one options correct type ('-1' negative marking) Q.2 to Q.7

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.8

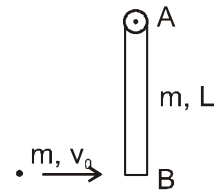
(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.9

(8 marks 6 min.) [08, 06]

- A loop of a string of mass per unit length μ and radius R is rotated about an axis passing through centre perpendicular to the plane with an angular velocity ω . A small disturbance is created in the loop having the same sense of rotation. The linear speed of the disturbance for a stationary observer is :
 (A) ωR (B) $2\omega R$ (C) $3\omega R$ (D) zero
- Four point masses are placed in a plane so that their centre of mass is at $(1, 1)$. Three of them are of mass m each and are placed at $(0, 0)$, $(2, 0)$ and $(0, 2)$ respectively. The fourth point of mass $2m$ is displaced from its initial position such that centre of mass of the system moves to $(2, 1)$. Then, the displacement of the fourth point mass is
 (A) parallel to X axis (B) inclined at an angle 45° with X axis
 (C) of magnitude $\frac{5}{2}$ units (D) of magnitude 5 units.
- A particle is projected from origin along the positive x-axis with a speed u . The acceleration ' a ' varies along x-axis as $a = -bx$, where b is a positive constant. Then:
 (A) the maximum displacement of the particle from the starting point is $\frac{u}{\sqrt{b}}$
 (B) the particle will oscillate about the origin
 (C) velocity is maximum at the origin
 (D) given data is insufficient to determine the exact motion of the particle.

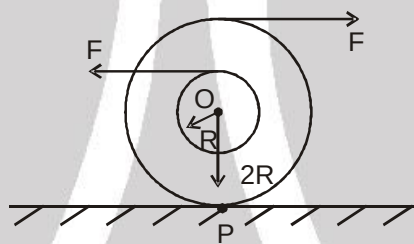
4. A thin uniform rod is free to rotate about a fixed smooth horizontal axis as shown. A point mass hits horizontally with velocity v_0 to the one end B of the rod. When it hits, it sticks to the rod, then :



- (A) Minimum value of v_0 for the rod to rotate by an angle $\frac{\pi}{2}$ is $2\sqrt{gL}$.
- (B) Angular acceleration of the rod when the rod is horizontal is $\frac{9g}{8L}$.
- (C) Force applied by the axis on the rod in the horizontal state is $5mg/16$
- (D) For a small value of v_0 the rod performs small oscillations with a period of $\frac{4}{3}\pi\sqrt{\frac{\ell}{g}}$

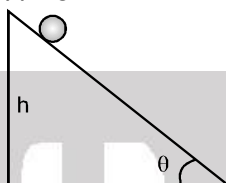
COMPREHENSION

In the given figure $F=10\text{N}$, $R=1\text{m}$ mass of the body is 2kg and moment of inertia of the body about an axis passing through O and perpendicular to plane of body is 4kgm^2 . O is the centre of mass of the body.



5. Choose the correct option(s).
- (A) Frictional force acting on the body if it performs pure rolling is $\frac{10}{3}\text{N}$
- (B) Acceleration of centre of mass is $\frac{5}{3}\text{ m/s}^2$
- (C) Acceleration of centre of mass is $\frac{10}{3}\text{ m/s}^2$
- (D) Frictional force acting on the body if it performs pure rolling is $\frac{5}{3}\text{N}$
6. Kinetic energy of the body :
- (A) At $t = 3\text{s}$, If body performs pure rolling is 75 J
- (B) At $t = 3\text{s}$, If ground is smooth is 115.5 Js
- (C) At $t = 3\text{s}$, If ground is smooth is 112.5 J
- (D) At $t = 3\text{s}$, If body performs pure rolling is $\frac{75}{2}\text{ J}$
7. Which of the following statement(s) are correct :
- (A) Moment of inertia about an axis passing through point P and perpendicular to the plane of body is 12 kg-m^2
- (B) Moment of inertia about an axis passing through point P and perpendicular to the plane of body is 6 kg-m^2
- (C) If surface is smooth, then at the given instant, net torque on body about any point will be FR
- (D) Moment of inertia about an axis passing through point P and perpendicular to the plane of body is 8 kg-m^2

8. A particle executes SHM on a straight line path. The amplitude of oscillation is 2 cm. When the displacement of the particle from the mean position is 1 cm, the numerical value of magnitude of acceleration is equal to the numerical value of magnitude of velocity. If the frequency of SHM is $\frac{\sqrt{n}}{2\pi}$ where n is an integer. Find n.
9. A uniform hollow cylinder, uniform solid cylinder, uniform hollow sphere and uniform solid sphere of same mass and radius R are released from rest from top of a fixed inclined plane as shown in figure. The object roll down the plane without slipping. Match I and II.

**Column I**

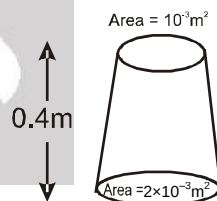
- (A) Hollow cylinder
(B) Solid cylinder
(C) Hollow sphere
(D) solid sphere

Column II

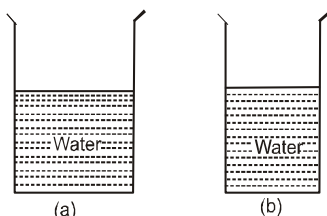
- (p) angular acceleration is maximum
(q) time taken to reach the bottom is maximum
(r) kinetic energy at the bottom of inclined plane is mgh.
(s) rotational kinetic energy is less than translational kinetic energy at the bottom of inclined plane
(t) rotational kinetic energy is equal to translational kinetic energy at the bottom of incline plane

DPP No. : B49 (JEE-MAIN)**Total Marks : 63****Single choice Objective ('-1' negative marking) Q.1 to Q.21****Max. Time : 42 min.****(3 marks 2 min.)****[63, 42]**

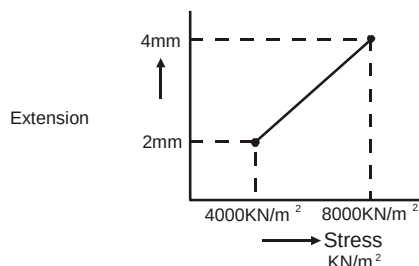
1. If pressure at half the depth of a lake is equal to $\frac{2}{3}$ pressure at the bottom of the lake then what is the depth of the lake :
(A) 10m (B) 20m (C) 60m (D) 30m
2. A uniformly tapering vessel is filled with a liquid of density 900 kg/m^3 . The force that acts on the base of the vessel due to the liquid is ($g = 10 \text{ ms}^{-2}$, $P_{\text{atm}} = 0$)
(A) 3.6 N
(B) 7.2 N
(C) 9.0 N
(D) 14.4 N
3. The pressure at the bottom of a tank containing a liquid does not depend on
(A) Acceleration due to gravity (B) Height of the liquid column
(C) Area of the bottom surface (D) Nature of the liquid
4. The value of g at a place decreases by 2%. The barometric height of mercury
(A) Increases by 2% (B) Decreases by 2%
(C) Remains unchanged (D) Sometimes increases and sometimes decreases
5. A barometer kept in a stationary elevator reads 76 cm. If the elevator starts accelerating up the reading will be :
(A) Zero (B) Equal to 76 cm
(C) More than 76 cm (D) Less than 76 cm



6. From the adjacent figure, the correct observation is

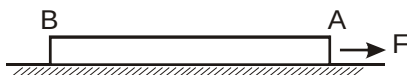


- (A) The pressure on the bottom of tank (a) is greater than at the bottom of (b).
 (B) The pressure on the bottom of the tank (a) is smaller than at the bottom of (b)
 (C) The pressure depend on the shape of the container
 (D) The pressure on the bottom of (a) and (b) is the same
7. Two solids A and B float in water. It is observed that A floats with half its volume immersed and B floats with $\frac{2}{3}$ of its volume immersed. Compare the densities of A and B
 (A) 4 : 3 (B) 2 : 3 (C) 3 : 4 (D) 1 : 3
8. A body is just floating on the surface of a liquid. The density of the body is same as that of the liquid. The body is slightly pushed down. What will happen to the body.
 (A) It will slowly come back to its earlier position
 (B) It will remain submerged, where it is left
 (C) It will sink
 (D) It will come out violently
9. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied
 (A) length 50 cm and diameter 0.5 mm (B) length 100 cm and diameter 1 mm
 (C) length 200 cm and diameter 2 mm (D) length 300 cm and diameter 3 mm
10. The length of a copper wire increases by 0.01 metre when loaded with 10kg weight. Another copper wire of the same length but double the radius is loaded with the same weight. The increase in length of the second wire in metres will be
 (A) 0.002 (B) 0.005 (C) 0.0025 (D) 0.01
11. Which of the following statements is true ?
 (A) Elasticity of water is more than that of air (B) Elasticity of water is less than that of air.
 (C) Both water and air, have equal elasticity (D) None of these.
12. A copper solid cube of 60 mm. side is subjected to a pressure of $2.5 \times 10^7 \text{ N/m}^2$. If the bulk modulus of copper is $1.25 \times 10^{11} \text{ N/m}^2$ then the change in the volume of the cube will be :
 (A) 43.2 mm^3 (B) 33.2 mm^3 (C) 23.2 mm^3 (D) 432 mm^3
13. In determination of young modulus of elasticity of wire, a force is applied and extension is recorded. Initial length of wire is '1 m'. The curve between extension and stress is depicted then young modulus of wire will be:



- (A) $2 \times 10^9 \text{ N/m}^2$ (B) $1 \times 10^9 \text{ N/m}^2$ (C) $2 \times 10^{10} \text{ N/m}^2$ (D) $1 \times 10^{10} \text{ N/m}^2$
14. Under elastic limit, stress is :
 (A) indirectly proportional to strain (B) directly proportional to strain
 (C) independent to strain (D) zero

15. A uniform rod is kept at smooth horizontal surface, a constant force is applied on the rod in horizontal direction at end 'A'. Find the ratio of energy stored per unit volume at end A to the energy stored per unit volume in the middle of rod.

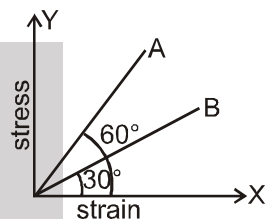


- (A) 2 (B) 4 (C) 8 (D) 10

16. The work done per unit volume in stretching a wire is
 (A) $(1/2)$ stress $\times$ strain (B) $(1/4)$ stress $\times$ elongation
 (C) $(1/2)$ stress $\times$ elongation force $\times$ elongation (D) force $\times$ elongation

17. The stress versus strain graphs for wires of two materials A and B are as shown in the figure. If Y_A and Y_B are the Young's moduli of the materials, then

- (A) $Y_B = 2Y_A$
 (B) $Y_A = Y_B$
 (C) $Y_B = 3Y_A$
 (D) $Y_A = 3Y_B$



18. If ρ is density of the material of a uniform rope of length ℓ and σ is the breaking stress. It is rotated in horizontal circular path then maximum angular velocity it can have for circular path :

- (A) $\frac{1}{\ell} \sqrt{\frac{2\sigma}{\rho}}$ (B) $\frac{1}{\ell} \sqrt{\frac{\sigma}{2\rho}}$ (C) $\frac{2}{\ell} \sqrt{\frac{2\sigma}{\rho}}$ (D) $\frac{2}{\ell} \sqrt{\frac{\sigma}{\rho}}$

19. Two steel wires, where one has twice diameter and three times the length of the other, are stretched by the same force. The ratio of the elastic strain energy stored in them is

- (A) 2 : 3 (B) 3 : 4 (C) 3 : 2 (D) 6 : 1

20. Three waveforms travelling along a straight line have the forms :

$$2A \sin\left(kx - \omega t + \frac{\pi}{3}\right), \sqrt{3}A \cos\left(kx - \omega t - \frac{\pi}{3}\right), 2\sqrt{3}A \cos\left(kx - \omega t + \frac{\pi}{3}\right)$$

the amplitude of the resulting waveform is

- (A) $(2 + 3\sqrt{3})A$ (B) $\sqrt{31}A$ (C) $\sqrt{19}A$ (D) $(2 - \sqrt{3})A$

21. A rubber band (two parallel strands of elastic material) has a spring constant of 10 Nm^{-1} . If the band cut in one place such that it now forms a single long strand of elastic material, then its new spring constant is:

- (A) 20 Nm^{-1} (B) 40 Nm^{-1} (C) 5 Nm^{-1} (D) 2.5 Nm^{-1}

DPP No. : B50 (JEE-ADVANCED)

Total Marks : 38

Max. Time : 25 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.7

(4 marks 2 min.) [20, 10]

Subjective Questions ('-1' negative marking) Q.8

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.9

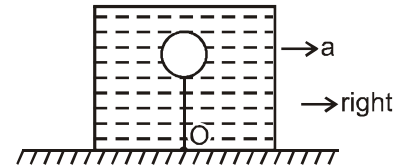
(8 marks 6 min.) [08, 06]

1. A uniform meter scale is supported from its 20 cm mark. A body suspended from 10 cm mark keeps the scale horizontal. However, the scale gets unbalanced if the body is completely immersed in water. To regain the balance the body is shifted to the 8 cm mark. Therefore, the specific gravity of the material of the body is.

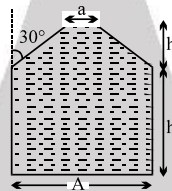
- (A) 5 (B) 6 (C) 7 (D) 4



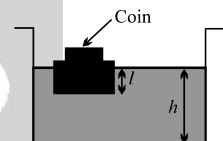
2. One end of light inelastic string is tied to a helium filled balloon and its other end is tied to bottom of a water filled container at point O. The container lies on a fixed horizontal surface and is pulled horizontally towards right with constant horizontal acceleration of magnitude a . Assuming no relative motion of balloon and water with respect to container, the string will be inclined with vertical line passing through O by an angle. (g is acceleration due to gravity)



- (A) $\theta = \tan^{-1} \frac{a}{g}$ and string will be on right of vertical line passing through O.
 (B) $\theta = \tan^{-1} \frac{g}{a}$ and string will be on right of vertical line passing through O.
 (C) $\theta = \tan^{-1} \frac{a}{g}$ and string will be on left of vertical line passing through O.
 (D) $\theta = \tan^{-1} \frac{g}{a}$ and string will be on left of vertical line passing through O.
3. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h . The upper part is a conical vessel of height h with base area ' A ' and top area ' a ' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height $2h$. Neglecting atmospheric pressure.

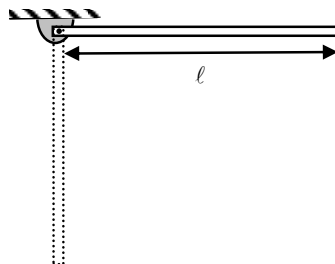


- (A) The force F exerted by the liquid on the base of the vessel is $2h\rho gA$
 (B) the pressure P at the base of the vessel is $2h\rho g$
 (C) the weight of the liquid W is greater than the force exerted by the liquid on the base
 (D) the walls of the vessel exert a downward force ($F-W$) on the liquid.
4. A wooden block, with a coin placed on its top, floats in water as shown in figure. The distances l and h are shown there. After some time the coin falls into the water. Then
- (A) l decreases (B) h increases
 (C) l increases (D) h decreases



COMPREHENSION

A thin uniform rod of length ℓ and mass m is freely pivoted about its end. The rod is initially held horizontally and released from rest. When the rod is vertical, an impulse J is applied to bring it to rest (this is in addition to any impulse provided by the pivot)



5. Choose the correct option(s).

- (A) The angular velocity of the rod at the moment it reaches the vertical position is $2\sqrt{\frac{3g}{\ell}}$
- (B) The angular velocity of the rod at the moment it reaches the vertical position is $\sqrt{\frac{3g}{\ell}}$
- (C) Normal reaction at hinge when rod becomes vertical is $\frac{5mg}{2}$
- (D) Normal reaction at hinge when rod becomes vertical is $\frac{3mg}{2}$

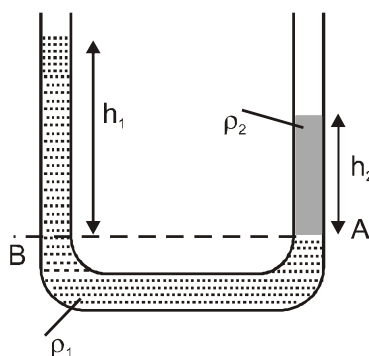
6. The impulse J required to bring it to rest when it reaches the vertical position :

- (A) Impulse is $m\sqrt{\frac{g\ell}{3}}$ when it is applied horizontally at distance ℓ from pivoted point
- (B) Impulse is $2m\sqrt{\frac{g\ell}{3}}$ when it is applied horizontally at distance $\frac{\ell}{2}$ from pivoted point.
- (C) Minimum impulse is $m\sqrt{\frac{g\ell}{2}}$ when it is applied at distance ℓ from pivoted point.
- (D) If rod is stopped by applying minimum impulse then the impulse imparted by pivot is $\frac{m\sqrt{g\ell}}{2\sqrt{3}}$

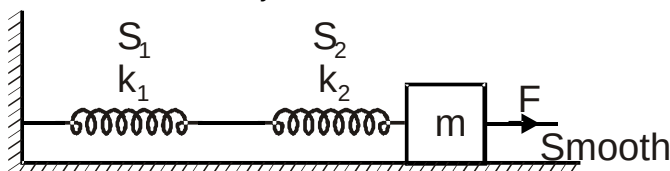
7. If impulse is due to the collision of a particle of mass m, which collides perpendicularly to the rod when it reaches the vertical position and sticks to it. If rod comes to rest after collision then :

- (A) Minimum required velocity of the particle at the time of collision is $\sqrt{\frac{g\ell}{4}}$
- (B) Minimum required velocity of the particle at the time of collision is $\sqrt{\frac{g\ell}{3}}$
- (C) If particle collides at distance greater than $\frac{\ell}{2}$ from pivoted point then velocity of the particle must be less than $2\sqrt{\frac{g\ell}{3}}$
- (D) If particle collides at distance greater than $\frac{\ell}{2}$ from pivoted point then velocity of the particle can be $\frac{5}{2}\sqrt{\frac{g\ell}{3}}$

8. Two immiscible liquids are poured in a U-tube having densities $\rho_1 = 1.0 \times 10^3 \text{ kg/m}^3$ and $\rho_2 = 3.0 \times 10^3 \text{ kg/m}^3$. Find the ratio of heights (of the liquids above their interface) $\frac{h_1}{h_2}$.



9. Initially springs are in natural length. On applying a variable force F on block towards right, elongation in spring S_1 is x_1 . If block moves slowly and floor is smooth then -



Column – I

- (A) Work done by S_2 on block
(B) Work done by S_2 on S_1
(C) Work done by F on block
(D) Work done by S_1 on wall

Column – II

- (p) $\frac{1}{2} \frac{K_1(K_1 + K_2)x_1^2}{K_2}$
(q) $-\frac{1}{2} \frac{K_1(K_1 + K_2)x_1^2}{K_2}$
(r) $\frac{1}{2} K_1 x_1^2$
(s) zero

DPP No. : B51 (JEE-ADVANCED)

Total Marks : 38

Max. Time : 28 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.6

(4 marks 2 min.) [16, 08]

Subjective Questions ('-1' negative marking) Q.7 to Q.8

(4 marks 5 min.) [08, 10]

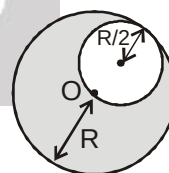
Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. A spherical shell of mass M and radius R filled completely with a liquid of same mass and set to rotate about a vertical axis through its centre has a moment of inertia I_1 about the axis. The liquid starts leaking out of the hole at the bottom. If moment of inertia of the system is I_2 when the shell is half filled and I_3 is the moment of inertia when entire water drained off, then :

- (A) $\frac{I_1}{I_2} \approx 1.5$ (B) $\frac{I_1}{I_2} \approx 0.67$ (C) $\frac{I_1}{I_3} \approx 1.6$ (D) $\frac{I_2}{I_3} \approx 1.4$

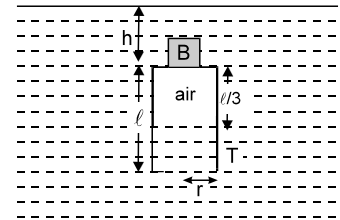
2. A uniform disc of mass m & radius R is pivoted at its centre O with its plane vertical as shown in figure. A circular portion of disc of radius $\frac{R}{2}$ is removed from it. The time period of small oscillations of remaining portion about O is -



- (A) $3\pi \sqrt{\frac{R}{g}}$ (B) $\pi \sqrt{\frac{13R}{g}}$ (C) $2\pi \sqrt{\frac{39R}{16g}}$ (D) $2\pi \sqrt{\frac{7R}{6g}}$

3. An elastic rod will change its length, if
(A) The rod is suspended at one end
(B) The rod is allowed to fall freely under gravity
(C) The rod is rotated about one end on a frictionless horizontal table
(D) The rod is given a horizontal acceleration by a force applied at one end

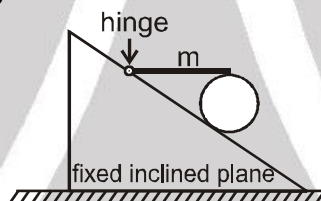
4. A light cylindrical tube 'T' of length ℓ and radius 'r' containing air is inverted in water (density d). One end of the tube is open and the other is closed. A block 'B' of density $2d$ is kept on the tube as shown in the figure. The tube stays in equilibrium in the position shown. (Assume the atmospheric pressure is to be P_0). Assume that density of air is very small than density of block and water. Pick up the correct statement(s).



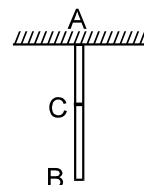
- (A) the volume of block B is $\frac{\pi r^2 \ell}{3}$
- (B) the volume of block B is $\frac{2\pi r^2 \ell}{3}$
- (C) the pressure of air trapped in the tube is $P_0 + d g \left(h + \frac{\ell}{3} \right)$
- (D) the pressure of air trapped in the tube is $P_0 + d g \left(h + \frac{2\ell}{3} \right)$

COMPREHENSION

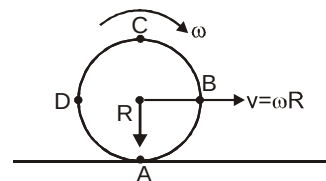
A horizontal uniform rod of mass 'm' has its left end hinged to the fixed incline plane, while its right end rests on the top of a uniform cylinder of mass 'm' which in turn is at rest on the fixed inclined plane as shown. The coefficient of friction between the cylinder and rod, and between the cylinder and inclined plane, is sufficient to keep the cylinder at rest.



5. Choose the correct option(s) for normal reaction
- (A) The magnitude of normal reaction exerted by the rod on the cylinder is $\frac{mg}{2}$
- (B) The magnitude of normal reaction exerted by the inclined plain on the cylinder is $\frac{3mg}{2}$
- (C) The magnitude of normal reaction exerted by the inclined plain on the cylinder is $\frac{5mg}{2}$
- (D) The magnitude of normal reaction exerted by the rod on the cylinder is $\frac{3mg}{2}$
6. Let mass of cylinder is doubled as a result friction between rod and cylinder is f_1 and between inclined plane and cylinder is f_2 . (suppose friction is still sufficient to keep cylinder at rest) choose the incorrect option(s)
- (A) $f_1 = f_2$ (B) $f_1 = 2f_2$ (C) $f_2 = 2f_1$ (D) $f_1 = \frac{5}{3}f_2$
7. A wire of uniform cross section is hanging vertically and due to its own weight its length changes. There is a point 'C' on the wire such that change in length AC is equal to the change in length BC. Points A, B & C are shown in figure. If $\frac{AC}{BC} = \sqrt{n} - 1$ (where 'n' is an integer). Find the value of $2n$.
8. A sample of a liquid has an initial volume of 1.5 L. The volume is reduced by 0.2 mL, when the pressure increases by 140 kPa. If the bulk modulus of the liquid is 1.05×10^n Pa, then find the value of 'n'.



9. The ring shown in figure is performing pure rolling on a rigid surface. Radius of ring is 'R' and angular velocity of ring is ' ω '. There are four points marked on the ring as shown. After time $t = \frac{\pi}{\omega}$. Match the columns.



Column I		Column II	
(P)	Displacement of A	(1)	$(\pi + 2) R$
(Q)	Displacement of B	(2)	$R\sqrt{\pi^2 + 4}$
(R)	Displacement of C	(3)	$(\pi - 2) R$
(S)	Displacement of D	(4)	$4R$

Codes :

	P	Q	R	S
(A)	1	2	4	3
(B)	2	1	2	3
(C)	2	3	2	1
(D)	4	3	2	1

DPP No. : B52 (JEE-MAIN)

Total Marks : 60

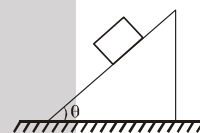
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

(3 marks 2 min.)

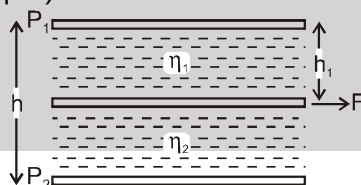
[60, 40]

1. A cubical block of side 'a' and density ' ρ ' slides over a fixed inclined plane with constant velocity ' v '. There is a thin film of viscous fluid of thickness 't' between the plane and the block. Then the coefficient of viscosity of the thin film will be : (Acceleration due to gravity is g)



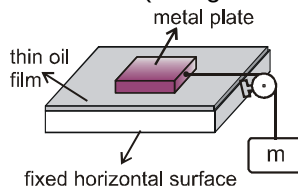
- (A) $\eta = \frac{\rho a g t \sin \theta}{v}$ (B) $\frac{\rho a g t^2 \sin \theta}{v}$ (C) $\frac{v}{\rho a g t \sin \theta}$ (D) none of these

2. A thin horizontal movable plate is separated from two fixed horizontal plates P_1 and P_2 by two highly viscous liquids of coefficient of viscosity η_1 and η_2 as shown, where $\eta_2 = 4\eta_1$. Area of contact of movable plate with each fluid is same. If the distance between two fixed plates is h, then the distance h_1 of movable plate from upper fixed plate such that the movable plate can be moved with a constant velocity by applying a minimum constant horizontal force F on movable plate is (assume velocity gradient to be uniform in each liquid).



- (A) $\frac{h}{4}$ (B) $\frac{h}{2}$ (C) $\frac{2h}{3}$ (D) $\frac{h}{3}$

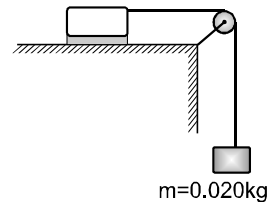
3. A rectangular metal plate has dimensions of 10 cm × 20 cm. A thin film of oil separates the plate from a fixed horizontal surface. The separation between the rectangular plate and the horizontal surface is 0.2 mm. An ideal string is attached to the plate and passes over an ideal pulley to a mass m. When $m = 125$ gm, the metal plate moves at constant speed of 5 cm/s across the horizontal surface. Then the coefficient of viscosity of oil in dyne-s/cm² is (Use $g = 1000$ cm/s²)



- (A) 5 (B) 25 (C) 2.5 (D) 50

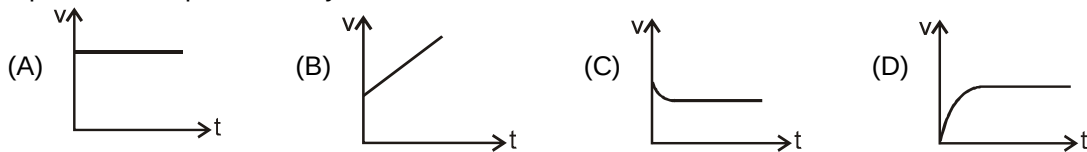


4. A metal block of area 0.10 m^2 is connected to a 0.020 kg mass via a string that passes over an ideal pulley as shown in figure. A liquid with a film thickness of 0.30 mm is placed between the block and the table. When system is released the block moves to the right with a small constant speed of 0.090 ms^{-1} , then coefficient of viscosity of liquid is :



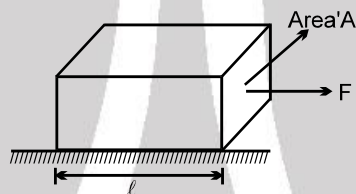
- (A) $\frac{10}{3} \times 10^{-3} \text{ Pa s}$ (B) $\frac{20}{3} \times 10^{-3} \text{ Pa s}$ (C) 10^{-2} Pa s (D) $\frac{40}{3} \times 10^{-3} \text{ Pa s}$
5. A sphere of mass m and radius r is projected in a gravity free space with speed v . If coefficient of viscosity of the medium in which it moves is $\frac{1}{6\pi}$, the distance travelled by the body before it stops is :
- (A) $\frac{mv}{2r}$ (B) $\frac{2mv}{r}$ (C) $\frac{mv}{r}$ (D) $\frac{mv}{4r}$
6. Two identical spherical drops of water are falling (vertically downwards) through air with a steady velocity of 5 cm/sec . If both the drops coalesce (combine) to form a new spherical drop, the terminal velocity of the new drop will be- (neglect buoyant force on the drops.)
- (A) $5 \times 2 \text{ cm/sec}$ (B) $5 \times \sqrt{2} \text{ cm/sec}$ (C) $5 \times (4)^{1/3} \text{ cm/sec}$ (D) $\frac{5}{\sqrt{2}} \text{ cm/sec}$
7. A uniform solid sphere of relative density 5 is released in water filled in a long vertical tube. Its terminal velocity achieved is V . If another uniform solid sphere of same material but double the radius is released in the same water then the terminal velocity achieved will be.
- (A) V (B) $4V$ (C) $V/4$ (D) $2V$
8. Two uniform solid balls of same density and of radii r and $2r$ are dropped in air and fall vertically downwards. The terminal velocity of the ball with radius r is 1 cm s^{-1} , then the terminal velocity of the ball of radius $2r$ will be (neglect buoyant force on the balls.)
- (A) 0.5 cm s^{-1} (B) 4 cm s^{-1} (C) 1 cm s^{-1} (D) 2 cm s^{-1}
9. When a ball is released from rest in a very long column of viscous liquid, its downward acceleration is 'a' (just after release). Then its acceleration when it has acquired two third of the maximum velocity :
- (A) $\frac{a}{3}$ (B) $\frac{2a}{3}$ (C) $\frac{a}{6}$ (D) none of these
10. A container filled with viscous liquid is moving vertically downwards with constant speed $3v_0$. At the instant shown, a sphere of radius r is moving vertically downwards (in liquid) has speed v_0 . The coefficient of viscosity is η . There is no relative motion between the liquid and the container. Then at the shown instant, the magnitude of viscous force acting on sphere is
- (A) $6\pi\eta r v_0$ (B) $12\pi\eta r v_0$ (C) $18\pi\eta r v_0$ (D) $24\pi\eta r v_0$
11. A rain drop of radius 0.3 mm falling vertically downwards in air has a terminal velocity of 1 m/s . The viscosity of air is $18 \times 10^{-5} \text{ poise}$. The viscous force on the drop is -
- (A) $101.73 \times 10^{-4} \text{ dyne}$ (B) $101.73 \times 10^{-5} \text{ dyne}$
(C) $16.95 \times 10^{-5} \text{ dyne}$ (D) $16.95 \times 10^{-4} \text{ dyne}$
12. Two copper balls of radius r and $2r$ are released at rest in a long tube filled with liquid of uniform viscosity. After some time when both the spheres acquire critical velocity (terminal velocity) then ratio of viscous force on the balls is :
- (A) $1 : 2$ (B) $1 : 4$ (C) $1 : 8$ (D) $1 : 18$

13. From amongst the following curves, which one shows the variation of the velocity v with time t for a small sized spherical body (release from rest) falling vertically downwards in a long column of a viscous liquid is best represented by :

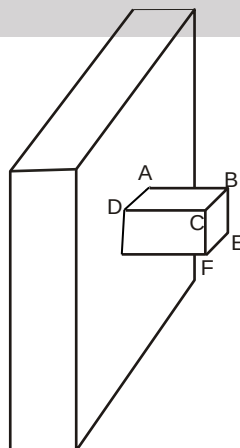


14. A solid sphere falls with a terminal velocity of 10 m/s in air. If it is allowed to fall in vacuum,
 (A) terminal velocity will be more than 10 m/s (B) terminal velocity will be less than 10 m/s
 (C) terminal velocity will be 10 m/s (D) there will be no terminal velocity

15. A block of mass ' M ' area of cross-section ' A ' & length ' ℓ ' is placed on smooth horizontal floor. A force ' F ' is applied on the block as shown. If ' y ' is young modulus of material, then total extension in the block will be:

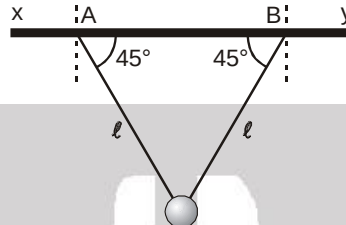


- (A) $\frac{F\ell}{Ay}$ (B) $\frac{F\ell}{2Ay}$ (C) $\frac{F\ell}{3Ay}$ (D) can not extend
16. A particle moves in a plane with a constant speed along a path $y = 2x^2 + 3x - 4$ When the particle is at $(0, -4)$ the direction along which it is moving is inclined to the X axis at an angle
 (A) 63° (B) 72° (C) 27° (D) 0°
17. Two particles A and B are moving in XY plane. Particle A moves along a line with equation $y = x$ while B moves along X axis such that their X coordinates are always equal. If B moves with a uniform speed 3 m/s, the speed of A is.
 (A) 3 m/s (B) $\frac{1}{3}$ m/s (C) $3\sqrt{2}$ m/s (D) $\frac{3}{\sqrt{2}}$ m/s
18. A steel rod is projecting out of a rigid wall. The shearing strength of steel is 345 MN/m^2 . The dimensions $AB = 5 \text{ cm}$, $BC = BE = 2 \text{ cm}$. The maximum load that can be put on the face ABCD is: (neglect bending of the rod) ($g = 10 \text{ m/s}^2$)



- (A) 3450 Kg (B) 1380 Kg (C) 13800 Kg (D) 345 Kg

19. Four uniform wires of the same material are stretched by the same force. The dimensions of wire are as given below. The one which has the minimum elongation has :
 (A) radius 3mm, length 3m (B) radius 0.5 mm, length 0.5 m
 (C) radius 2mm, length 2m (D) radius 3mm, length 2m
20. Two light strings, each of length ℓ , are fixed at points A and B on a fixed horizontal rod xy. A small bob is tied by both strings and in equilibrium, the strings are making angle 45° with the rod. If the bob is slightly displaced normal to the plane of the strings and released then period of the resulting small oscillation will be :



(A) $2\pi\sqrt{\frac{2\sqrt{2}\ell}{g}}$

(B) $2\pi\sqrt{\frac{\sqrt{2}\ell}{g}}$

(C) $2\pi\sqrt{\frac{\ell}{g}}$

(D) $2\pi\sqrt{\frac{\ell}{\sqrt{2}g}}$

DPP No. : B53 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. A man is rowing a boat of mass m with a constant velocity ' v_0 ' in a river the contact area of boat is ' A ' and coefficient of viscosity is η . The depth of river is ' D '. The force required to row the boat is (Assume that velocity gradient is constant, given that $v_0 = 100$ m/s, $\eta = 10^{-1}$ poise = 10^{-2} N-sec-m $^{-2}$, $D = 4$ m and $A = 4$ m 2)

(A) 1 N

(B) 0.1 N

(C) 100 N

(D) 10 N

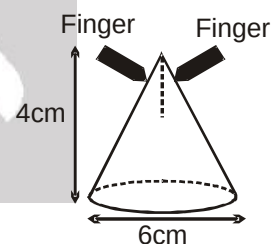
2. With two fingers, you hold a cone motionless upside down, as shown in figure. The mass of the cone is ($m = 2$ kg), and the coefficient of static friction between your fingers and the cone is ($\mu = 1$). The minimum normal force (in Newton) that must be applied with each finger in order to hold up the cone is :

(A) 25 N

(B) 75 N

(C) 50 N

(D) 100 N



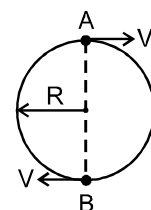
3. Two bodies A and B are moving with same constant speed v in clockwise direction in a horizontal circle of radius R and are initially diagonally opposite. The particle B now achieves a tangential acceleration a m/s 2 . Then :

(A) they collide after time $\sqrt{\frac{\pi R}{a}}$

(B) they collide after time $\sqrt{\frac{2\pi R}{a}}$

(C) relative velocity just before collision is $\sqrt{\pi a R}$

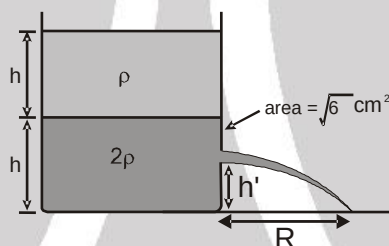
(D) relative velocity just before collision is $\sqrt{2\pi a R}$



4. A wave pulse moving in the positive x-direction along the x-axis is represented by the wavefunction $y(x, t) = \frac{2.0}{(x - 3.0t)^2 + 1}$, where x and y are in centimeters and t is in seconds. Then
- (A) The speed of particle at time $t = 1$ sec. and $x = 3$ cm is zero.
 (B) The speed of particle at time $t = 1$ sec. and $x = 3$ cm is 2 cm/s.
 (C) The speed of the pulse is 3.0 cm/s
 (D) The speed of the pulse is 0.33 cm/s
5. If $\vec{A} \times \vec{B} = \vec{C}$, then choose the correct option : [$\vec{A}$ and $\vec{B}$ are non zero vectors]
- (A) $\vec{C}$ is perpendicular to $(\vec{A} + \vec{B})$ (B) $\vec{C}$ is perpendicular to $(\vec{A} - \vec{B})$
 (C) $\vec{C}$ is perpendicular to $(\vec{A} \times \vec{B})$ (D) $\vec{C}$ is perpendicular to $\vec{A}$ and $\vec{B}$

COMPREHENSION :

A fixed cylindrical tank having large cross-section area is filled with two liquids of densities ρ and 2ρ and in equal volumes as shown in the figure. A small hole of area of cross-section $a = \sqrt{6} \text{ cm}^2$ is made at height h' ($h' < h$) from the bottom.



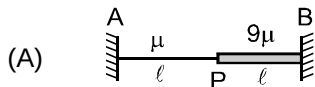
6. Choose the correct option(s)
- (A) Height h' from the bottom of cylinder, it should be punctured for R to be maximum is $\frac{3h}{4}$.
 (B) Value of maximum R is $\frac{3h}{2}$
 (C) Height h' from the bottom of cylinder, it should be punctured for R to be maximum is $\frac{h}{2}$.
 (D) Value of maximum R is $\frac{3h}{4}$
7. Choose the correct option(s) as per above comprehension.
- (A) For $h' = \frac{h}{2}$, area of cross section of stream of liquid just before it hits the ground is $\sqrt{3} \text{ cm}^2$
 (B) For $h' = \frac{h}{2}$, area of cross section of stream of liquid just before it hits the ground is 2 cm^2
 (C) Time required by efflux to hit the ground is $\sqrt{\frac{2h'}{g}}$
 (D) Time required by efflux to hit the ground is $\sqrt{\frac{h'}{g}}$
8. For the initial liquid stream, at the shown instant, Choose correct options :
- (A) velocity of efflux is $v = \sqrt{\frac{(6h - 4h')g}{2}}$
 (B) velocity of efflux is $v = \sqrt{3gh}$
 (C) For $h' = \frac{h}{2}$, angle with the surface at which stream of liquid hits the ground is $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$
 (D) For $h' = \frac{h}{2}$, angle with the surface at which stream of liquid hits the ground is $\tan^{-1}(\sqrt{2})$



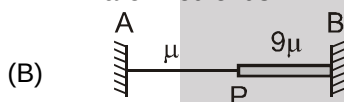
9. A 900 kg elevator hangs by a steel cable for which the allowable stress is $1.15 \times 10^8 \text{ N/m}^2$. Minimum diameter required if the elevator accelerates upward at 1.5 m/s^2 is $\frac{6 \times 10^{-2}}{\sqrt{n\pi}} \text{ m}$ (Take $g = 10 \text{ m/s}^2$, here 'n' is an positive integer). Find 'n'.

10. Match the column :

Column-I



Two strings each of length ℓ and linear mass density μ and 9μ are joined together and system is oscillated such that joint P is node. T is tension in the strings. A and B are fixed ends.



Two strings each of length ℓ and linear mass density μ and 9μ are joined together and system is oscillated such that joint P is antinode. T is tension in each string. A and B are fixed ends.



P is the mid-point of the string fixed at both ends. T is tension in the string and μ is its linear mass density.



T is the tension in the string fixed at A and B is free end. P is mid-point. μ is its the linear mass density.

Column-II

(p) Speed of component travelling wave is portion AP will be $\sqrt{\frac{T}{\mu}}$

(q) Speed of component travelling wave in the portion AP will be more than that in portion BP.

(r) Frequency of oscillation of the system AB can be $\frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$

(s) Frequency of oscillation of the system AB can be $\frac{1}{4\ell} \sqrt{\frac{T}{\mu}}$

(t) Wavelength of the wave in the portion PB can be $\frac{2\ell}{3}$

DPP No. : B54 (JEE-MAIN)

Total Marks : 60

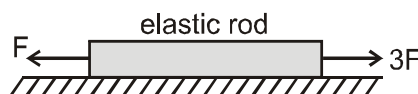
Max. Time : 40 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.20

(3 marks 2 min.)

[60, 40]

1. A uniform elastic rod of cross-section area A, natural length L and Young's modulus Y is placed on a smooth horizontal surface. Now two horizontal forces (of magnitude F and 3F) directed along the length of rod and in opposite direction act at two of its ends as shown. After the rod has acquired steady state, the extension of the rod will be



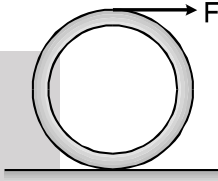
(A) $\frac{2F}{YA}L$

(B) $\frac{4F}{YA}L$

(C) $\frac{F}{YA}L$

(D) $\frac{3F}{2YA}L$

2. The equation of a wave travelling on a string is $y = 4 \sin \frac{\pi}{2} \left(8t - \frac{x}{8} \right)$. If x and y are in cm. then velocity of wave is -
 (A) 64 cm/sec in $-x$ direction (B) 32 cm/sec in $-x$ direction
 (C) 32 cm/sec in $+x$ direction (D) 64 cm/sec in $+x$ direction
3. The equation of a progressive wave is given by $y = a \sin (628t - 31.4x)$. If the distance are expressed in cms and time in seconds, then the wave length will be -
 (A) 314 cm (B) 628 cm (C) 0.2 cm (D) 400 cm
4. The displacement y (in cm) produced by a simple harmonic wave is $y = \frac{10}{\pi} \sin \left(200\pi t - \frac{\pi x}{17} \right)$. The periodic time and maximum velocity of the particles in the medium will respectively be -
 (A) 10^{-3} sec and 330 m/sec (B) 10^{-4} sec and 20 m/sec
 (C) 10^{-2} sec and 20 m/sec (D) 10^{-2} sec and 200 m/sec
5. The density of the material of a wire used in sonometer is $75 \times 10^{-2} \text{ kg/m}^3$. If the stress on the wire is $3.0 \times 10^4 \text{ N/m}^2$, the speed of transverse wave in the wire will be -
 (A) 100 m/s (B) 200 m/s (C) 300 m/s (D) 400 m/s
6. Two small boats are 10m apart on a lake. Each pops up and down with a period of 4.0 seconds due to wave motion on the surface of water. When one boat is at its highest point, the other boat is at its lowest point. Both boats are always within a single cycle of the waves. The speed of the waves is
 (A) 2.5 m/s (B) 5.0 m/s (C) 14 m/s (D) 40 m/s
7. A transverse periodic wave on a string with a linear mass density of 0.200 kg/m is described by the following equation $y = 0.05 \sin(420t - 21.0 x)$ where x and y are in metres and t is in seconds. The tension in the string is equal to :
 (A) 32 N (B) 42 N (C) 66 N (D) 80 N
8. Which of the following function correctly represent the travelling wave equation for finite values of x and t :
 (A) $y = x^2 - t^2$ (B) $y = \cos x^2 \sin t$
 (C) $y = \log(x^2 - t^2) - \log(x - t)$ (D) $y = e^{2x} \sin t$
9. When a harmonic wave is propagating through a medium, the displacement ' y ' of a particle of the medium is represented by $y = 10 \sin \frac{2\pi}{5} (1800 t - x)$. The time period will be
 (A) $\frac{1}{360}$ s (B) $\frac{1}{36}$ s (C) 36 s (D) 360 s
10. A transverse wave described by equation $y = 0.02 \sin(x + 30t)$ (where x and t are in metres and sec. respectively) is travelling along a wire of area of cross-section 1 mm^2 and density 8000 kg/m^3 . What is the tension in the string ?
 (A) 20 N (B) 7.2 N (C) 30 N (D) 14.4 N
11. The motion represented by, $x = A \cos \omega t + B \cos 2\omega t + C \cos^2 2\omega t$ is not:
 (A) oscillatory (B) periodic
 (C) simple harmonic (D) with constant amplitude
12. A transverse periodic wave on a string with linear mass density of 0.2 kg/m^3 is described by equation.
 $y = 0.01 \sin \{220 t - 11x\}$
 where x and y are in metres and t in seconds. The tension in the string is equal to :
 (A) 32 N (B) 66 N (C) 80 N (D) 100 N

13. Two particles of medium disturbed by the wave propagation are at $x_1 = 0$ and $x_2 = 1\text{cm}$. The respective displacements (in cm) of the particles can be given by the equations :
 $y_1 = 2\sin 3\pi t$, $y_2 = 2\sin(3\pi t - \pi/8)$
 The wave speed is :
 (A) 16 cm/sec (B) 24 cm/sec (C) 12 cm/sec (D) 8 cm/sec.
14. The displacement from the position of equilibrium of a point 4 cm from a source of sinusoidal oscillations is half the amplitude at the moment $t = T/6$ (T is the time period). Assume that the source was at mean position at $t = 0$. The wavelength of the running wave is :
 (A) 0.96 m (B) 0.48 m (C) 0.24 m (D) 0.12 m
15. A uniform thin ring of mass m and radius R rolls on a horizontal rough surface without slipping due to an applied force 'F'. The friction force acting on ring is :
 (A) $\frac{F}{3}$ (B) $\frac{2F}{3}$
 (C) $\frac{F}{4}$ (D) Zero
- 
16. Two waves of same frequency but of amplitude a and $2a$ respectively superimpose over each other. The intensity at a point where the phase difference is $\frac{3\pi}{2}$, will be proportional to :
 (A) $9a^2$ (B) $3a^2$ (C) a^2 (D) $5a^2$
17. Spacing between two successive nodes in a standing wave on a string is x . If frequency of the standing wave is kept unchanged but tension in the string is doubled, then new spacing between successive nodes will become :
 (A) $x/\sqrt{2}$ (B) $\sqrt{2}x$ (C) $x/2$ (D) $2x$
18. A sine wave of wavelength λ is travelling in a medium. The minimum distance between the two particles, always having same speed, is -
 (A) $\lambda/4$ (B) $\lambda/3$ (C) $\lambda/2$ (D) λ
19. There are some points in a stationary waves which -
 (A) Are never at rest (B) Are always in motion
 (C) Are at rest twice in each cycle (D) Are at rest once in each cycle
20. Length of a string tied to two rigid supports is 40 cm. Maximum length (wavelength in cm) of a stationary wave produced on it, is -
 (A) 20 (B) 80 (C) 40 (D) 120

DPP No. : B55 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.10

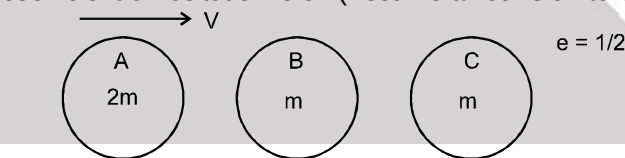
(8 marks 6 min.) [08, 06]

1. A standing wave pattern is formed on a string. One of the waves is given by equation $y_1 = a \cos(\omega t - kx + \pi/3)$ then the equation of the other wave such that at $x = 0$ a node is formed.
- (A) $y_2 = a \sin(\omega t + kx + \frac{\pi}{3})$ (B) $y_2 = a \cos(\omega t + kx + \frac{\pi}{3})$
 (C) $y_2 = a \cos(\omega t + kx + \frac{2\pi}{3})$ (D) $y_2 = a \cos(\omega t + kx + \frac{4\pi}{3})$

2. Which of the following statements for a rigid object undergoing pure translational motion is **true**?
- (A) If an object receives an impulse its kinetic energy must change.
 (B) An object's kinetic energy can change without the object receiving any impulse.
 (C) An object can receive a net impulse without any work being done on it.
 (D) A force may do work on an object without delivering any impulse.
3. A solid sphere moves at a terminal velocity of 20 m/s in air at a place where $g = 9.8 \text{ m/s}^2$. The sphere is taken in a gravity free hall having air at the same pressure and pushed down at a speed of 20 m/s.
- (A) its initial acceleration will be 9.8 m/s^2 downward
 (B) its initial acceleration will 9.8 m/s^2 upward
 (C) the magnitude of acceleration will decrease as the time passes
 (D) it will eventually stop
4. Consider the wave represented by, $y = \cos (500 t - 70 x)$ where y is in mm, x in m and t in seconds. Which of the following is/are true ?
- (A) the wave is a standing wave
 (B) the speed of the wave is $50/7 \text{ ms}^{-1}$
 (C) the frequency of oscillation is $500 \times 2 \pi \text{ Hz}$
 (D) two nearest points in the same phase have separation $20 \pi/7 \text{ cm}$.
5. A 20 gm particle is subjected to two simple harmonic motions
 $x_1 = 2 \sin 10 t$,
 $x_2 = 4 \sin (10 t + \frac{\pi}{3})$. where x_1 & x_2 are in metre & t is in sec.
- (A) The displacement of the particle at $t = 0$ will be $2\sqrt{3} \text{ m}$.
 (B) Maximum speed of the particle will be $20\sqrt{7} \text{ m/s}$.
 (C) Magnitude of maximum acceleration of the particle will be $200\sqrt{7} \text{ m/s}^2$.
 (D) Energy of the resultant motion will be 50 J.

COMPREHENSION

Three balls A, B and C of identical size having uniform mass distribution are placed on a smooth horizontal surface. Given that $m_A = 2m$, $m_B = m$ and $m_C = m$. Ball A collides with initial velocity V with B as shown in figure. The coefficient of resitution is e . (Assume all collision to be head on)



6. For the coefficient of resitution $e = \frac{1}{2}$
- (A) Speed of C after collision of B and C for first time is $\frac{3V}{4}$.
 (B) Speed of A after all the collisions end is $\frac{3V}{8}$
 (C) Speed of C after collision of B and C for first time is $\frac{3V}{8}$.
 (D) Speed of A after all the collisions end is $\frac{3V}{4}$

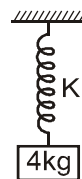
7. Choose the correct option(s)

- (A) For $e = \frac{1}{2}$, the ratio of speed of C with A $\left(\text{i.e. } \frac{V_c}{V_A} \right)$ when all collision ends is 2.
- (B) For $e = \frac{1}{2}$, the ratio of speed of C with A $\left(\text{i.e. } \frac{V_c}{V_A} \right)$ when all collision ends is 4.
- (C) For $e = 1$, the ratio of speed of C with A $\left(\text{i.e. } \frac{V_c}{V_A} \right)$ when all collision ends is 4.
- (D) For $e = 1$, the ratio of speed of C with A $\left(\text{i.e. } \frac{V_c}{V_A} \right)$ when all collision ends is 12.

8. For $e = 1/2$, choose correct option :

- (A) Ratio of final kinetic energy of particle C to particle A after final collision is 2.
- (B) Ratio of K.E. of particle B to A after second collision is greater than 1
- (C) Relative velocity of A w.r.t. B after second collision is $v/4$
- (D) Percentage change in kinetic energy of A after all collisions is nearly 76 percent

9. A body of mass 4 kg is suspended from a spring of spring constant 400 N/m. Another body of mass 4 kg moving vertically upward with 2 m/s hits it & gets embedded in it. If amplitude is $\frac{1}{x} \sqrt{\frac{3}{4}}$ m. find 'x'.



10. A particle of mass m moves as shown in figures of column-I. In cases A, B and C mass is attached at one end of a light string and the other end of string is fixed. 'a' is the initial acceleration of the particle. Match the column.

Column – I	Column – II
(A) conical pendulum	(p) $F > mg$
(B) released from rest	(q) $F < mg$
(C) fixed point vertical velocity is just sufficient to complete the circle	(r) $a > g$
(D) Initial normal reaction is F fixed smooth hemisphere	(s) $a < g$ (t) motion of the particle is periodic



DPP No. : B56 (JEE-ADVANCED)

Total Marks : 42

Max. Time : 27 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.8

(4 marks 2 min.) [24, 12]

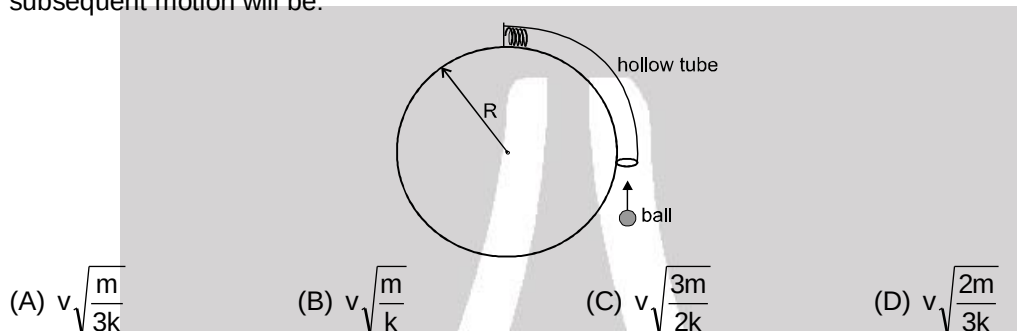
Subjective Questions ('-1' negative marking) Q.9

(4 marks 5 min.) [04, 05]

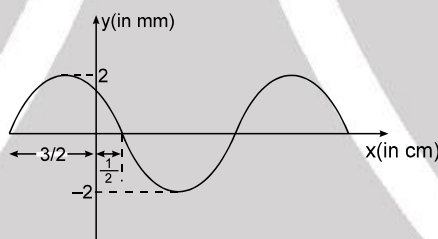
Match the Following (no negative marking) Q.10

(8 marks 6 min.) [08, 06]

1. A hollow tube of mass m and small cross sectional radius is fitted on a horizontal disc of mass $M = 2m$ and radius R as shown in the figure. The whole system is placed on a smooth horizontal table. A light spring (spring constant k) is fitted at the closed end of the tube. A small ball of mass m enters tangentially to the disc into the open end of the tube with a speed v . The disc can freely rotate about a fixed vertical axis passing through the center of the disc. The maximum compression of the spring in the subsequent motion will be:

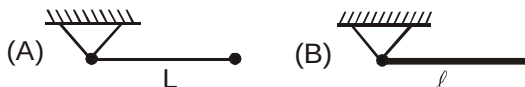


2. A standing wave pattern of maximum amplitude 2mm is obtained in a string whose shape at $t = 0$ is represented in the graph.



If the speed of the travelling wave in the string is 5 cm/s then component waves are :

- (A) $\cos\left(\frac{\pi}{2}x + \frac{3\pi}{4} + \frac{5\pi}{2}t\right)$ & $\sin\left(\frac{\pi}{2}x + \frac{3\pi}{4} - \frac{5\pi}{2}t\right)$
 (B) $\sin\left(\frac{\pi}{2}x + \frac{3\pi}{4} + \frac{5\pi}{2}t\right)$ & $\cos\left(\frac{\pi}{2}x + \frac{3\pi}{4} - \frac{5\pi}{2}t\right)$
 (C) $\cos\left(\frac{\pi}{2}x + \frac{3\pi}{4} + \frac{5\pi}{2}t\right)$ & $\cos\left(\frac{\pi}{2}x + \frac{3\pi}{4} - \frac{5\pi}{2}t\right)$
 (D) $\sin\left(\frac{\pi}{2}x + \frac{3\pi}{4} + \frac{5\pi}{2}t\right)$ & $\sin\left(\frac{\pi}{2}x + \frac{3\pi}{4} - \frac{5\pi}{2}t\right)$
3. A simple pendulum A and a homogeneous rod B hinged at its end are released from horizontal position as shown in figure.

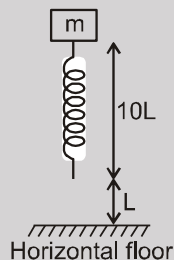


- (A) The times period of swing of simple pendulum is greater than that of rod if $L = \ell$
 (B) The time period of swing of simple pendulum is equal to that of rod if $L = \ell$
 (C) The time period of swing of simple pendulum is greater than that of rod if $L = \frac{2\ell}{3}$
 (D) The time period of swing of simple pendulum is equal to that of rod if $L = \frac{2\ell}{3}$.

4. A thin rod of mass m and length ℓ is free to rotate on a smooth horizontal plane about its one fixed end. When it is at rest, it receives a horizontal impulse J at its other end, at angle of 37° with the length. Immediately after impact :
- (A) Angular momentum of the rod is $0.6 J \ell$
- (B) Angular velocity of the rod is $\frac{1.8J}{m\ell}$
- (C) Kinetic energy of the rod is $\frac{0.54J^2}{m}$
- (D) Linear velocity of the centre of mass of the rod is $\frac{9}{10} \frac{J}{m}$
5. The length, tension, diameter and density of a wire B are double than the corresponding quantities for another stretched wire A. Then (both are fixed at the ends)
- (A) Fundamental frequency of B is $\frac{1}{2\sqrt{2}}$ times that of A.
- (B) The velocity of wave in B is $\frac{1}{\sqrt{2}}$ times that of velocity in A.
- (C) The fundamental frequency of A is equal to the third overtone of B.
- (D) The velocity of wave in B is half that of velocity in A.

COMPREHENSION

A small block of mass m is fixed at upper end of a massless vertical spring of spring constant $K = \frac{4mg}{L}$ and natural length ' $10L$ '. The lower end of spring is free and is at a height L from fixed horizontal floor as shown. The spring is initially unstressed and the spring-block system is released from rest in the shown position.



6. Choose the correct option(s) :
- (A) At the instant speed of block is maximum, the magnitude of force exerted by spring on the block is mg .
- (B) As the block is coming down, the maximum speed attained by the block is $\frac{3}{2}\sqrt{gL}$
- (C) Till the block reaches its lowest position for the first time, the time duration for which the spring remains compressed is $\frac{\pi}{4}\sqrt{\frac{L}{g}} + \sqrt{\frac{L}{4g}} \sin^{-1} \frac{1}{3}$
- (D) None of these
7. When free end of spring just touches the ground, the velocity of the block at that instant is ' v '. then which of the following is (are) true :
- (A) The magnitude of velocity ' v ' is $\sqrt{2gL}$.
- (B) Block will regain the velocity of magnitude ' v ', when compression in spring is $\frac{L}{2}$
- (C) Block will reach the velocity of magnitude ' v ', twice in a cycle.
- (D) Block will reach the velocity of magnitude ' v ', four times in a cycle.

8. When spring just touches the ground, take that instant as $t = 0$ and velocity of block at that instant as v . Then the time ' t ' at which block will have the same magnitude of velocity ' v ' is(are) given by :

Given :

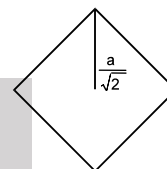
(1) At time instant t_0 , block first time reaches its mean position.

(2) T = time period of S.H.M.

- (A) $t = 2t_0$, (B) $t = \frac{T}{2}$ (C) $t = \frac{T}{2} + 2t_0$ (D) $t = \frac{T}{2} - t_0$

9. A uniform square plate of side ' a ' is hinged at one of its corners. It is suspended such that it can rotate about horizontal axis. If its time period of

small oscillation about its equilibrium position is $2\pi\sqrt{\frac{2\sqrt{n}a}{mg}}$. (Where ' m ' and ' n ' are integer) Find the minimum value of ' $m + n$ '.



10. For a particle executing SHM along a straight line, match the statements in column-I with statement in column-II. (Note that displacement given in column-I is to be measured from mean position).

Column-I

- (A) Velocity-displacement graph may be
(B) Acceleration-velocity graph may be
(C) Acceleration-displacement graph will be
(D) Acceleration-time graph will be

Column-II

- (p) straight line
(q) circle
(r) ellipse
(s) sinusoidal curve

DPP No. : B57 (JEE-MAIN)

Total Marks : 61

Max. Time : 40 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.19

(3 marks 2 min.)

[57, 38]

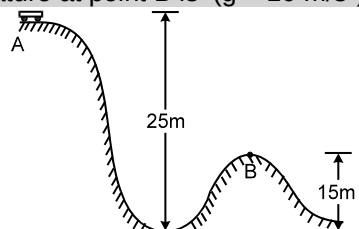
One or more than one options correct type ('-1' negative marking) Q.20 (4 marks 2 min.)

[04, 02]

1. A body is thrown with the velocity of 10 m/s at an angle of 45° to the horizontal. The radius of curvature of its trajectory in $t = \frac{1}{\sqrt{2}}$ sec after the body began to move is:

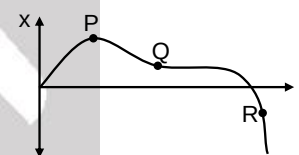
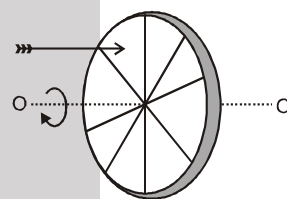
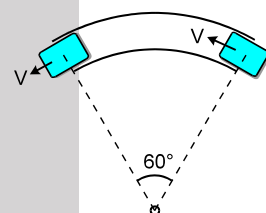
- (A) 0 m (B) 2.5 m (C) 5 m (D) none of these

2. Figure shows the roller coaster track. Each car will start from rest at point A and will roll with negligible friction. It is important that there should be at least some small positive normal force exerted by the track on the car at all points, otherwise the car would leave the track. With the above fact, the minimum safe value for the radius of curvature at point B is ($g = 10 \text{ m/s}^2$) :



- (A) 20 m (B) 10 m (C) 40 m (D) 25 m

3. ✎ A pilot plane of total mass M is taking a circular loop of radius r in a horizontal plane at a height where acceleration due to gravity is g_0 . The speed of the plane is constant and equal to v .
- (A) force exerted by air on the plane is mg_0
- (B) force exerted by air on the plane is $\frac{mv^2}{r}$
- (C) force exerted by air on the plane is $mg_0 + \frac{mv^2}{r}$
- (D) force exerted by air on the plane is $\sqrt{(mg_0)^2 + \left(\frac{mv^2}{r}\right)^2}$
4. A car moves around a curve at a constant speed. When the car goes around the arc subtending 60° at the centre, then the ratio of magnitude of instantaneous acceleration to average acceleration over the 60° arc is :
- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$
- (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{3}$
5. A wheel has eight equally spaced, very thin spokes of length 30 cm. It is rotating about axis OO' which is perpendicular to its plane and passing through centre, with 2.5 rps. We want to pass a 20 cm long arrow parallel to the axis and without hitting any spoke. What should be the minimum velocity of the arrow ?
- (A) 3 m/sec (B) 4 m/sec (C) 5 m/sec (D) 6 m/sec
6. A 25 kg uniform solid sphere with a 20 cm radius is suspended by a vertical wire such that the point of suspension is vertically above the centre of the sphere. A torque of 0.10 N-m is required to rotate the sphere through an angle of 1.0 rad and then maintain the orientation. If the sphere is then released, its time period of the oscillation will be :
- (A) π second (B) $\sqrt{2} \pi$ second (C) 2π second (D) 4π second
7. In the following displacement (x) vs time (t) graph, at which among the point P, Q and R is the objects speed increasing ?
- (A) R only (B) P only
- (C) Q and R only (D) P, Q, R
8. A box when dropped from a certain height reaches the ground with a speed v . when it slides from rest from the same height down a rough inclined plane inclined at an angle 45° to the horizontal, it reaches the ground with a speed $v/3$. The coefficient of sliding friction between the box and the plane is (acceleration due to gravity is 10 m/s^2)
- (A) $8/9$ (B) $1/9$ (C) $2/3$ (D) $1/3$
9. In stationary waves all particle between two nodes pass through the mean position -
- (A) At different times with difference velocities (B) At different times with same velocity
- (C) At the same time with equal velocity (D) At the same time with difference velocities
10. In a stretched string -
- (A) only even harmonics are produced
- (B) only odd harmonics are produced
- (C) even as well as odd harmonics are produced
- (D) neither even nor odd harmonics are produced



11. A long wire PQR is made by joining two wires PQ and QR of equal radii. PQ has a length 4.8 m and mass 0.06 kg. QR has length 2.56 m and mass 0.2 kg. The wire PQR is under a tension of 80 N. A sinusoidal wave pulse of amplitude 3.5 cm is sent along the wire PQ from the end P. No power is dissipated during the propagation of wave pulse. Find amplitude (in mm) of reflected pulse from junction Q.
(A) 15 (B) 3.5 (C) 25 (D) 30
12. Equation of a standing wave is generally expressed as $y = 2A \sin \omega t \cos kx$. In the equation, quantity ω/k represents
(A) the transverse speed of the particles of the string.
(B) the speed of either of the component waves.
(C) the speed of the standing wave.
(D) a quantity that is independent of the properties of the string.
13. Standing waves are produced in a 10 m long stretched string. If the string vibrates in 5 segment and the wave velocity is 20 m/s, the frequency is -
(A) 2 Hz (B) 4 Hz (C) 5 Hz (D) 10 Hz
14. If you set up the seventh harmonic on a string fixed at both ends, how many nodes and antinodes are set up in it -
(A) 8, 7 (B) 7, 7 (C) 8, 9 (D) 9, 8
15. The fundamental frequency in a stretched string is 100 Hz. To double the frequency, the tension in it must be -
(A) $T_2 = 4T_1$ (B) $T_2 = \frac{T_1}{4}$ (C) $T_2 = T_1$ (D) $T_2 = 2T_1$
16. The equation of a stationary wave is $y = 0.8 \cos \left(\frac{\pi x}{20} \right) \sin 200 \pi t$, where x is in cm and t is in sec. The separation between consecutive nodes will be -
(A) 20 cm (B) 10 cm (C) 40 cm (D) 30 cm
17. The amplitude of a particle due to superposition of following S.H.Ms. Along the same line is
 $X_1 = 2 \sin 50 \pi t$; $X_2 = 10 \sin (50 \pi t + 37^\circ)$
 $X_3 = -4 \sin 50 \pi t$; $X_4 = -12 \cos 50 \pi t$
 (A) $4\sqrt{2}$ (B) 4 (C) $6\sqrt{2}$ (D) none of these
18. The time period and the amplitude of a simple pendulum are 4 seconds and 0.20 meter respectively. If the displacement is 0.1 meter at time $t = 0$, the equation of its displacement is represented by
 (A) $x = 0.2 \sin(0.5\pi t)$ (B) $x = 0.1 \sin \left(0.5\pi t + \frac{\pi}{6} \right)$
 (C) $x = 0.1 \sin \left(\pi t + \frac{\pi}{6} \right)$ (D) $x = 0.2 \sin \left(0.5\pi t + \frac{\pi}{6} \right)$
19. **Statement-1 (S-1)** : A string is carrying sinusoidal wave. Consider a small segment of string then total energy of this small segment is conserved.
Statement-2 (S-2) : A standing wave pattern is formed in a string. The power transfer through a point (other than node and antinode) is zero always.
 Choose the correct option
 (A) S-1 True, S-2 True (B) S-1 True, S-2 False
 (C) S-1 False, S-2 True (D) S-1 False, S-2 False

20. Two simple pendulum of length 100cm and 121cm length start swinging together.
 (A) they will swing together again when the pendulum of length 100cm makes 10 oscillations
 (B) they will swing together again when the pendulum of length 121cm makes 11 oscillations
 (C) they will swing together again when the pendulum of length 100cm makes 11 oscillations
 (D) they will swing together again when the pendulum of length 121cm makes 10 oscillations

DPP No. : B58 (JEE-MAIN)

Total Marks : 60

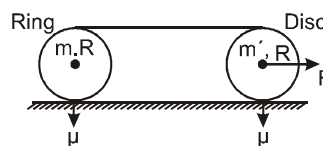
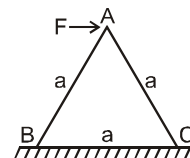
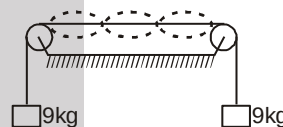
Max. Time : 40 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.20

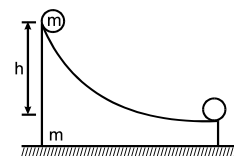
(3 marks 2 min.)

[60, 40]

1. A wire of density $9 \times 10^3 \text{ kg/m}^3$ is stretched between two clamps 1 m apart and is subjected to an extension of $4.9 \times 10^{-4} \text{ m}$. What will be the lowest frequency of transverse vibrations in the wire ? ($Y = 9 \times 10^{10} \text{ N/m}^2$)
 (A) 35 Hz (B) 43 Hz (C) 40 Hz (D) 50 Hz
2. The length of the wire shown in figure between the pulleys is 1.5 m and its mass is 12.0 g. The frequency of vibration with which the wire vibrates in three loops forming antinode at the mid point of the wire is - (Take $g = 9.8 \text{ m/s}^2$)
 (A) 210 Hz (B) 140 Hz (C) 70 Hz (D) none of these
3. Amplitude of a travelling wave on a string is 1mm. If linear mass density of string is $10^{-4} \text{ kg m}^{-1}$, tension in the string is 1N and frequency of vibration is 10Hz, then average power needed to maintain such waves in string is : ($\pi^2 = 10$)
 (A) $3 \times 10^{-5} \text{ W}$ (B) $2 \times 10^{-5} \text{ W}$ (C) $4 \times 10^{-5} \text{ W}$ (D) 10^{-5} W
4. A standing wave $y = A \sin \left(\frac{20}{3} \pi x \right) \cos (1000 \pi t)$ is maintained in a taut string where y and x are expressed in meters. The distance between the successive points oscillating with the amplitude $A/2$ across a node is equal to
 (A) 2.5cm (B) 25cm (C) 5cm (D) 10cm
5. $x = x_1 + x_2$ (where $x_1 = 4 \cos \omega t$ and $x_2 = 3 \sin \omega t$) is the equation of motion of a particle along x-axis. The phase different between x_1 and x is :
 (A) 37° (B) 53° (C) 90° (D) none of these
6. A string fixed at both ends has consecutive standing wave modes for which the distances between adjacent nodes are 18 cm and 16 cm respectively. The length of the string is -
 (A) 144 cm (B) 152 cm (C) 176 cm (D) 200 cm.
7. A horizontal force F is applied at the top of an equilateral triangular block having mass m. The minimum coefficient of friction required to topple the block before translation will be
 (A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$
8. A disc and a ring having different masses and same radius are connected with an ideal string as shown in the figure. Both are placed on rough surface of coefficient of friction μ . A force F is applied on the centre of disc horizontally. Assume that initially both the bodies were at rest and they would be rolling without slipping when they start moving. Then frictional force acting on the ring is.
 (A) $\frac{2F}{3}$ (B) $\frac{F}{3}$ (C) zero (D) None of these



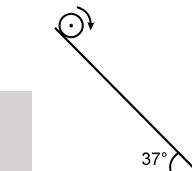
9. A hollow sphere is released from the top of a wedge as shown in the figure. There is no friction between the wedge & the ground. There is sufficient friction to provide pure rolling. The velocity of the sphere just before it leaves the wedge will be:



[Assume: Masses of the wedge and sphere are equal & $h \gg R$ to be the radius of sphere]

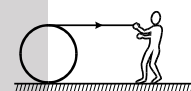
- (A) $\sqrt{\frac{3gh}{7}}$ (B) $\sqrt{\frac{3}{4}gh}$ (C) $\sqrt{\frac{5}{7}gh}$ (D) $\sqrt{\frac{6}{5}gh}$

10. A solid cylinder having radius 0.4 m, initially rotating (at $t = 0$) with $\omega_0 = 54$ rad/sec is placed on a rough inclined plane with $\theta = 37^\circ$ having friction coefficient $\mu = 0.5$. The time taken by the cylinder to start pure rolling is:



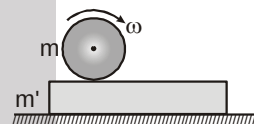
- (A) 5.4 sec (B) 1.4 sec
(C) 1.2 sec (D) 1.8 sec

11. A man pulls a solid cylinder (initially at rest) horizontally by a massless string as shown. The string is wrapped on the cylinder and the cylinder performs pure rolling (that is, rolling without slipping). Mass of the cylinder is 100 kg, radius is π metre & tension in string is 100 N. Then the angular speed of the cylinder after one revolution will be :



- (A) 4 rad/sec (B) $\frac{4}{\sqrt{3}}$ rad/sec (C) $\frac{4}{3}$ rad/sec (D) none of these

12. A uniform solid cylinder is given an angular speed ω and placed on a rough plate of negligible thickness. The horizontal surface below the plate is smooth. Then the angular speed of the cylinder when it starts pure rolling on the plate will be: [Assume sufficient length of plate]

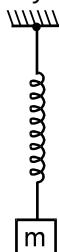


- (A) $\frac{\omega}{2}$ (B) $\frac{\omega}{3}$ (C) $\frac{2\omega}{3}$ (D) none of these

13. A uniform ball of mass 'm' rolls without sliding on a fixed horizontal surface. The velocity of the lowest point of the ball with respect to the centre of the ball is V. The total kinetic energy of the ball is :

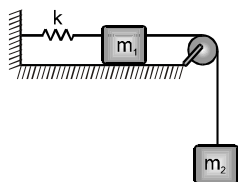
- (A) $\frac{7}{10} mv^2$ (B) mv^2 (C) $\frac{1}{5} mv^2$ (D) $\frac{4}{3} mv^2$

14. A block of mass 'm' is suspended from a spring and executes vertical SHM of time period T as shown in figure. The amplitude of the SHM is A and spring is never in compressed state during the oscillation. The magnitude of minimum force exerted by spring on the block is

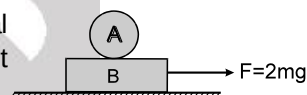
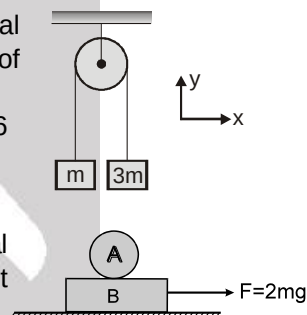
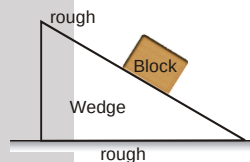


- (A) $mg - \frac{4\pi^2}{T^2} mA$ (B) $mg + \frac{4\pi^2}{T^2} mA$ (C) $mg - \frac{\pi^2}{T^2} mA$ (D) $mg + \frac{\pi^2}{T^2} mA$

15. m_1 & m_2 are connected with a light inextensible string with m_1 lying on smooth table and m_2 hanging as shown in figure. m_1 is also connected to a light spring which is initially unstretched and the system is released from rest.



- (A) system performs SHM with angular frequency given by $\sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}}$
- (B) system performs SHM with angular frequency given by $\sqrt{\frac{k}{m_1 + m_2}}$.
- (C) tension in string will be zero when the system is released.
- (D) maximum displacement of m_1 will be $\frac{m_2 g}{k}$.
16. A body of mass 6 kg is acted upon by a force which causes a displacement in it given by $x = \frac{t^2}{4}$ metre where t is the time in second. The work done by the force is 2 seconds is:
 (A) 12 J (B) 9 J (C) 6 J (D) 3 J
17. When a block is placed on a wedge as shown in figure, the block starts sliding down and the wedge also start sliding on ground. All surfaces are rough. The centre of mass of (wedge + block) system will move
 (A) leftward and downward. (B) right ward and downward.
 (C) leftward and upwards. (D) only downward.
18. In the figure shown pulley and string are massless. The blocks move in vertical plane due to gravity. If the magnitude of acceleration of centre of mass of blocks is a (in m/s^2) then value of $2a$ is (Take $g = 10 \text{ m/s}^2$)
 (A) 3 (B) 2.5 (C) 5 (D) 6
19. In the figure shown a block B of mass m can slide on a fixed horizontal smooth plane. A solid sphere A of radius r of the same mass rolls without sliding on the block B. Find the angular acceleration of the sphere.
 (A) $\frac{10g}{9r}$ (B) $\frac{5g}{r}$ (C) $\frac{15g}{r}$ (D) None of these
20. In Quincke's tube a detector detects minimum intensity. Now one of the tube is displaced by 5 cm. During displacement detector detects maximum intensity 10 times, then finally a minimum intensity (when displacement is complete). The wavelength of sound is:
 (A) 10/9 cm (B) 1 cm (C) 1/2 cm (D) 5/9 cm



DPP No. : B59 (JEE-MAIN)

Total Marks : 60

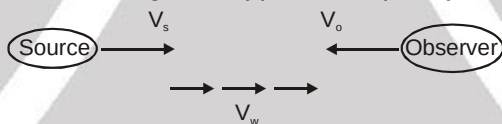
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

(3 marks 2 min.)

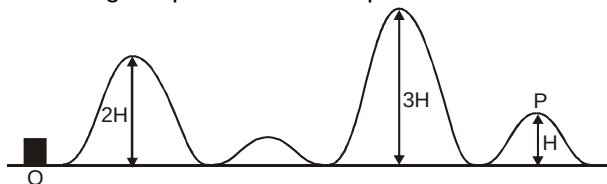
[60, 40]

1. A string of length 1.5 m with its two ends clamped is vibrating in fundamental mode. Amplitude at the centre of the string is 4 mm. Minimum distance between the two points having amplitude 2 mm is :
 (A) 1 m (B) 75 cm (C) 60 cm (D) 50 cm
2. Which of the following travelling wave will produce standing wave, with node at $x = 0$, when superimposed on $y = A \sin(\omega t - kx)$
 (A) $A \sin(\omega t + kx)$ (B) $A \sin(\omega t + kx + \pi)$
 (C) $A \cos(\omega t + kx)$ (D) $A \cos(\omega t + kx + \pi)$
3. A string vibrates in 5 segments to a frequency of 480 Hz. The frequency that will cause it to vibrate in 2 segments will be
 (A) 96 Hz (B) 192 Hz (C) 1200 Hz (D) 2400 Hz
4. A wave travels on a light string. The equation of the wave is $Y = A \sin(kx - \omega t + 30^\circ)$. It is reflected from a heavy string tied to an end of the light string at $x = 0$. If 64% of the incident energy is reflected the equation of the reflected wave is
 (A) $Y = 0.8 A \sin(kx - \omega t + 30^\circ + 180^\circ)$
 (B) $Y = 0.8 A \sin(kx + \omega t + 30^\circ + 180^\circ)$
 (C) $Y = 0.8 A \sin(kx + \omega t - 30^\circ)$
 (D) $Y = 0.8 A \sin(kx + \omega t + 30^\circ)$
5. The speed of the sound in still air is C . Wind is blowing at a speed v_w as shown. The source of actual frequency f_0 is moving as shown in fig. The apparent frequency as observed by an observer is



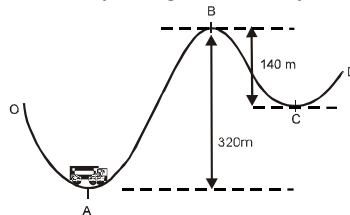
- (A) $f_0 \frac{(C - v_w - v_o)}{(C - v_w + v_s)}$ (B) $f_0 \frac{(C - v_w - v_o)}{(C - v_w - v_s)}$
 (C) $f_0 \frac{(C + v_w - v_o)}{(C + v_w - v_s)}$ (D) $f_0 \frac{(C + v_w + v_o)}{(C + v_w - v_s)}$

6. On a smooth curved track shown in the diagram, a small particle is projected from the point O such that it just manages to reach the point P. Assume that particle always remain contact with track. The velocity of the particle on reaching the point P will be equal to :



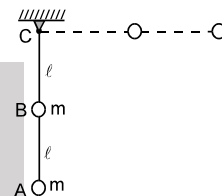
- (A) zero (B) $\sqrt{2gH}$ (C) $\sqrt{6gH}$ (D) $\sqrt{4gH}$

7. Track OABCD (as shown in figure) is smooth. What minimum speed has to be given to a particle lying at point A, so that it can reach point C ? (take $g = 10 \text{ m/s}^2$)



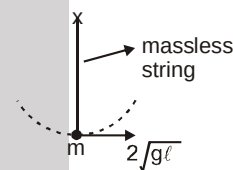
- (A) 60 m/s (B) 100 m/s (C) 70 m/s (D) 80 m/s

8. A weightless rod of length 2ℓ carries two equal masses 'm', one tied at lower end A and the other at the middle of the rod at B. The rod can rotate in vertical plane about a fixed horizontal axis passing through C. The rod is released from rest in horizontal position. The speed of the mass B at the instant rod, become vertical is :



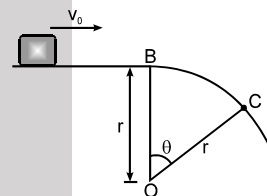
- (A) $\sqrt{\frac{3g\ell}{5}}$ (B) $\sqrt{\frac{4g\ell}{5}}$ (C) $\sqrt{\frac{6g\ell}{5}}$ (D) $\sqrt{\frac{7g\ell}{5}}$

9. A mass 'm' is tied by a massless string. It is given horizontal velocity $2\sqrt{g\ell}$. What will be velocity of the particle at its highest position .



- (A) Zero (B) $\frac{2\sqrt{2}}{3\sqrt{3}} \sqrt{\ell g}$
(C) $\sqrt{\ell g}$ (D) $\sqrt{\frac{\ell g}{2}}$

10. A small block slides with velocity $0.5\sqrt{gr}$ on the horizontal frictionless surface as shown in the Figure. The block leaves the surface at point C. The angle θ in the Figure is:

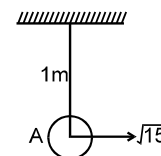


- (A) $\cos^{-1}(4/9)$ (B) $\cos^{-1}(3/4)$
(C) $\cos^{-1}(1/2)$ (D) none of the above

11. Amplitude of simple pendulum is 60° . Find the tension in string when string makes an angle of 30° with vertical.

- (A) $mg \frac{\sqrt{3}}{2}$ (B) $(3\sqrt{3} - 2) mg$ (C) $mg \left(\frac{3\sqrt{3}}{2} - 1 \right)$ (D) none of these

12. In the figure shown find out the maximum height reached by the particle. ($g = 10 \text{ m/s}^2$)



- (A) 0.5 m (B) 0.75 m
(C) 1 m (D) none of these

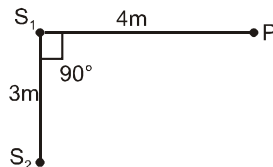
13. The sound intensity is 0.008 W/m^2 at a distance of 10 m from an isotropic point source of sound. The power of the source is :

- (A) 2.5 watt (B) 0.8 watt (C) 8 watt (D) 10 watt

14. A point source of power 50π watts is producing sound waves of frequency 1875Hz. The velocity of sound is 330m/s, atmospheric pressure is $1.0 \times 10^5 \text{ Nm}^{-2}$, density of air is 1.0 kgm^{-3} . Then pressure amplitude at $r = \sqrt{330} \text{ m}$ from the point source is (using $\pi = 22/7$) :

- (A) 5 Nm^{-2} . (B) 10 Nm^{-2} . (C) 15 Nm^{-2} . (D) 20 Nm^{-2} .

15. S_1 and S_2 are two coherent sources of sound of frequency 110Hz each. They have no initial phase difference. The intensity at a point P due to S_1 is I_0 and due to S_2 is $4I_0$. If the velocity of sound is 330 m/s then the resultant intensity at P is



- (A) I_0 (B) $9I_0$ (C) $3I_0$ (D) $8I_0$

16. State True or false

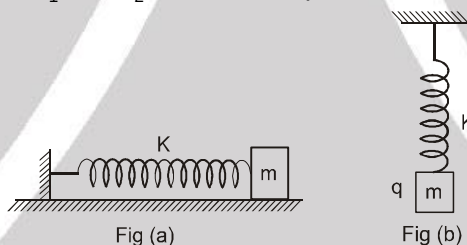
S_1 : If pressure of a gas is increased keeping the temperature constant the speed of sound wave in the gas increases.

S_2 : In a travelling sinusoidal sound wave the phase difference between displacement wave and pressure wave is $\frac{\pi}{2}$.

S_3 : The apparent frequency is v_1 , when a source approaches a stationary observer with speed u and is v_2 when the observer approaches the same stationary source with the same speed u . Then $v_2 < v_1$, if $u < v$ where v is the speed of sound.

- (A) T F T (B) F T T (C) F F T (D) F T F

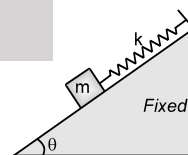
17. A particle of mass m is performing simple harmonic motion as shown in figure (a) and (b). In figure (b), the charge q of particle is such that $qE = mg$. If their velocities are same at mean position and let A_1 and A_2 be their amplitudes and T_1 and T_2 be their time periods then.



- (A) $A_1 = A_2$, $T_1 = T_2$ (B) $A_1 > A_2$, $T_1 = T_2$
(C) $A_1 < A_2$, $T_1 = T_2$ (D) $A_1 = A_2$, $T_1 > T_2$

18. In the figure shown, the time period and the amplitude respectively when m is released from rest when the spring is relaxed is: (the inclined plane is smooth)

- (A) $2\pi\sqrt{\frac{m}{k}}$, $\frac{mg \sin \theta}{k}$ (B) $2\pi\sqrt{\frac{m \sin \theta}{k}}$, $\frac{2mg \sin \theta}{k}$
(C) $2\pi\sqrt{\frac{m}{k}}$, $\frac{mg \cos \theta}{k}$ (D) none of these



19. A particle performs SHM of amplitude A and angular frequency $\omega = \frac{2\pi}{T}$. The speed of the particle at time $\frac{T}{4}$ after crossing the extreme position is

- (A) 0 (B) $\frac{\sqrt{3}}{2}\omega A$ (C) $\frac{\omega A}{2}$ (D) ωA

20. A loaded vertical spring executes SHM with period of 4s. The difference between the kinetic energy and the potential energy of this system oscillates with a period of

- (A) 8 s (B) 1 s (C) 2s (D) 4s



DPP No. : B60 (JEE-ADVANCED)

Total Marks : 50

Max. Time : 31 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.10

(4 marks 2 min.) [32, 16]

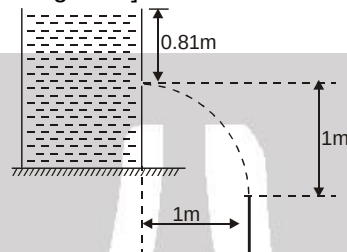
Subjective Questions ('-1' negative marking) Q.11

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.12

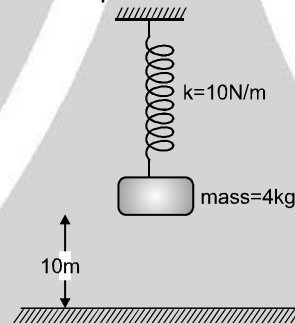
(8 marks 6 min.) [08, 06]

1. For the arrangement shown in the figure. The time interval in seconds after which the water jet ceases to cross the wall is (Area of the cross section of the tank $A = \sqrt{5} \text{ m}^2$ and area of the orifice $a = 4 \text{ cm}^2$) [Assume that the container remaining fixed]



- (A) 1000 (B) 2000 (C) 100 (D) 10000

2. Initially a small block is in equilibrium and at a height 10m from ground as shown. At $t = 0$, block explodes in two equal parts. One part collides with the ground at $t = 1$ sec and sticks to ground at the position just below the initial equilibrium position and another part remains held with spring,



- (A) then the amplitude of oscillations of the block held by spring is 3 meter
 (B) then the amplitude of oscillations of the block held by spring is 5 meter
 (C) then the amplitude of oscillations of the block held by spring is 2 meter
 (D) then the amplitude of oscillations of the block held by spring is 7 meter

3. Which of the following combinations can give standing wave.

- (A) $y_1 = A \sin^2(\omega t - kx)$; $y_2 = -A \sin^2(\omega t + kx)$
 (B) $y_1 = A \sin(kx - \omega t)$; $y_2 = A \cos(\omega t + kx)$
 (C) $y_1 = 2A \cos^2(\omega t - kx + \pi)$; $y_2 = A[\sin 2(\omega t + kx) - 1]$
 (D) $y_1 = A \sin(kx - \omega t + 30^\circ)$; $y_2 = A \cos(\omega t + kx - 60^\circ)$

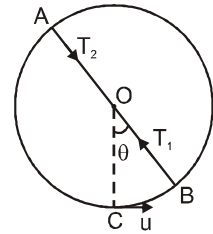
4. The equation of a practical travelling wave is /are

- (A) $A \tan(\omega t - kx)$ (B) $A \sin^2(\omega t - kx)$
 (C) $A \sin(\omega t - kx) \cos(\omega t - kx)$ (D) none

5. A particle of mass m performs SHM along a straight line with frequency f and amplitude A .

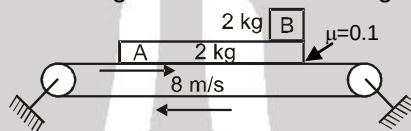
- (A) The average kinetic energy of the particle is zero.
 (B) The average potential energy is $m\pi^2 f^2 A^2$.
 (C) The frequency of oscillation of kinetic energy is $2f$.
 (D) Velocity function leads acceleration by $\pi/2$.

6. ✎ A heavy particle hanging vertically from a point by a light inextensible string of length ℓ is started so as to make a complete revolution in a vertical plane. Then :
- (A) Sum of tension at the ends of any diameter is constant
 (B) Sum of tension at the ends of any diameter is independent on angle θ with vertical.
 (C) Tension does no work
 (D) $T_1 + T_2 = \frac{m}{\ell} (2u^2 - 4g\ell)$

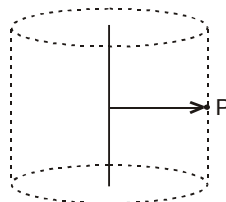


COMPREHENSION

A long horizontal conveyer belt moves with a constant velocity of 8 m/s. Two blocks A and B each of mass 2 kg are placed gently on the belt with B on A. coefficient of kinetic friction b/w A and belt is 0.1. There is no friction between A and B. Length of A is 4 m and length of B is very small ($g = 10 \text{ m/s}^2$).



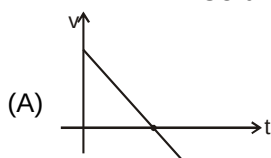
7. ✎ Choose the correct options :
- (A) The time when B falls off A. Initially B is on the right end of A is 2 second
 (B) The time when B falls off A. Initially B is on the right end of A is 4 second
 (C) The velocity of A when B falls off A is 4 m/s
 (D) The velocity of A when B falls off A is 6 m/s
8. ✎ Choose the correct options:
- (A) The coefficient of friction between the block B and belt is 0.4, then the separation between the two blocks when B comes to rest w.r.t. belt is 2 meter
 (B) The coefficient of friction between the block B and belt is 0.4, then the separation between the two blocks when B comes to rest w.r.t. belt is 1 meter
 (C) Just after the block B falls of A, then the acceleration of block A is 1 m/s^2 .
 (D) Just after the block B falls of A, then the acceleration of block A is 2 m/s^2 .
9. ✎ Just before block B falls of A. then
- (A) distance travelled by B w.r.t. ground 4m
 (B) distance travelled by B w.r.t. to ground is zero.
 (C) distance travelled by B w.r.t. to A is 4m.
 (D) distance travelled by B w.r.t A is zero
10. If the length of Block A is 20m. then
- (A) B falls off A after 4.5s
 (B) B falls off A after $2\sqrt{5}$ s.
 (C) when B falls off A, A has velocity 8 m/s
 (D) when B falls off A, then A has velocity $2\sqrt{5}$ m/s
11. ✎ A straight line source of sound of length $L = 10 \text{ m}$, emits a pulse of sound that travels radially outward from the source. What is the power (in mW) intercepted by an acoustic detector of surface area 1.2 cm^2 , located at a perpendicular distance 7m from the source. The waves reach perpendicularly at the surface of the detector. The total power emitted by the source in the form of sound is $2.2 \times 10^4 \text{ W}$. (Use $\pi = 22/7$)



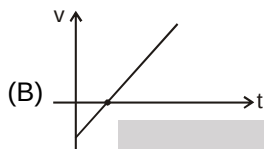
12. The velocity time graph for a particle moving along a straight line is given in each situation of column-I. In the time interval $\infty > t > 0$, match the graph in column-I with corresponding statements in column-II.

Column-I

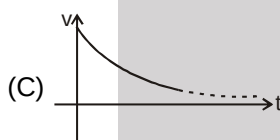
Column-II



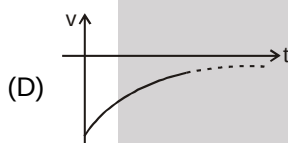
(p) speed of particle is continuously decreasing.



(q) magnitude of acceleration of particle is decreasing with time.



(r) direction of acceleration of particle does not change.



(s) magnitude of acceleration of particle does not change.

DPP No. : B61 (JEE-ADVANCED)

Total Marks : 50

Max. Time : 31 min.

Single choice Objective ('-1' negative marking) Q.1 to Q.2

(3 marks 2 min.) [06, 04]

One or more than one options correct type ('-1' negative marking) Q.3 to Q.10

(4 marks 2 min.) [32, 16]

Subjective Questions ('-1' negative marking) Q.11

(4 marks 5 min.) [04, 05]

Match the Following (no negative marking) Q.12

(8 marks 6 min.) [08, 06]

- A small ball of mass m is released from rest at a height h_1 above ground at time $t = 0$. At time $t = t_0$, the ball again comes to rest at a height h_2 above ground. Consider the ground to be perfectly rigid and neglect air friction. In the time interval from $t = 0$ to $t = t_0$, pick up the incorrect statement.

(A) Work done by gravity on ball is $mg(h_1 - h_2)$

(B) Work done by ground on ball for duration of contact is $mg(h_2 - h_1)$.

(C) Average acceleration of the ball is zero.

(D) Net work done on the ball by all forces except gravity is $mg(h_2 - h_1)$.
- A ball of mass 1.6 kg is projected with a velocity of 20 m/s at an angle of 37° above the horizontal. After 1.2 sec., gravitational field vanishes and a force of constant magnitude is applied after that, force being always perpendicular to the direction of motion till it strikes the ground. When it strikes the ground it is moving vertically. Choose the incorrect option ($g = 10 \text{ m/s}^2$) :

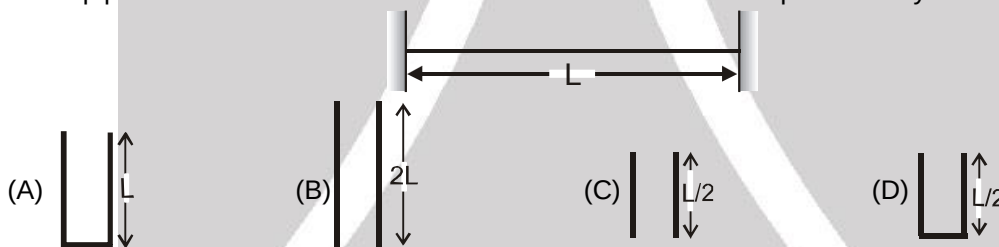
(A) Initially path is parabolic and later on it becomes hyperbolic

(B) The radius of the circle will be 7.2 m and constant magnitude of force applied is $\frac{512}{9} \text{ N}$

(C) The speed during circular motion will be 16 m/s

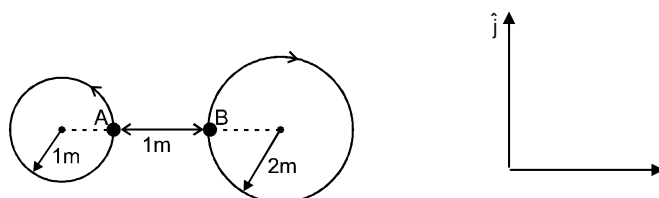
(D) The time it takes to strike the ground is less than that it would have taken in projectile motion

3. In a standing wave on a string.
 (A) In one time period all the particles are simultaneously at rest twice.
 (B) All the particles must be at their positive extremes simultaneously once in one time period
 (C) All the particles may be at their positive extremes simultaneously once in a time period.
 (D) All the particles are never at rest simultaneously.
4. A simple pendulum of length 2m with a bob of mass M oscillates with an angular amplitude of $\frac{\pi}{6}$ radians then (use $\sqrt{g} = \pi$) :
 (A) tension in the string is $mg \cos 15^\circ$ at angular displacement of 15°
 (B) rate of change of speed at angular displacement of 15° is $g \sin 15^\circ$
 (C) tension in the string is greater than $mg \cos 15^\circ$ at angular displacement of 15°
 (D) frequency of oscillation is $\frac{1}{2\sqrt{2}} \text{ sec}^{-1}$.
5. At the closed end of an organ pipe :
 (A) the displacement amplitude is zero
 (B) the displacement amplitude is maximum
 (C) the pressure amplitude is zero
 (D) the pressure amplitude is maximum
6. Figure shows a stretched string of length L and pipes of length L, 2L, L/2 and L/2 in options (A), (B), (C) and (D) respectively. The string's tension is adjusted such that the speed of waves on the string equals the speed of sound waves in the air. The fundamental mode of oscillation is then set up on the string. In which pipe will the air column be in the resonance with the sound produced by the string ?

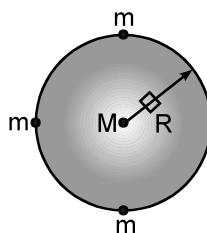


COMPREHENSION

Two particles are moving in different circles in same plane with different angular velocities as shown in figure. At $t = 0$, initial positions of particles A and B are shown by dots on the respective circles. Initial distance between particles is 1m. Particle A move anticlockwise in the first circle whereas B moves clockwise in the second circle. Angle described (rotated) by A and B in time 't' are $\theta_A = \left(\frac{\pi}{2}t\right)$ and $\theta_B = (\pi t)$ respectively. Here θ is in radian and t is in second. Radius of each circle is shown in diagram.



7. At time $t = 1$ sec, choose the correct options:
- (A) The magnitude of acceleration of A with respect to B is $\frac{\pi^2}{4} \sqrt{65} \frac{\text{m}}{\text{sec}^2}$
- (B) The magnitude of acceleration of A with respect to B is $\frac{\pi^2}{4} \sqrt{7} \frac{\text{m}}{\text{sec}^2}$
- (C) The magnitude of velocity of A with respect to B is $\frac{\pi}{2} \sqrt{17} \text{ m/s}$
- (D) The magnitude of velocity of A with respect to B is $\frac{\pi}{3} \sqrt{17} \text{ m/s}$
8. Choose the correct statements :
- (A) At time $t = 2$ second, the angular velocity of the particle A with respect to the particle B is $\frac{5\pi}{6} \text{ rad/sec}$
- (B) At time $t = 2$ second, the angular velocity of the particle A with respect to the particle B is $\frac{2\pi}{3} \text{ rad/sec}$
- (C) At $t = 4$ second particles retained their initial configuration of position and velocity simultaneously
- (D) At $t = 6$ second particles retained their initial configuration of position and velocity simultaneously
9. At $t = 3\text{s}$, which of the following is/are correct ?
- (A) velocity of A w.r.t. B is $\frac{\pi}{2} \hat{i} + 2\pi \hat{j}$
- (B) Velocity of A w.r.t. B is $\frac{\pi}{2} \hat{i} - 2\pi \hat{j}$
- (C) Distance between A & B is $\sqrt{37} \text{ m}$
- (D) Angular velocity of B w.r.t. A is $\frac{23\pi}{74}$
10. If the velocity of both the particles are equal in magnitude through out the motion. The path of 'A' & 'B' are circular as given. Which of the following is/are correct ?
- (A) They are having equal kinetic energy.
- (B) Distance covered by both the particles are same at any instant of time.
- (C) Angular velocity of 'B' is twice of 'A'
- (D) When particle 'B' reaches its initial position, displacement of both the particles are equal at that instant of time.
11. A solid cylinder of mass $M = 1 \text{ kg}$ & radius $R = 0.5 \text{ m}$ is pivoted at its centre & has three particles of mass $m = 0.1 \text{ kg}$ mounted at its perimeter as shown in the figure. The system is originally at rest. then the angular speed of the cylinder, when it has swung through 90° in anticlockwise direction is $\sqrt{n} \text{ r/sec}$. then find the value of n .



12. Consider incompressible and non-viscous liquid in a container. Density of liquid is ρ and acceleration due to gravity is g and h represents the vertical separation between two points. All points considered in column I are inside the liquid. Match the statements given in column-I with corresponding all possible conditions given in column-II.

Column-I

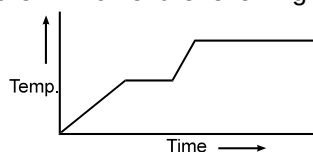
- (A) Pressure difference between two distinct points is ρgh where two points are at same vertical line at a separation h .
 (B) Pressure difference between any pair of two distinct points on same horizontal level is zero.
 (C) Pressure difference between any pair of two distinct points on same horizontal level is non-zero.
 (D) Pressure difference between any two distinct direction with points on same vertical line is zero.

Column-II

- (p) container is stationary.
 (q) container is accelerating in horizontal direction
 (r) container is falling freely.
 (s) container is accelerating up in vertical an acceleration $< g$.
 (t) container is rotating about a vertical axis passing through its symmetry.

DPP No. : B62 (JEE-ADVANCED)**Total Marks : 42****Max. Time : 27 min.****Single choice Objective ('-1' negative marking) Q.1 to Q.2****(3 marks 2 min.) [06, 04]****One or more than one options correct type ('-1' negative marking) Q.3 to Q.8****(4 marks 2 min.) [24, 12]****Subjective Questions ('-1' negative marking) Q.9****(4 marks 5 min.) [04, 05]****Match the Following (no negative marking) Q.10****(8 marks 6 min.) [08, 06]**

1. A nail is located at a certain distance vertically below the point of suspension of a simple pendulum. The pendulum bob is released from the position where the string makes an angle of 90° from the vertical. If the distance (in metre) of the nail from the point of suspension is r such that the bob will just perform revolution with the nail as centre. Assume the length of pendulum to be $\ell = 10$ m. Then r is ($g = 10 \text{ m/s}^2$)
 (A) 2m (B) 4m (C) 6m (D) None of these
2. River is flowing with constant speed $v_0 = 4 \text{ m/s}$. At the centre of river's width we have a fixed pole. A swimmer swim around the pole in a circular path of radius 2m in such a way that it's position vector from pole rotates at a constant rate $\frac{5}{2} \text{ rad/sec}$. Then the ratio of maximum to minimum speed of swimmer in still river is
 (A) 9 (B) 18 (C) 15 (D) None of these
3. A vessel is partly filled with a liquid. Co-efficients of volumetric expansion of material of the vessel and liquid are γ_V & γ_L respectively. If the system is heated then volume unoccupied by the liquid will necessarily :
 (A) increase if $\gamma_V > \gamma_L$ (B) decrease if $\gamma_V < \gamma_L$
 (C) remain unchanged if $\gamma_V = \gamma_L$ (D) increase if $\gamma_V = \gamma_L$
4. Heat is supplied to a certain homogeneous sample of matter at a uniform rate. Its temperature is plotted against time as shown in the figure. Which of the following conclusions can be drawn?



- (A) its specific heat capacity is greater in the solid state than in the liquid state.
 (B) its specific heat capacity is greater in the liquid state than in the solid state.
 (C) its latent heat of vaporization is greater than its latent heat of fusion.
 (D) its latent heat of vaporization is smaller than its latent heat of fusion.

5. A monatomic & a diatomic gas both at the N.T.P. having same no. of moles are compressed adiabatically to half of initial volume :
 (A) Increase in temp. will be more for the diatomic gas
 (B) Increase in temp. will be more for the monatomic gas
 (C) Increase in temp. will be same for both the gases.
 (D) Increase in pressure will be more for the monatomic gas.
6. The amplitude of a particle executing SHM about O is 10 cm. Then:
 (A) when the K.E. is 0.64 of its maximum K.E. its displacement is 6 cm from O.
 (B) when the displacement is 5 cm from O its K.E. is 0.75 times its maximum K.E.
 (C) Its total energy of SHM at any point is equal to its maximum K.E.
 (D) Its speed is half the maximum speed when its displacement is half the maximum displacement.

COMPREHENSION

The second overtone of an open pipe A and a closed pipe B have the same frequencies at a given temperature. Both pipes contain air.

7. Choose the correct options :
 (A) The ratio of fundamental frequency of A to the fundamental frequency of B is 5 : 3
 (B) The ratio of fundamental frequency of A to the fundamental frequency of B is 2 : 3
 (C) The ratio of the length of open pipe to closed pipe is 6 : 5
 (D) The ratio of the length of open pipe to closed pipe is 3 : 4
8. Choose the correct statements :
 (A) The ratio of the first overtone of A to first overtone of B is 10 : 9
 (B) The ratio of the first overtone of A to first overtone of B is 5 : 6
 (C) Ratio of 3rd harmonic of A to 5th harmonic of B is 1 : 1
 (D) Ratio of 3rd harmonic of A to 5th harmonic of B is 1 : 2
9. 4th harmonic of an organ pipe has frequency 33 Hz lesser than 5th harmonic of same organ pipe. Find difference of wavelength (in cm) in the given two cases (Assume velocity of sound in air = 330 m/s). Neglect end correction.
10. **Match the column :**
 In a spring block system on a horizontal smooth surface. K = spring constant, A = amplitude, m = mass of the block. Match the entries of column I with the entries of column II

Column I

- (A) If mass of the block is doubled (keeping K, A unchanged)
 (B) If the amplitude of oscillation is doubled (keeping K, m unchanged)
 (C) If force constant is doubled (keeping m, A unchanged)
 (D) If another spring of same force constant is attached parallel to the previous one (keeping m, A unchanged)

Column II

- (p) time period increases
 (q) time period decreases
 (r) energy of oscillation increases
 (s) energy of oscillation decreases
 (t) energy of oscillation remains constant

DPP No. : B63 (JEE-MAIN)

Total Marks : 60

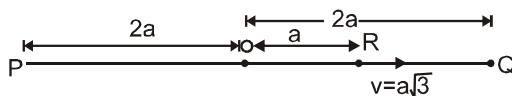
Single choice Objective ('-1' negative marking) Q.1 to Q.20

Max. Time : 40 min.

(3 marks 2 min.)

[60, 40]

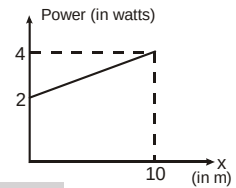
1. A point source is emitting sound in all directions. The ratio of distance of two points from the point source where the difference in loudness levels is 3 dB is: ($\log_{10} 2 = 0.3$)
- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$
2. If the frequency of a wave is increased by 25 %, then the change in its wavelength will be: (medium not changed)
- (A) 20 % increase (B) 20 % decrease (C) 25 % increase (D) 25 % decrease
3. A particle performs S.H.M. of amplitude A along a straight line. When it is at a distance $\frac{\sqrt{3}}{2} A$ from mean position, its kinetic energy gets increased by an amount $\frac{1}{2} m \omega^2 A^2$ due to an impulsive force. Then its new amplitude becomes:
- (A) $\frac{\sqrt{5}}{2} A$ (B) $\frac{\sqrt{3}}{2} A$ (C) $\sqrt{2} A$ (D) $\sqrt{5} A$
4. A particle of mass 10 gm is placed in a potential field given by $V = (50x^2 + 100)$ J/kg. The frequency of oscillation in cycle/sec is :
- (A) $\frac{10}{\pi}$ (B) $\frac{5}{\pi}$ (C) $\frac{100}{\pi}$ (D) $\frac{50}{\pi}$
5. A particle performs S.H.M. of amplitude A along a straight line. At two positions of the particle, the ratio of speeds is 1/2 and that of potential energy is 2/1 also. Then one position of them (distance from mean position) nearest to the extreme is :
- (A) $\frac{A}{\sqrt{3}}$ (B) $\frac{\sqrt{3}}{7} A$ (C) $\frac{\sqrt{6}}{7} A$ (D) $\frac{\sqrt{3}}{2} A$
6. Which of the following is correct about a SHM, along a straight line?
- (A) Ratio of acceleration to velocity is constant.
 (B) Ratio of acceleration to potential energy is constant.
 (C) Ratio of acceleration to displacement from the mean position is constant.
 (D) Ratio of acceleration to kinetic energy is constant.
7. A particle of mass m is performing SHM along line PQ with amplitude 2a with mean position at O. At $t = 0$ particle is at point R (OR = a) and is moving towards Q with speed $v = a\sqrt{3}$ m/sec. The equation can be expressed by :



- (A) $x = a (\sqrt{3} \sin t + \cos t)$ (B) $x = 2a (\sqrt{3} \sin t + \cos t)$
 (C) $x = 2a (\sin t + \sqrt{3} \cos t)$ (D) $x = a (\sin t + \sqrt{3} \cos t)$

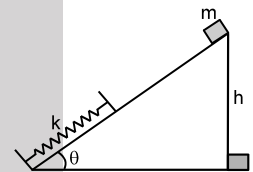
8. At $t = 0$, a particle of mass m starts moving from rest due to a force. $\vec{F} = F_0 \sin(\omega t) \hat{i}$
- (A) Particle performs SHM about its initial position of rest.
 (B) Particle performs SHM with initial position as an extreme position with angular frequency ω .
 (C) At any instant, distance moved by the particle equals its displacement from the initial position.
 (D) Initially velocity of particle increases with time but after time $t = 2\pi/\omega$ it becomes constant.

9. A particle A of mass $\frac{10}{7}$ kg is moving in the positive direction of x . Its initial position is $x = 0$ & initial velocity is 1 m/s. The velocity at $x = 10$ is: (use the graph given)



- (A) 4 m/s
 (B) 2 m/s
 (C) $3\sqrt{2}$ m/s
 (D) $100/3$ m/s

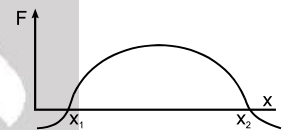
10. A body of mass m released from a height h on a smooth fixed inclined plane that is shown in the figure. The following can be true about the velocity of the block.



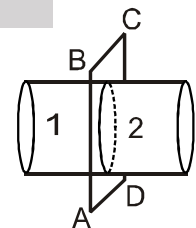
- (A) v is highest when it just touches the spring
 (B) v is highest when it compresses the spring by some amount
 (C) v is highest when the spring comes back to natural position
 (D) none of these
11. A particle is moving in a conservative force field from point A to point B. U_A and U_B are the potential energies of the particle at points A and B and W_C is the work done by conservative force in process of taking the particle from A to B :
- (A) $W_C = U_B - U_A$ (B) $W_C = U_A - U_B$ (C) $U_B \geq U_A$ (D) $U_B > U_A$

12. The potential energy of a particle varies with x according to the relation $U(x) = x^2 - 4x$. The point $x = 2$ is a point of :
- (A) stable equilibrium (B) unstable equilibrium
 (C) neutral equilibrium (D) none of above

13. The force acting on a body moving along x axis varies with the position of the particle as shown in the fig. The body is in stable equilibrium at
- (A) $x = x_1$ (B) $x = x_2$
 (C) both x_1 and x_2 (D) neither x_1 nor x_2

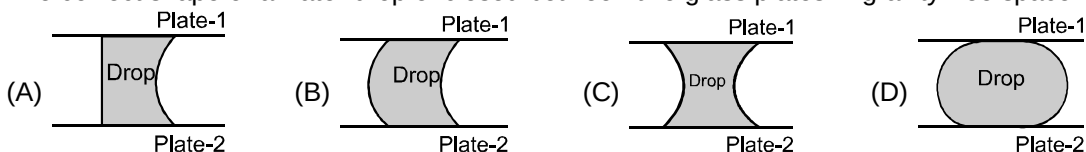


14. A waterjet of radius R is shown in the figure. The force between the parts 1 and part 2 at the section ABCD due to the surface tension is : (Assume that, T is surface tension)

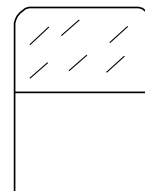


- (A) $2\pi RT$
 (B) $\pi R^2 T$
 (C) $2\pi RT + \pi R^2 T$
 (D) zero

15. The correct shape of a water drop enclosed between two glass plates in gravity free space will be -



16. Consider a U-shaped frame with a sliding wire of length ℓ and mass 'm' on its arm. It is dipped in a soap solution, taken out and placed in vertical position as shown in figure. Choose minimum value of m so that wire does not descend : (Surface tension of soap solution is S)



(A) $\frac{2S\ell}{g}$ (B) $\frac{S\ell}{g}$ (C) $\frac{S\ell}{2g}$ (D) $\frac{S\ell}{4g}$

17. A spherical drop of water has 1mm radius. If the surface tension of the water is 50×10^{-3} N/m, then the difference of pressure between inside and outside the spherical drop is :

(A) 25 N/m² (B) 10000 N/m² (C) 100 N/m² (D) 50 N/m²

18. A soap bubble of water has 1mm radius. If the surface tension of the soap is 50×10^{-3} N/m, then the difference of pressure between inside and outside the spherical bubble is :

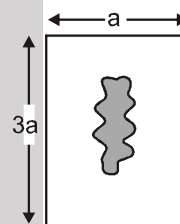
(A) 25 N/m² (B) 100 N/m² (C) 200 N/m² (D) 50 N/m²

19. If a capillary tube is dipped in water in a state of weightlessness, the water will

- (A) not rise
(B) rise
(C) rise to such a height that the excess of pressure balances the weight of water
(D) rise to the full length to the tube irrespective of its length

20. A rectangular blade of negligible weight floats on a water surface of surface tension $\frac{1}{7} \times 10^{-1}$ N/m. Its internal and external perimeter is 6a and 8a respectively then minimum force to lift up blade from water surface is $x \times 10^{-2}$ newton then x is (Take a = 10 cm)

- (A) 2 (B) 4
(C) 3 (D) None of these



ANSWERS

9. 1 DPP No. : B3

9. 0 DPP No. : B4

8. 16 N DPP No. : B6

11. 4 DPP No. : B7

10. 25 DPP No. : B9

9. 8 DPP No. : B10

9. 2 DPP No. : B12

9. 9 DPP No. : B13

9. 0 DPP No. : B15

9. 2 DPP No. : B17

9. 2 DPP No. : B18

9. 2 DPP No. : B20

9. 24 m/sec DPP No. : B22

8. 35 9. 4 DPP No. : B24

9. 10 DPP No. : B26

11. 3 DPP No. : B28

10. 3 DPP No. : B30

9. 17 DPP No. : B31

10. 5 DPP No. : B33

9. 20 DPP No. : B35

8. $n = 2$ DPP No. : B37

10. 3 DPP No. : B38

8. 3 DPP NO. : B40

9. 25 DPP NO. : B41

8. 24 N DPP NO. : B43

10. 29 DPP NO. : B45

8. 7 9. 17 DPP NO. : B46

8. 3 DPP NO. : B48

8. 3 DPP NO. : B50

7. 4 8. 9 DPP NO. : B51

9. 10 DPP NO. : B53

9. 5 DPP NO. : B55

9. 5 DPP NO. : B56

11. 6 DPP NO. : B60

11. 5 DPP NO. : B61

9. 50 DPP NO. : B62

